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(54) **WEARABLE PUMP ASSEMBLY**

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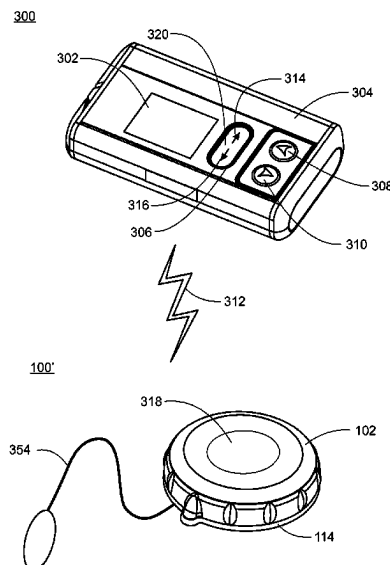
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(57) **ABSTRACT**

A wearable infusion pump assembly includes a reusable
housing assembly, and a disposable housing assembly
including a reservoir for receiving an infusible fluid. A
releasable engagement assembly is configured to allow the
reusable housing assembly to releasably engage the dispos-
able housing assembly. A detachable external infusion set is
configured to deliver the infusible fluid to a user.

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100

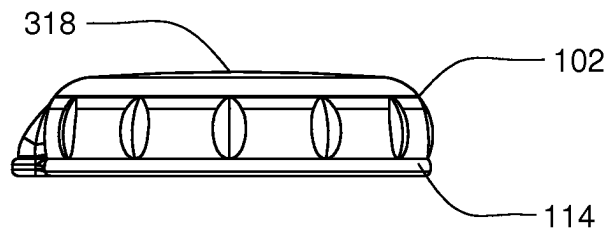


FIG. 1

100

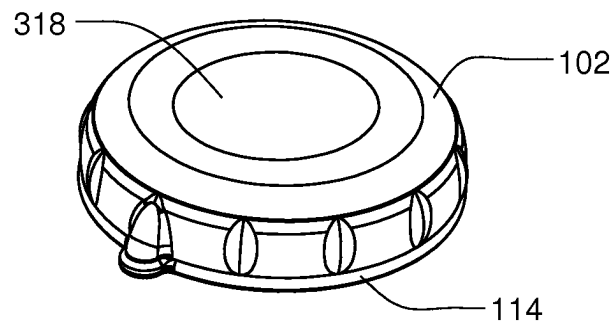


FIG. 2

100

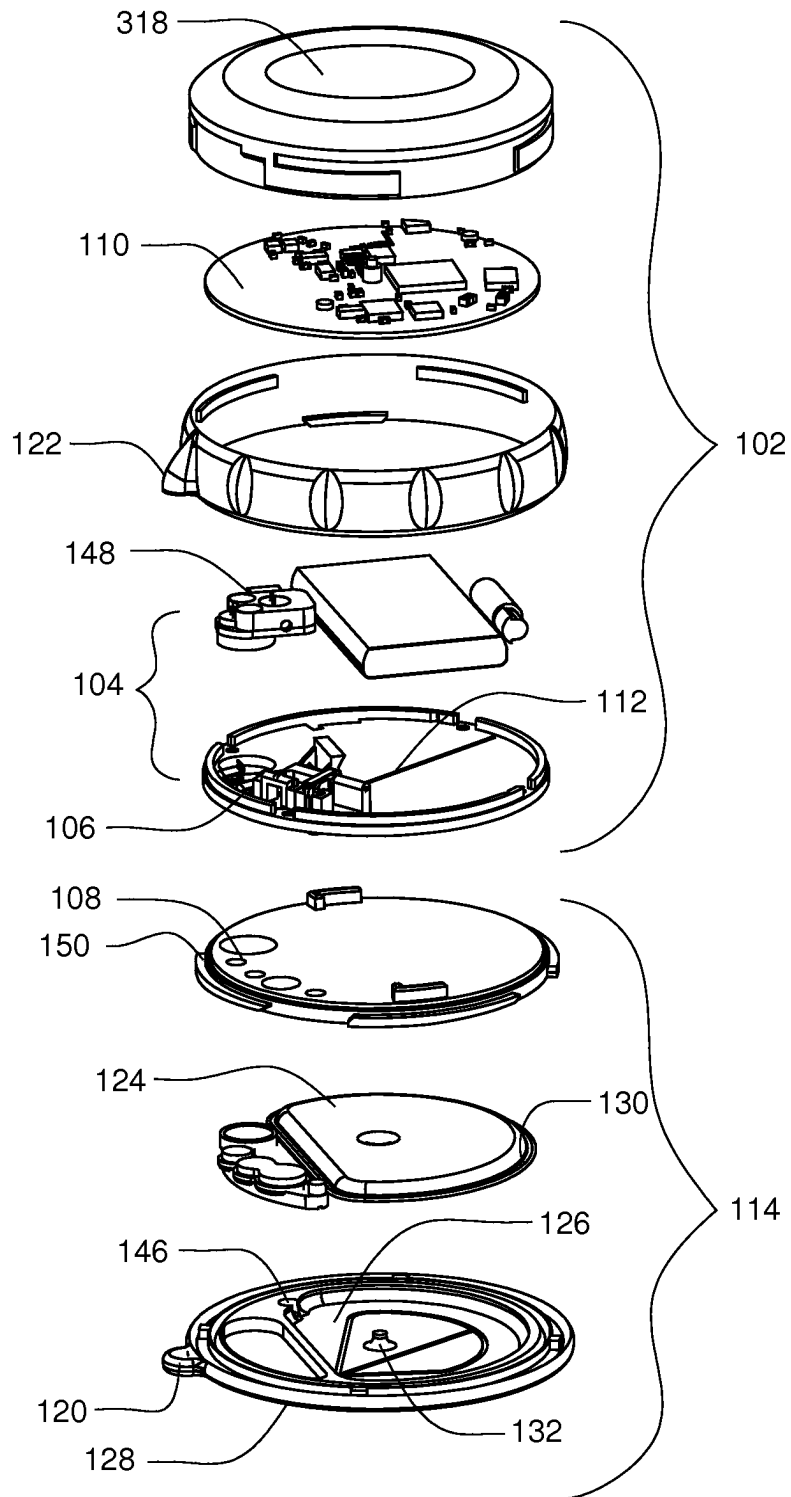


FIG. 3

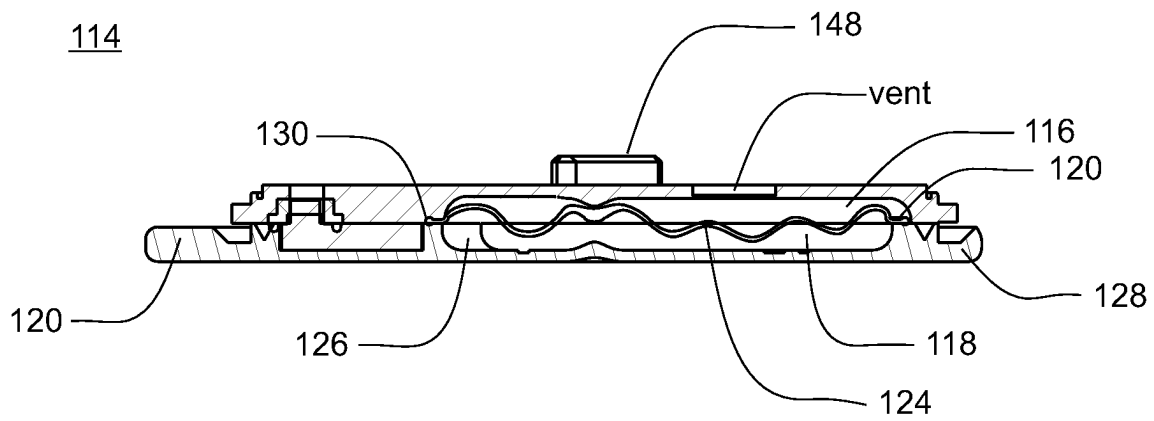
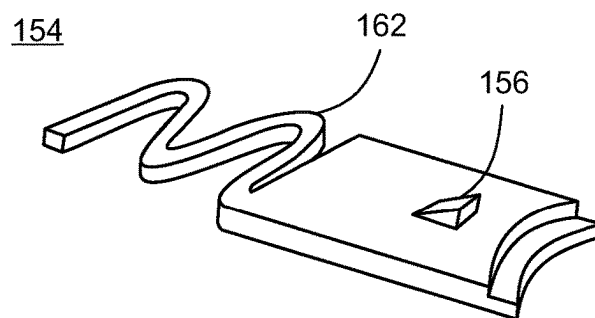
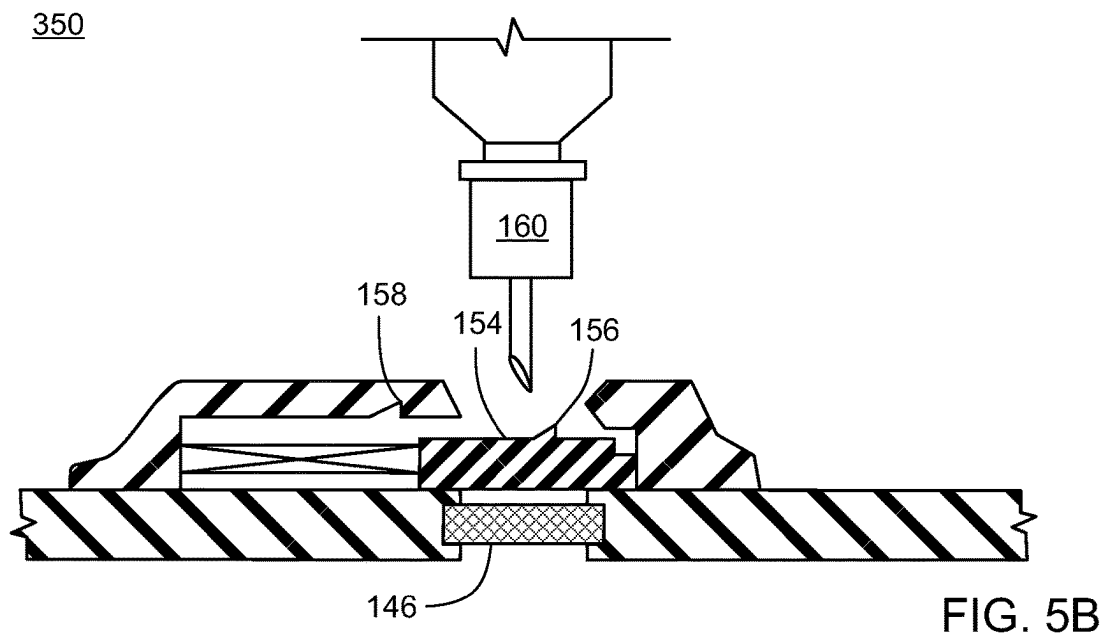
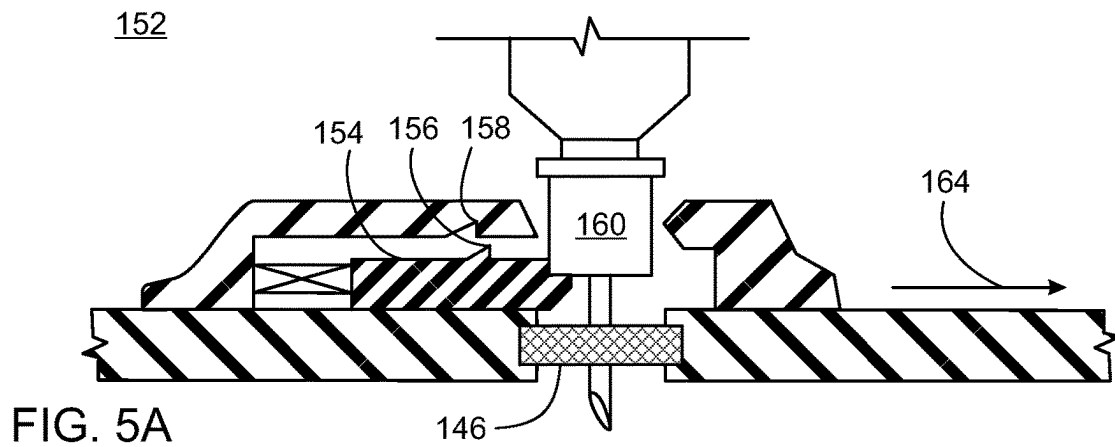


FIG. 4



166

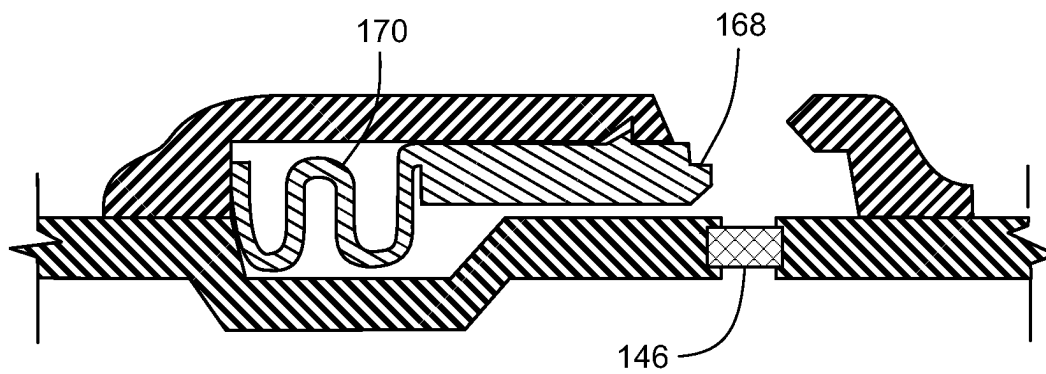


FIG. 6A

172

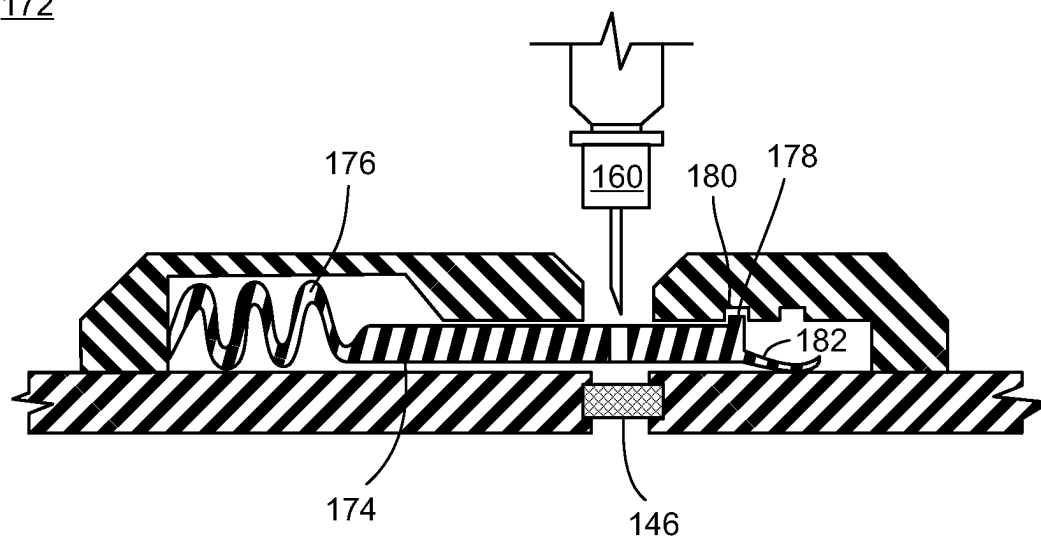


FIG. 6B

184

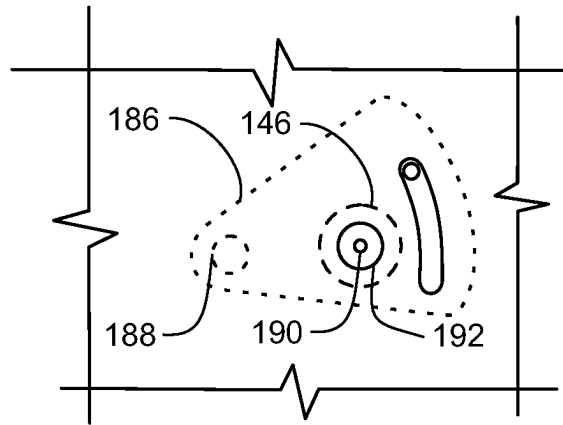


FIG. 7A

184

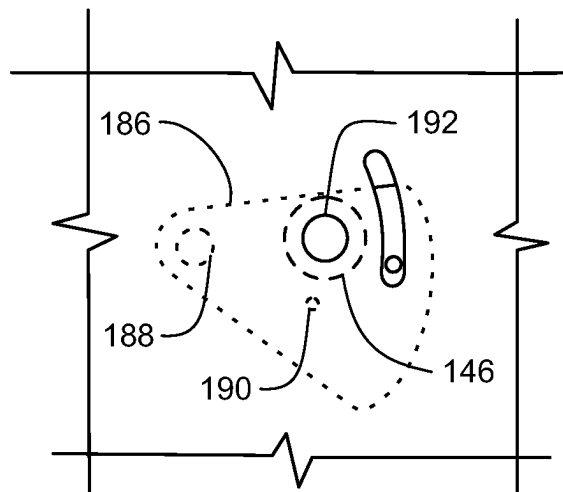


FIG. 7B

194

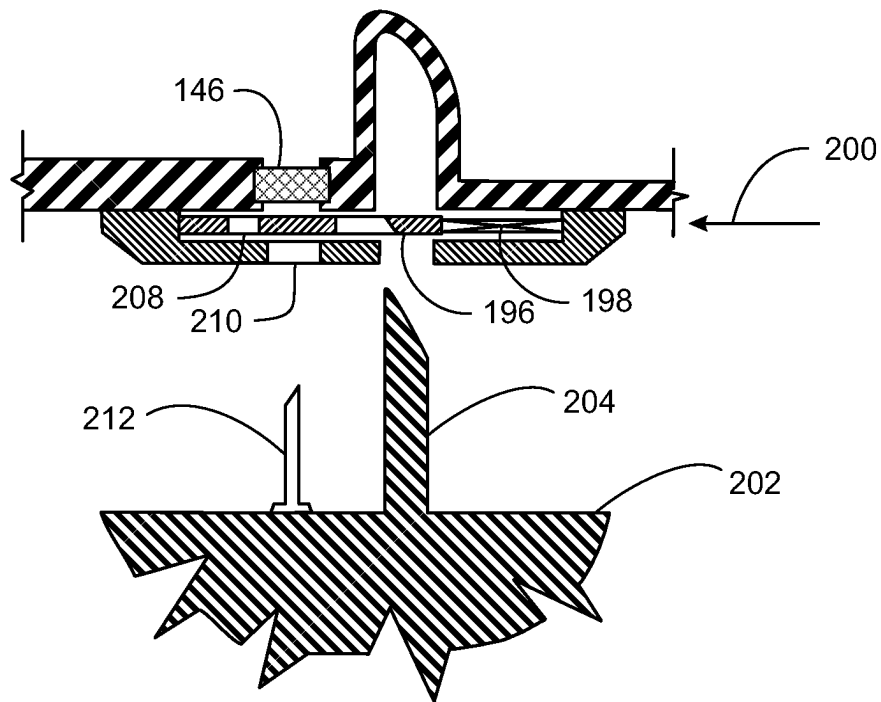


FIG. 8A

194

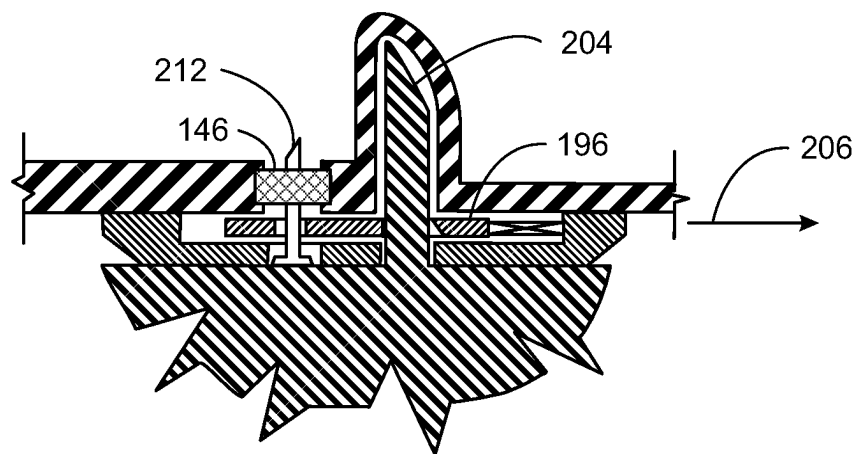


FIG. 8B

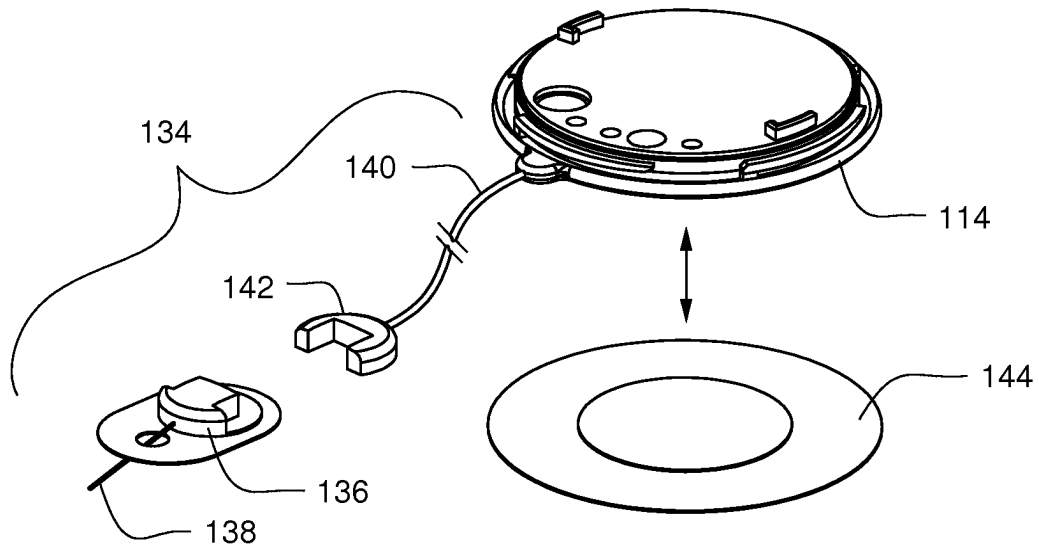


FIG. 9

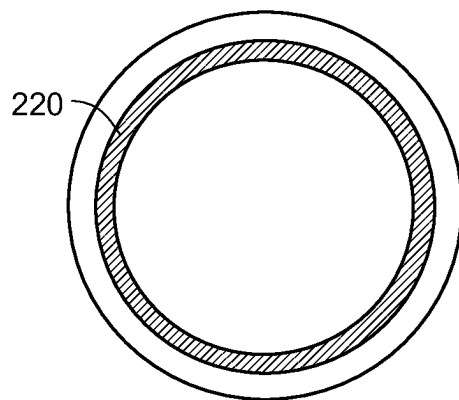


FIG. 10A

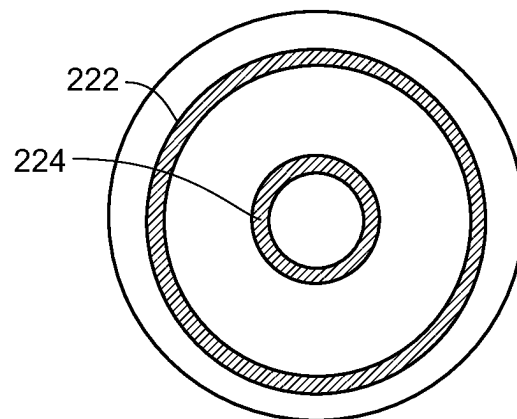


FIG. 10B

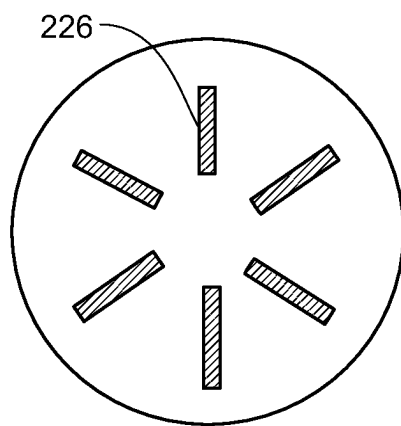


FIG. 10C

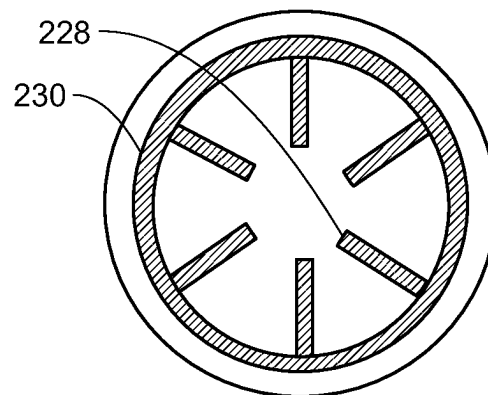


FIG. 10D

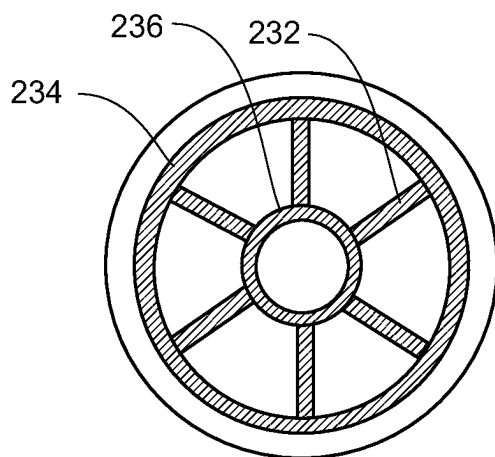


FIG. 10E

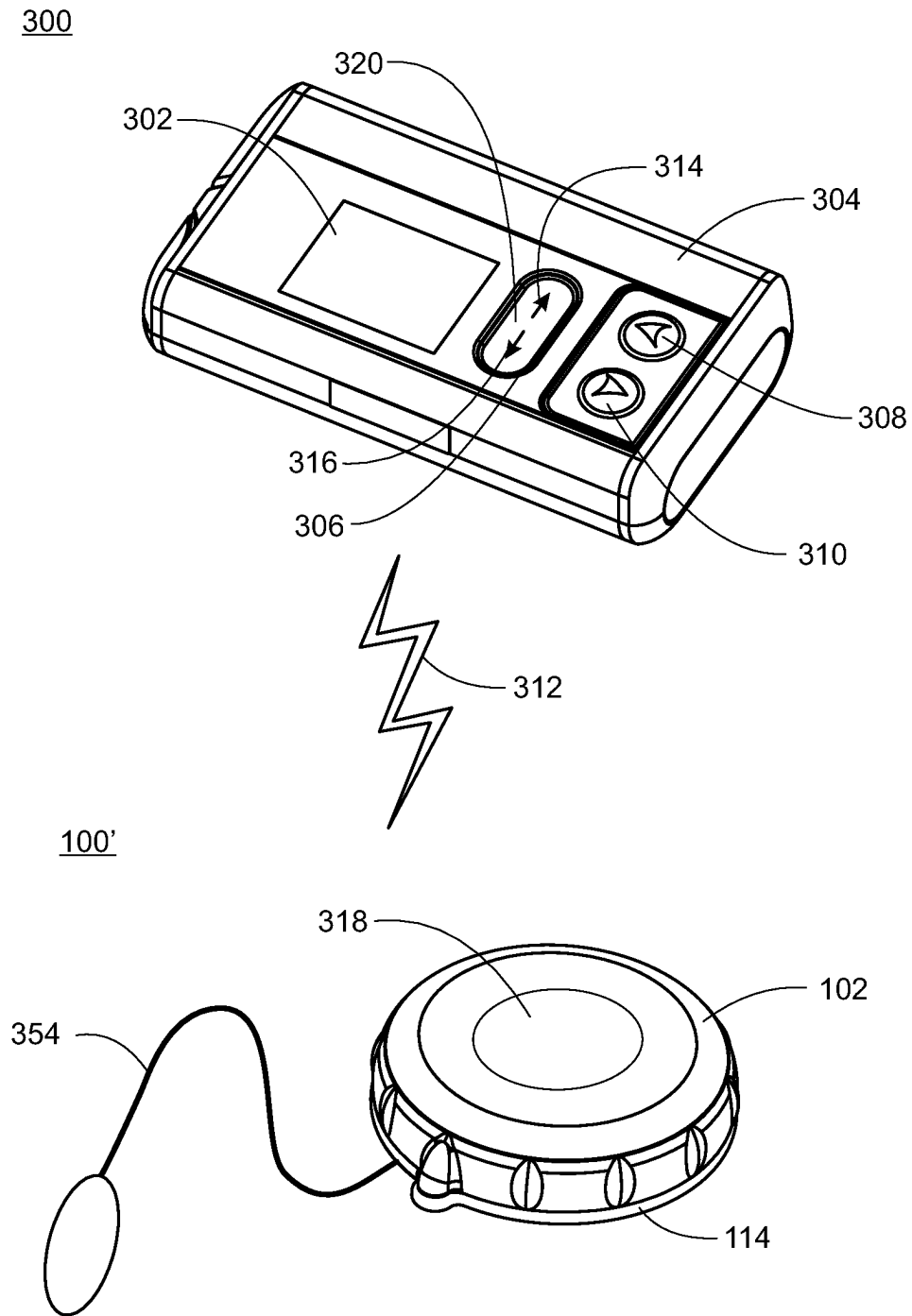


FIG. 11A

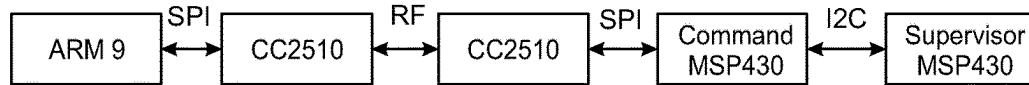


FIG. 11B

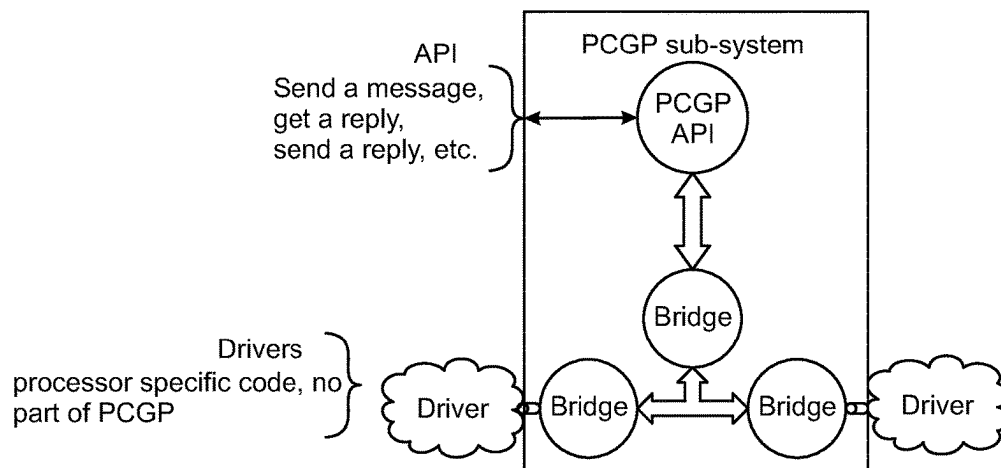


FIG. 11C

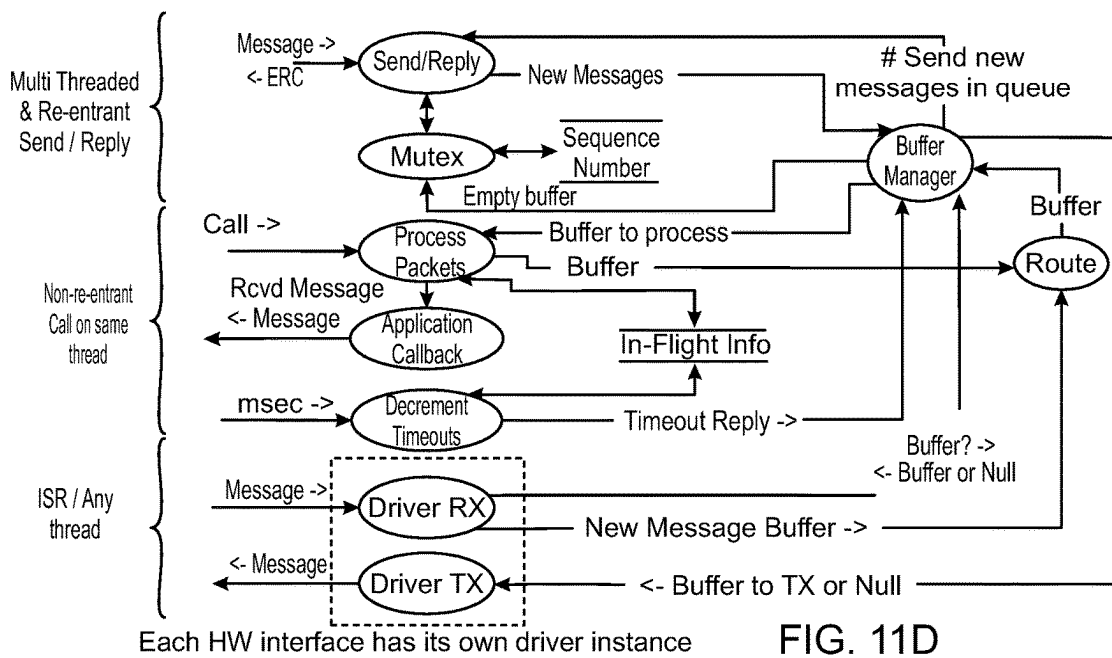


FIG. 11D

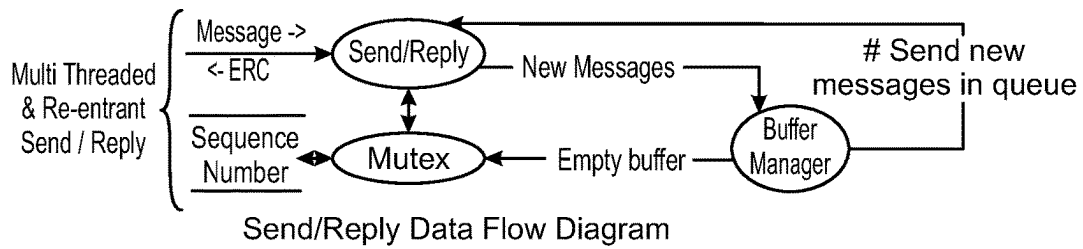


FIG. 11E

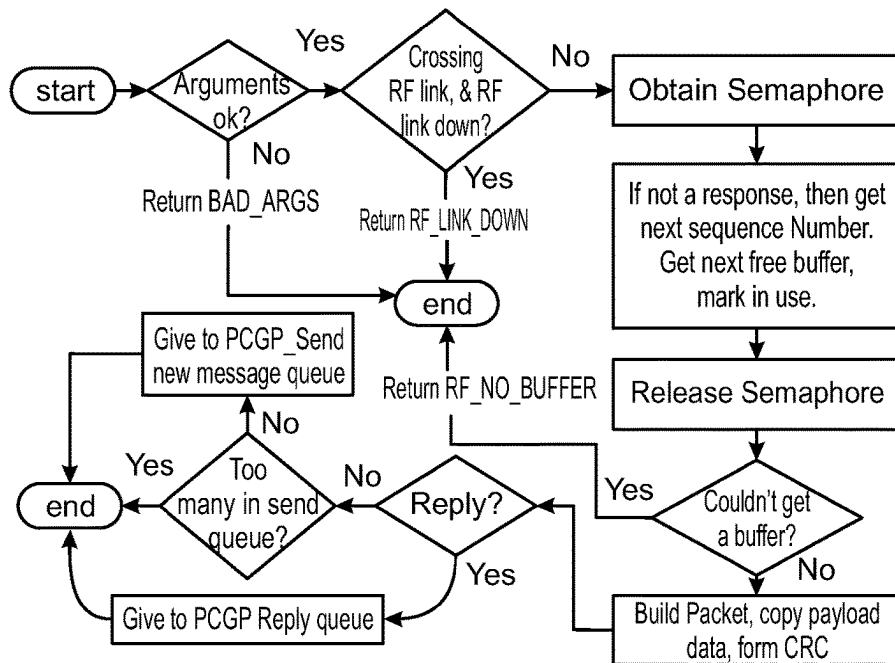


FIG. 11F

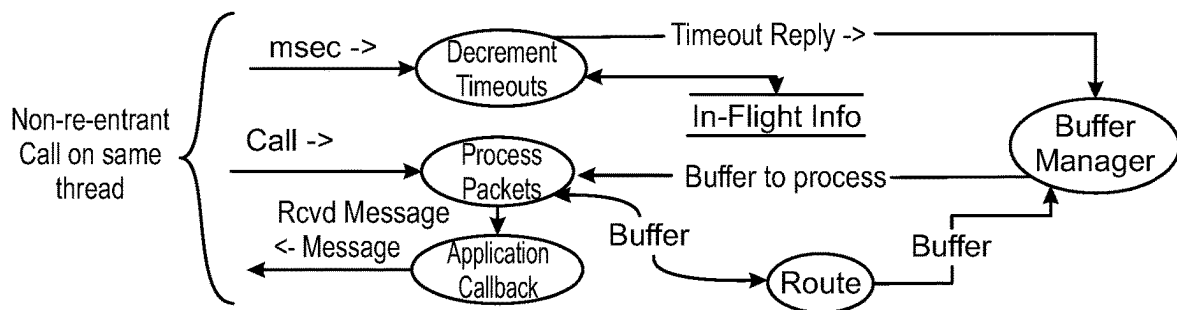


FIG. 11G

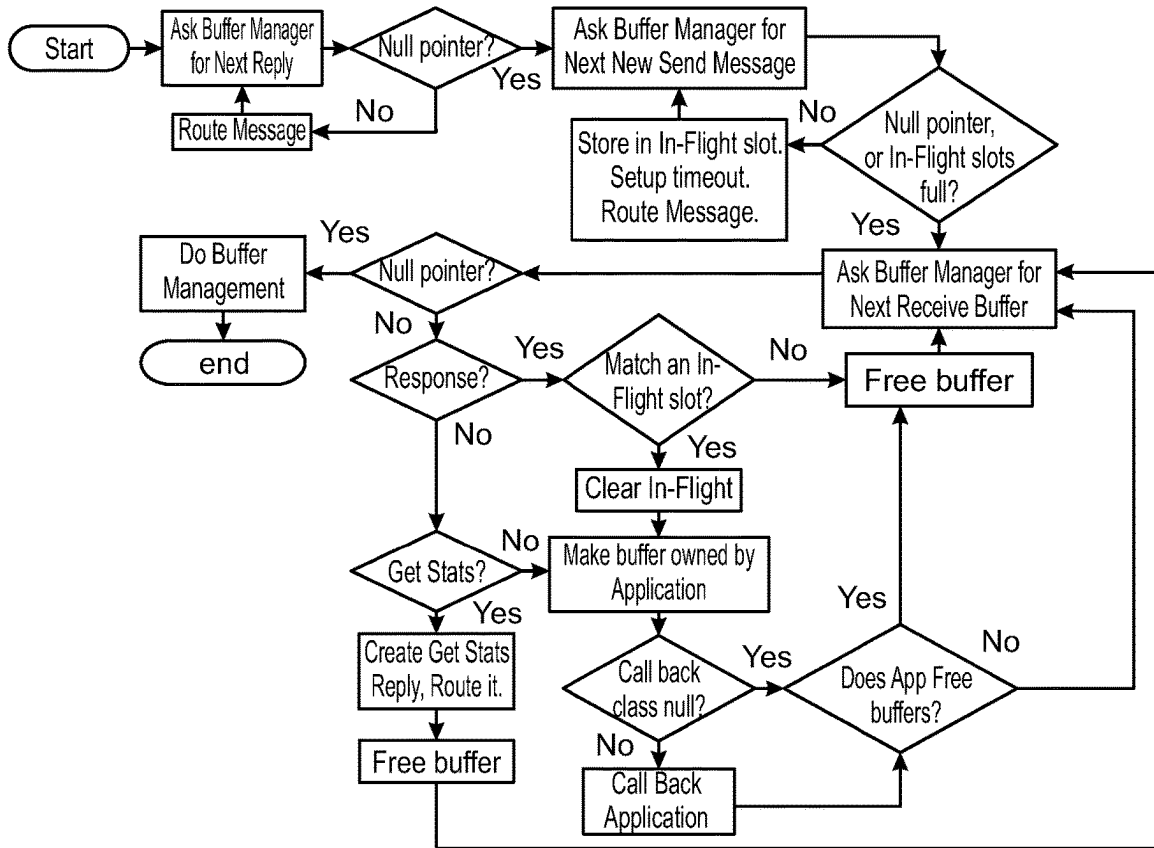


FIG. 11H

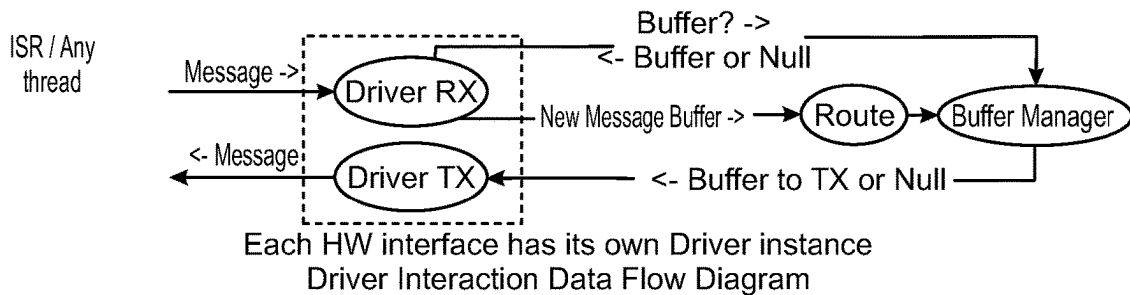


FIG. 11I

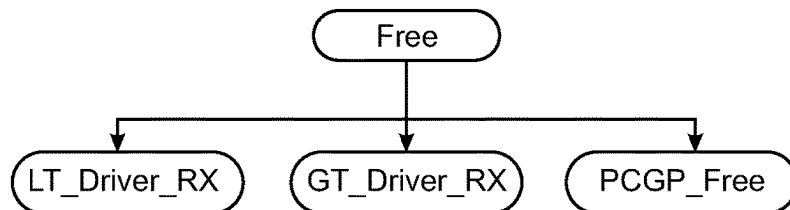


FIG. 11J

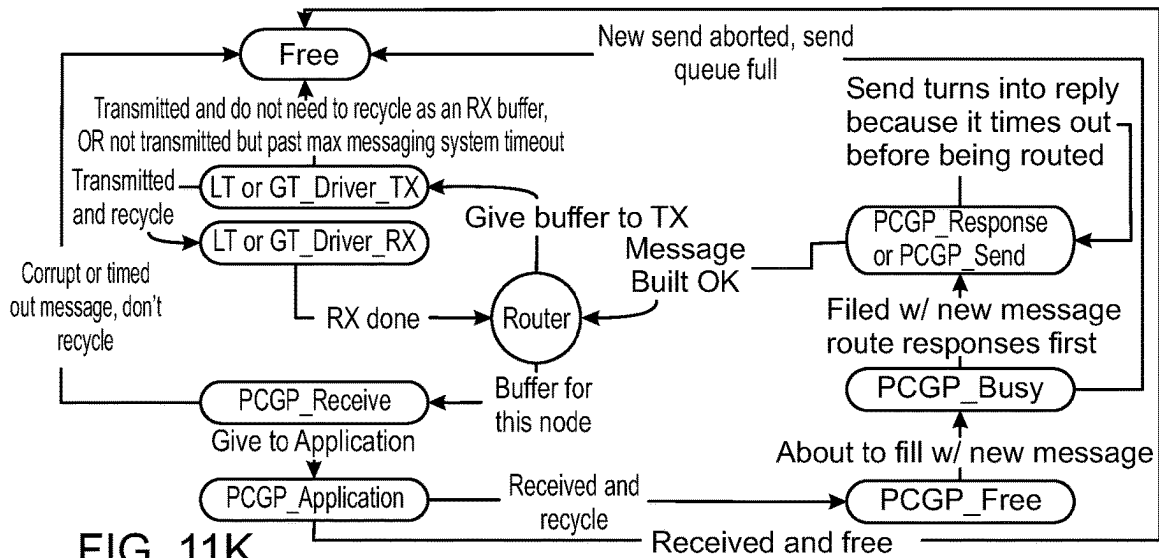


FIG. 11K

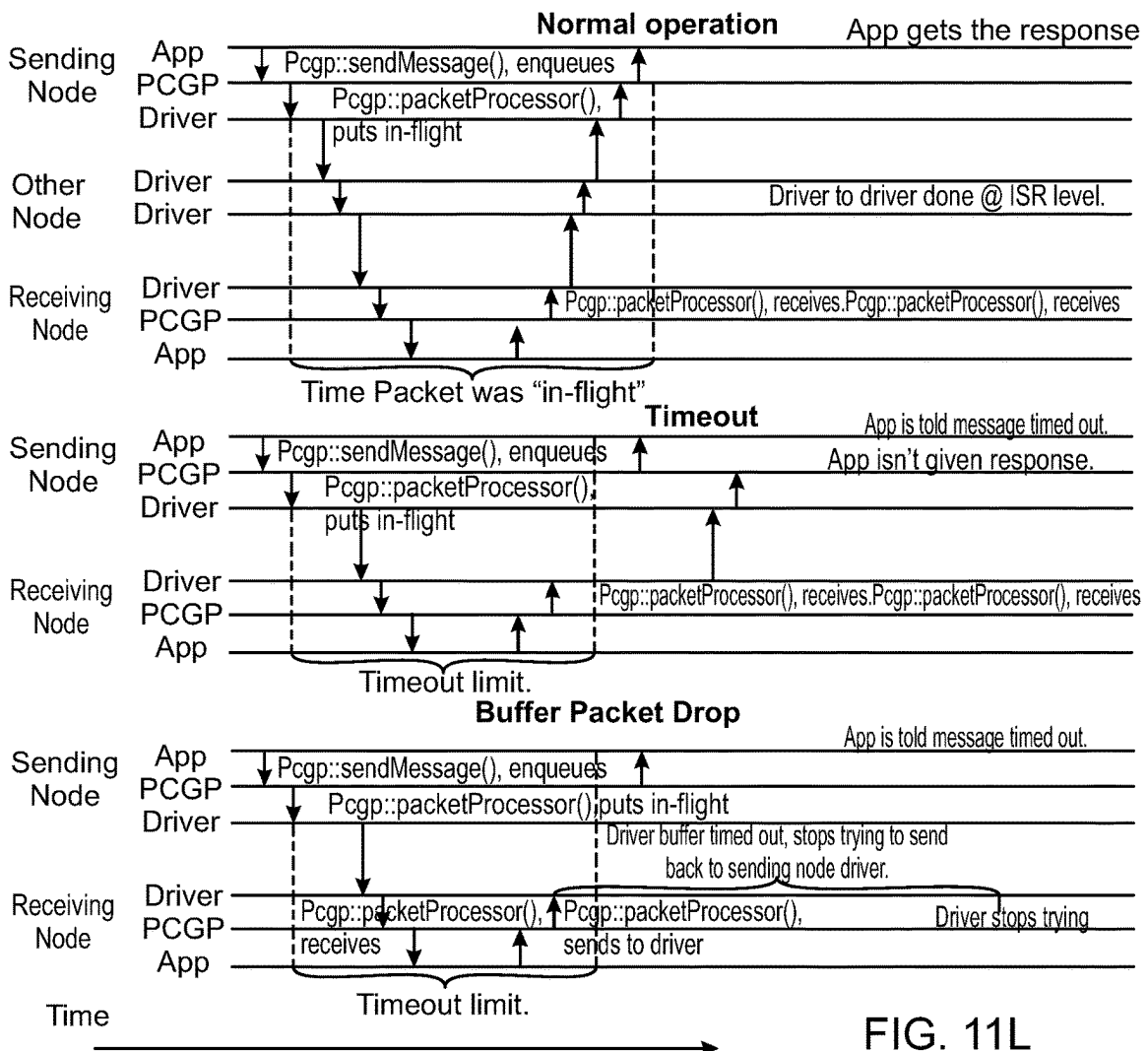


FIG. 11L

FIG. 11M

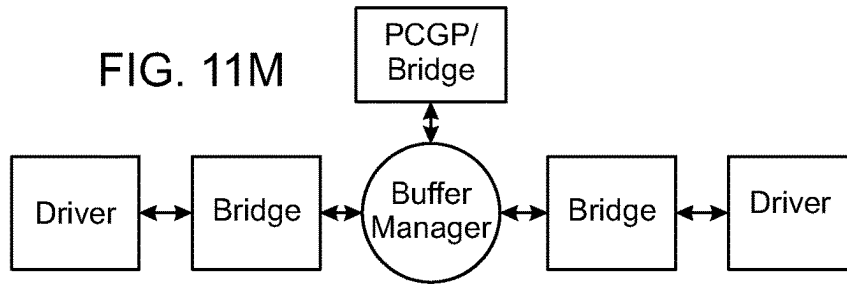


FIG. 11N

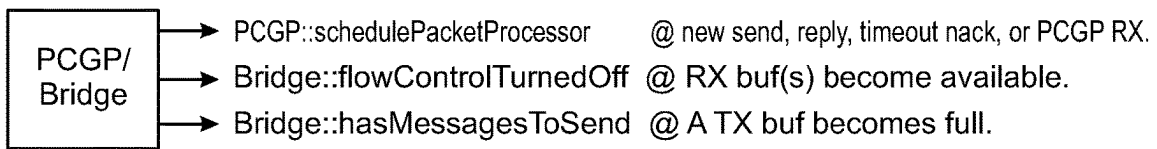


FIG. 11O

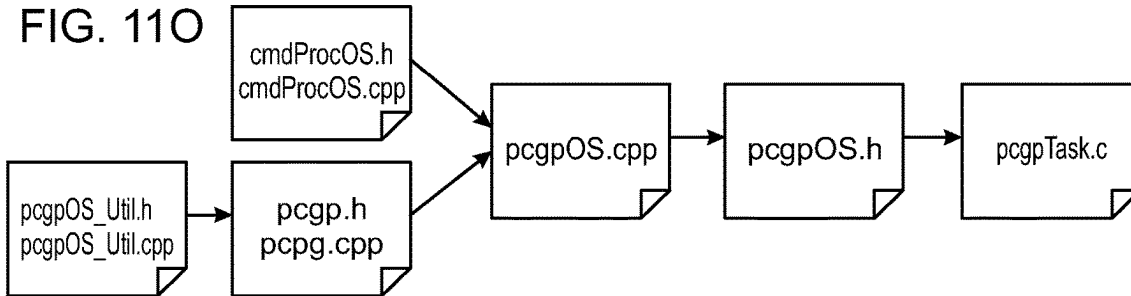
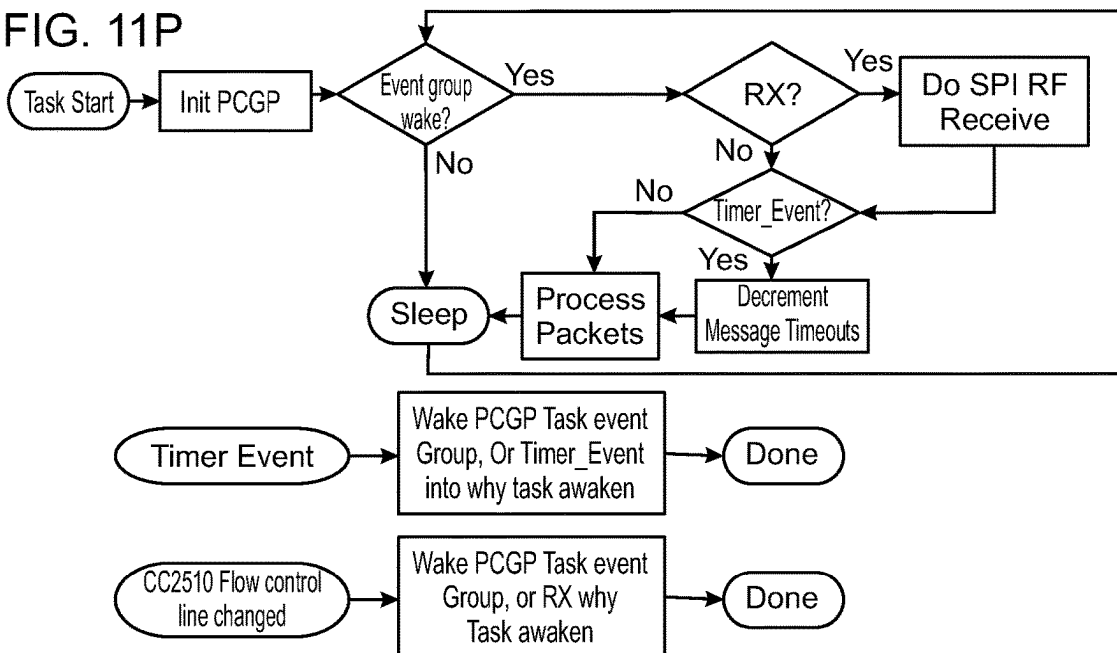
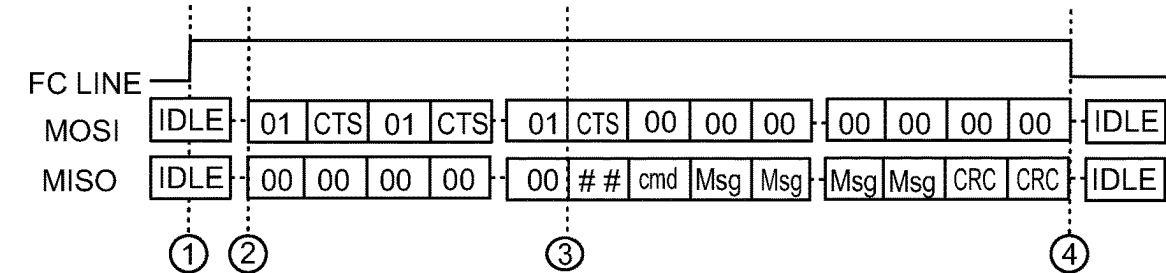


FIG. 11P

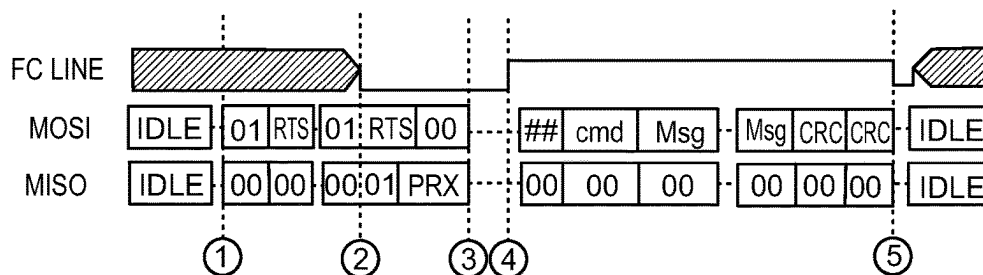




----- Indicates a time lapse

- ① Slave raises FlowControl line indicating a packet is pending
- ② Master begins sending 1 byte Clear To Send commands
- ③ Slave responds with the # of bytes being sent, Msg appended command & the Msg
- ④ The transaction is complete and the Slave lowers the FlowControl line

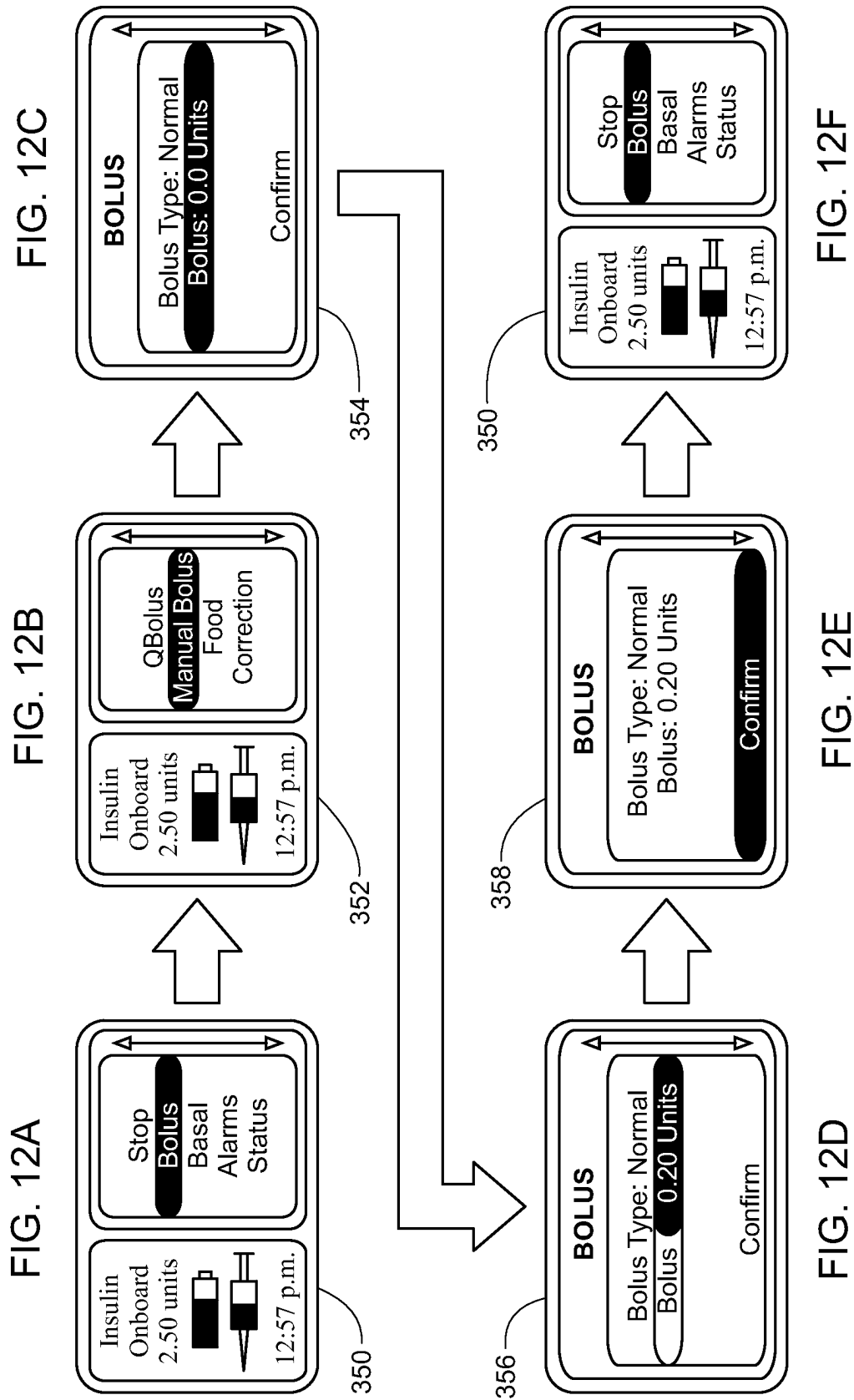
FIG.11Q



--- Indicates a time

- ① Master begins sending single byte RTS commands (state of Flow Control line is ignored)
- ② Slave sends a Prepare for RX command indicating it is setting up dma to receive
- ③ Master stops clocking bytes and waits for the Flow Control Signal
- ④ Slave asserts the Flow Control line indicating the RX dma is ready to receive
- ⑤ The transaction is complete.

FIG.11R



400

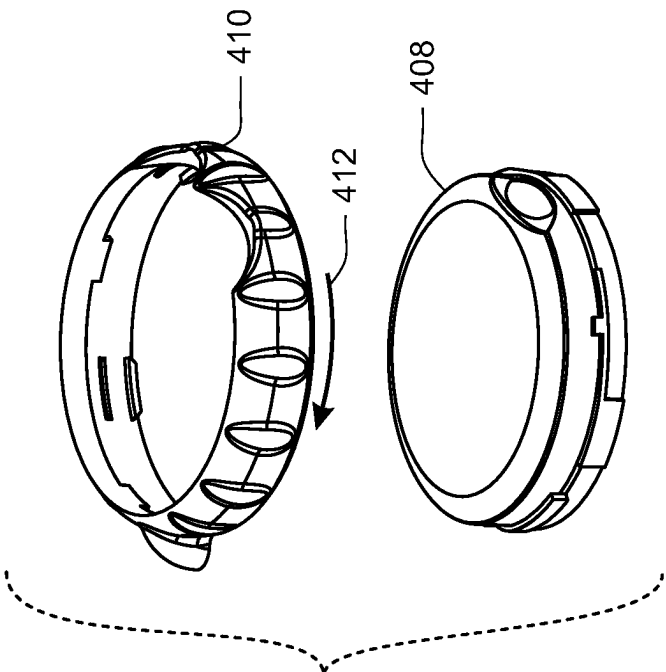
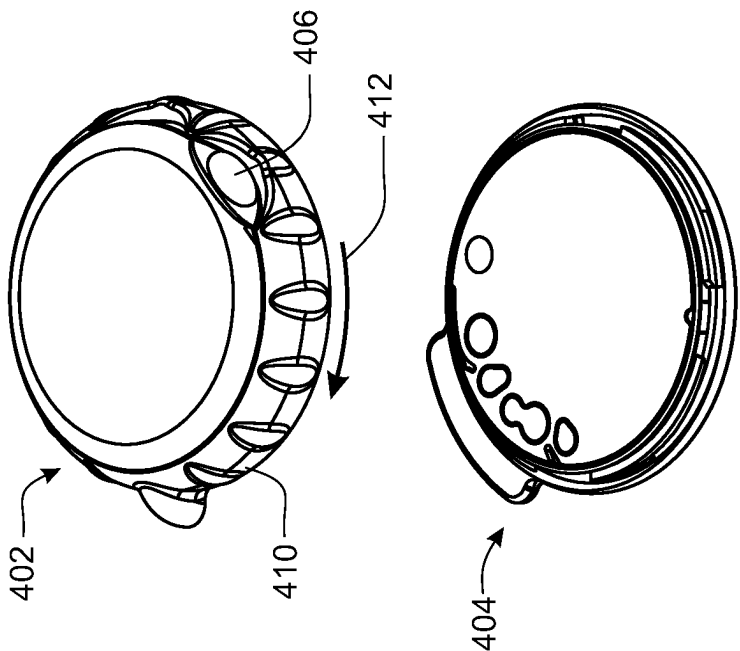


FIG. 13



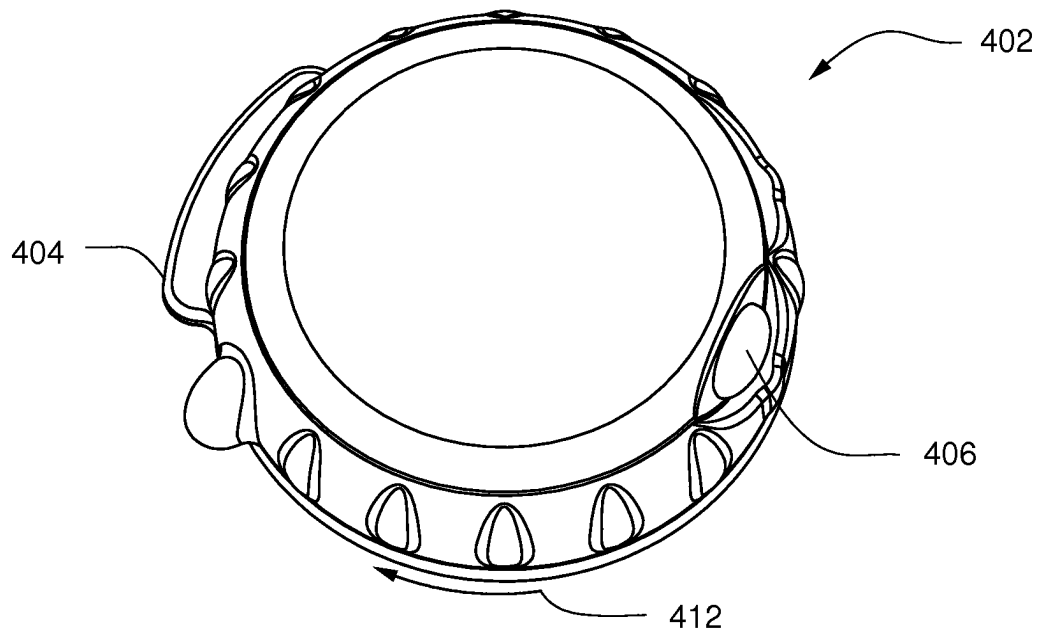


FIG. 14

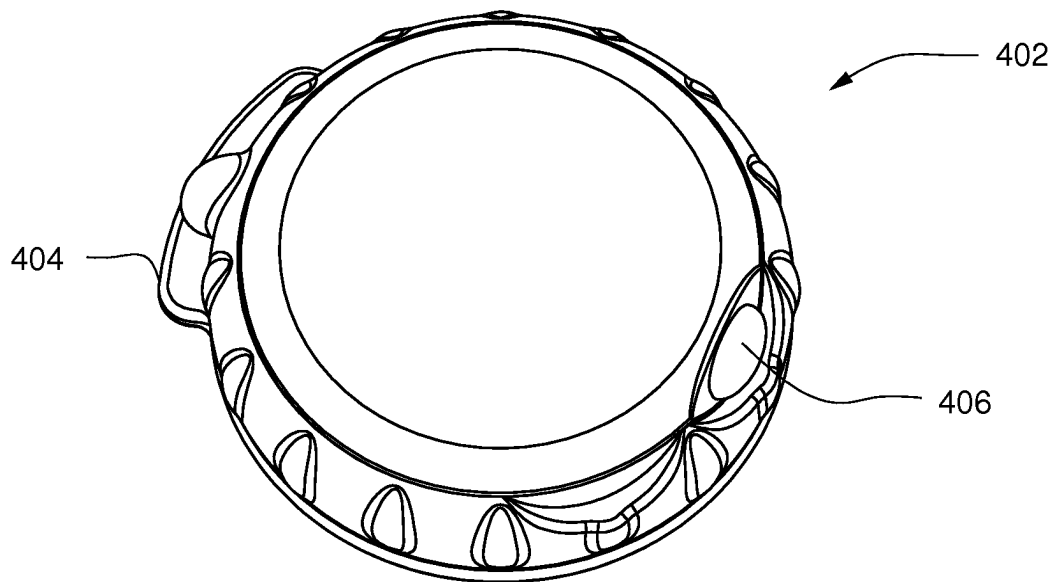


FIG. 15

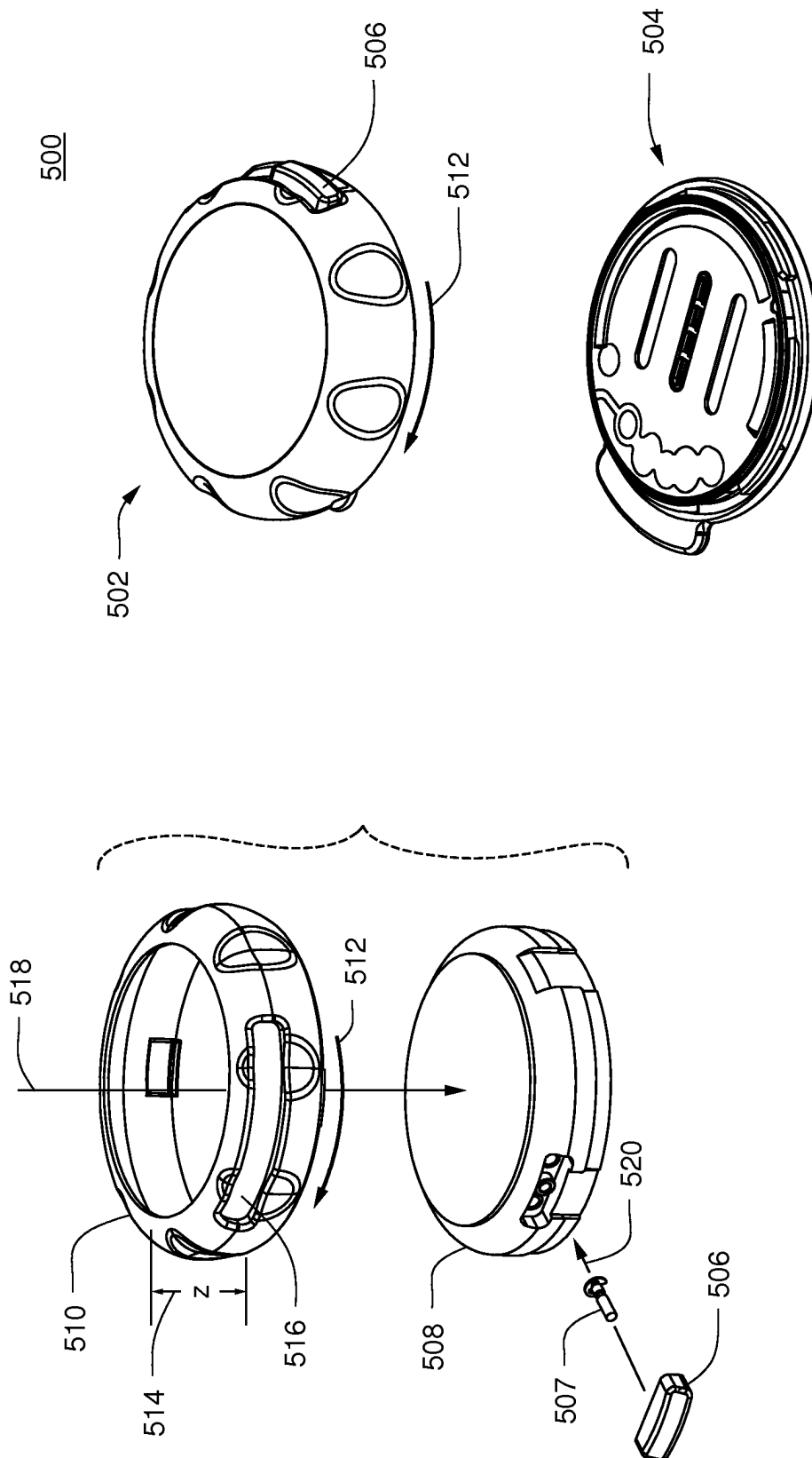


FIG. 16

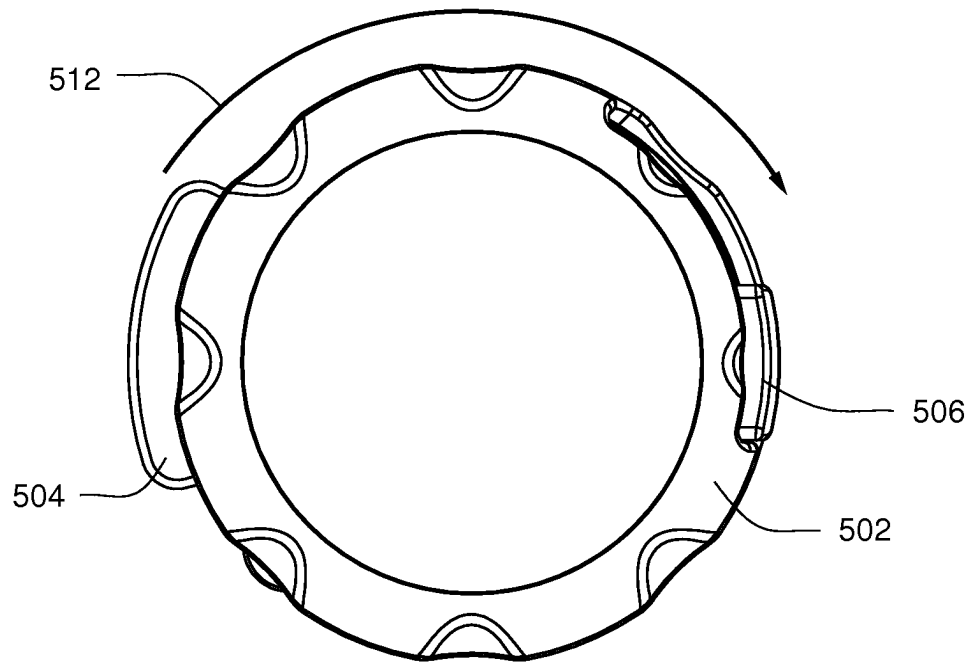


FIG. 17

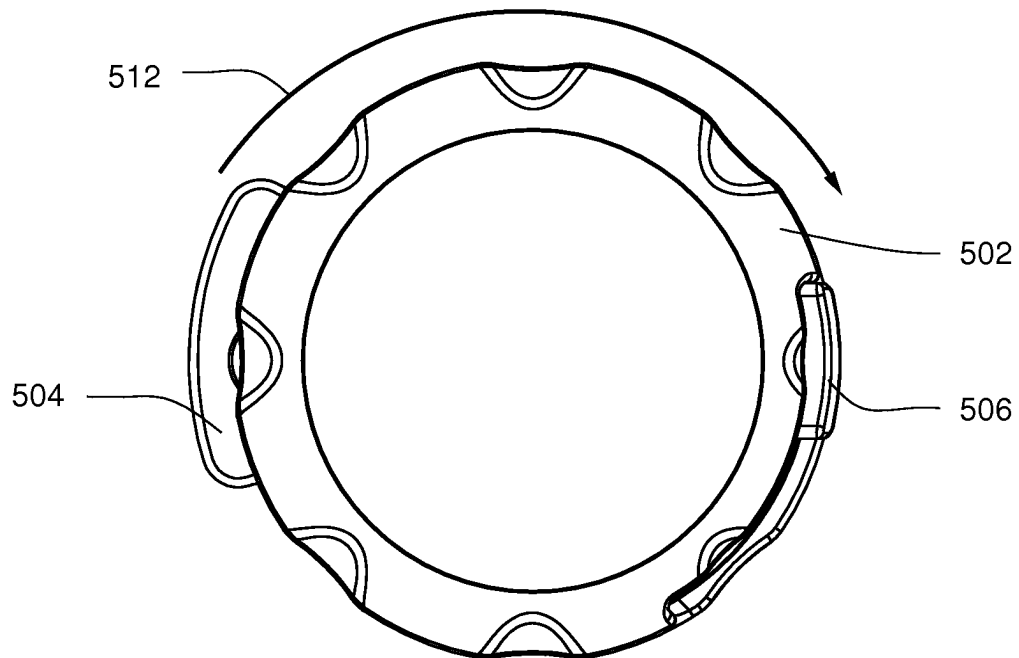


FIG. 18

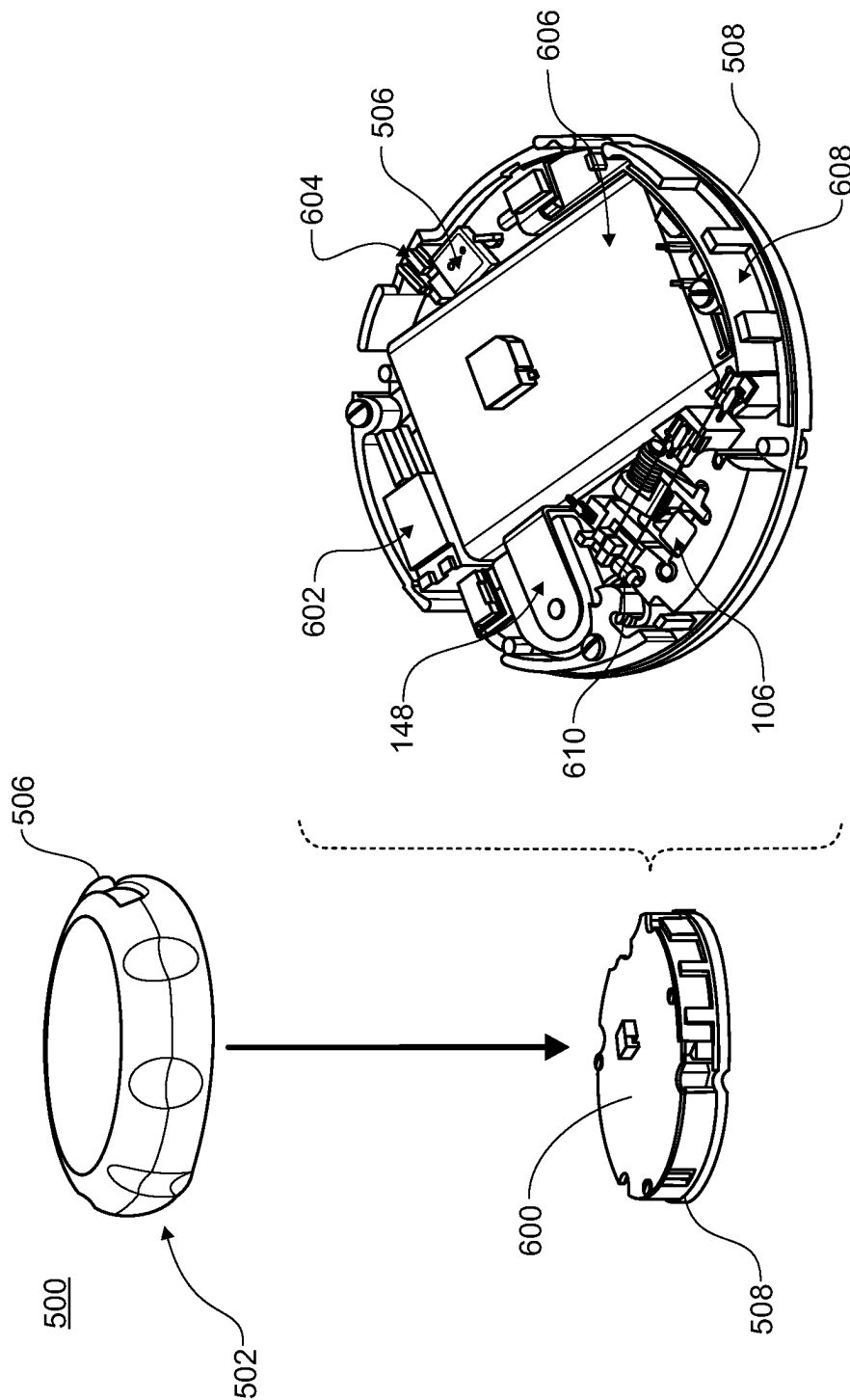


FIG.19B

FIG.19A

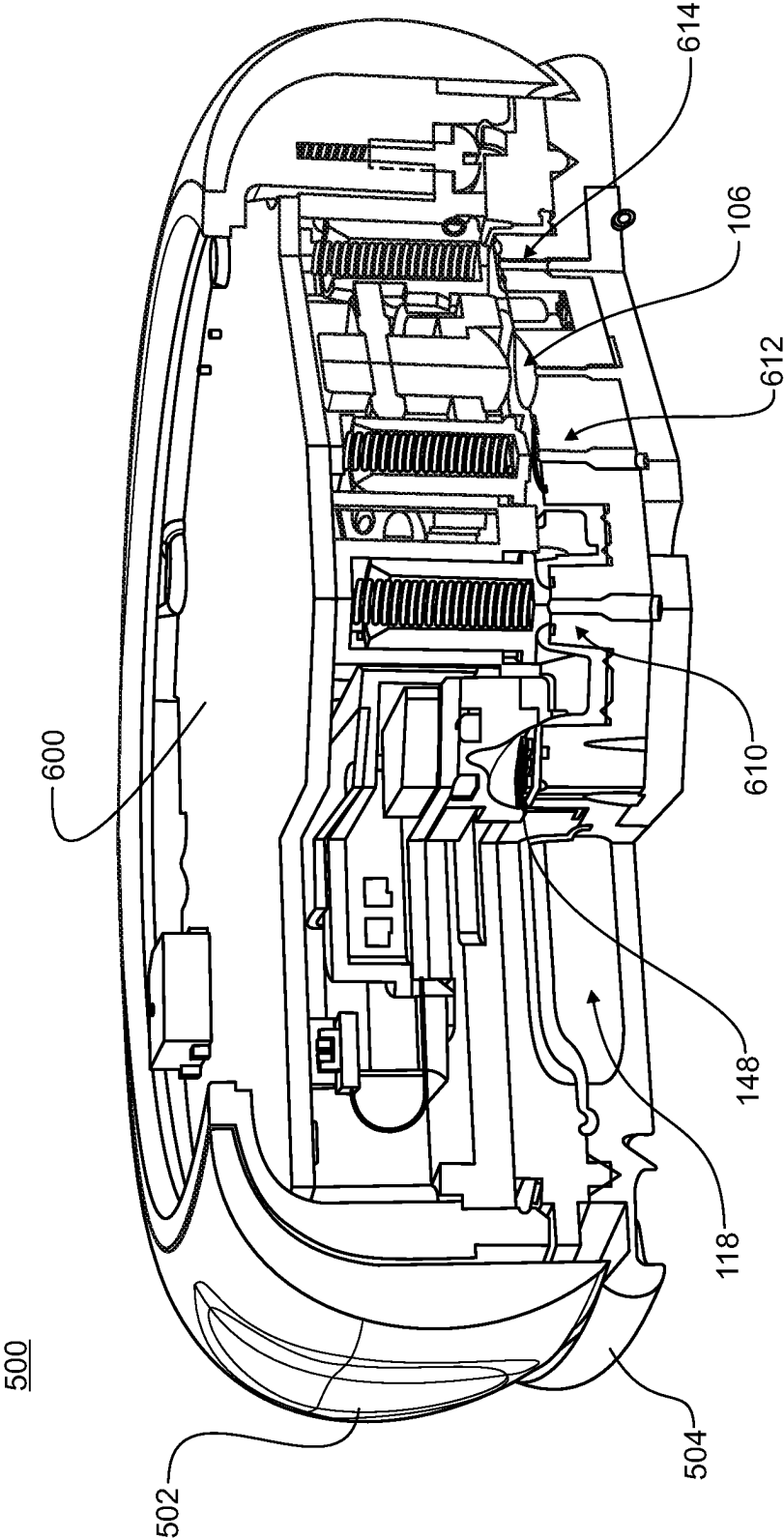


FIG. 20

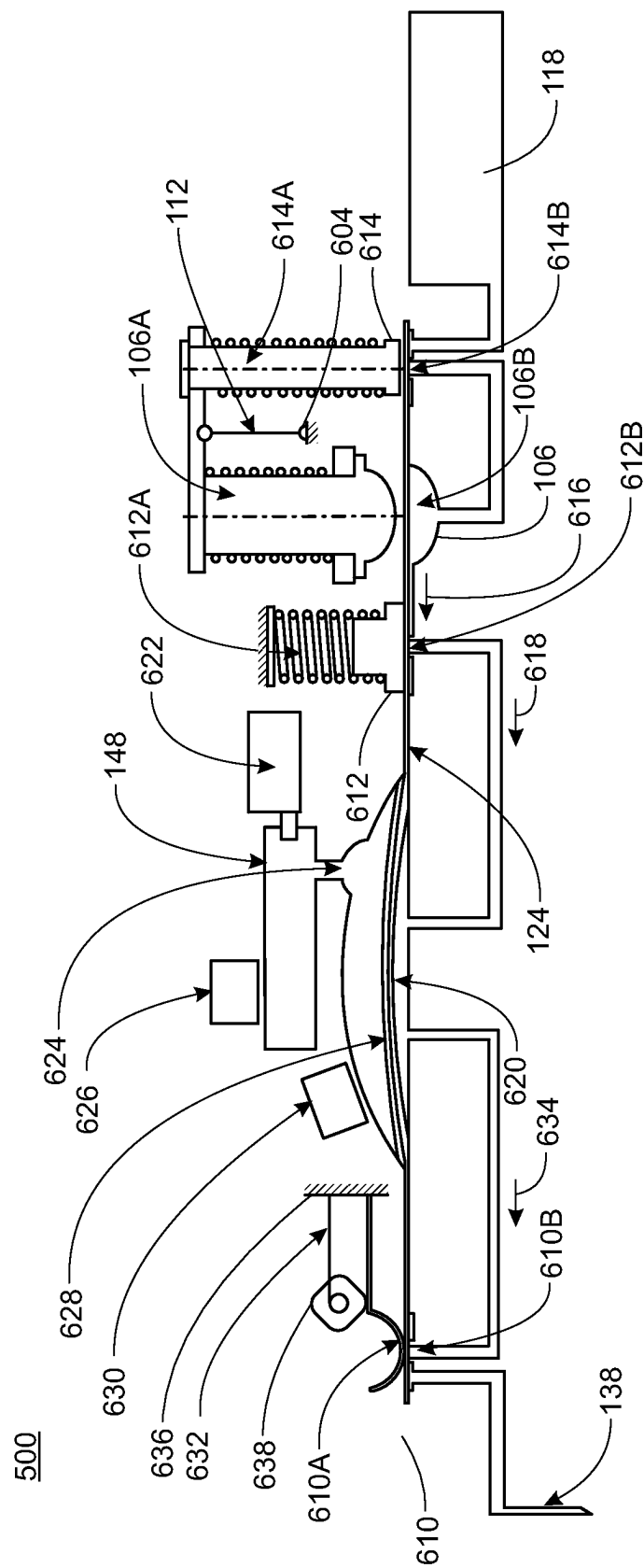


FIG. 21

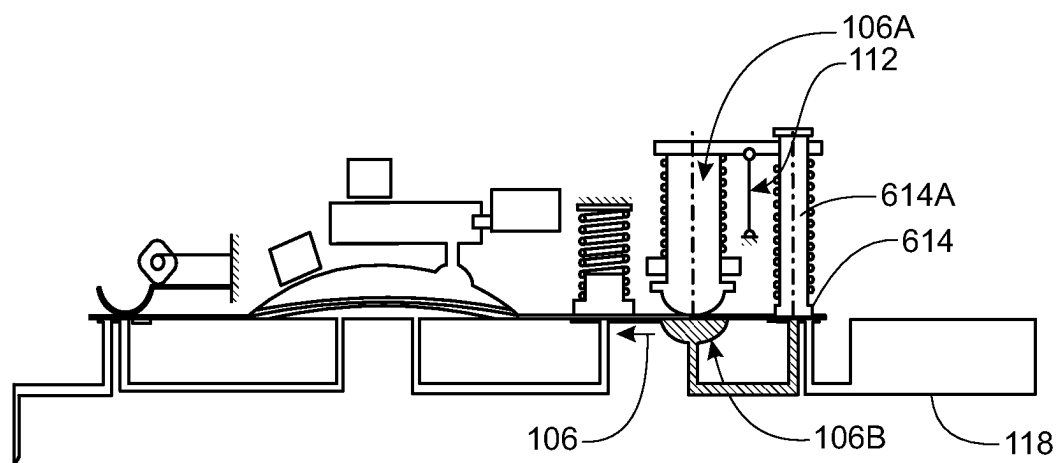


FIG. 22A

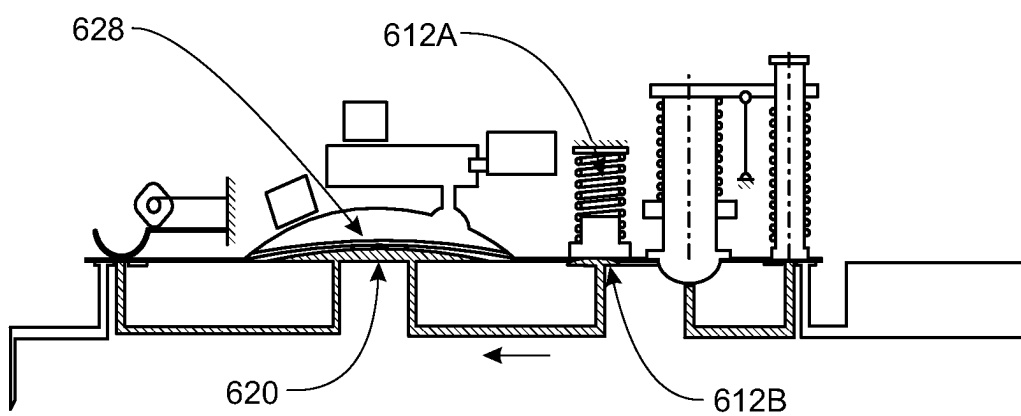


FIG. 22B

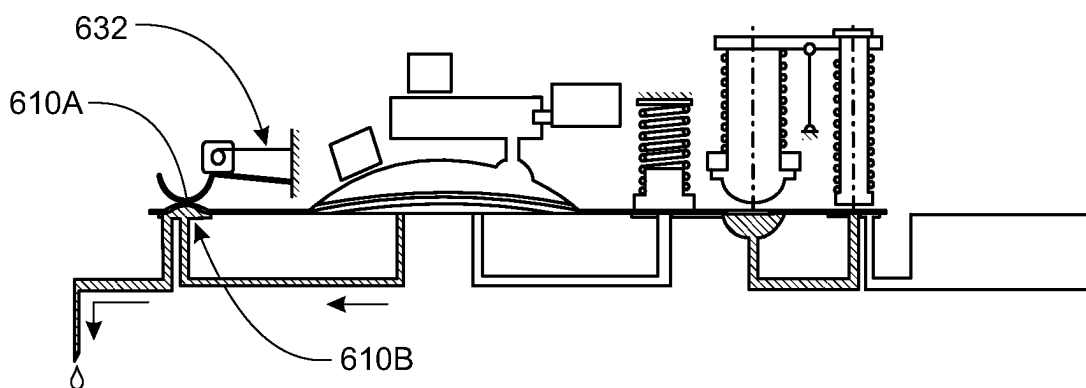


FIG. 22C

500

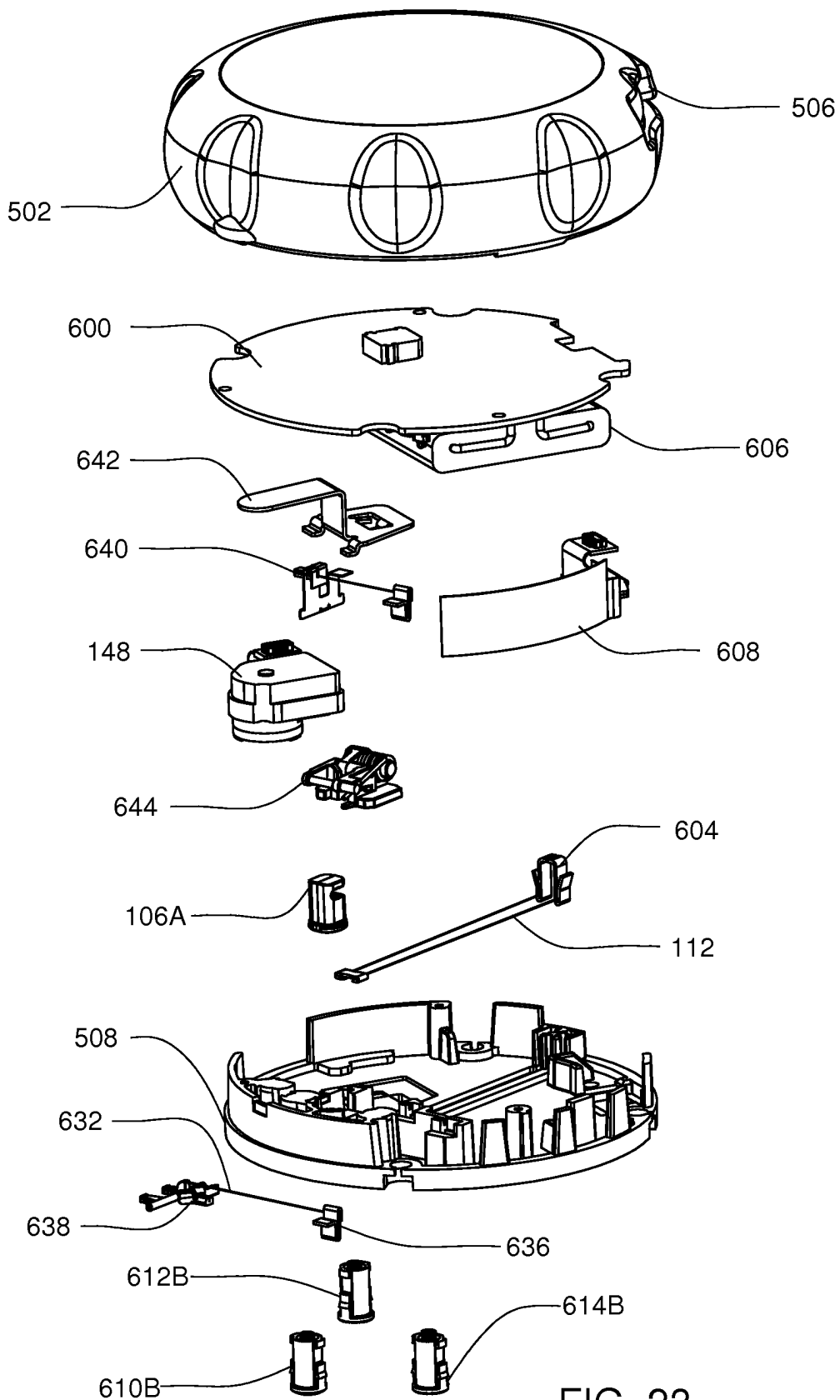


FIG. 23

106

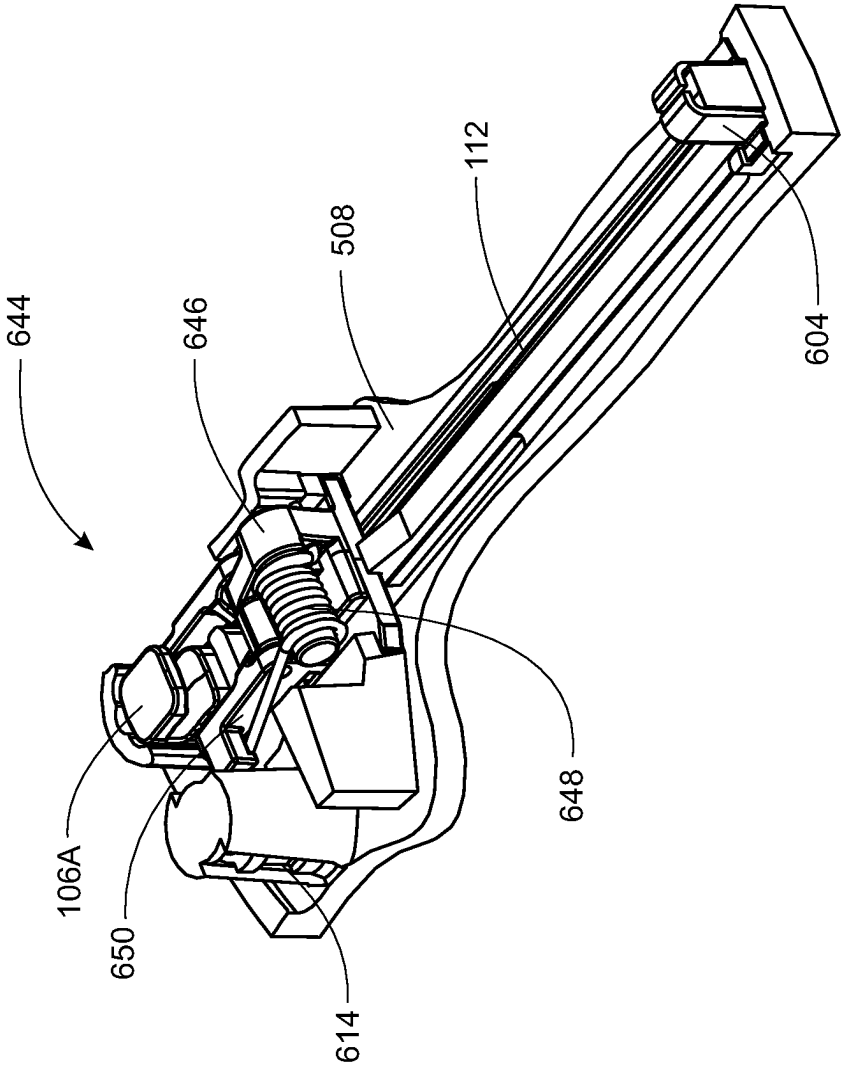


FIG. 24

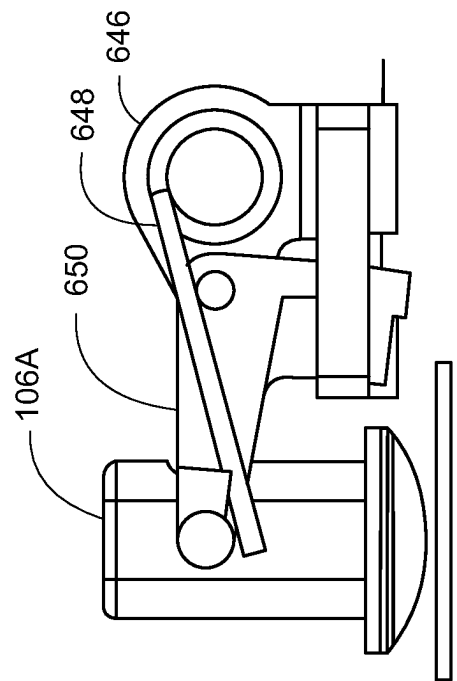


FIG. 25B

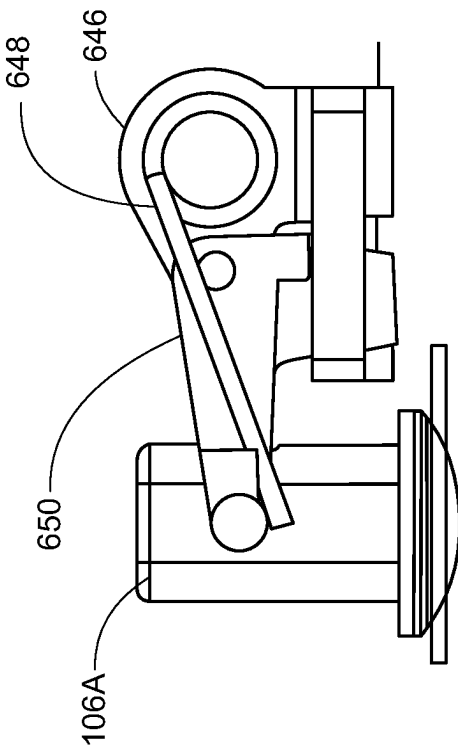


FIG. 25D

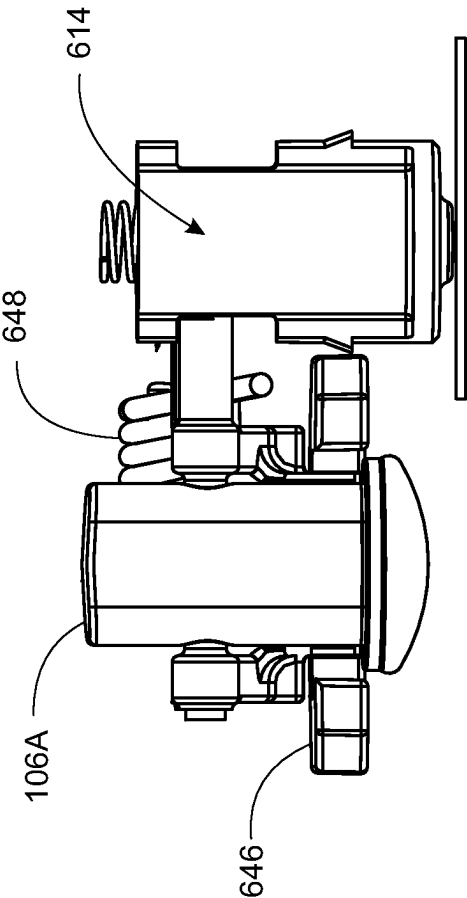


FIG. 25A

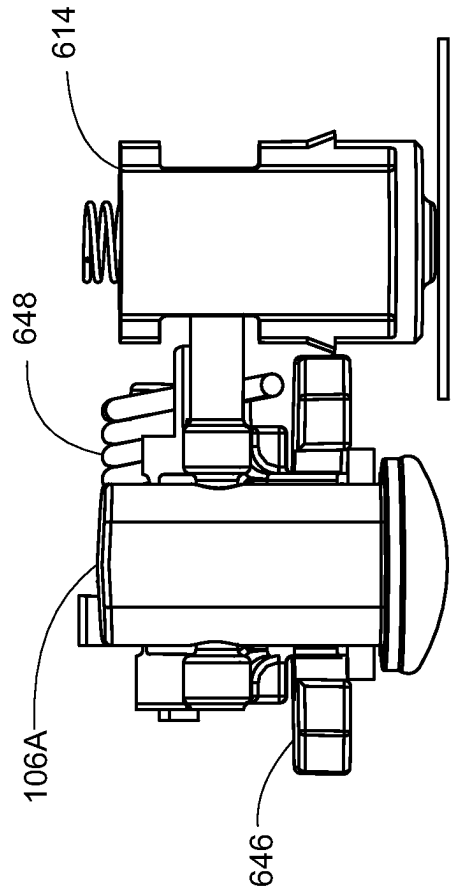


FIG. 25C

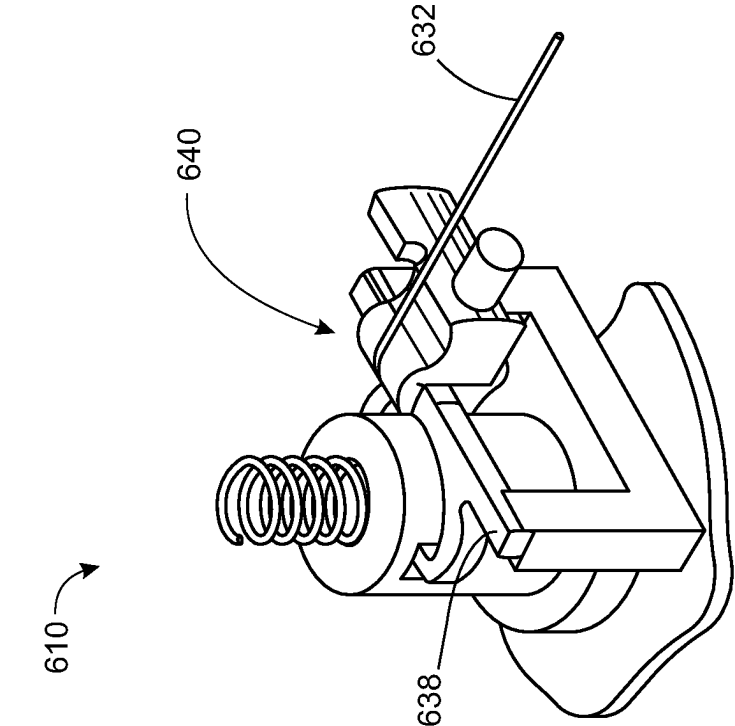
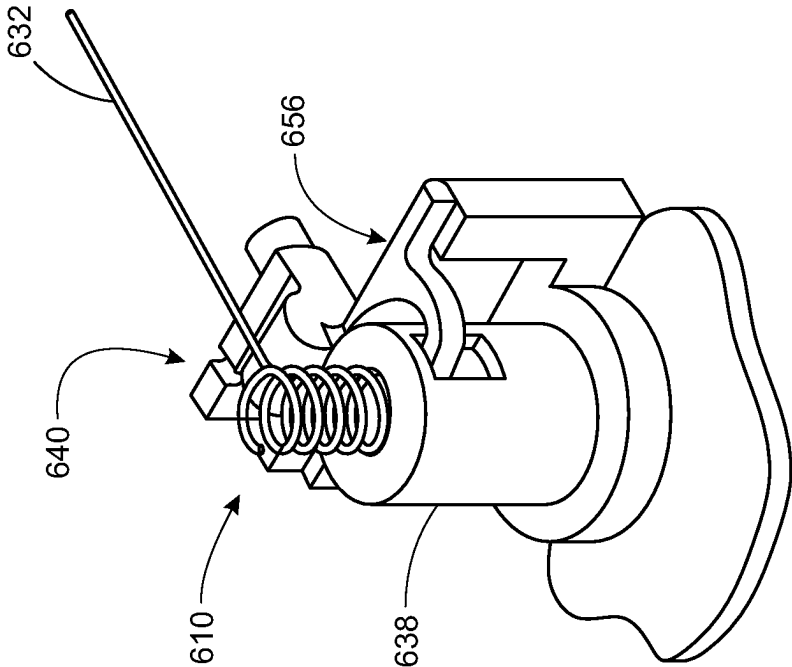


FIG. 26B



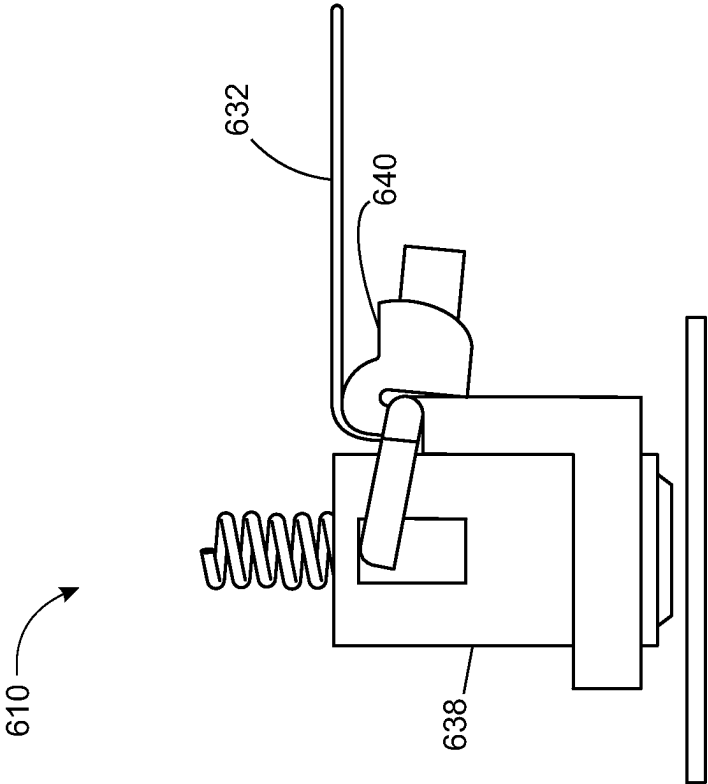


FIG. 27B

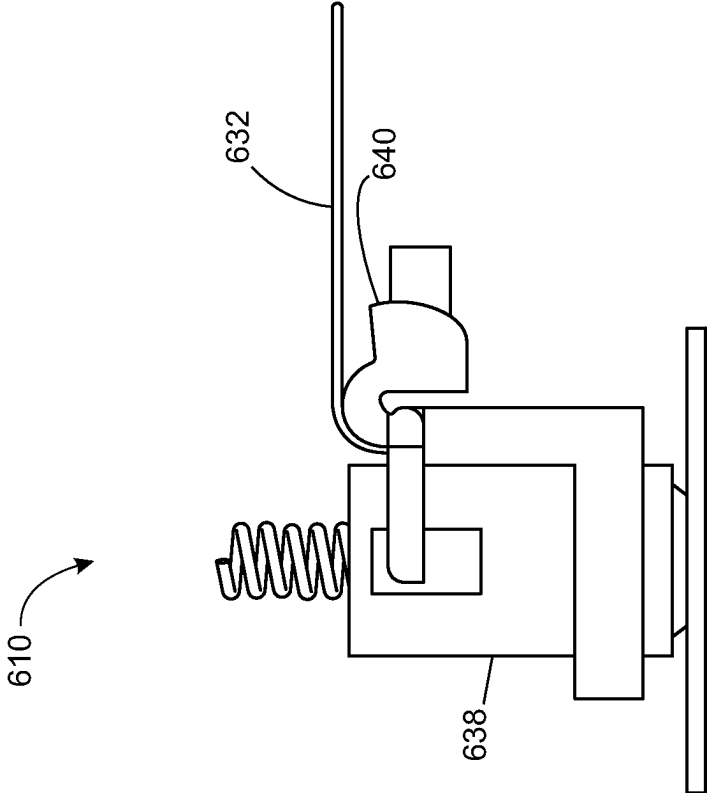


FIG. 27A

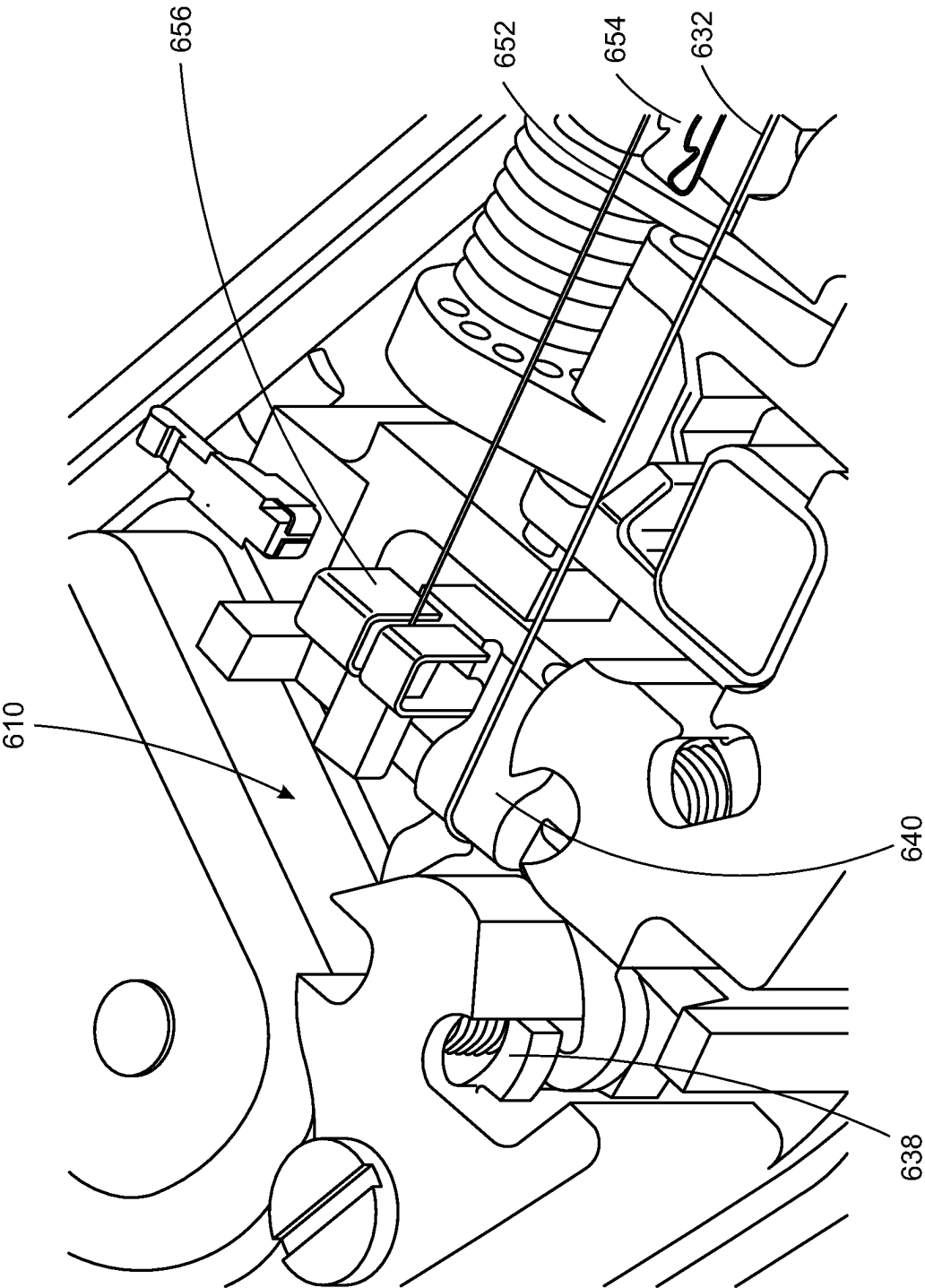


FIG. 28A

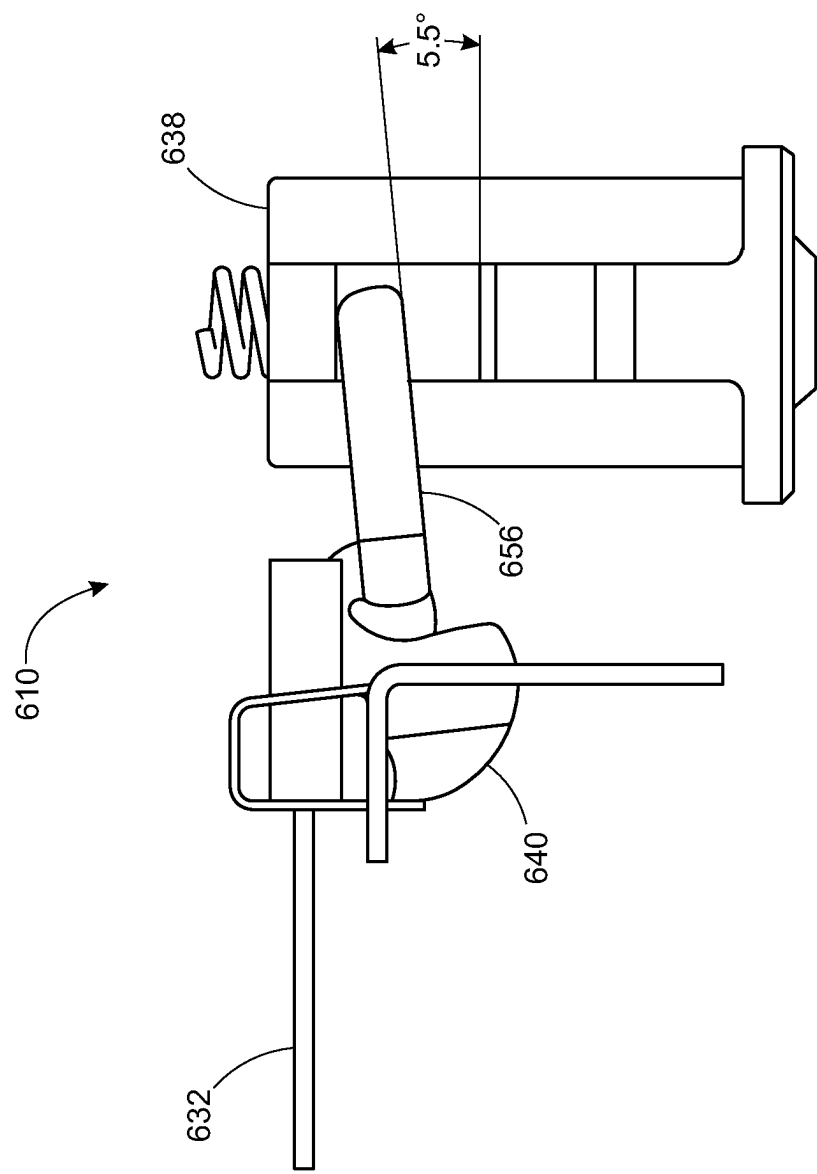


FIG. 28B

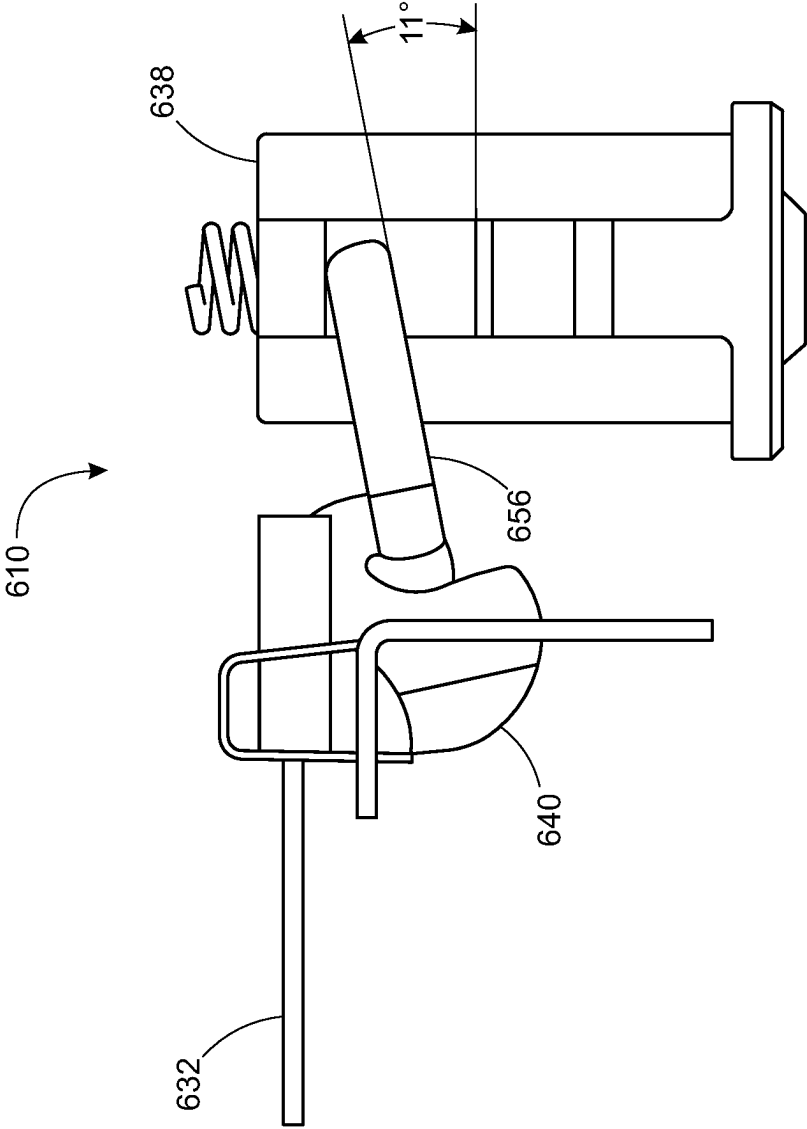


FIG. 28C

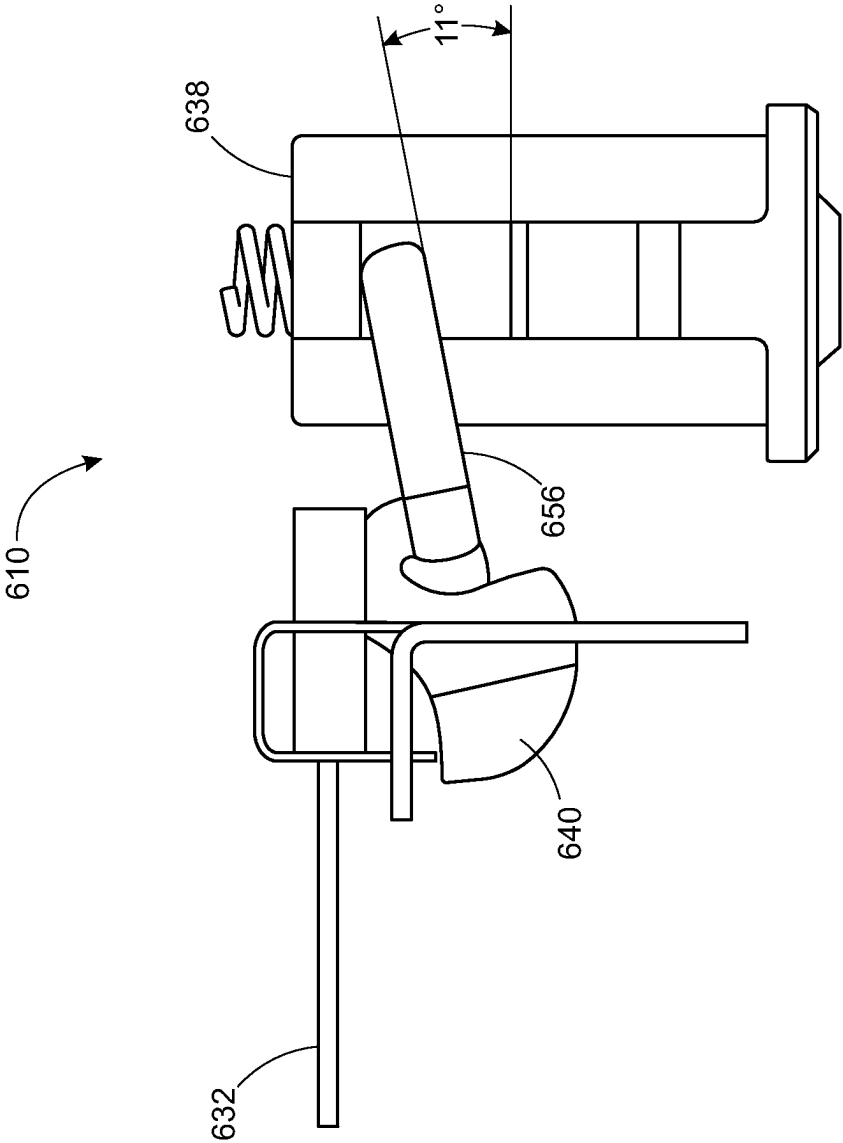


FIG. 28D

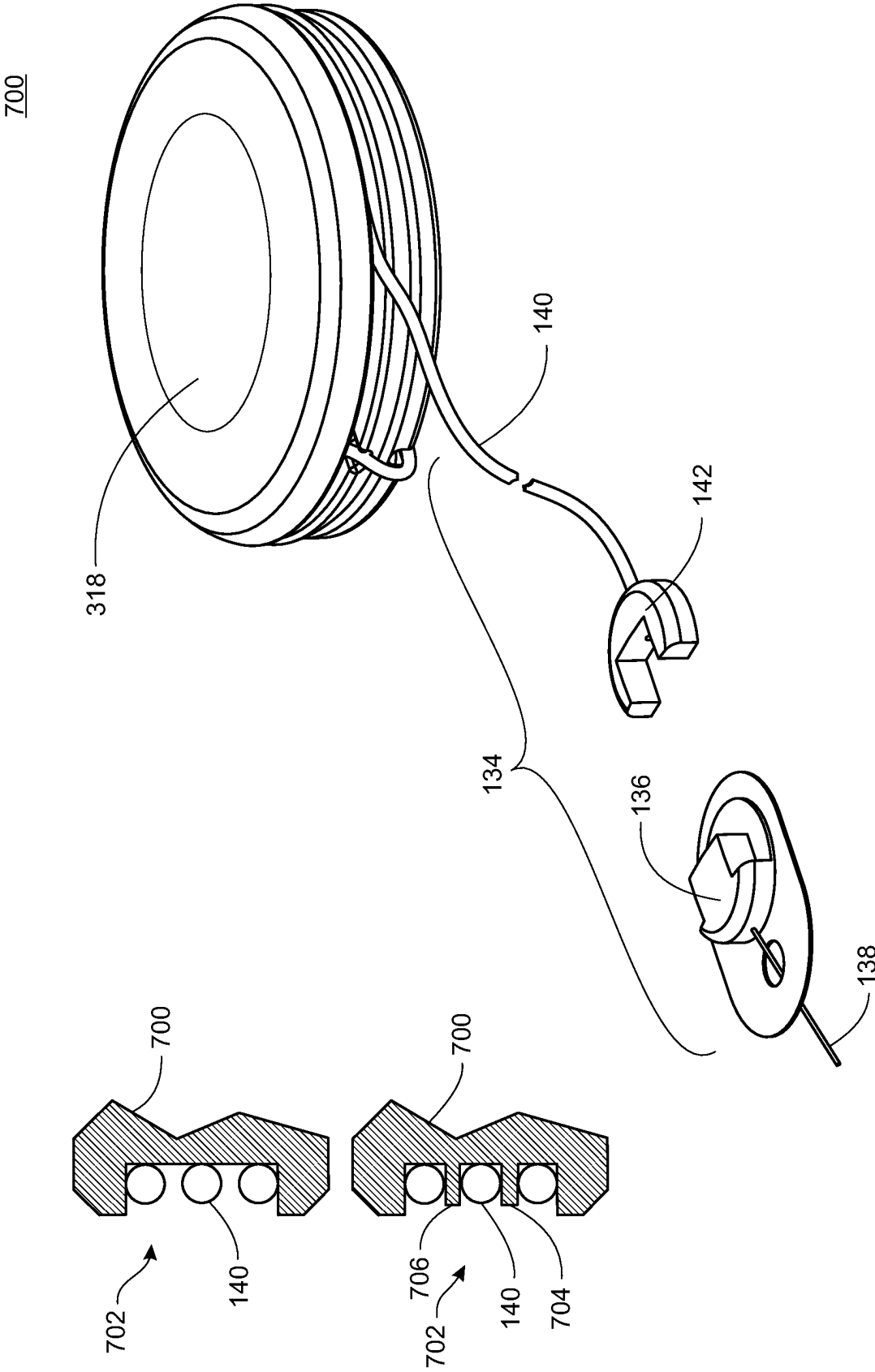


FIG. 29

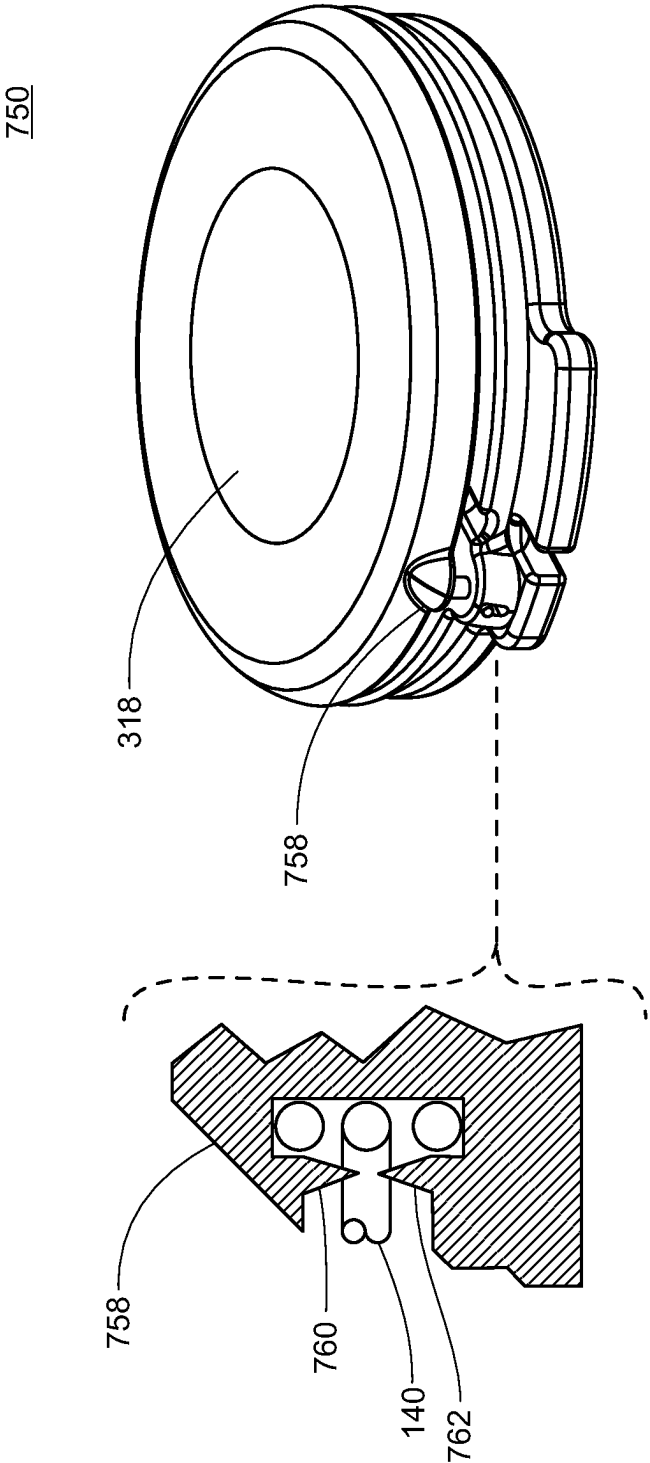


FIG. 30

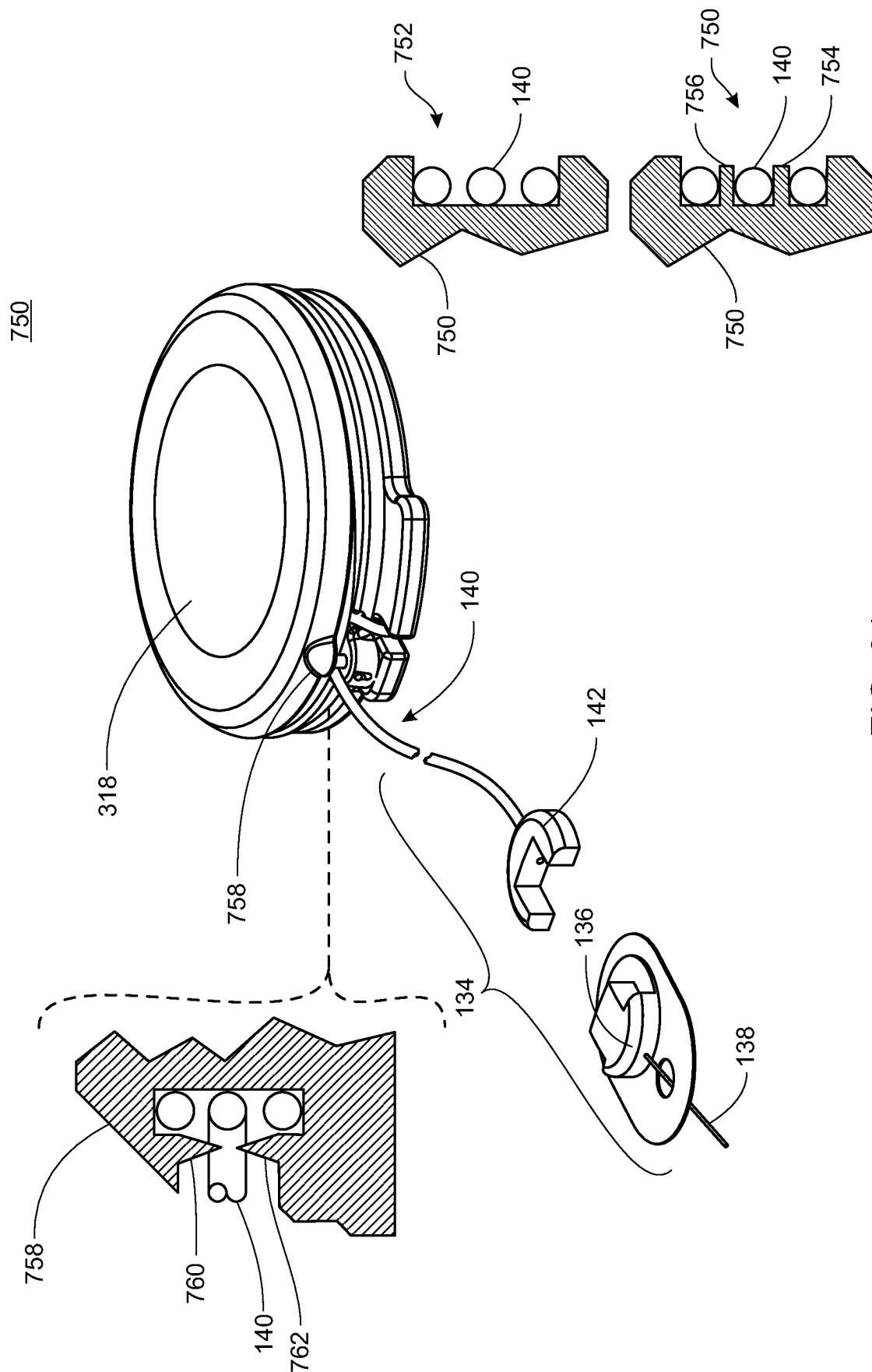


FIG. 31

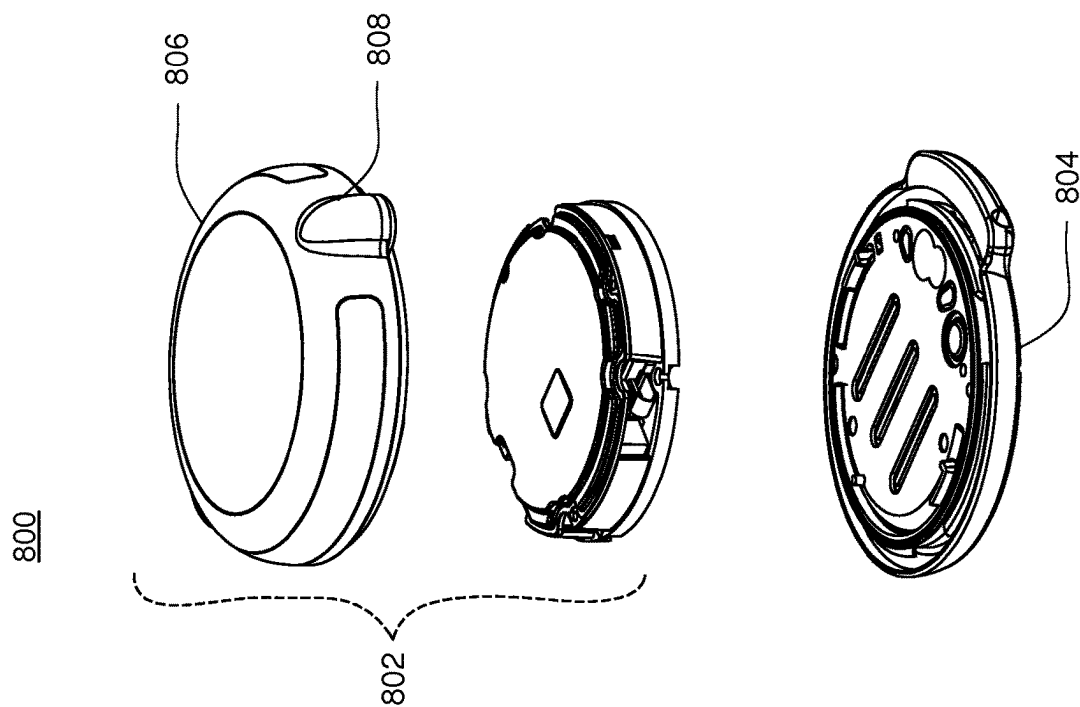


FIG. 32

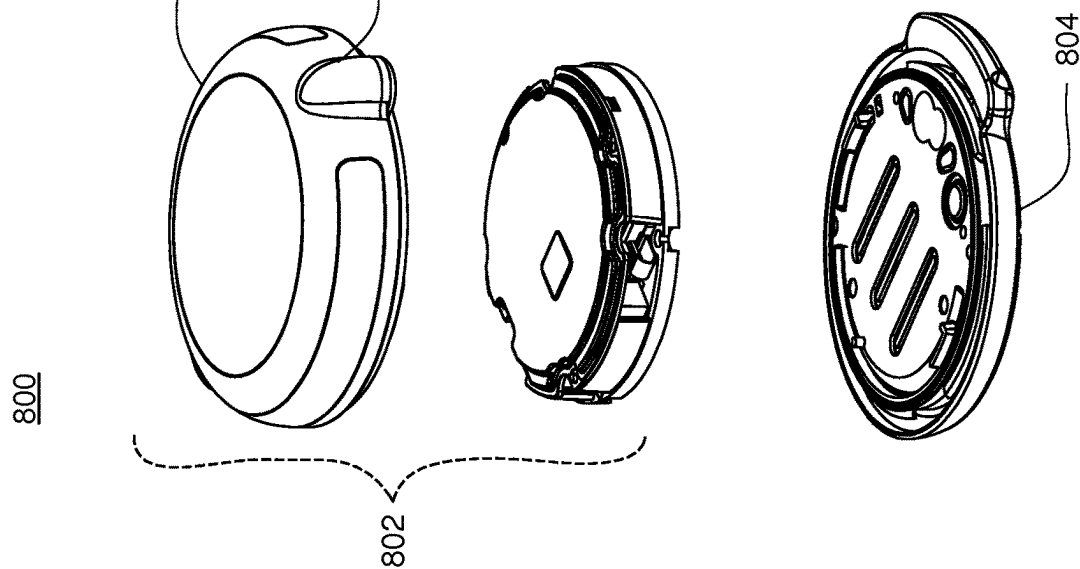


FIG. 33

800

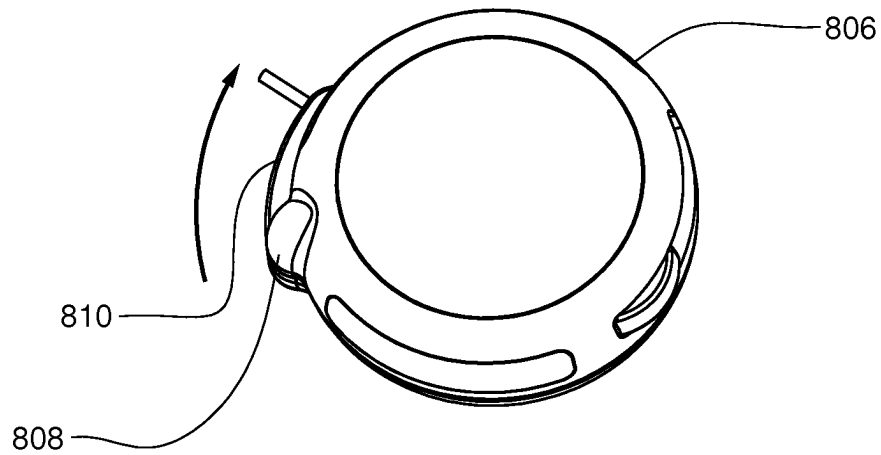


FIG. 34A

800

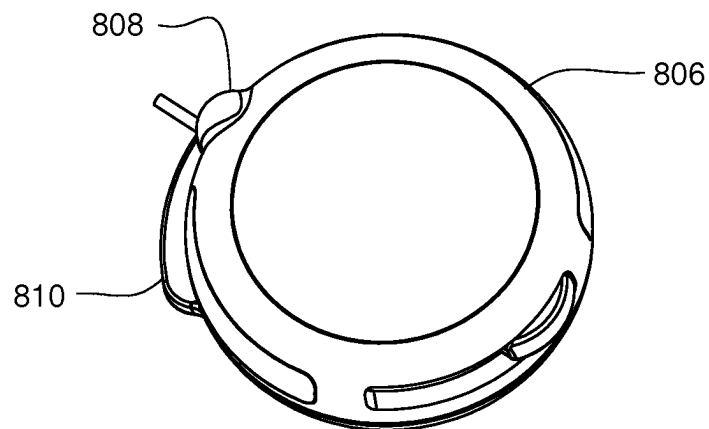


FIG. 34B

802

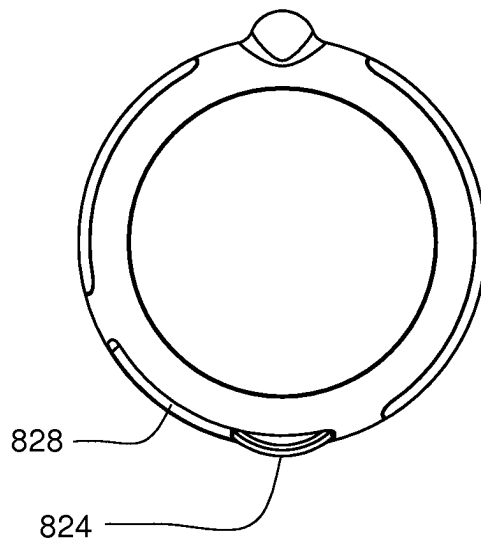


FIG. 35A

802

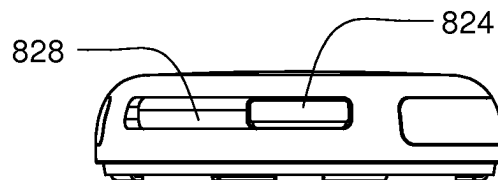


FIG. 35B

802

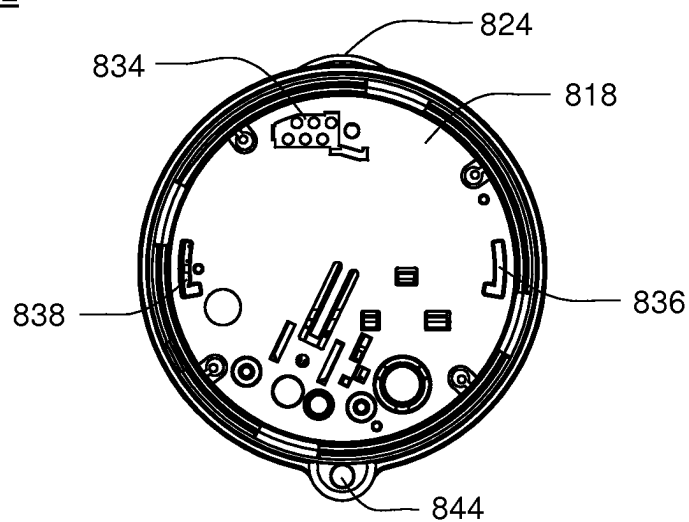


FIG. 35C

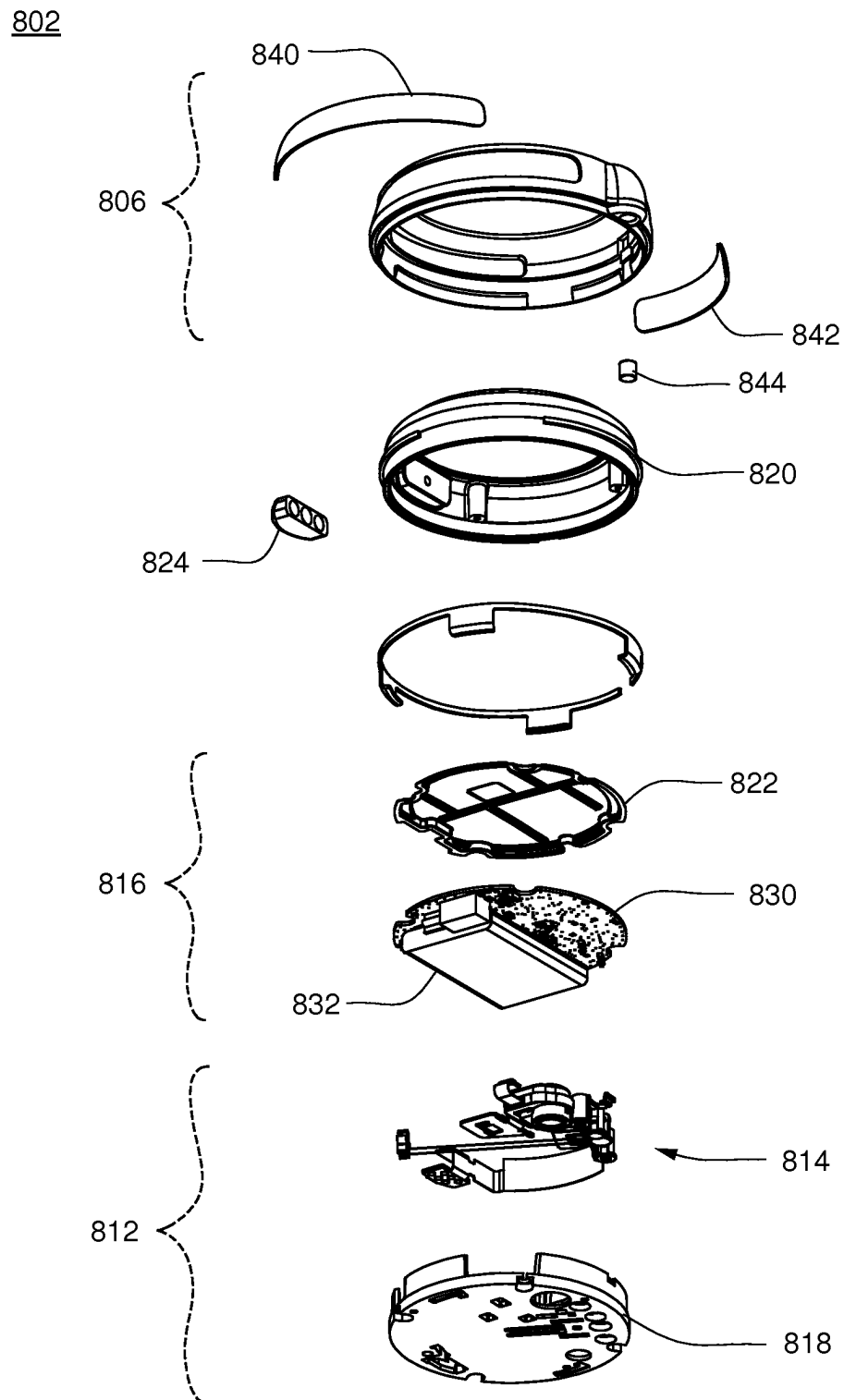


FIG. 36

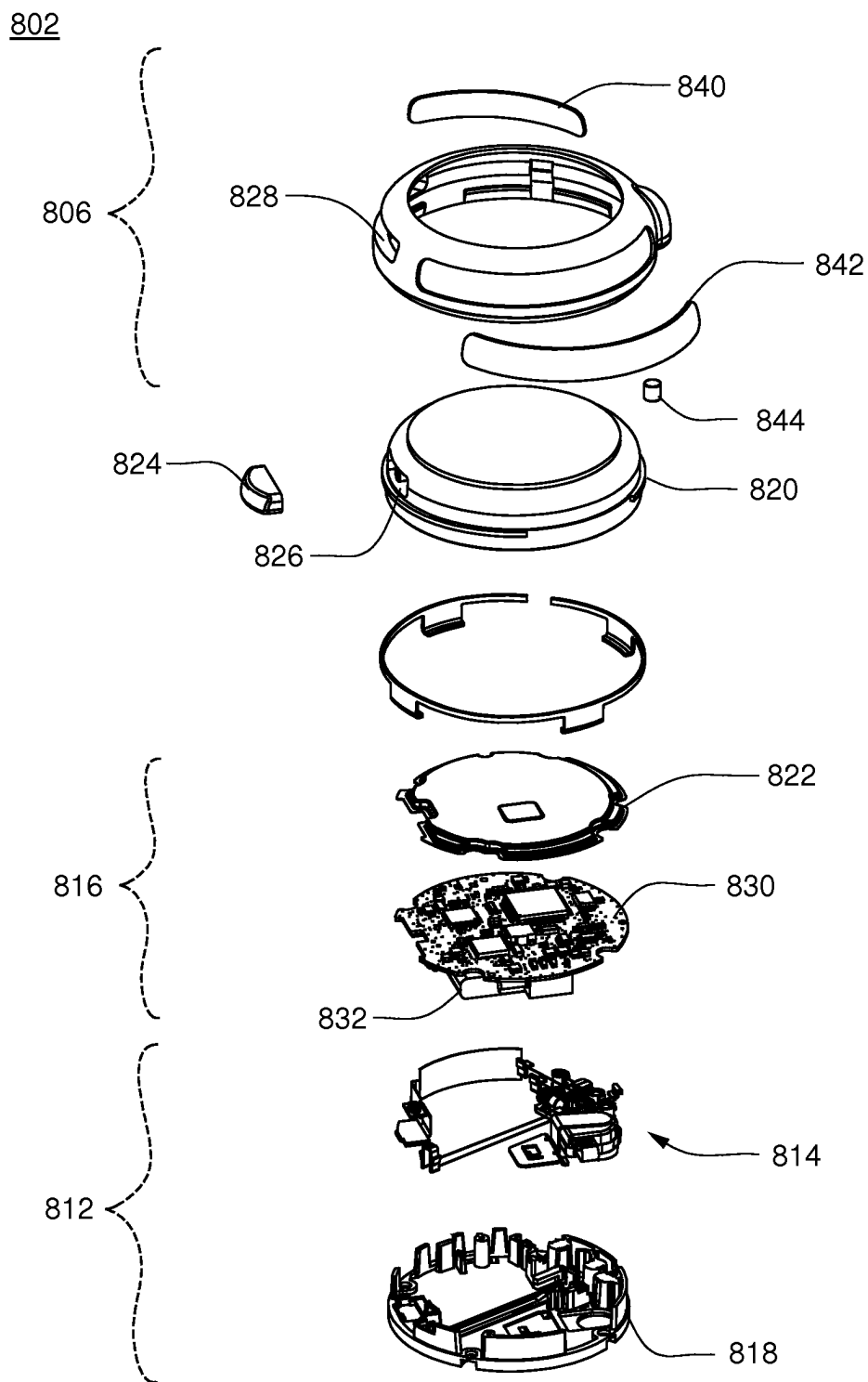


FIG. 37

802

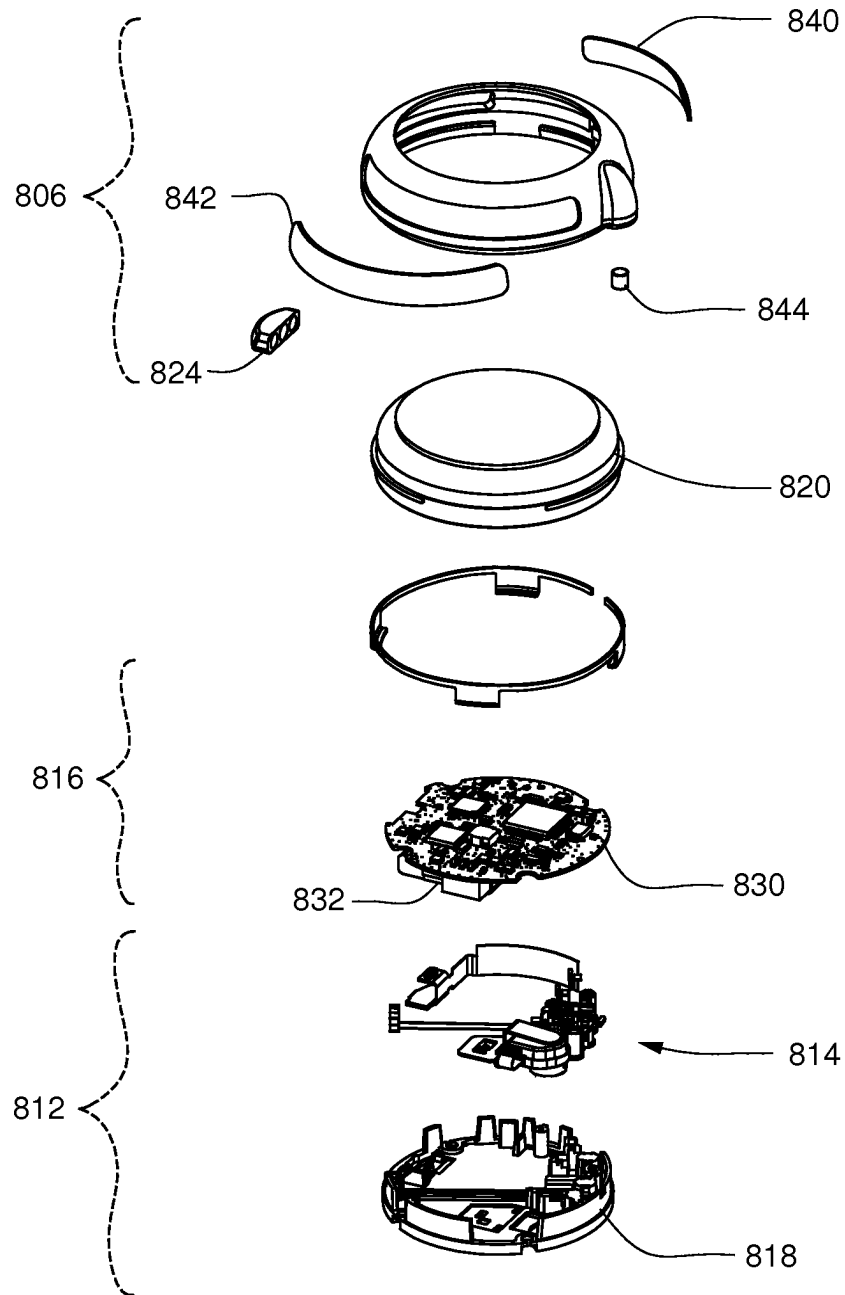


FIG. 38A

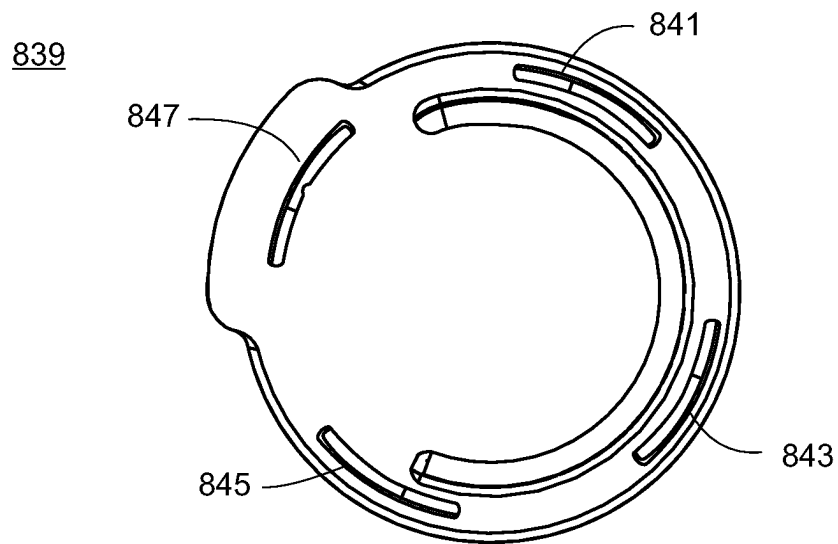


FIG. 38D



FIG. 38C

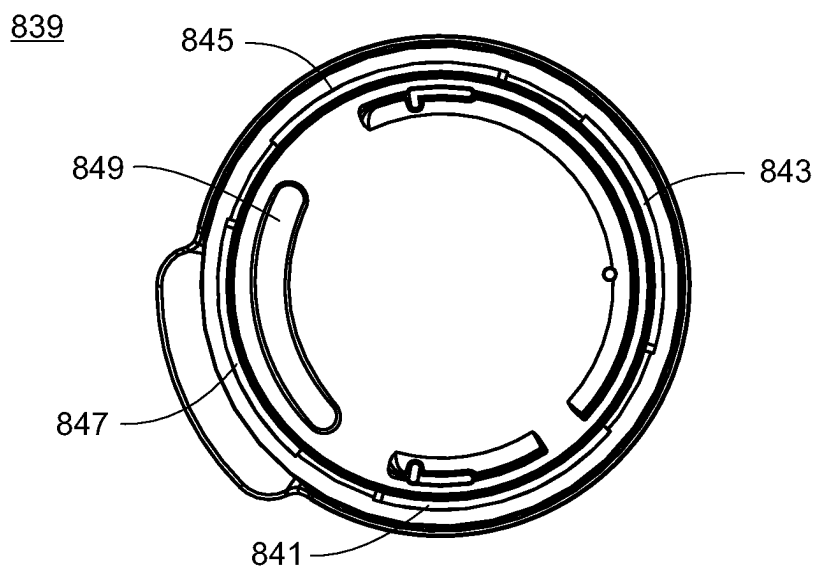


FIG. 38B

816

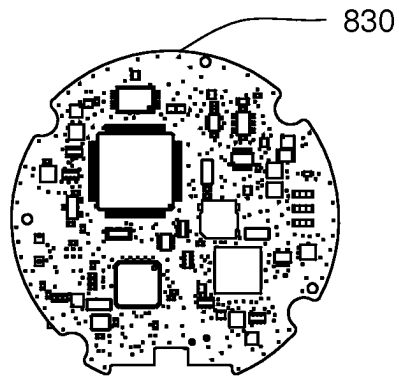


FIG. 39A

816

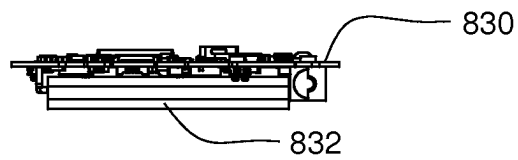


FIG. 39B

816

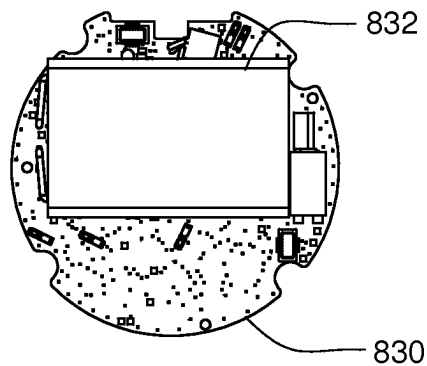


FIG. 39C

818

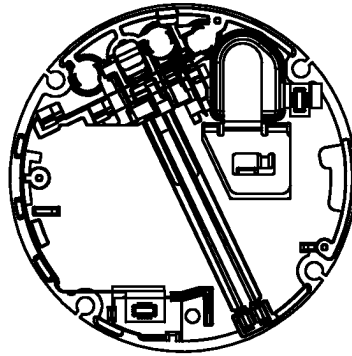


FIG. 40A

818



FIG. 40B

818

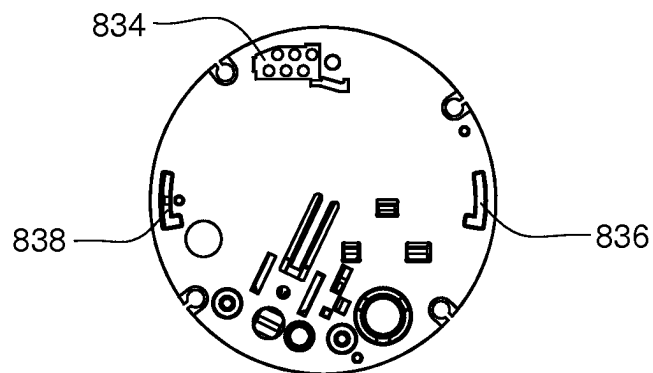


FIG. 40C

818

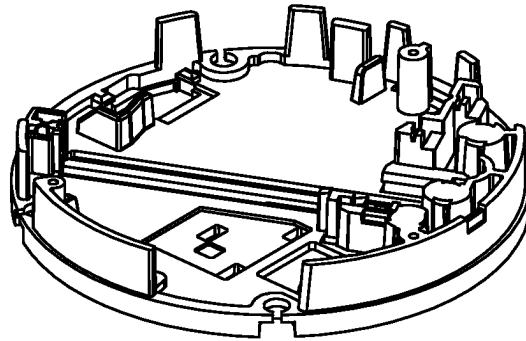


FIG. 41A

818

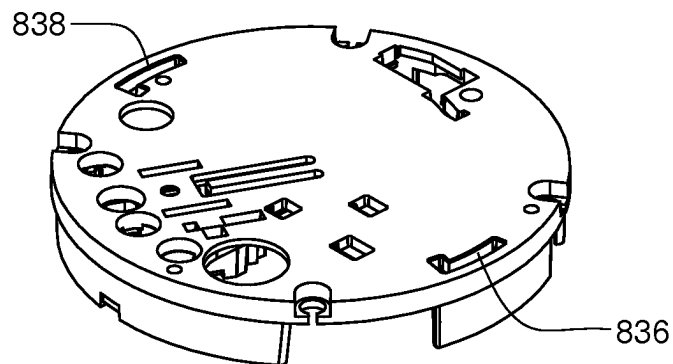


FIG. 41B

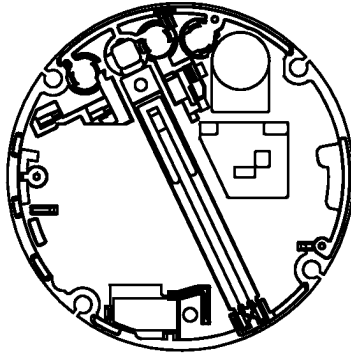


FIG. 42A



FIG. 42B

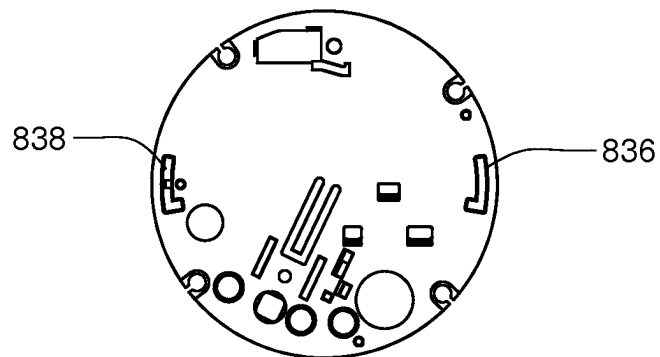


FIG. 42C

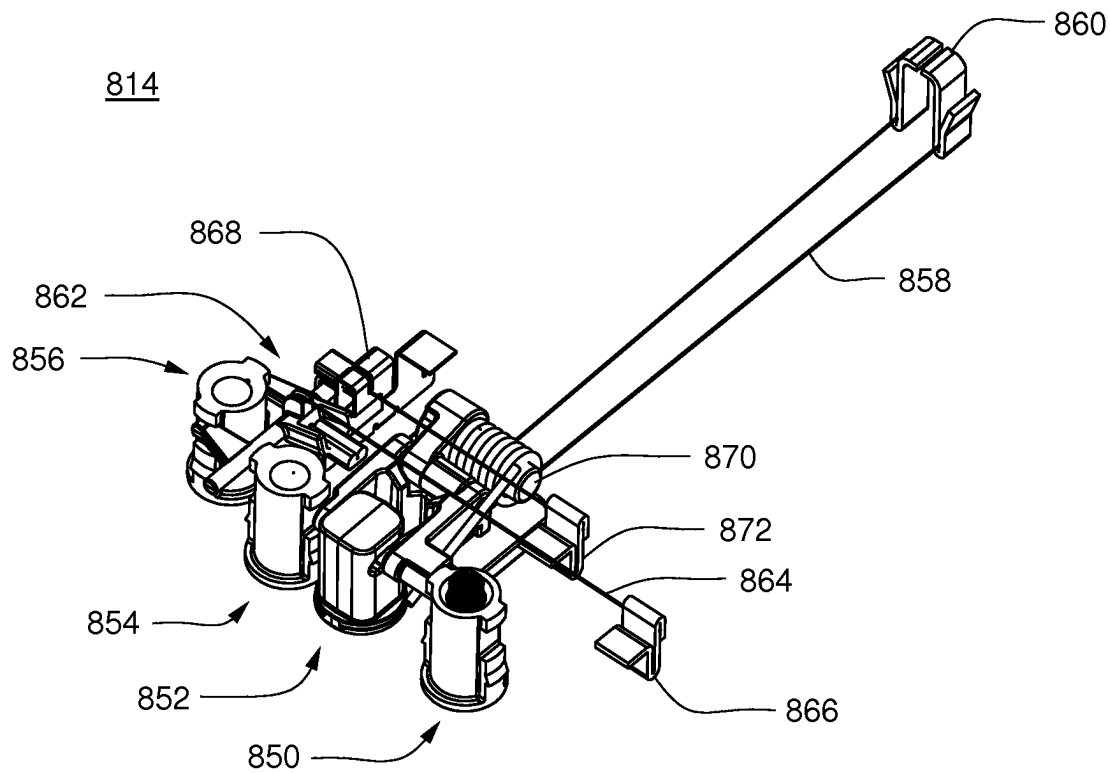


FIG. 43A

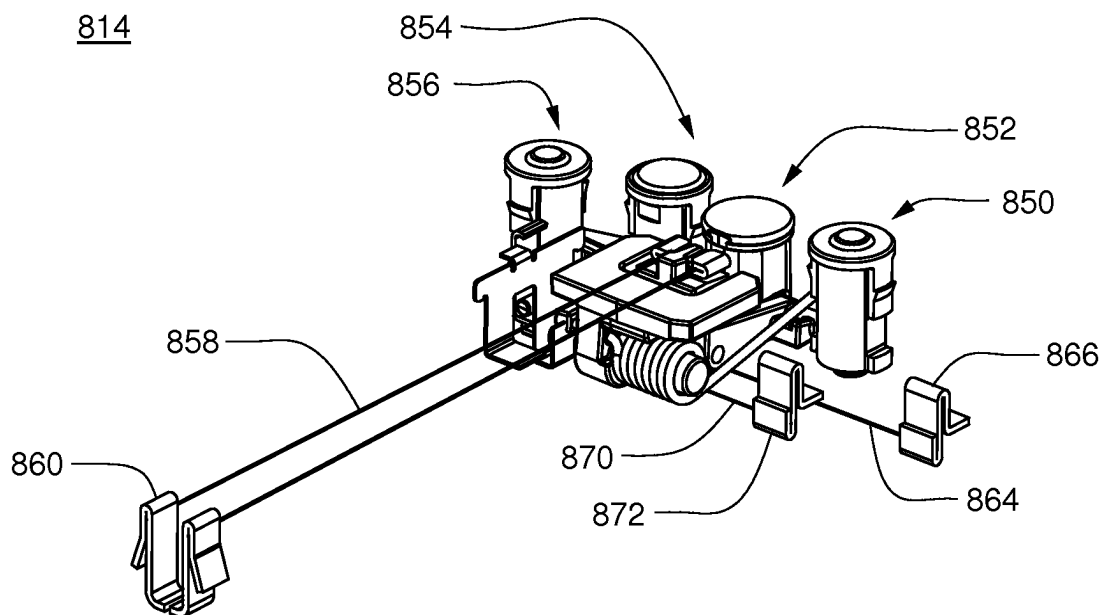


FIG. 43B

814

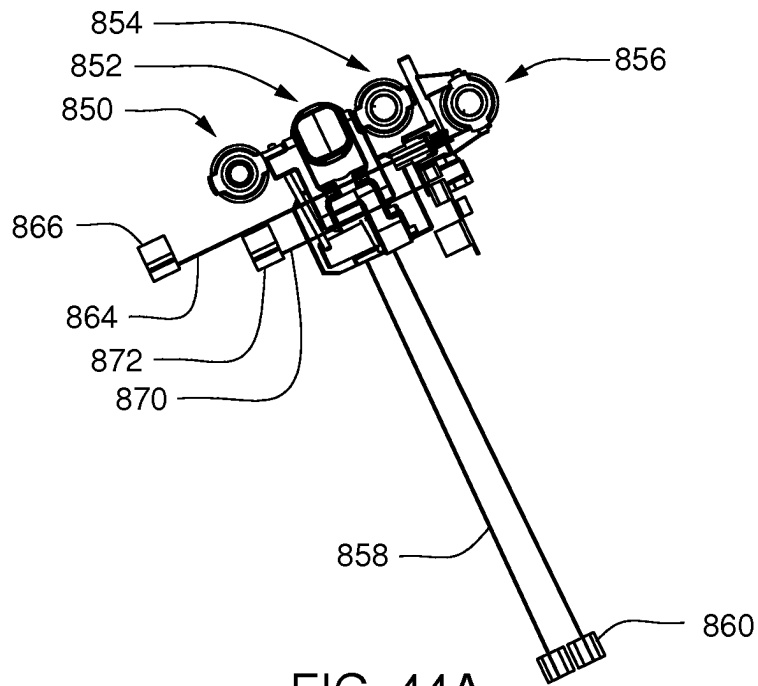


FIG. 44A

814

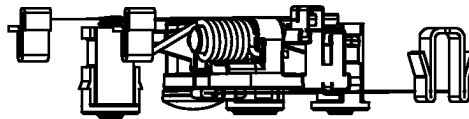


FIG. 44B

814

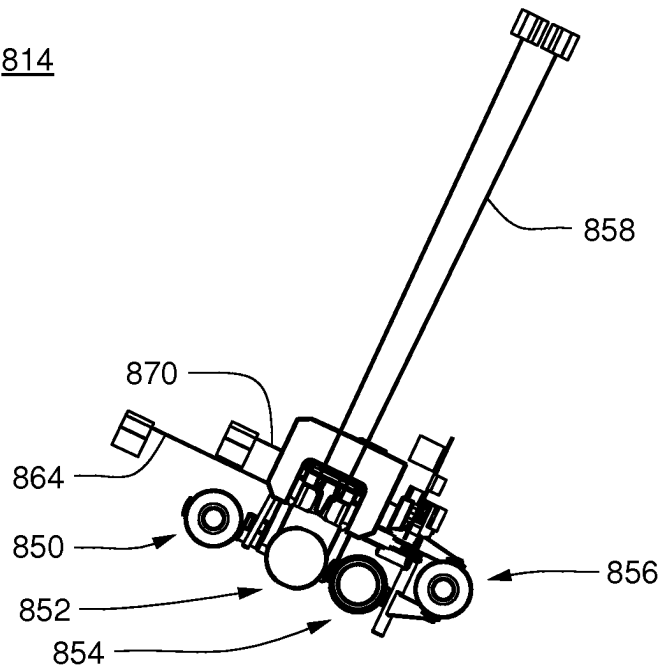
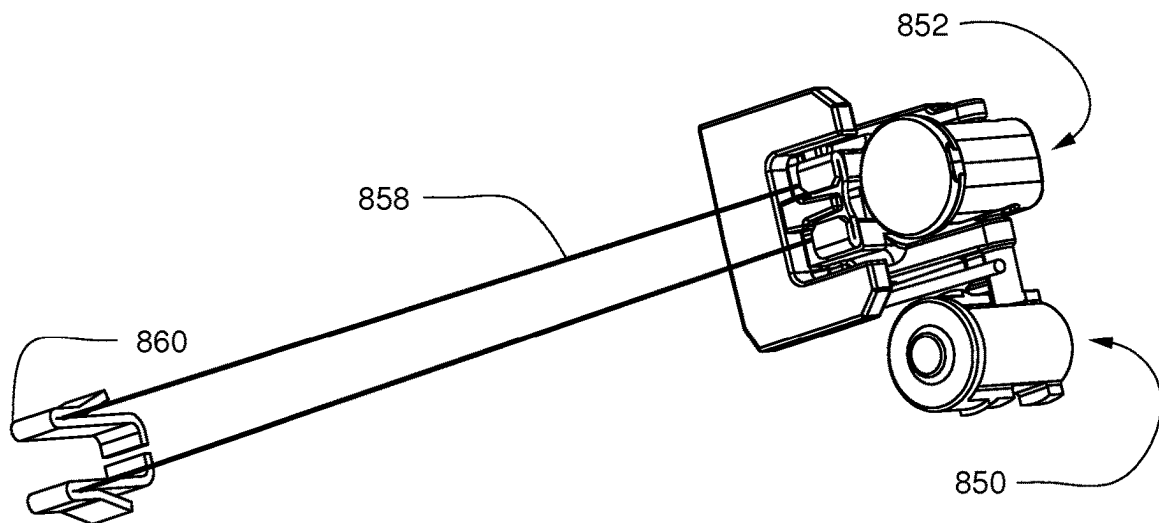
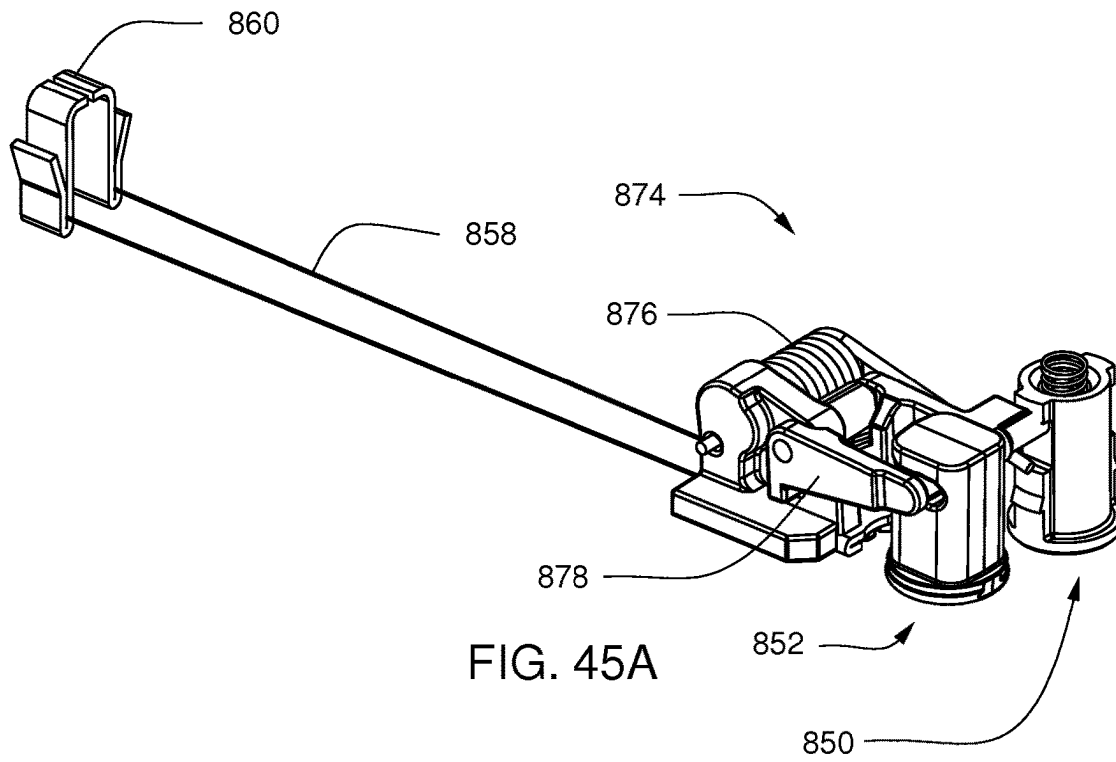
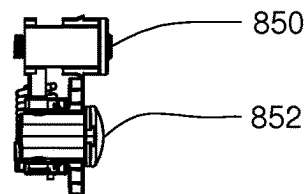
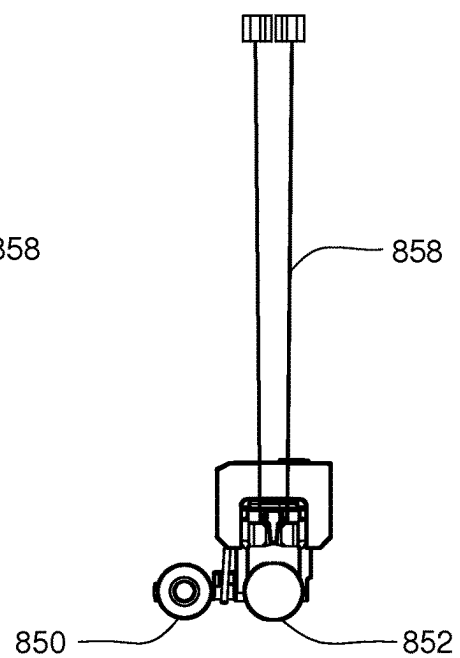
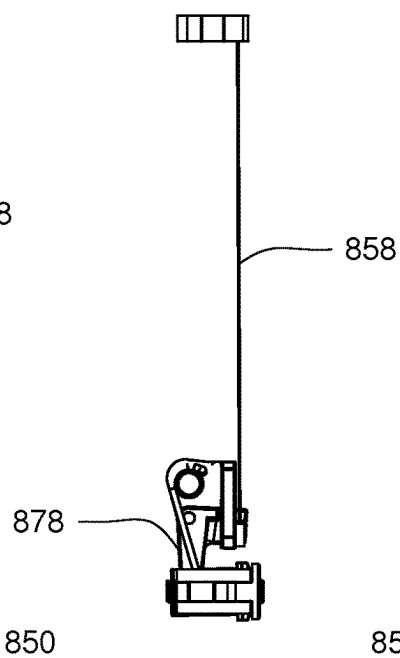
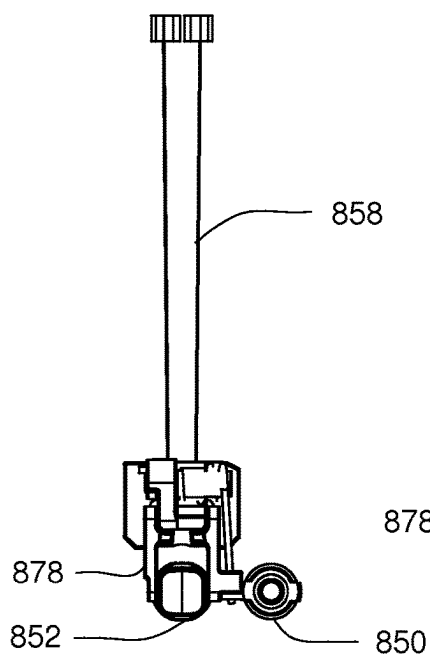
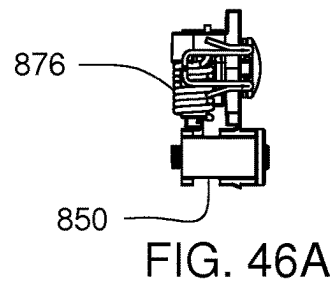


FIG. 44C





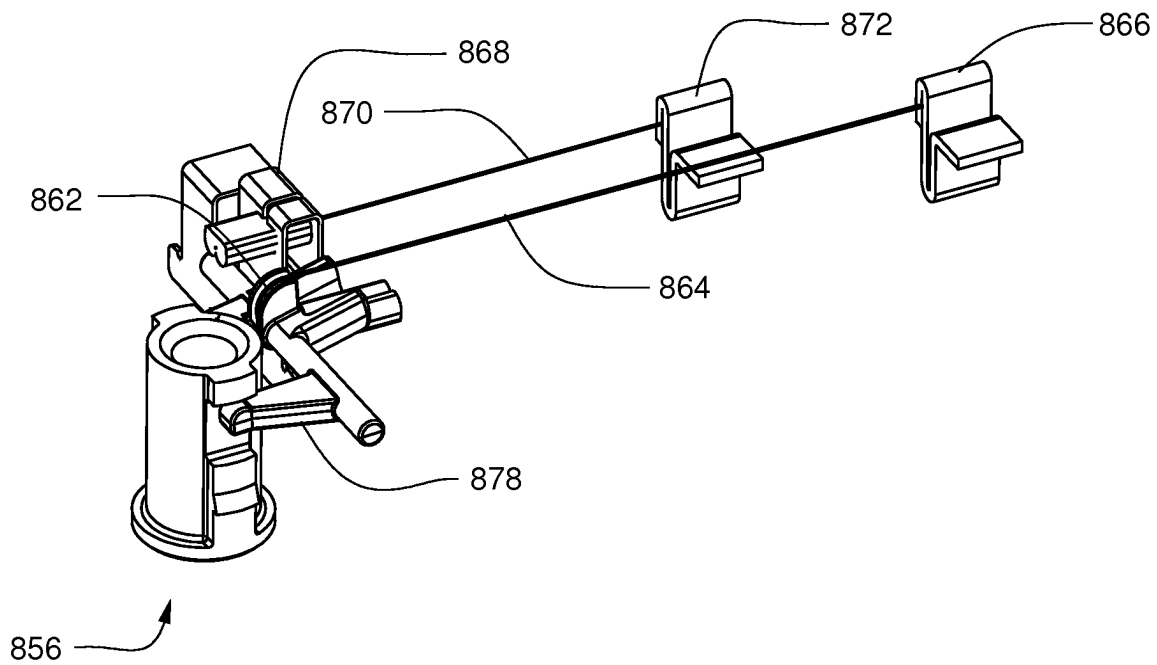


FIG. 47A

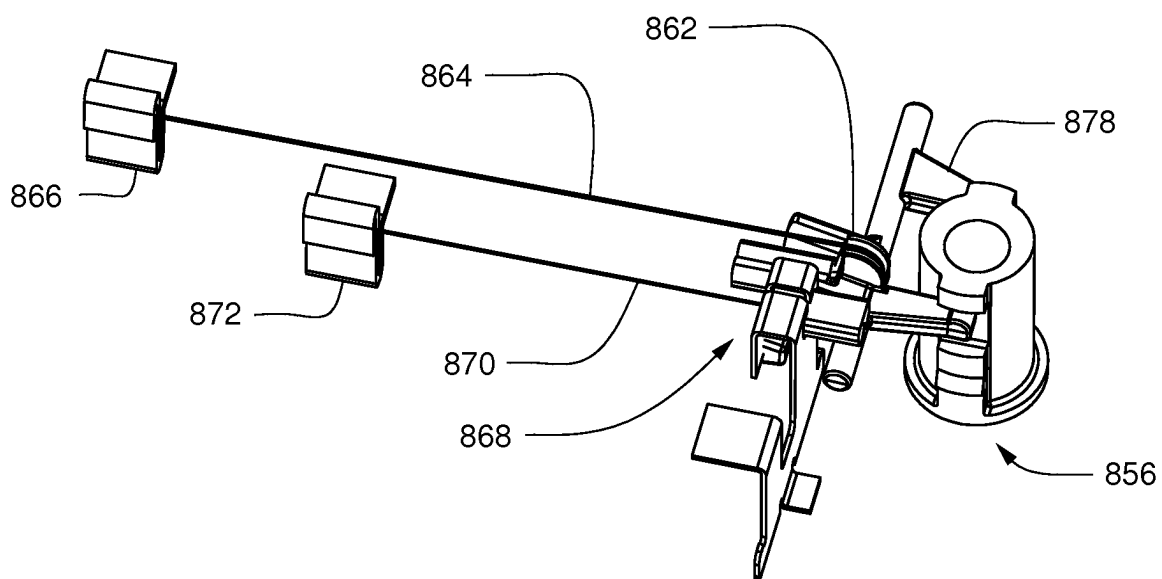


FIG. 47B

804

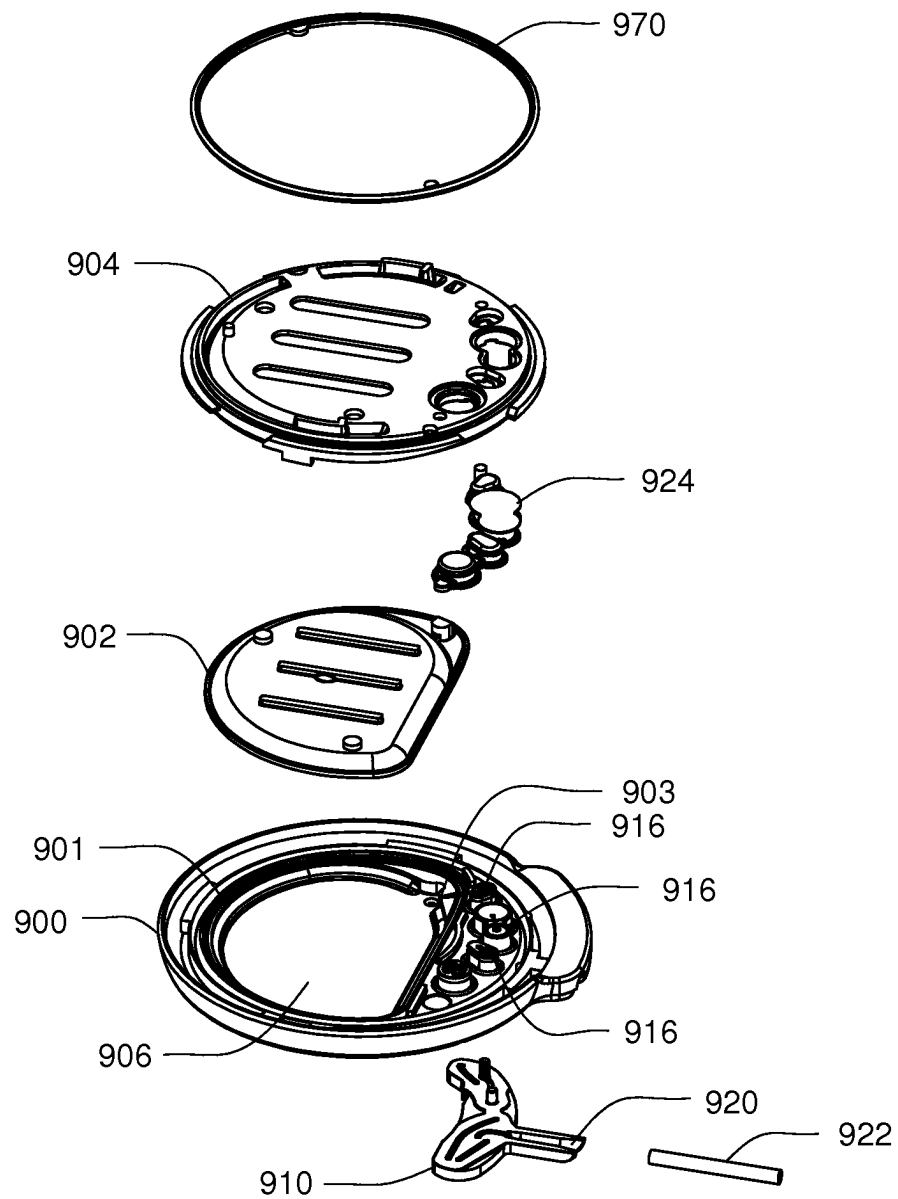


FIG. 48

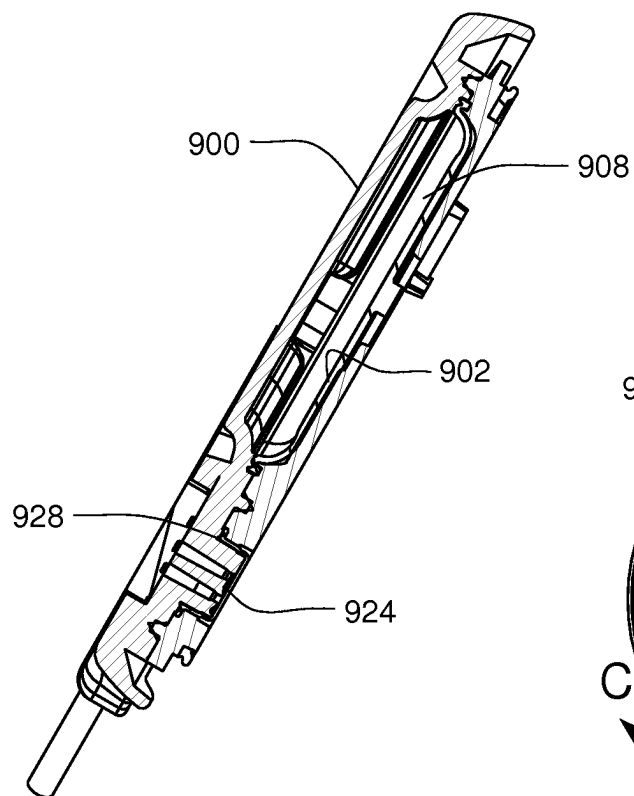


FIG. 49B

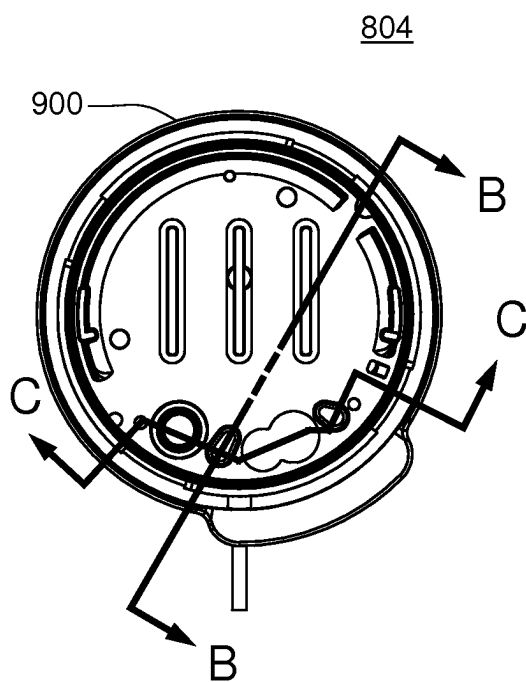


FIG. 49A

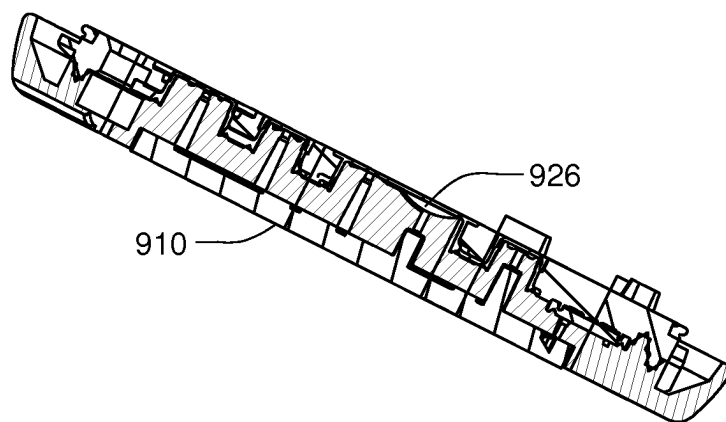


FIG. 49C

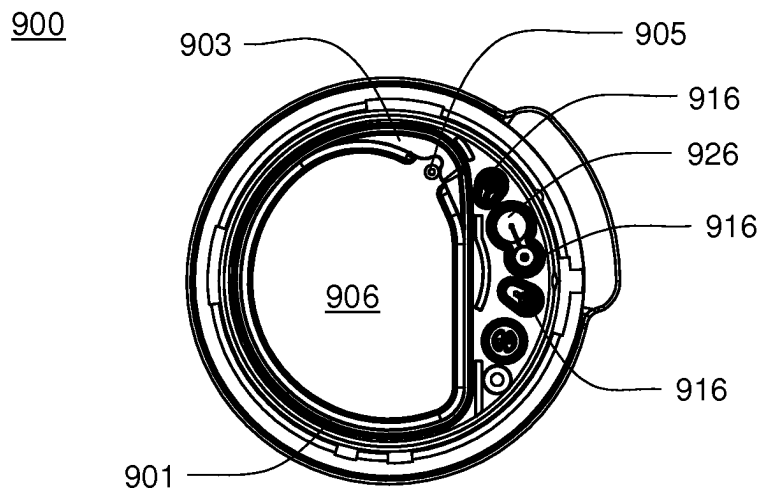


FIG. 50A

900

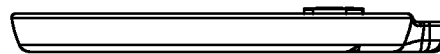


FIG. 50B

900

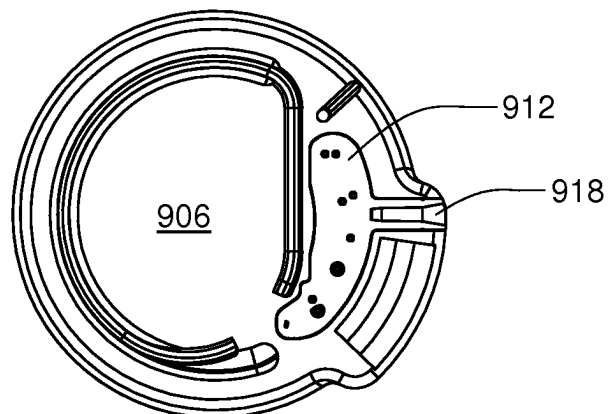


FIG. 50C

910

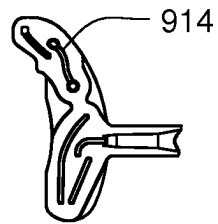


FIG. 51A

910

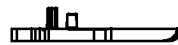


FIG. 51B

910

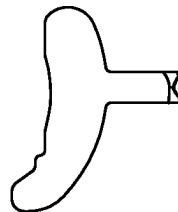


FIG. 51C

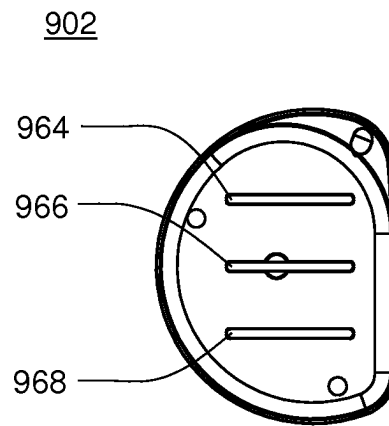


FIG. 52A

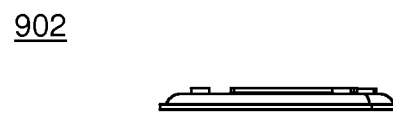


FIG. 52B

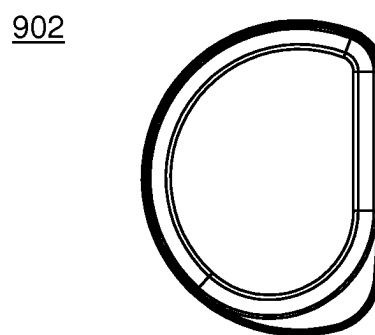


FIG. 52C

904

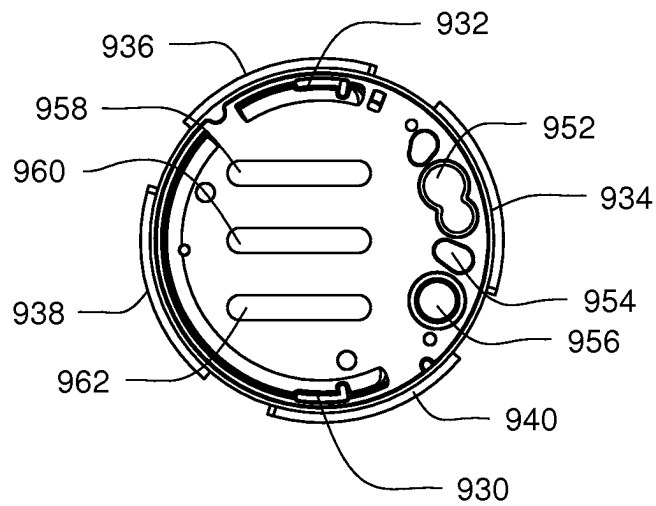


FIG. 53A

904

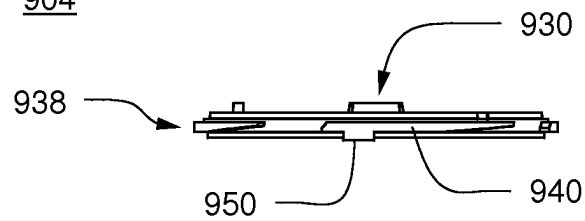


FIG. 53B

904

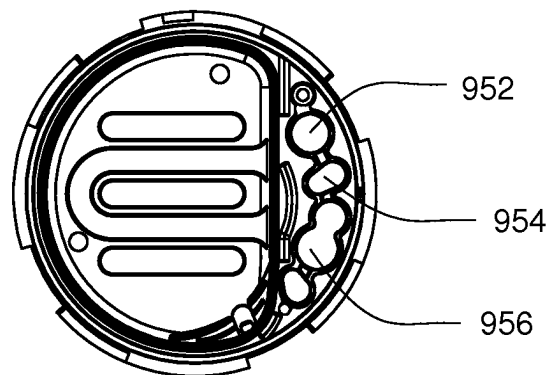


FIG. 53C

924



FIG. 54A

924



FIG. 54B

924



FIG. 54C

806

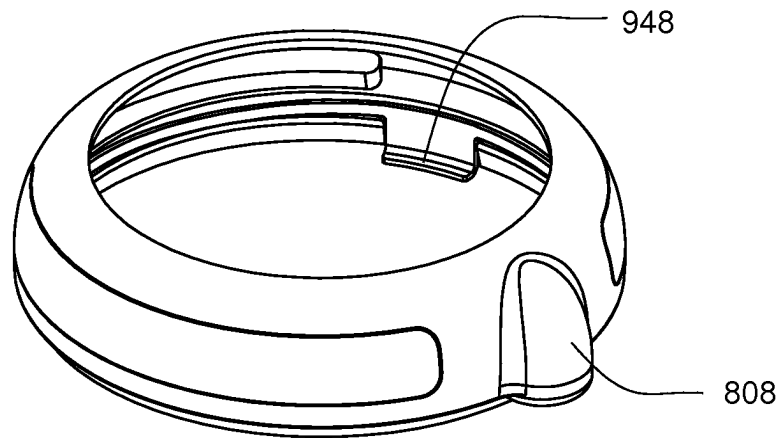


FIG. 55A

806

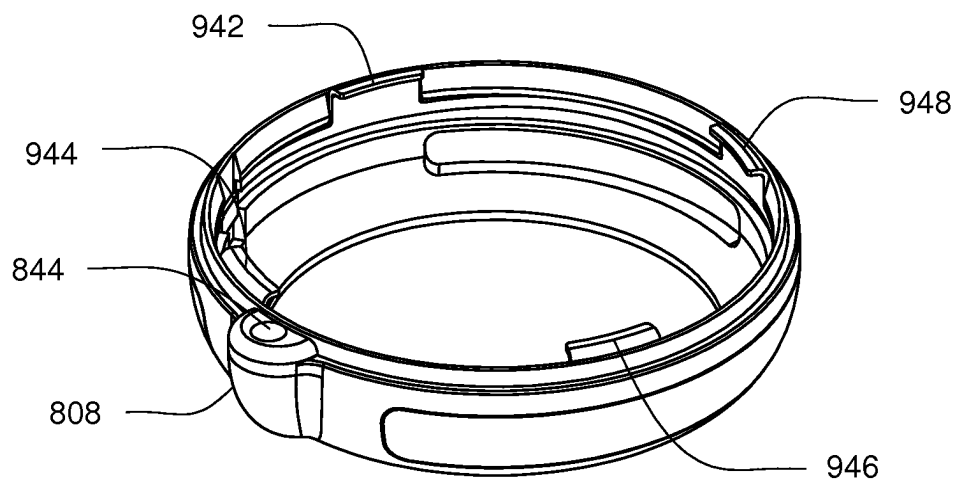


FIG. 55B

806

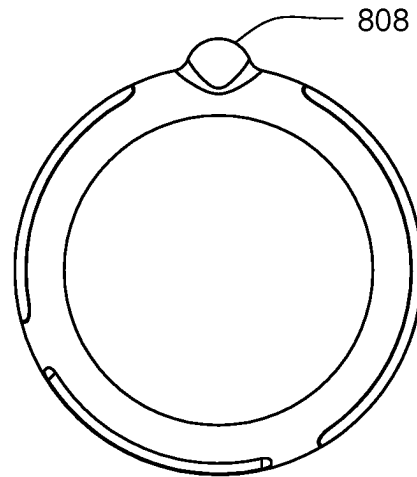


FIG. 56A

806

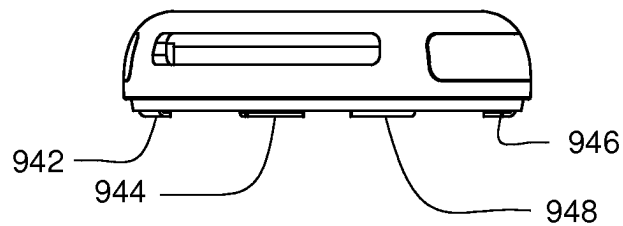


FIG. 56B

806

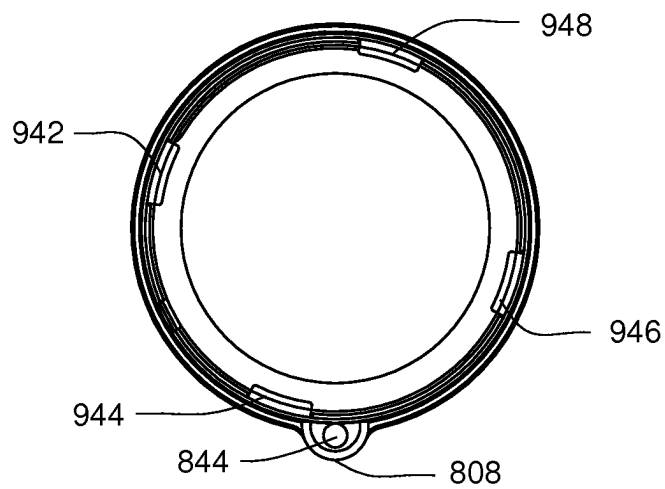


FIG. 56C

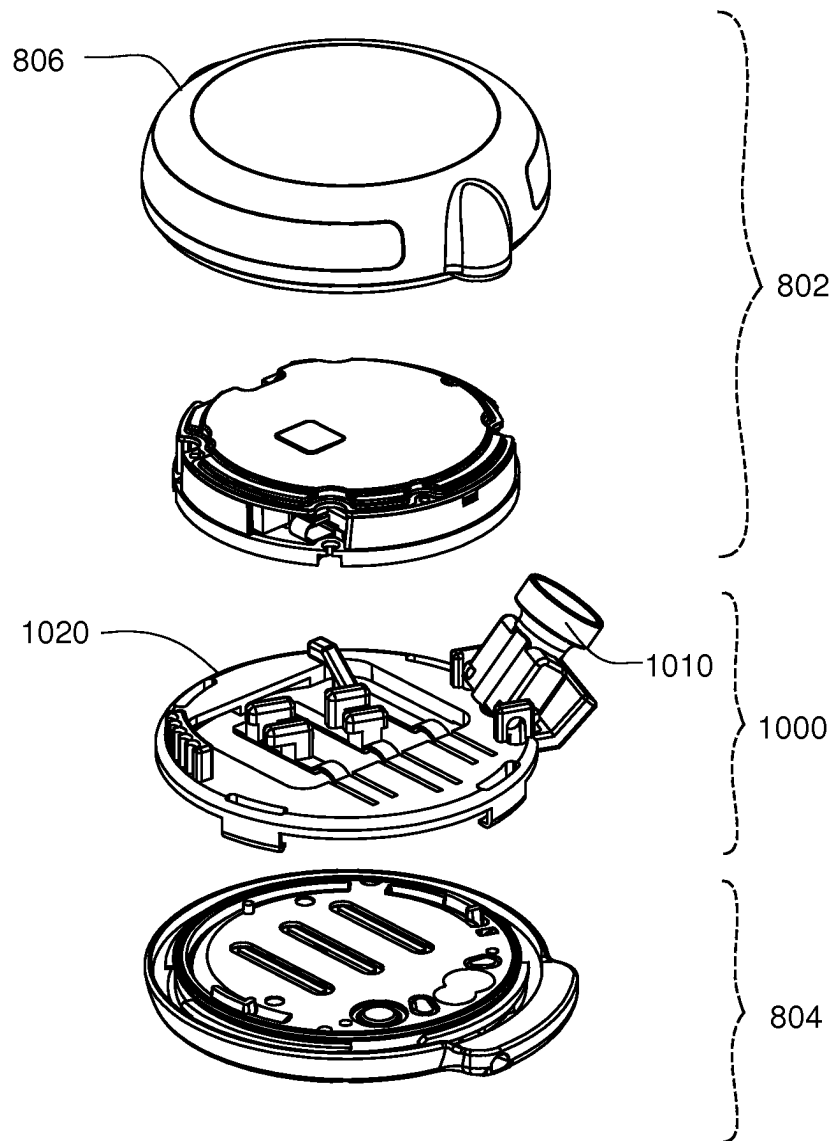


FIG. 57

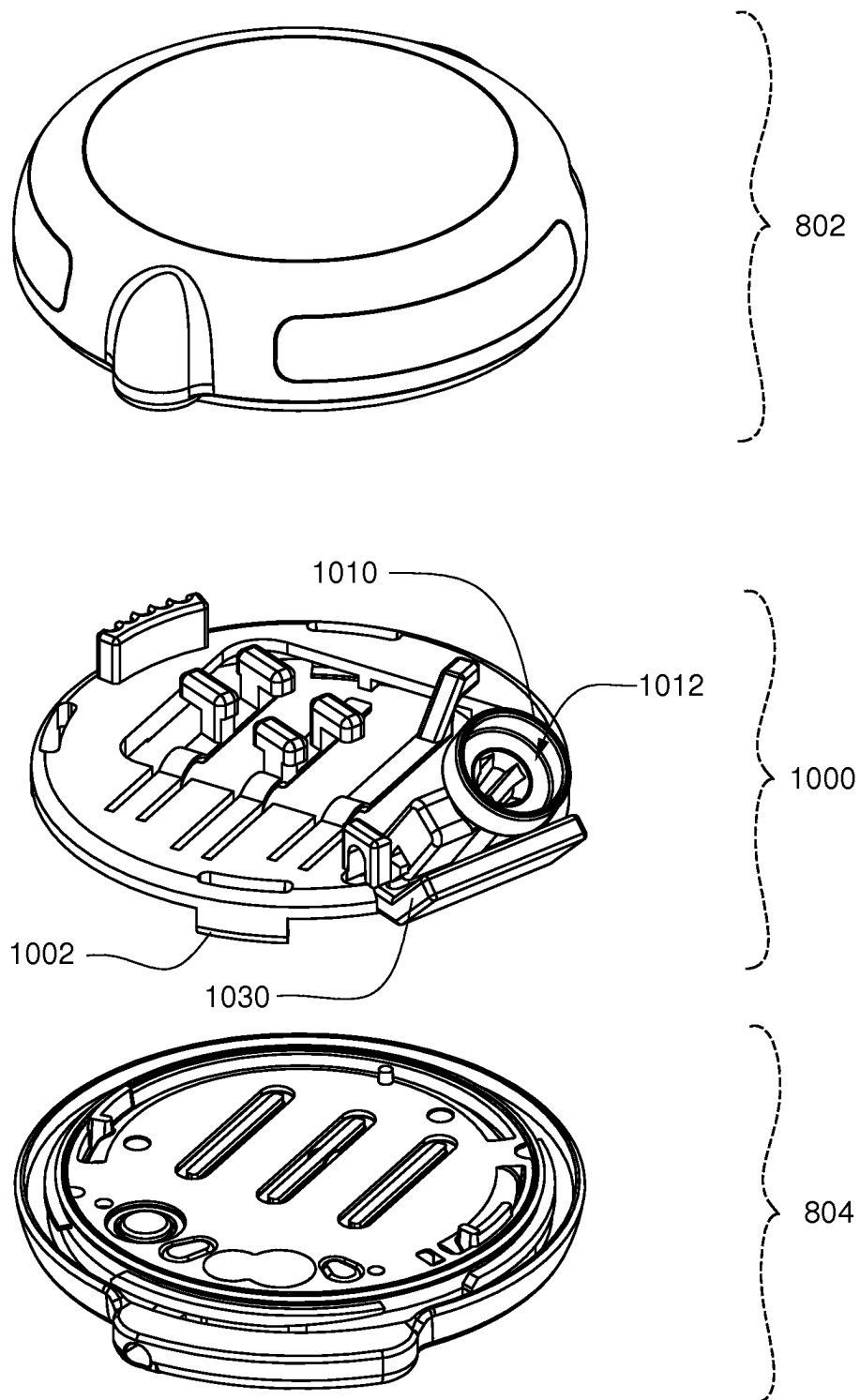


FIG. 58

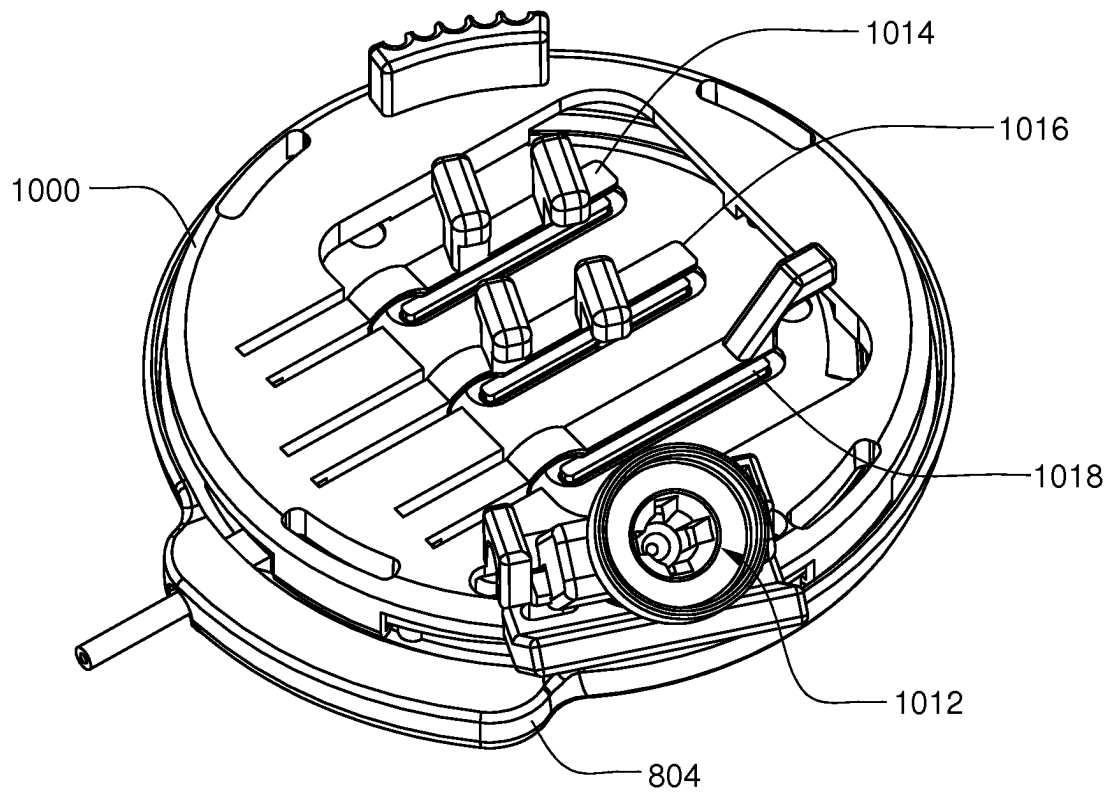


FIG. 59

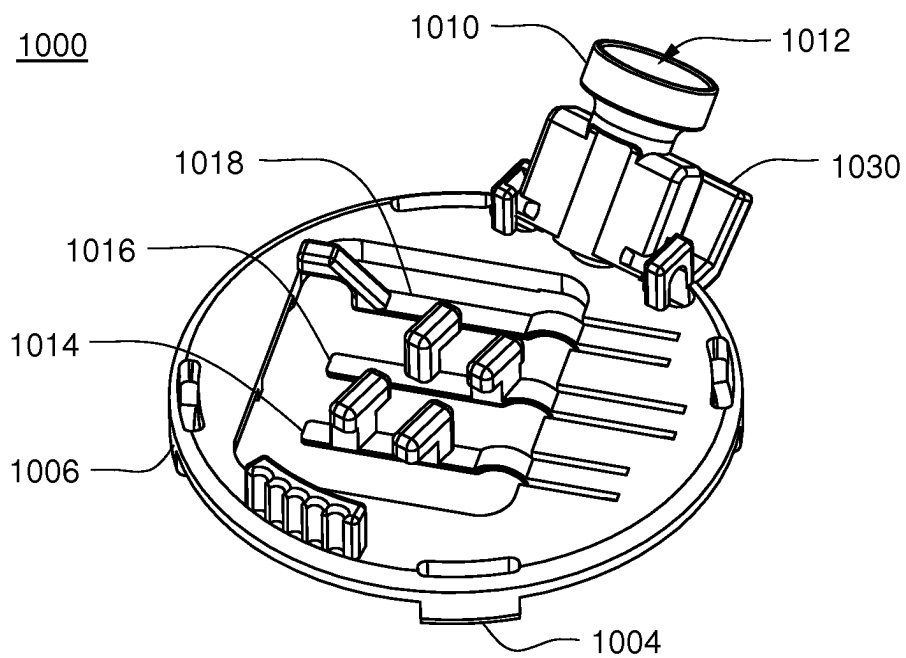


FIG. 60

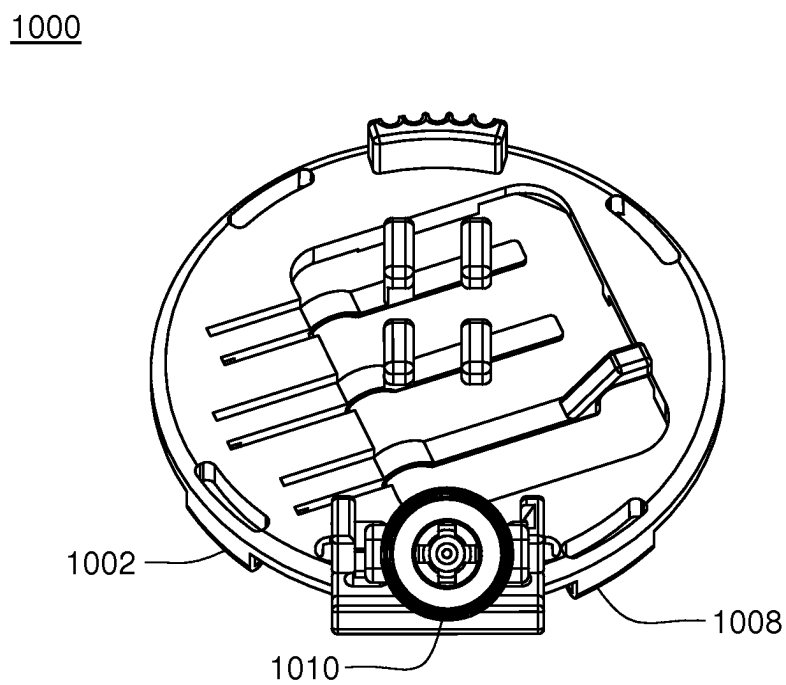


FIG. 61

1000

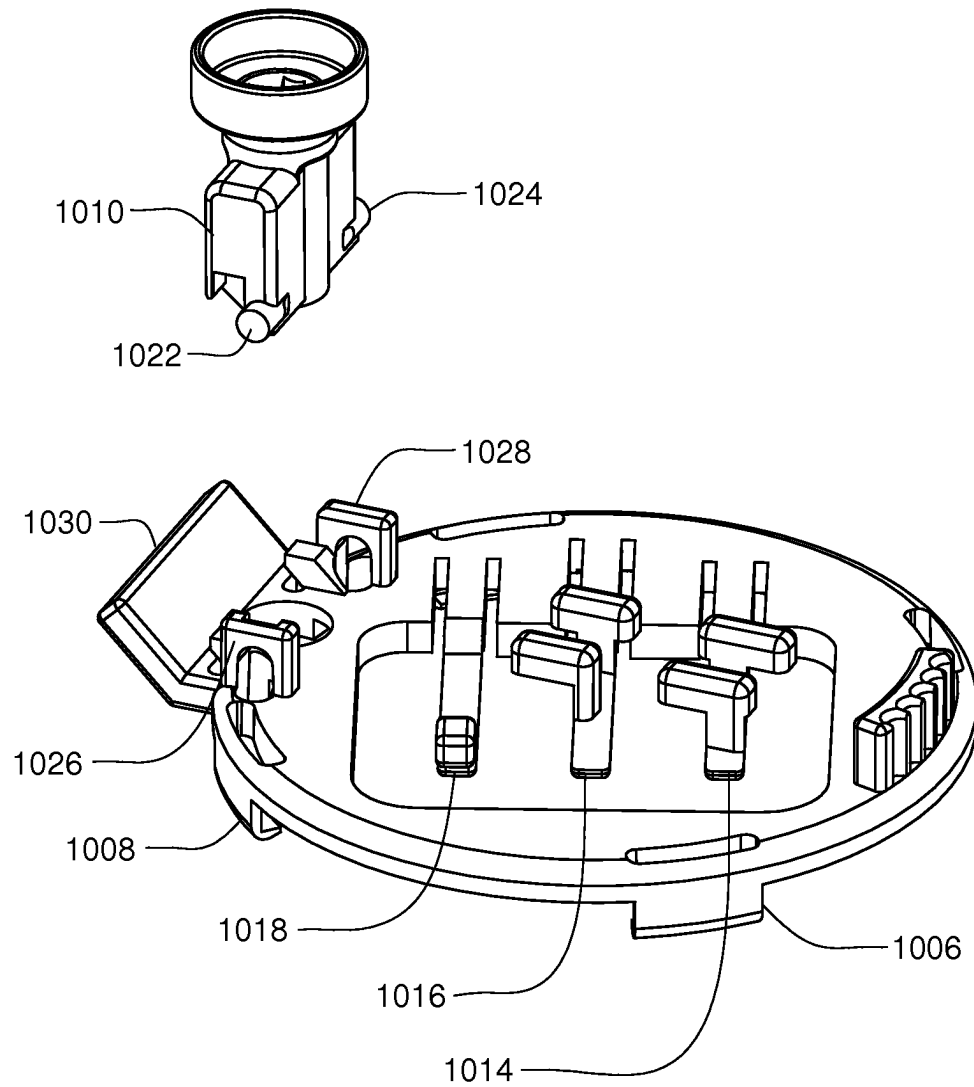


FIG. 62

1000

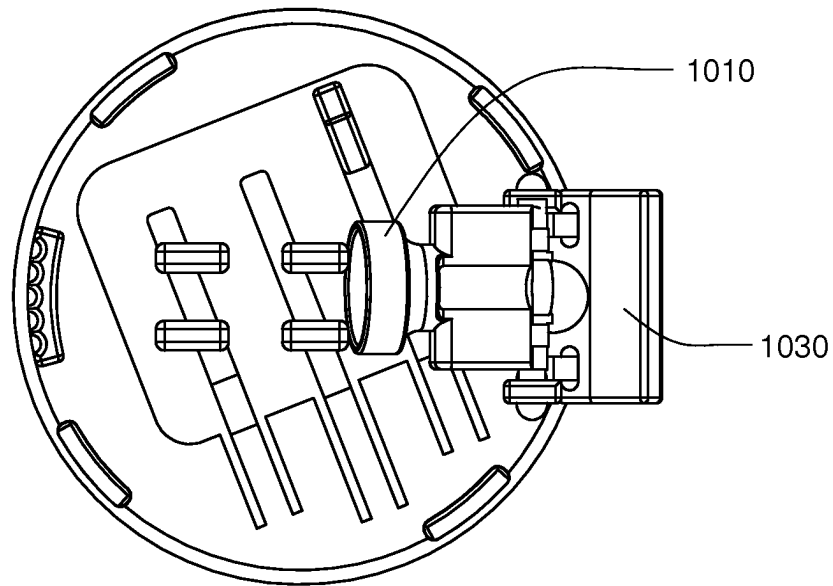


FIG. 63

1000

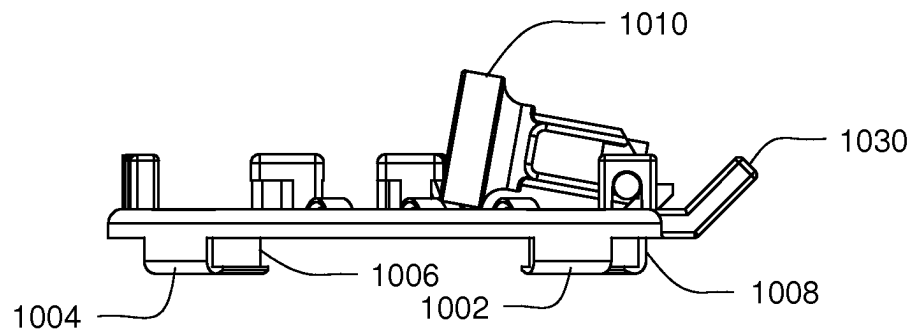


FIG. 64

1050

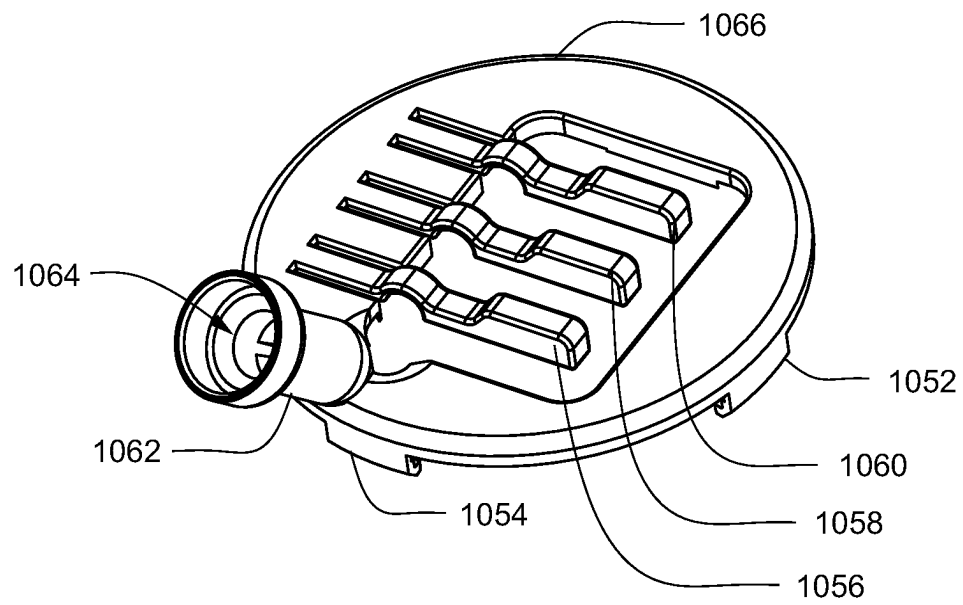


FIG. 65

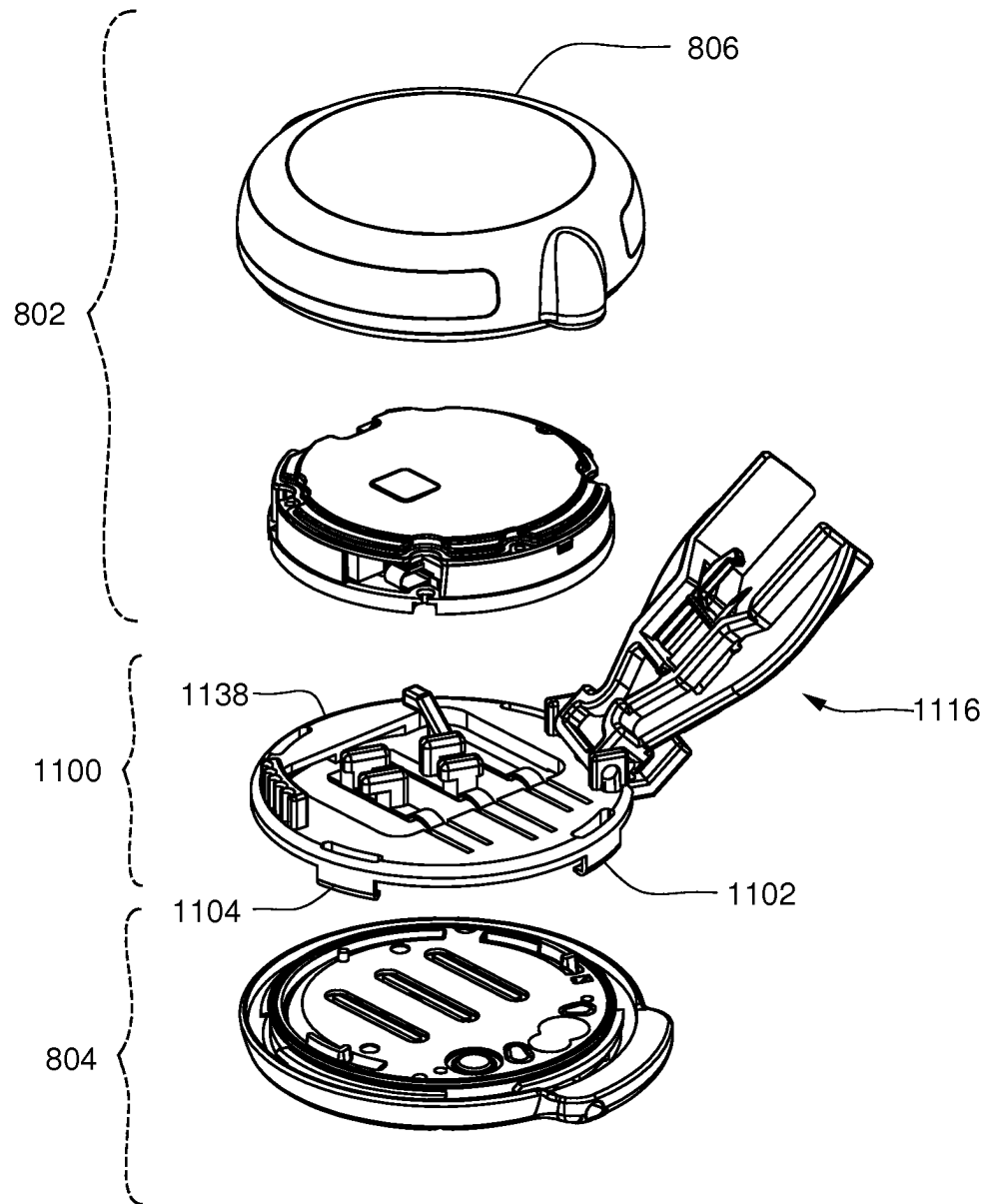


FIG. 66

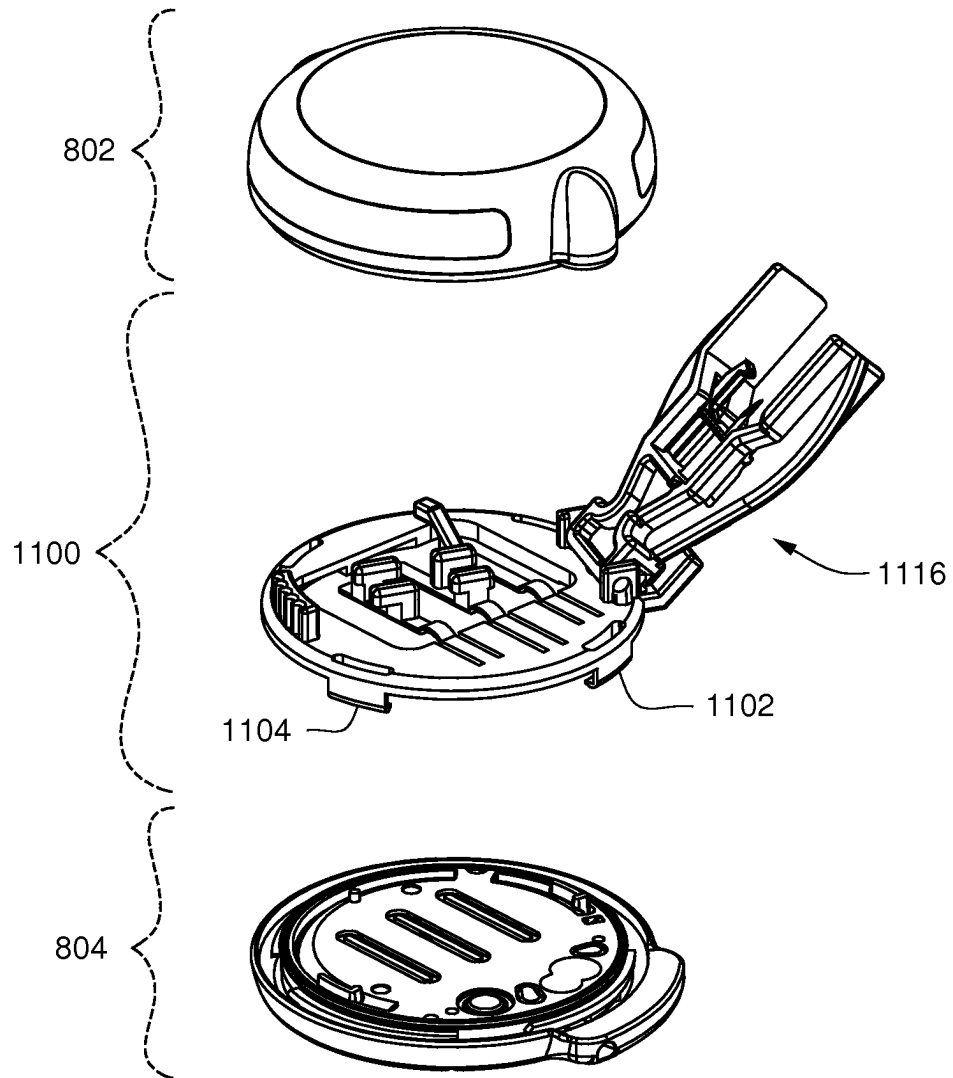


FIG. 67

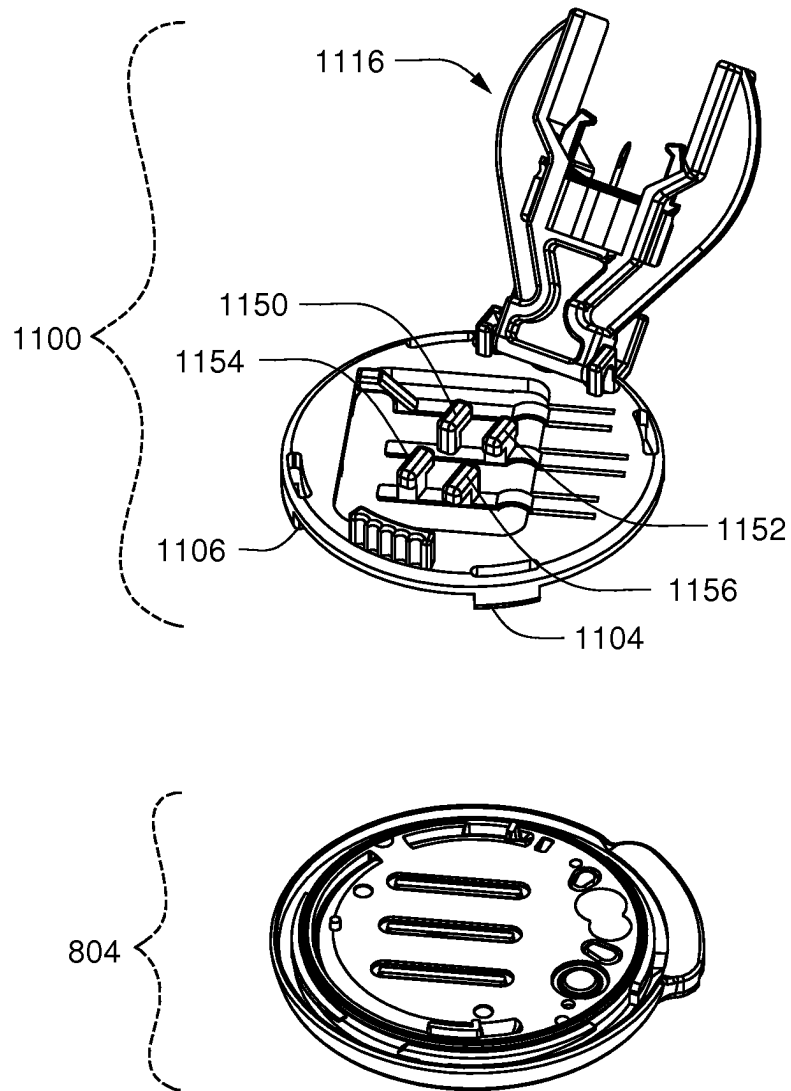


FIG. 68

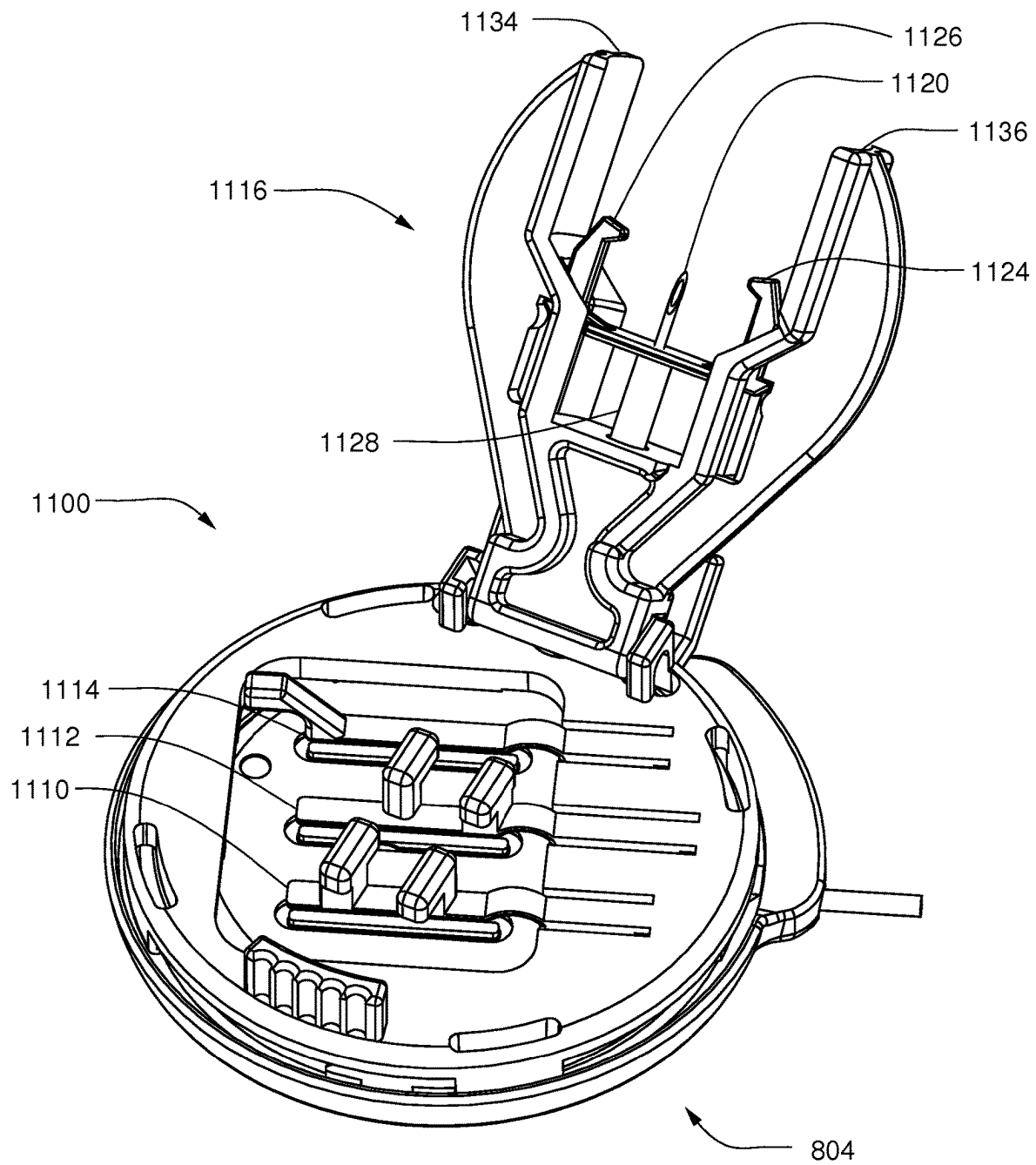


FIG. 69

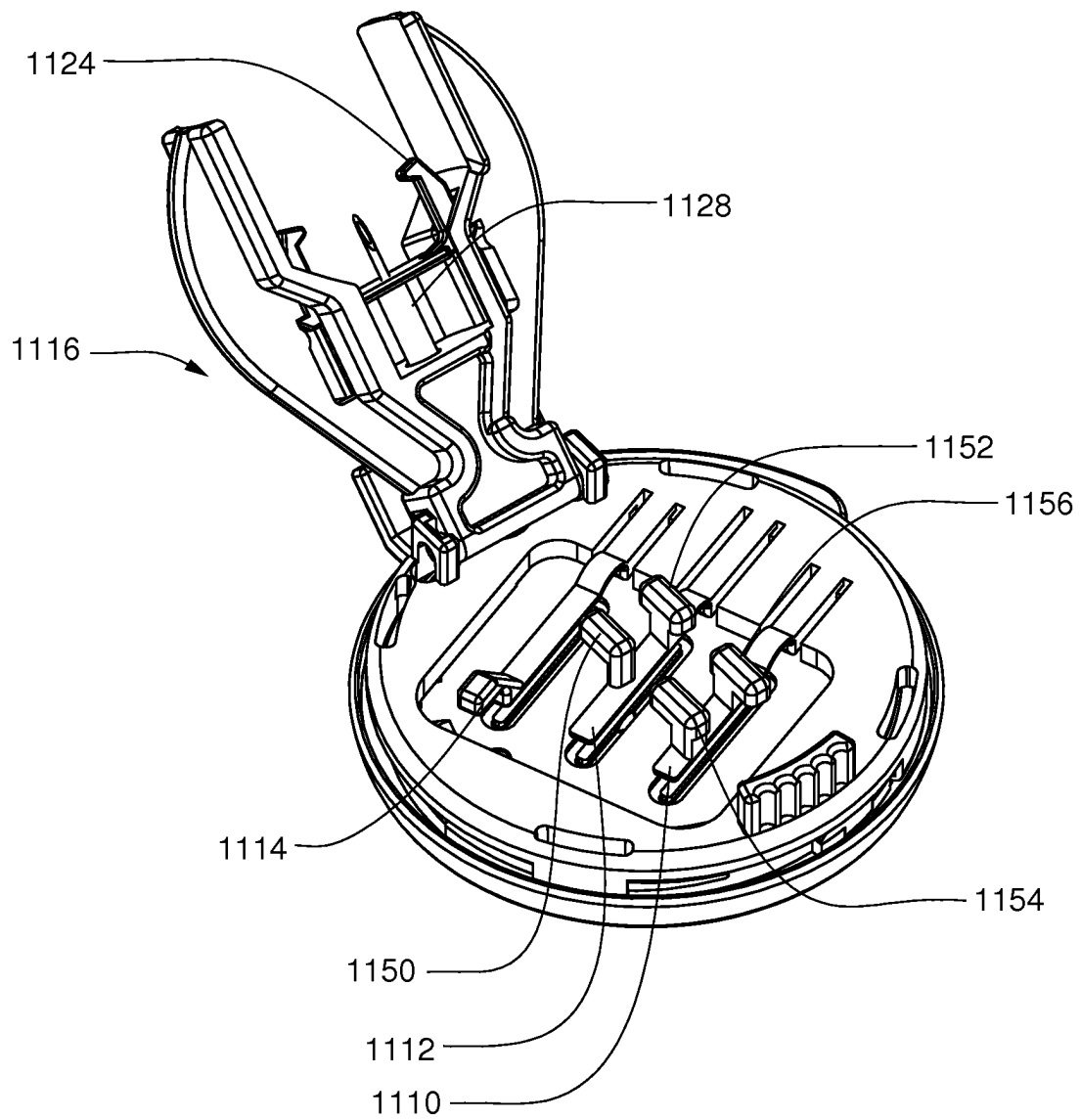


FIG. 70

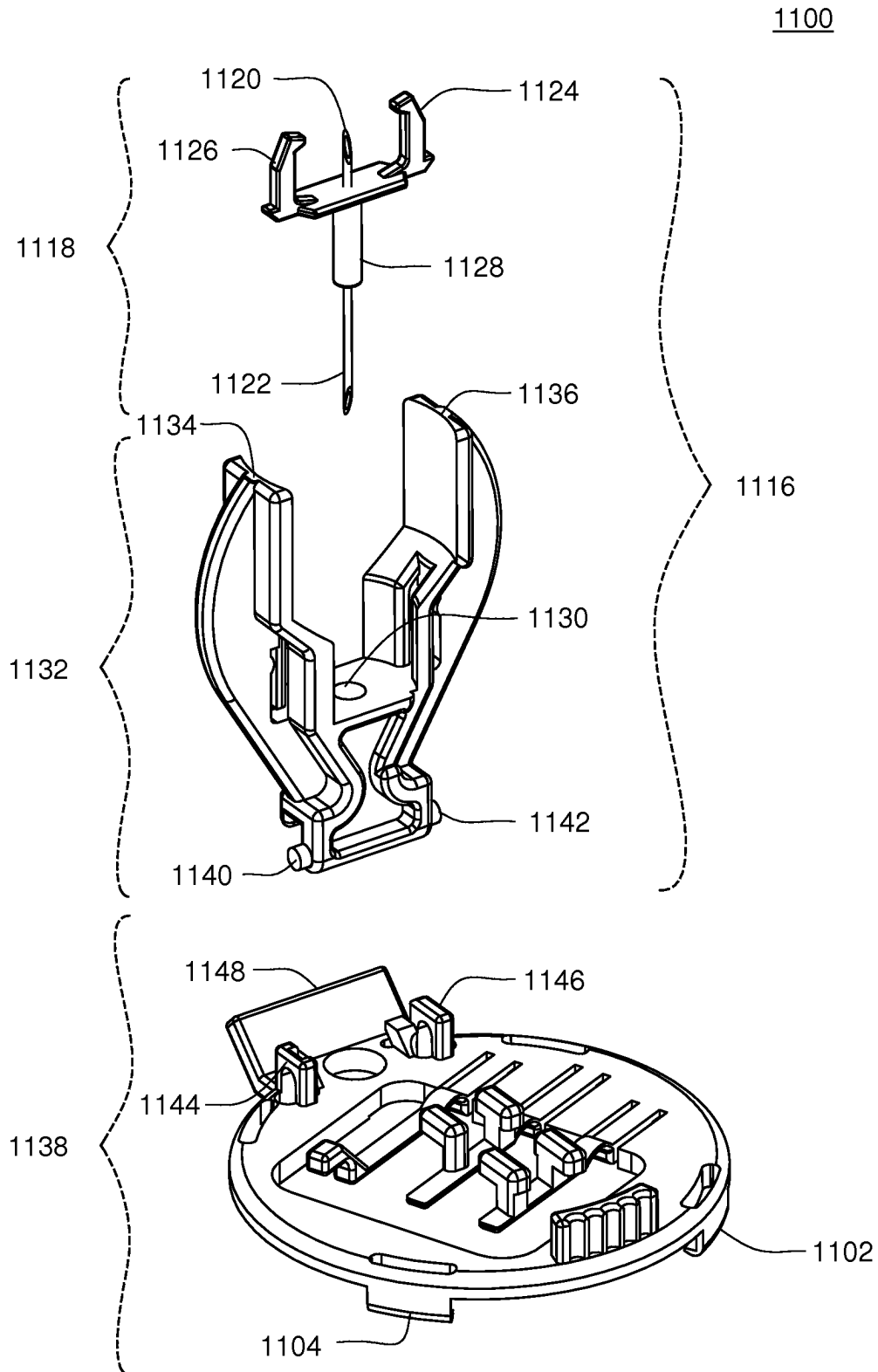


FIG. 71

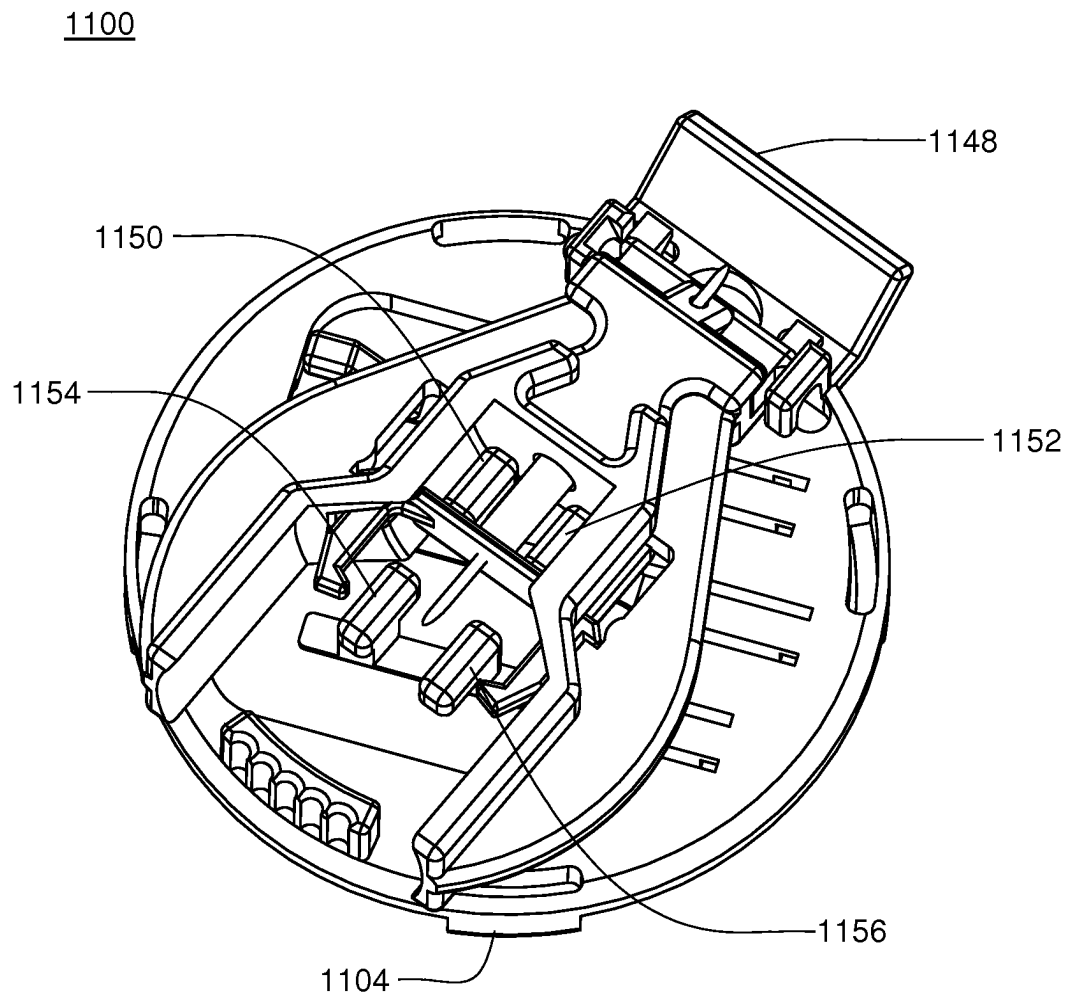


FIG. 72

1100

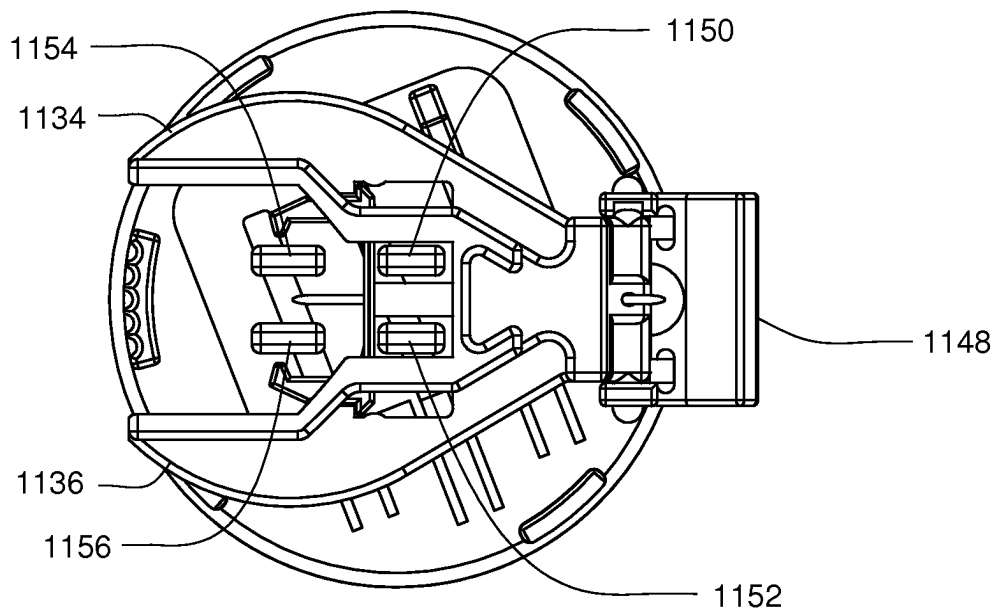


FIG. 73

1100

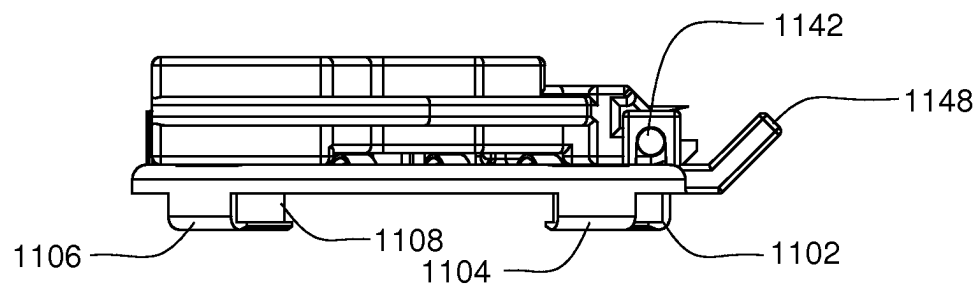


FIG. 74

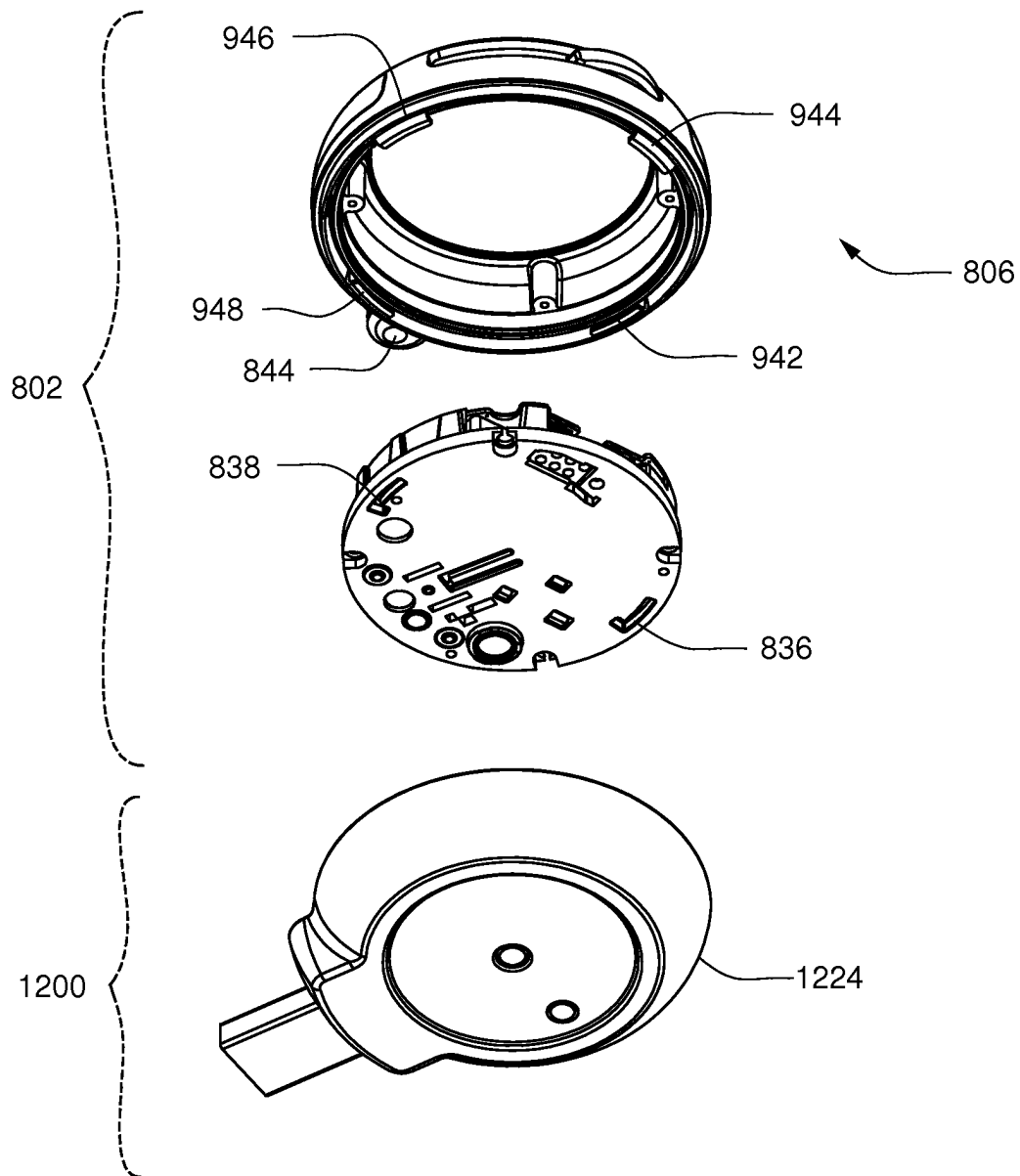


FIG. 75

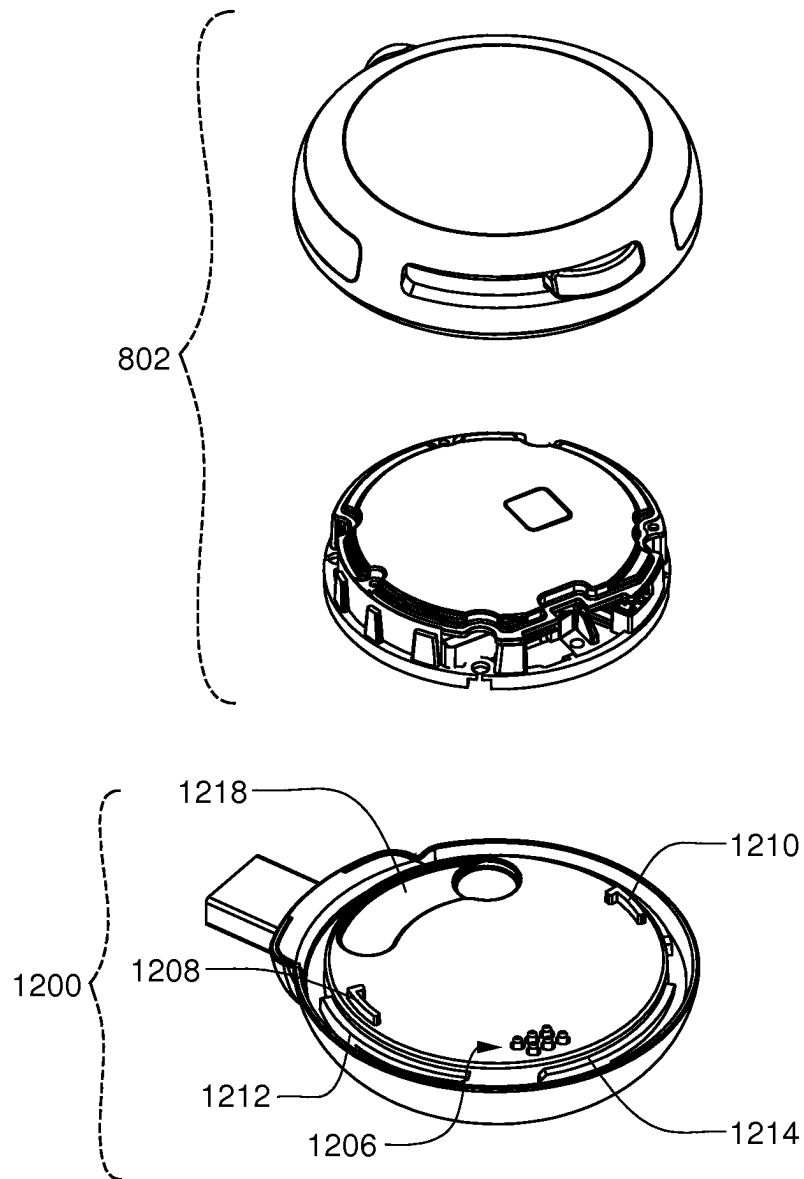


FIG. 76

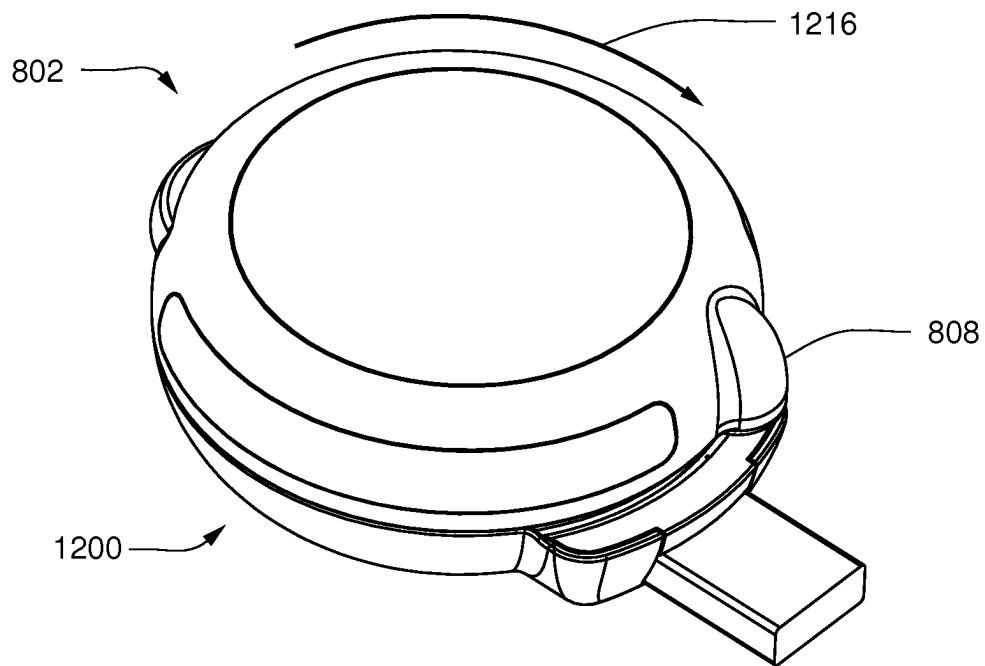


FIG. 77

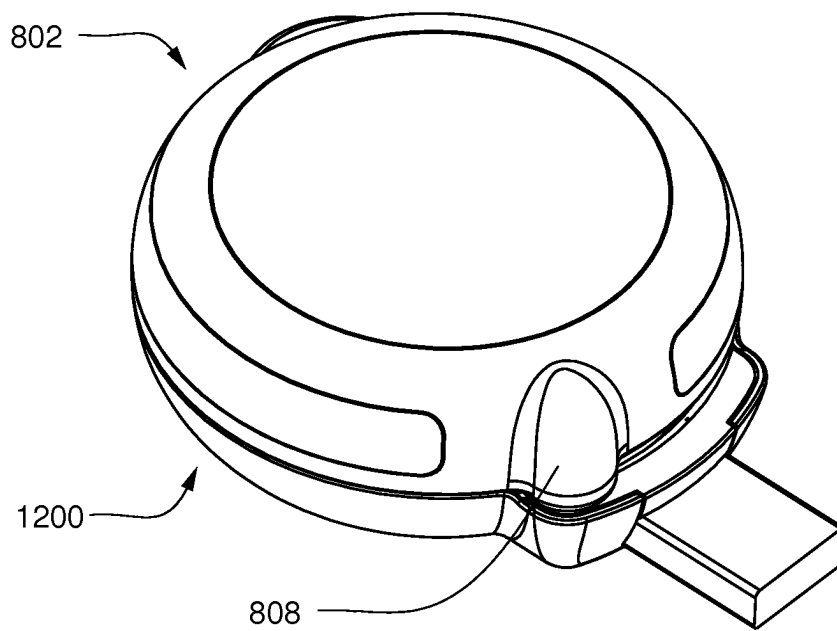


FIG. 78

1200

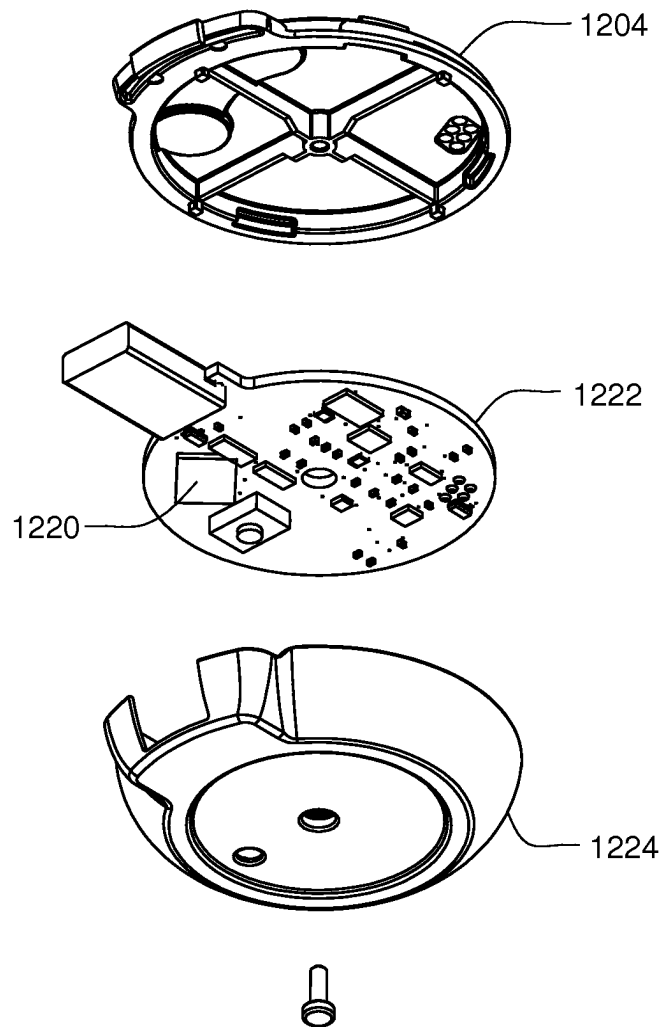


FIG. 79

1200

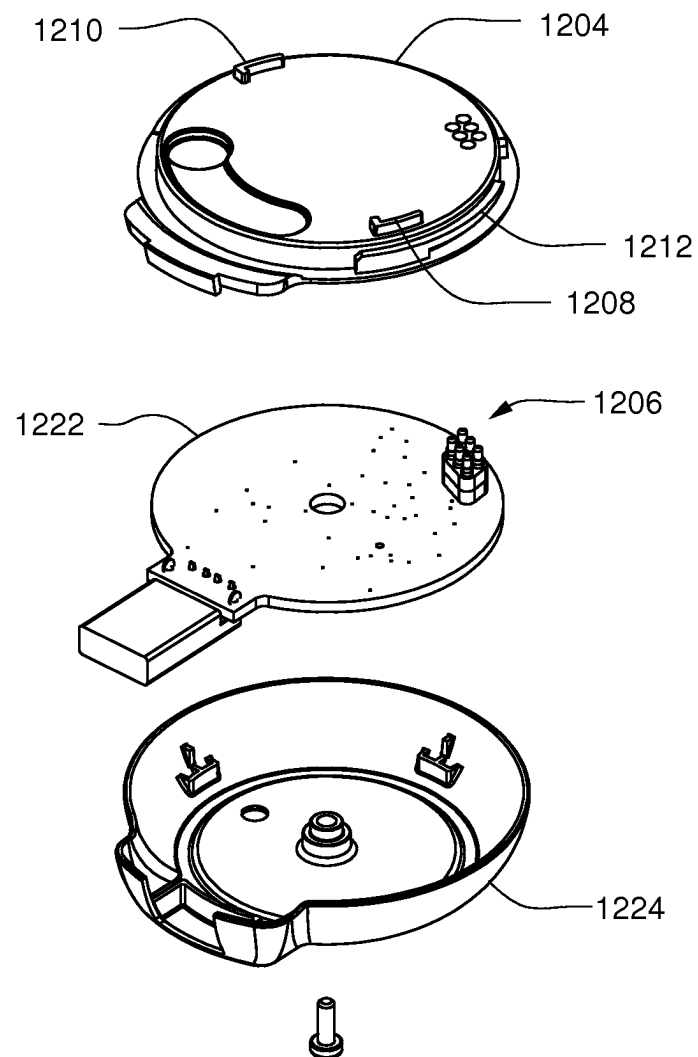


FIG. 80

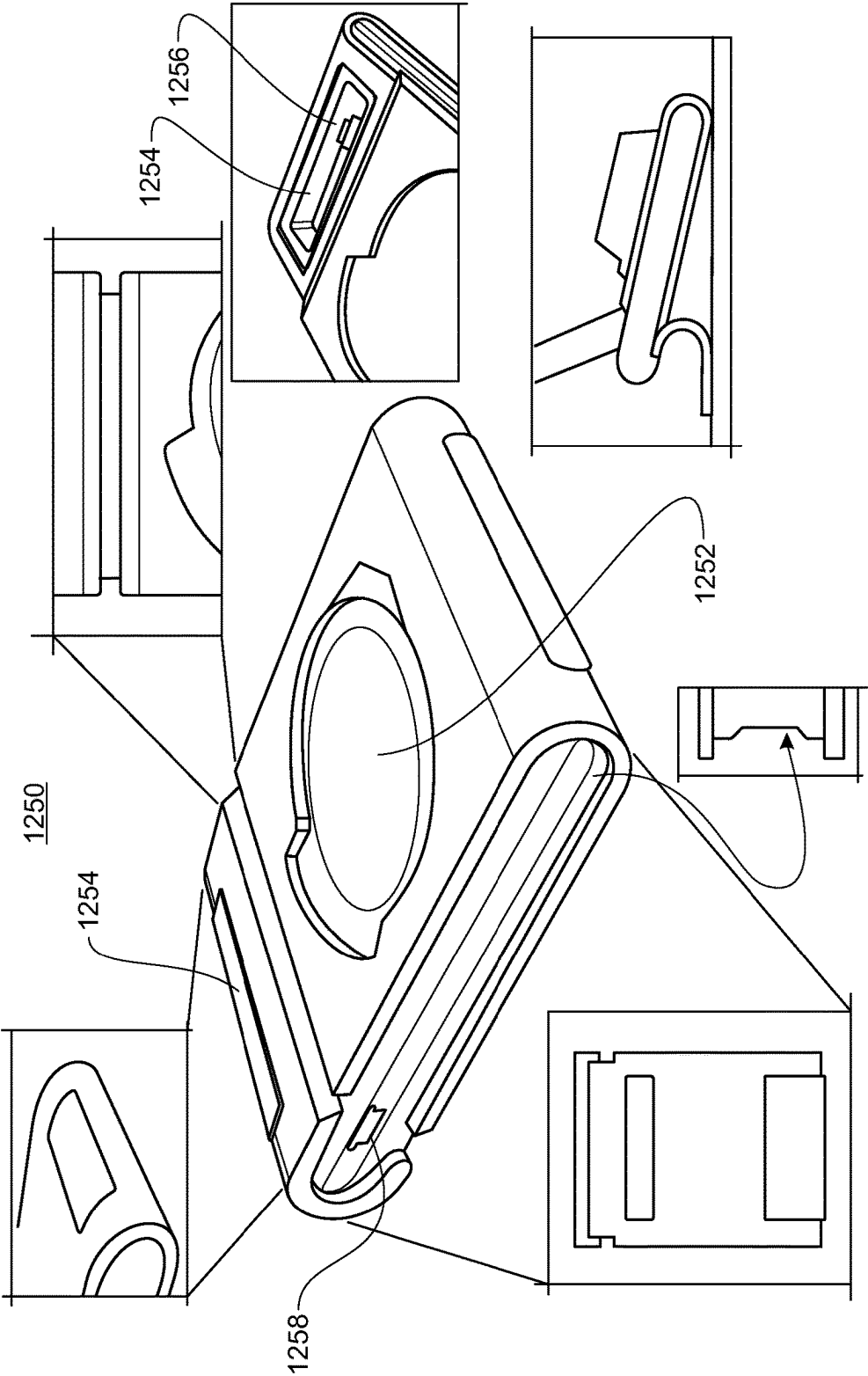


FIG. 81

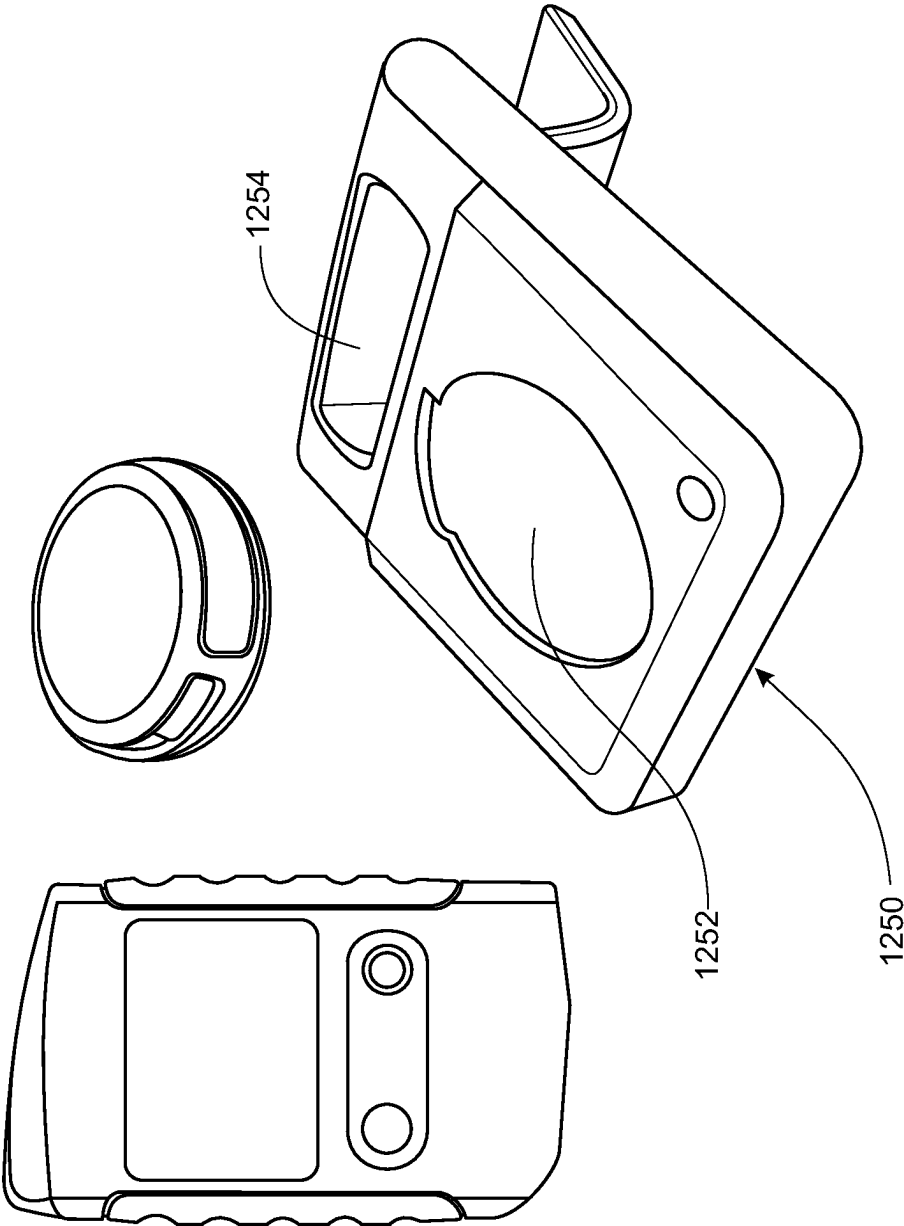


FIG.82

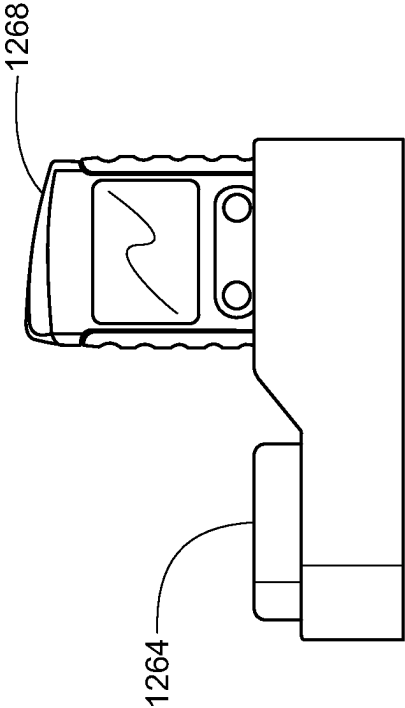


FIG. 83B

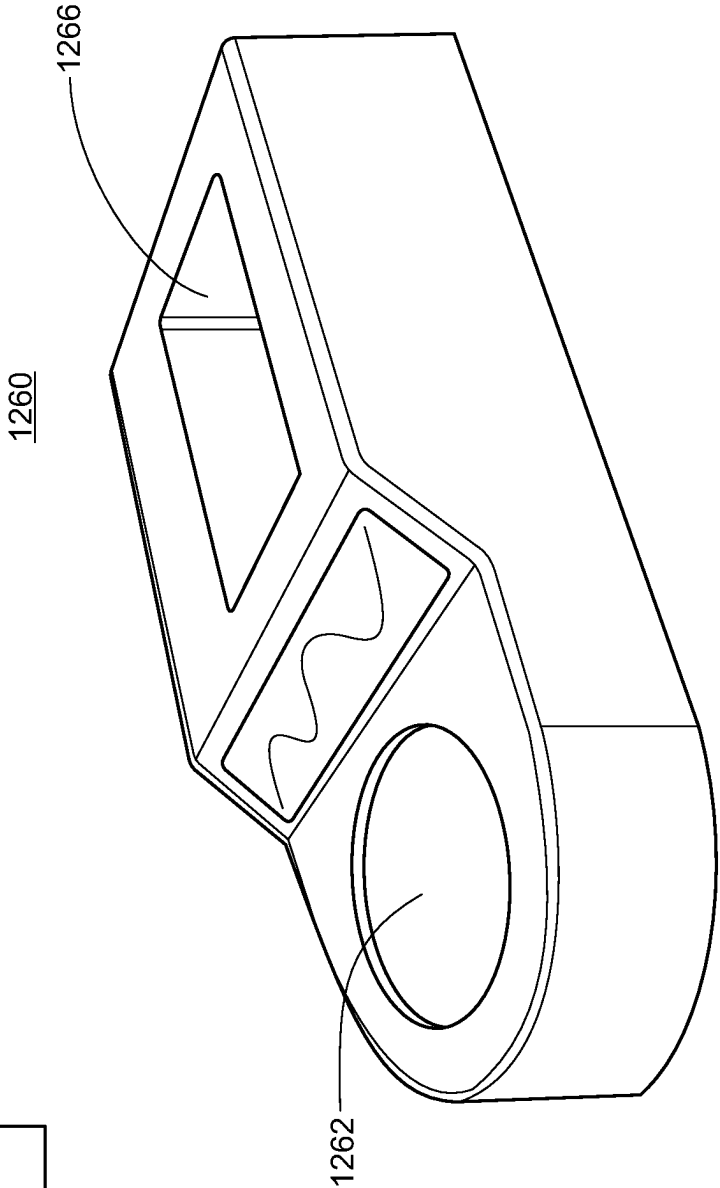


FIG. 83A

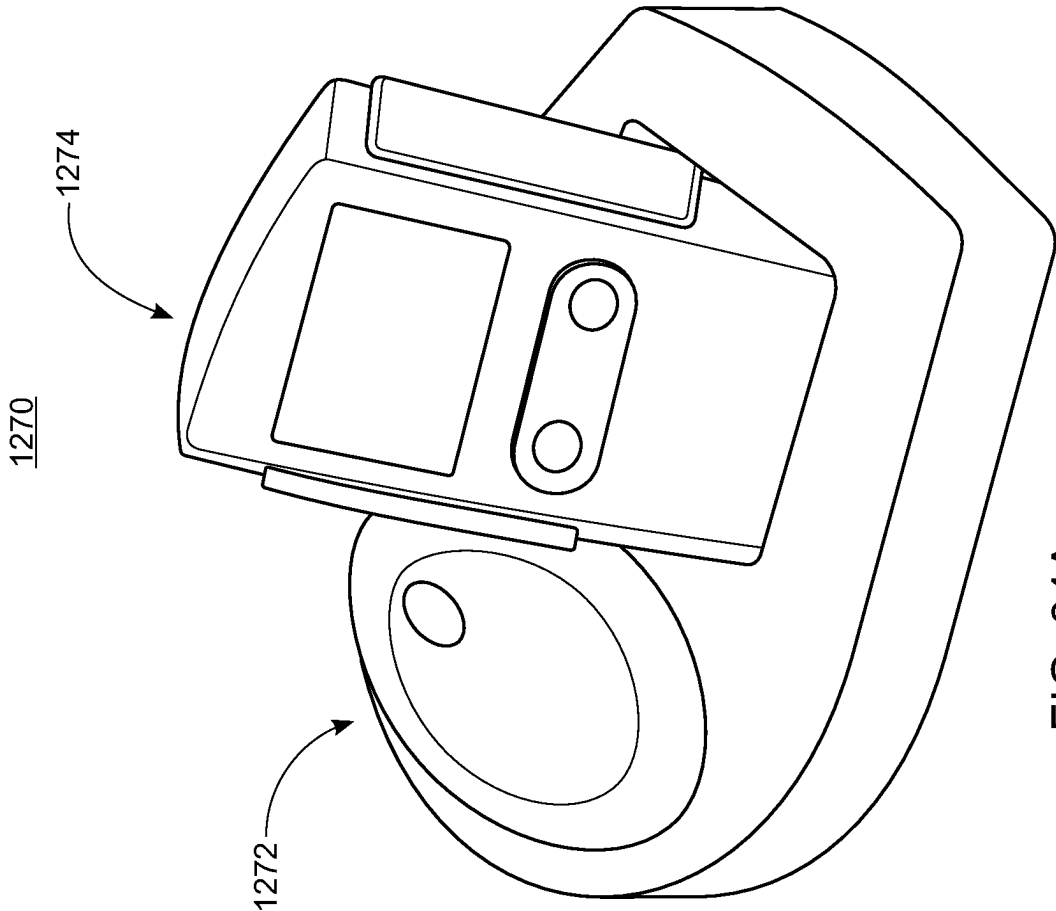


FIG. 84A

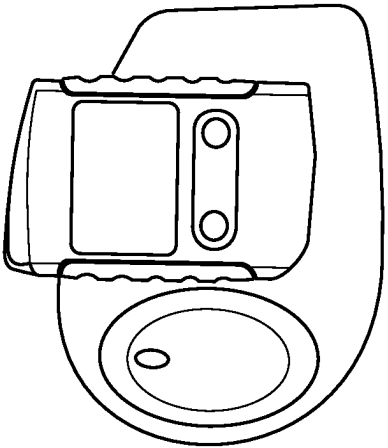


FIG. 84B

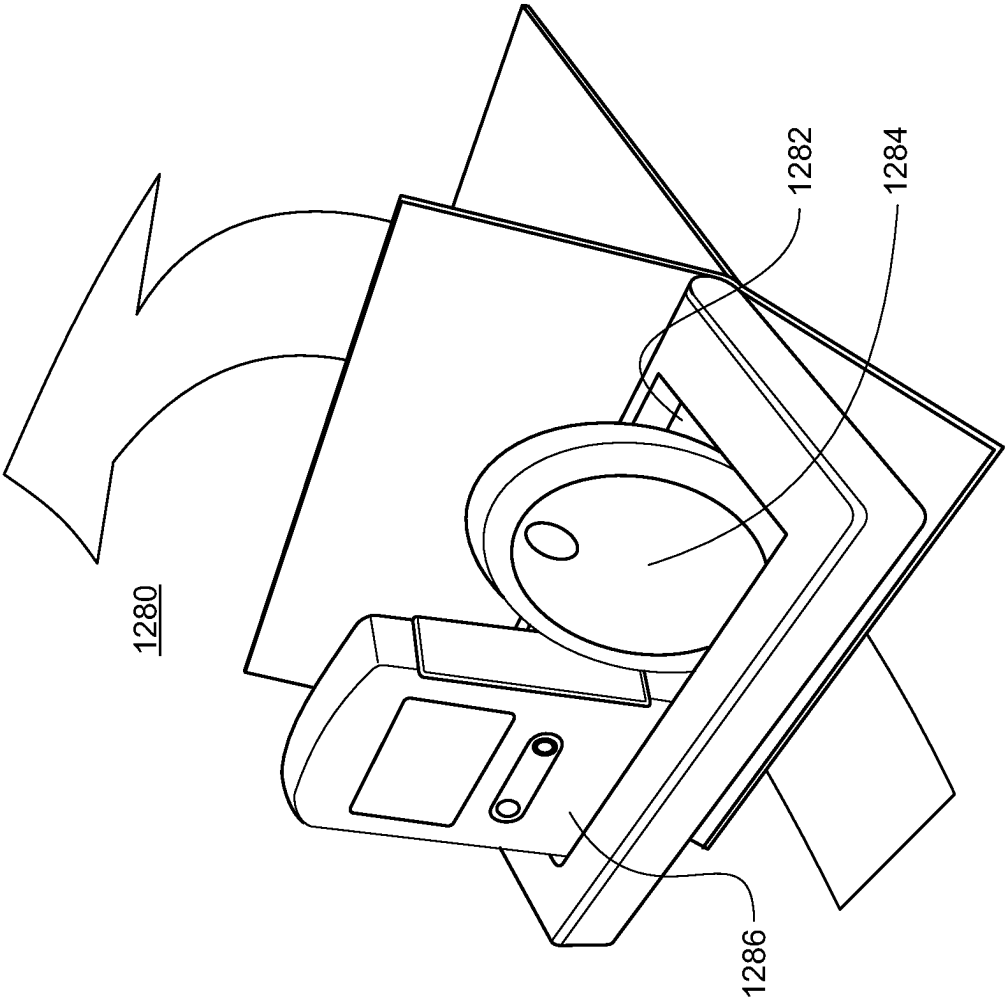


FIG. 85A

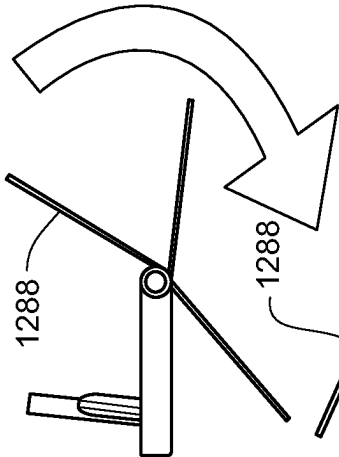


FIG. 85B



FIG. 85C

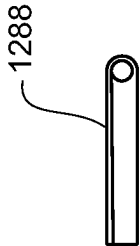


FIG. 85D

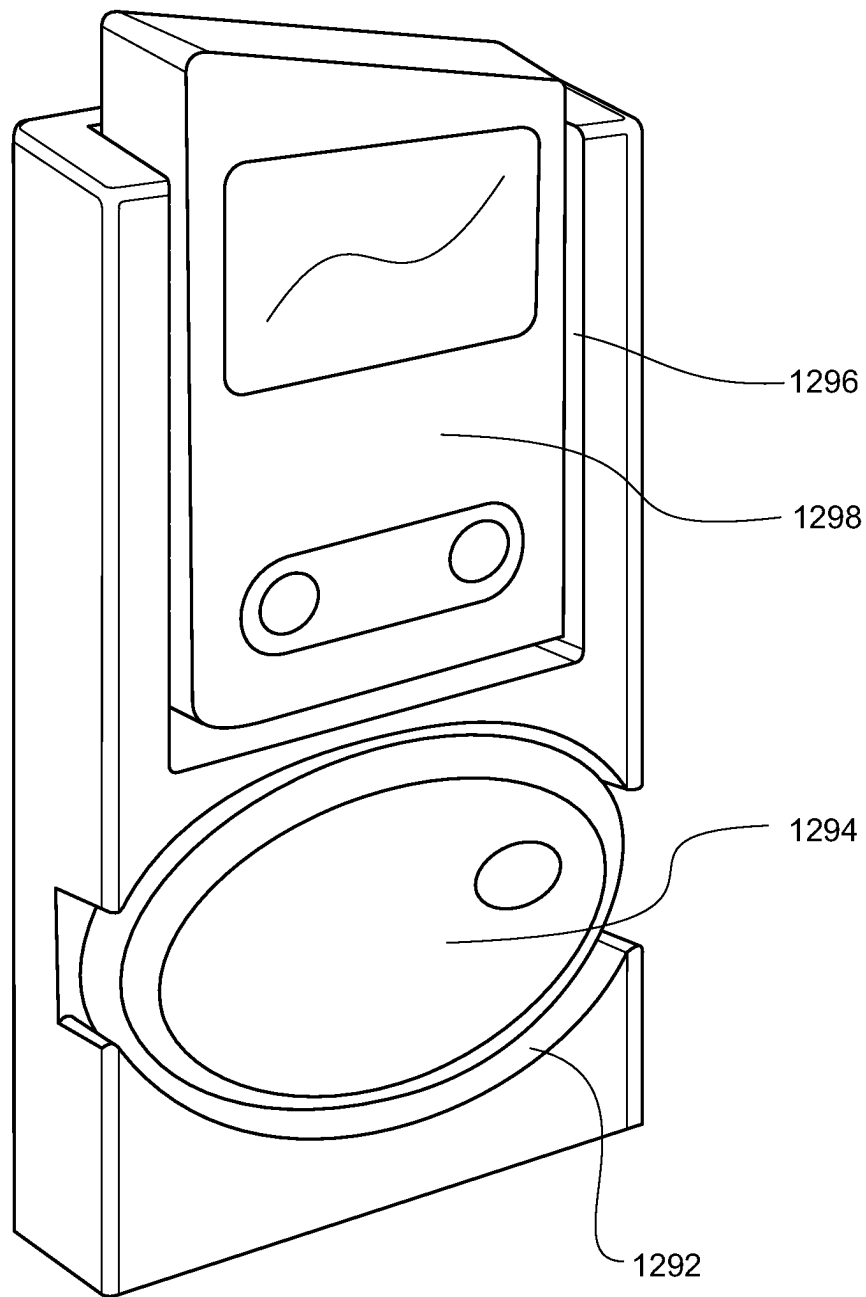
1290

FIG. 86

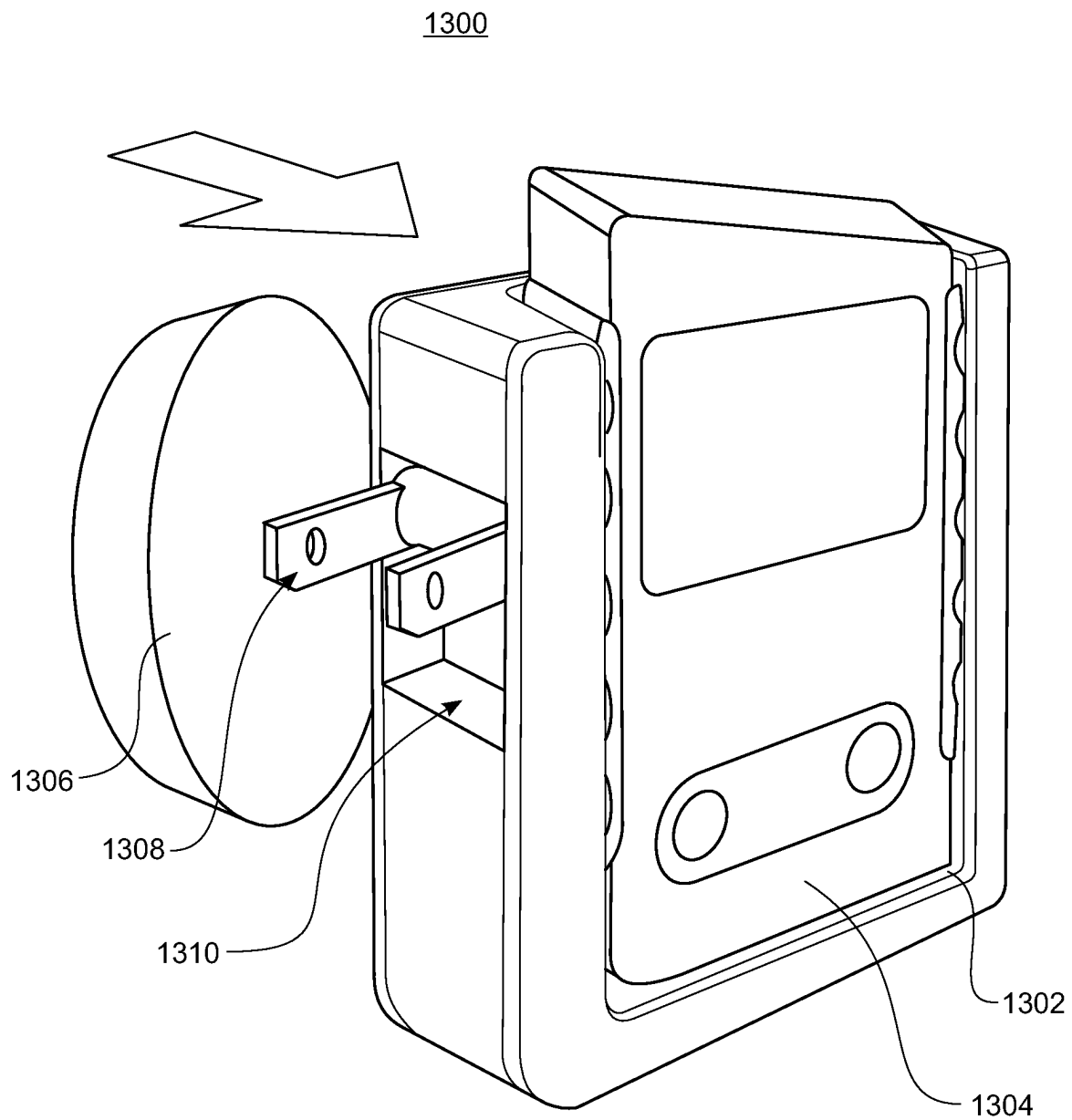


FIG. 87

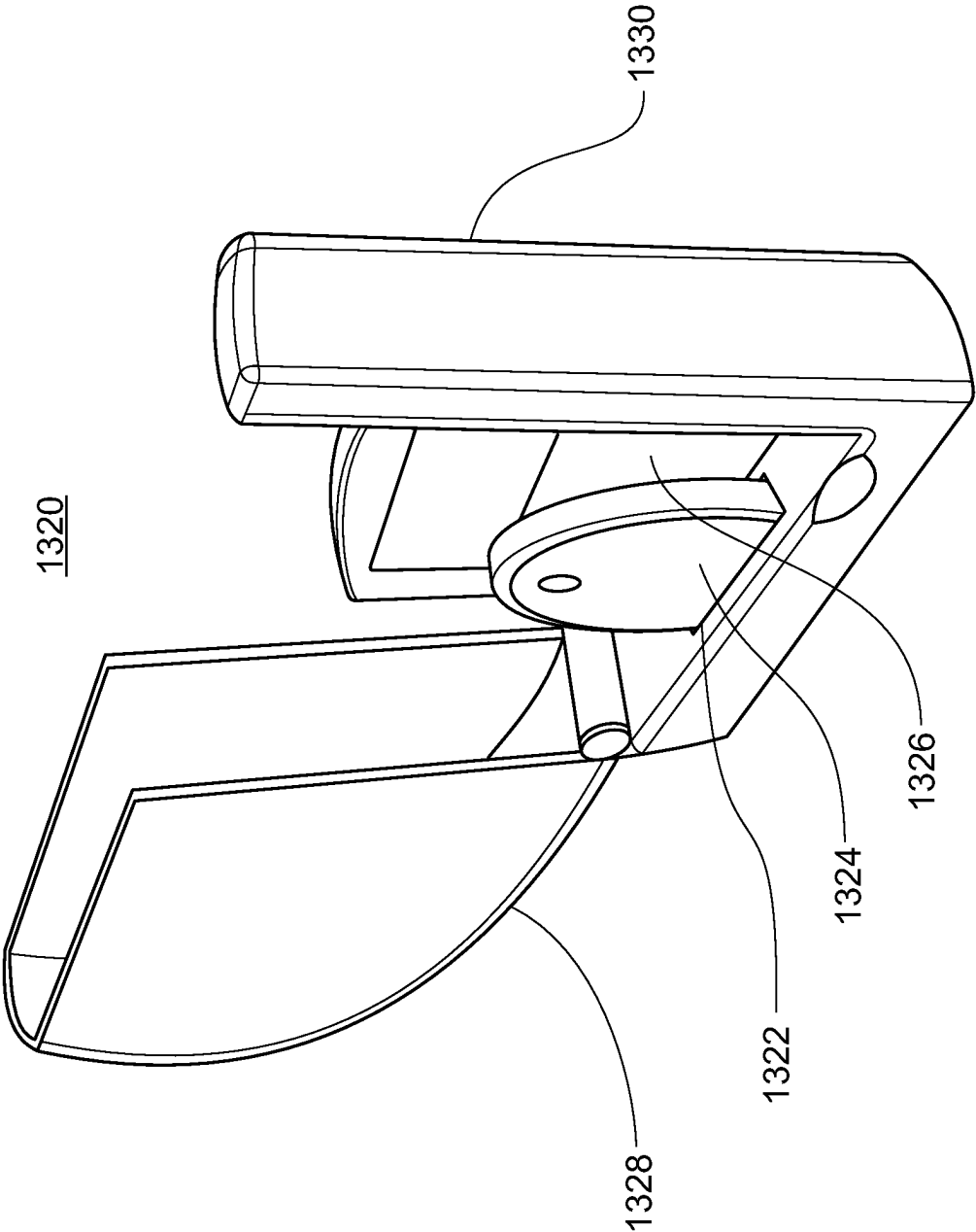


FIG. 88

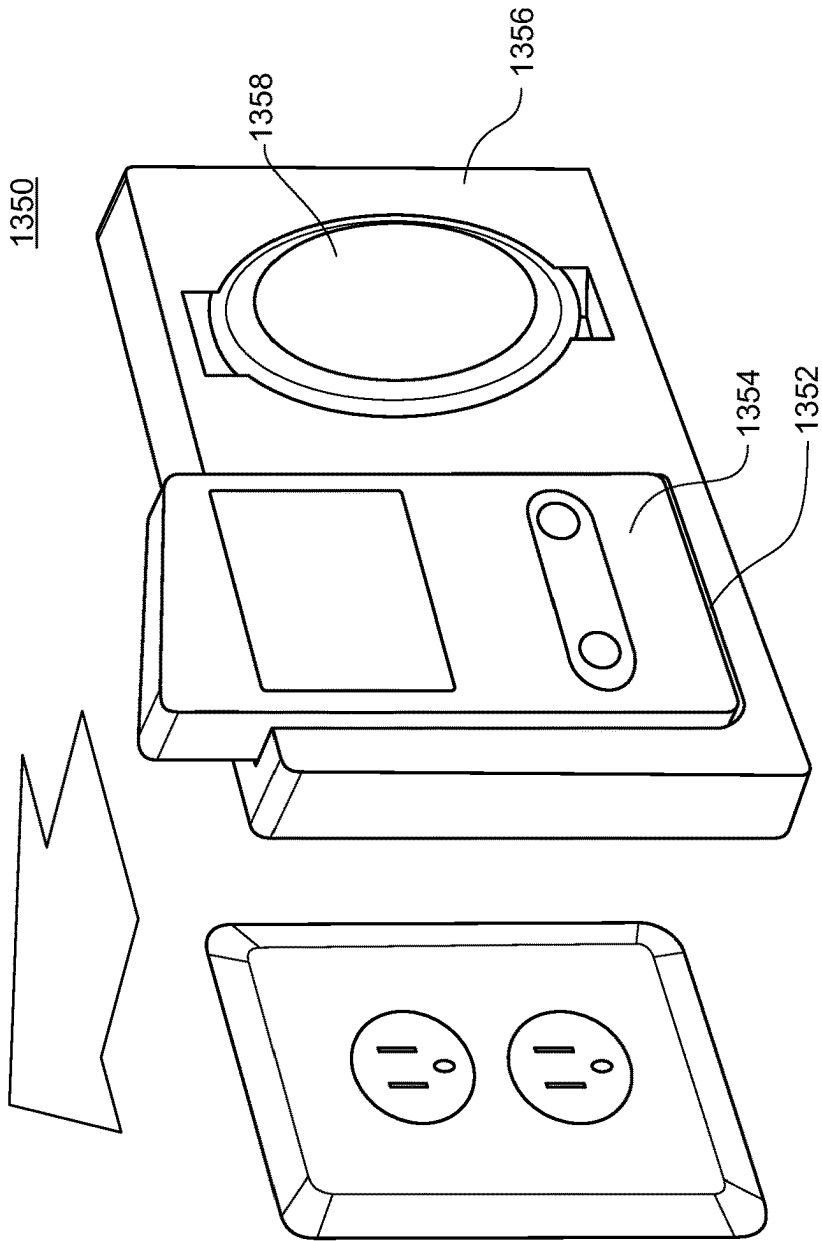


FIG. 89A

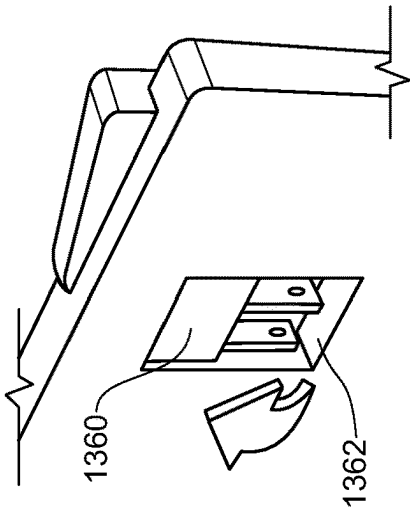


FIG. 89B

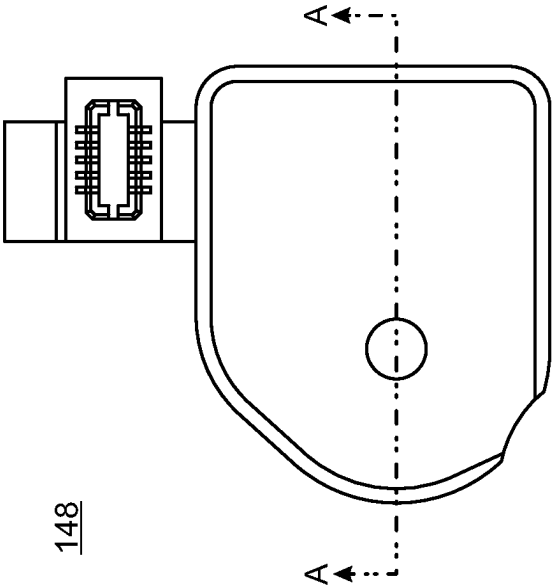


FIG. 90A

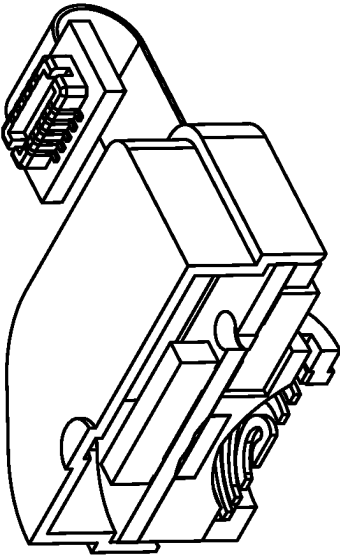
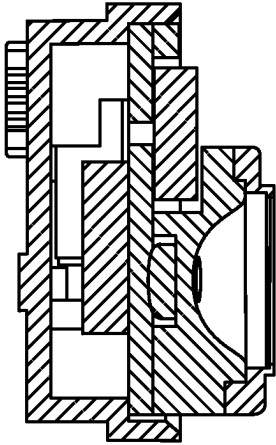


FIG. 90B



Section A-A

FIG. 90C

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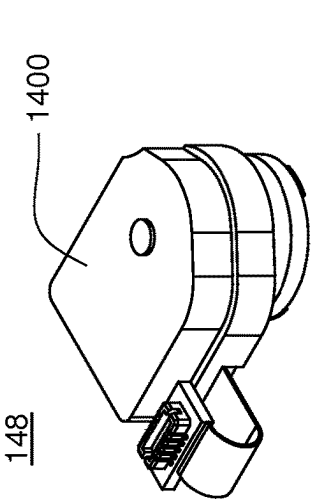


FIG. 91A

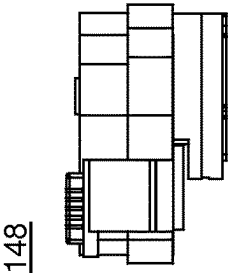


FIG. 91D

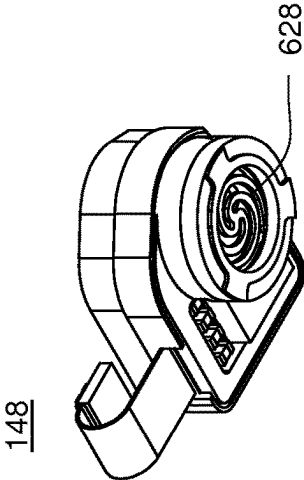


FIG. 91G

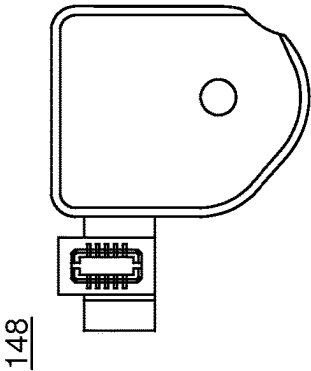


FIG. 91B

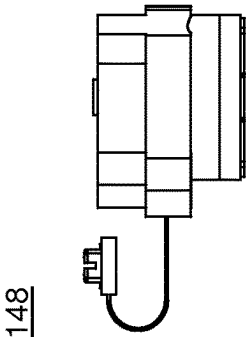


FIG. 91E

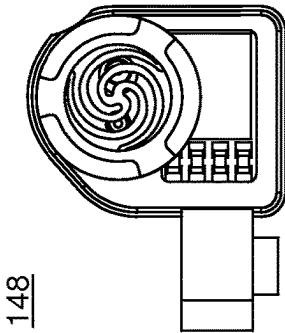


FIG. 91H

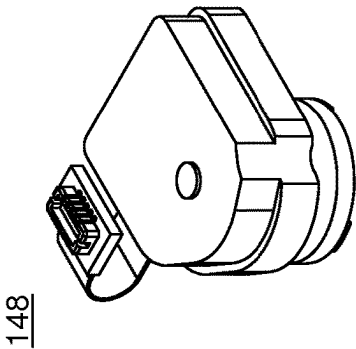


FIG. 91C

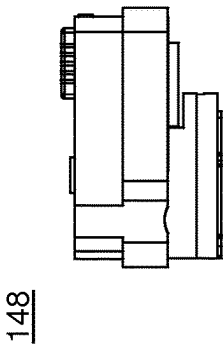


FIG. 91F

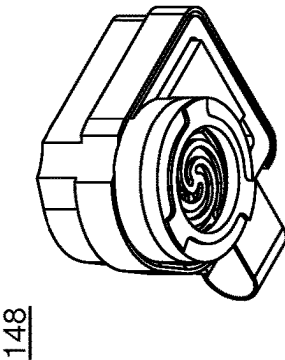


FIG. 91I

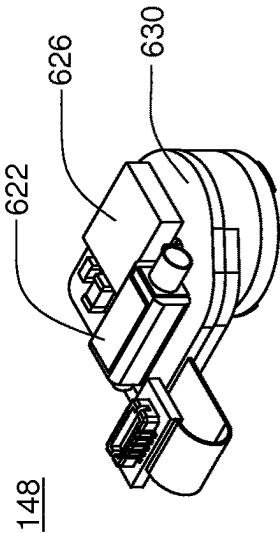


FIG. 92A

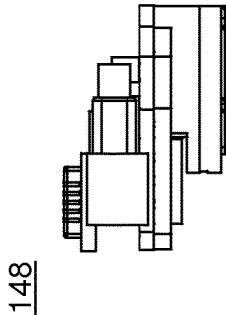


FIG. 92D

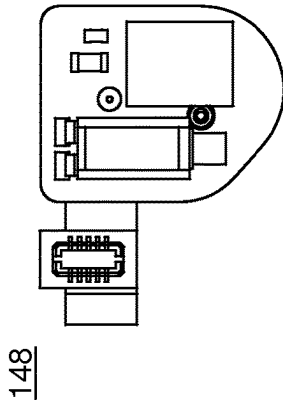


FIG. 92B

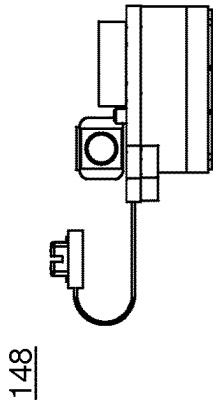


FIG. 92E

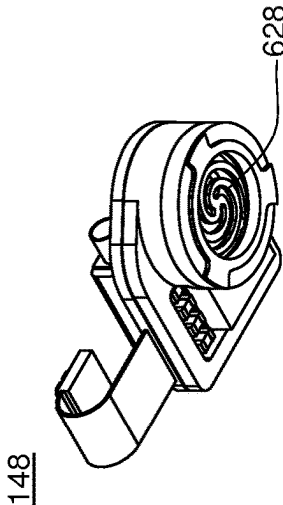


FIG. 92G

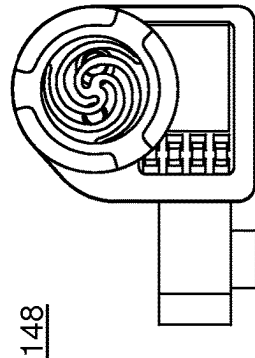


FIG. 92H

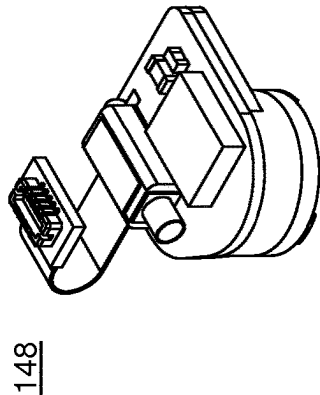


FIG. 92C

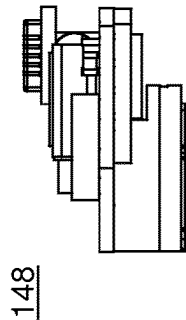


FIG. 92F

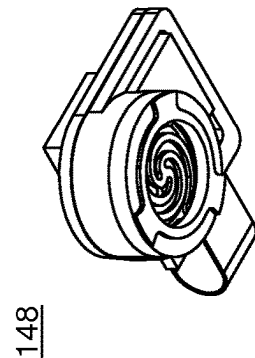


FIG. 92I

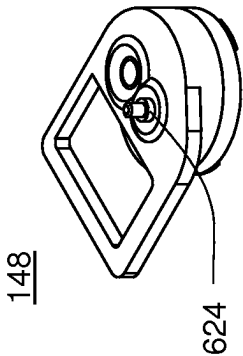


FIG. 93A

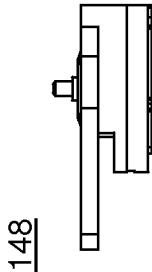


FIG. 93D

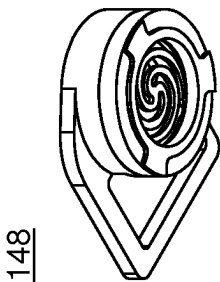


FIG. 93G

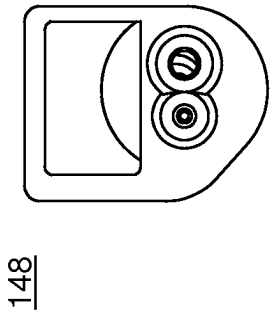


FIG. 93B

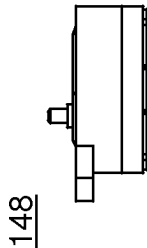


FIG. 93E

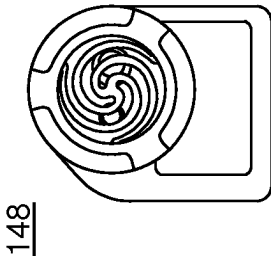


FIG. 93H

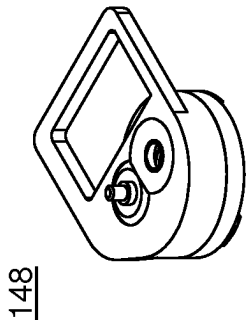


FIG. 93C

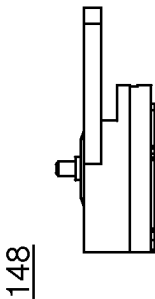


FIG. 93F

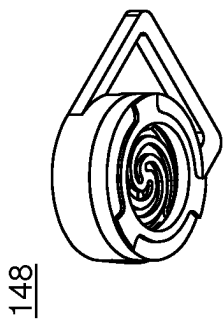


FIG. 93I

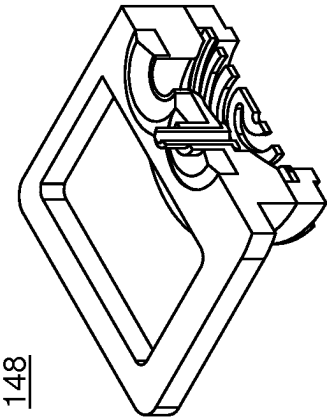


FIG. 94A

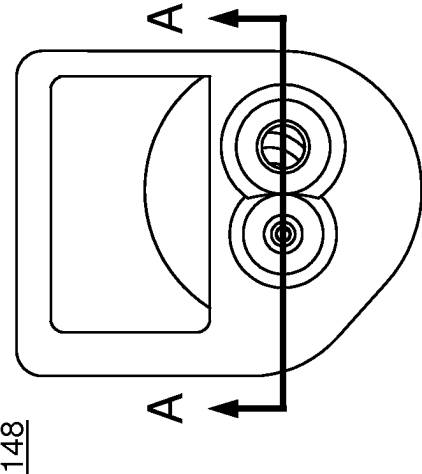


FIG. 94B

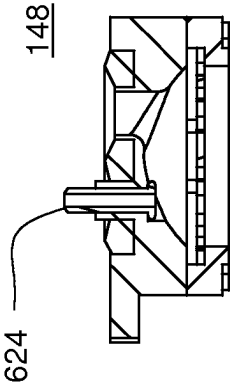


FIG. 94E

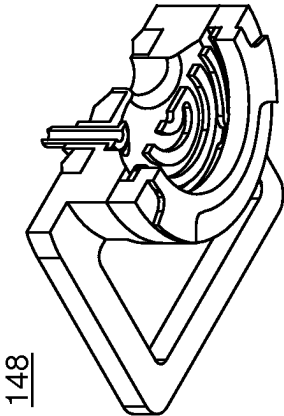


FIG. 94D

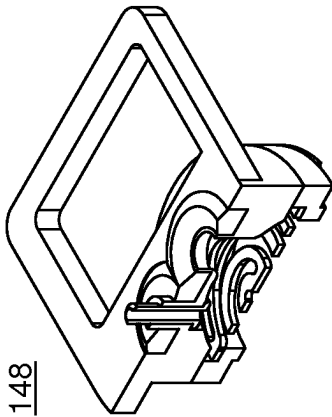


FIG. 94C

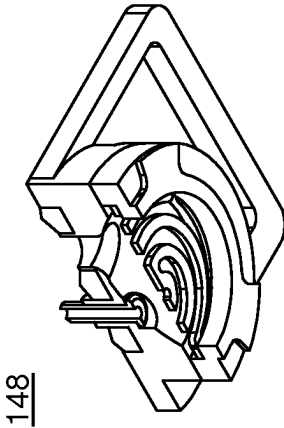


FIG. 94F

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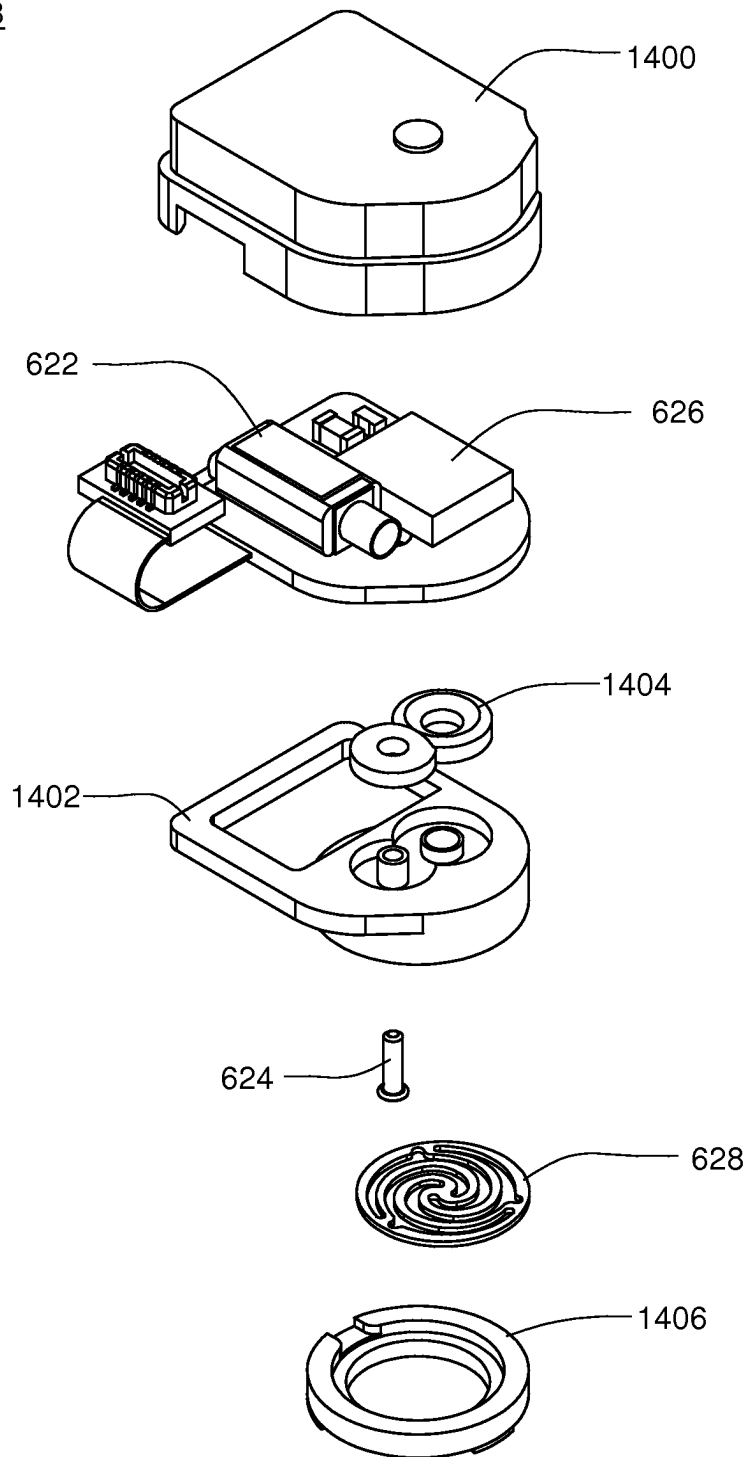


FIG. 95

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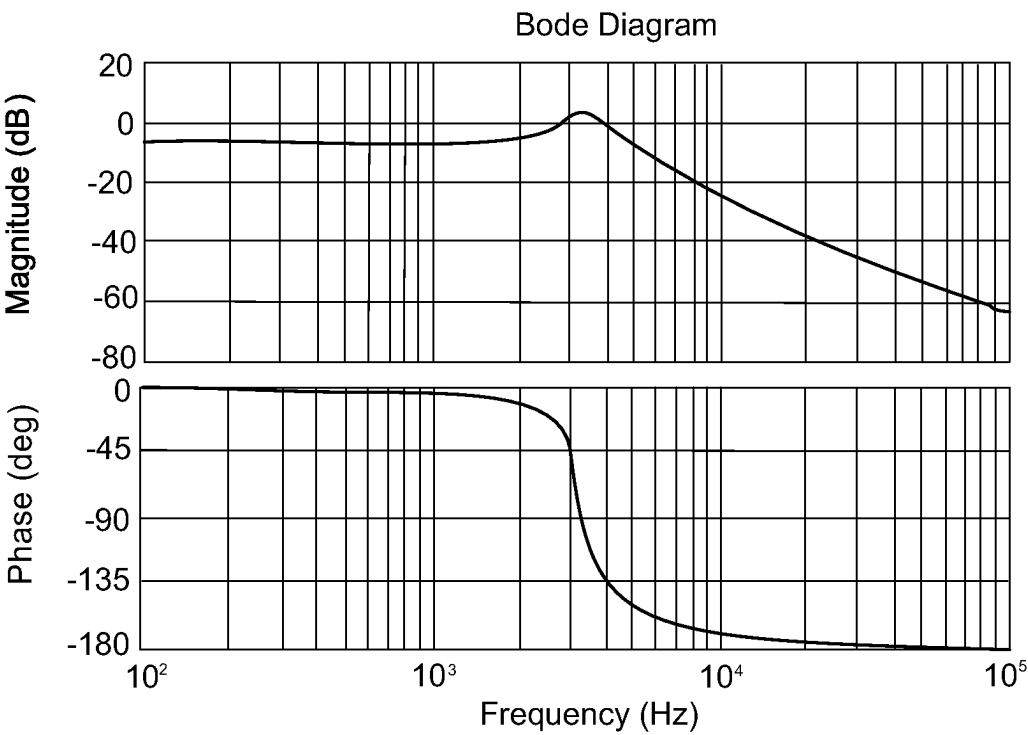
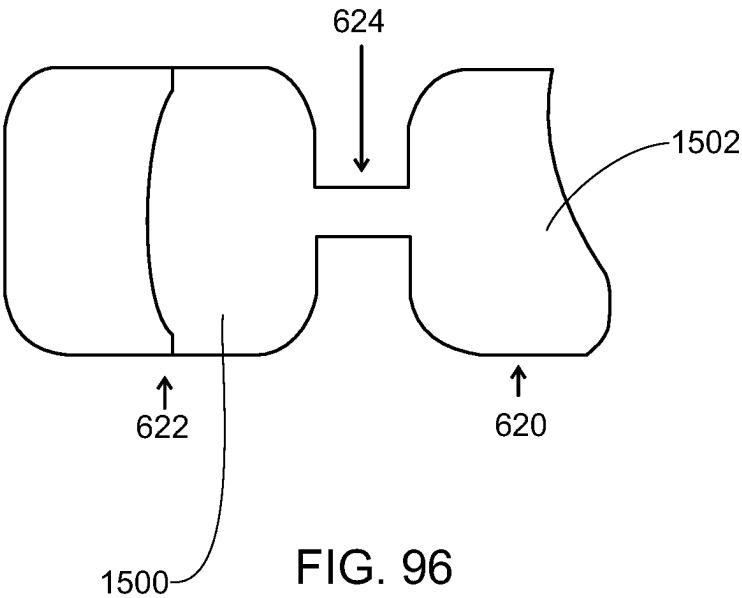


FIG. 97

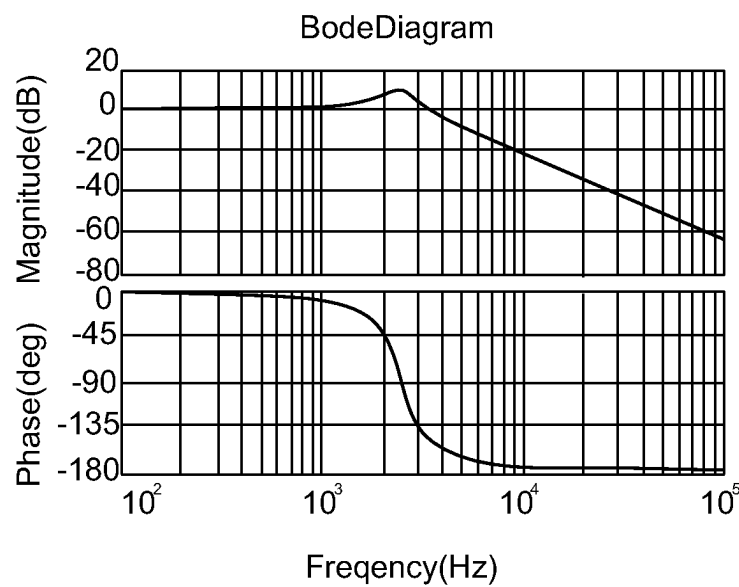


FIG. 98

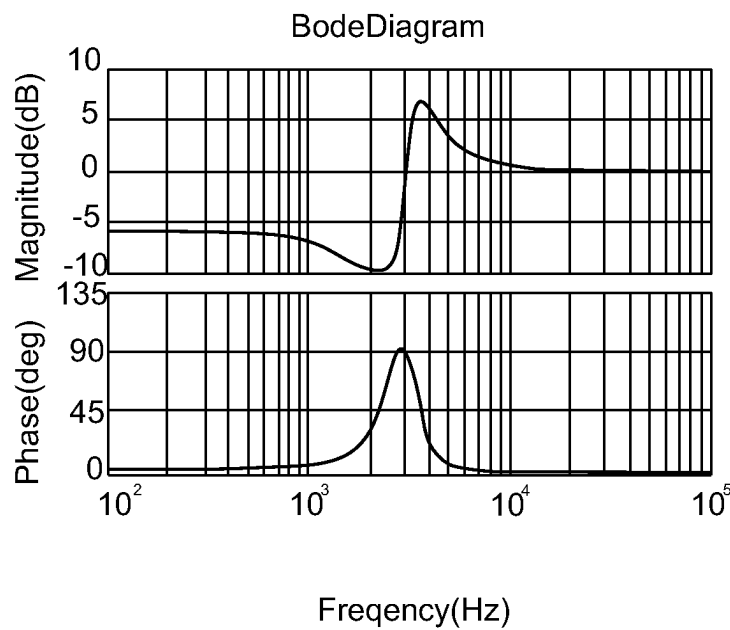


FIG. 99

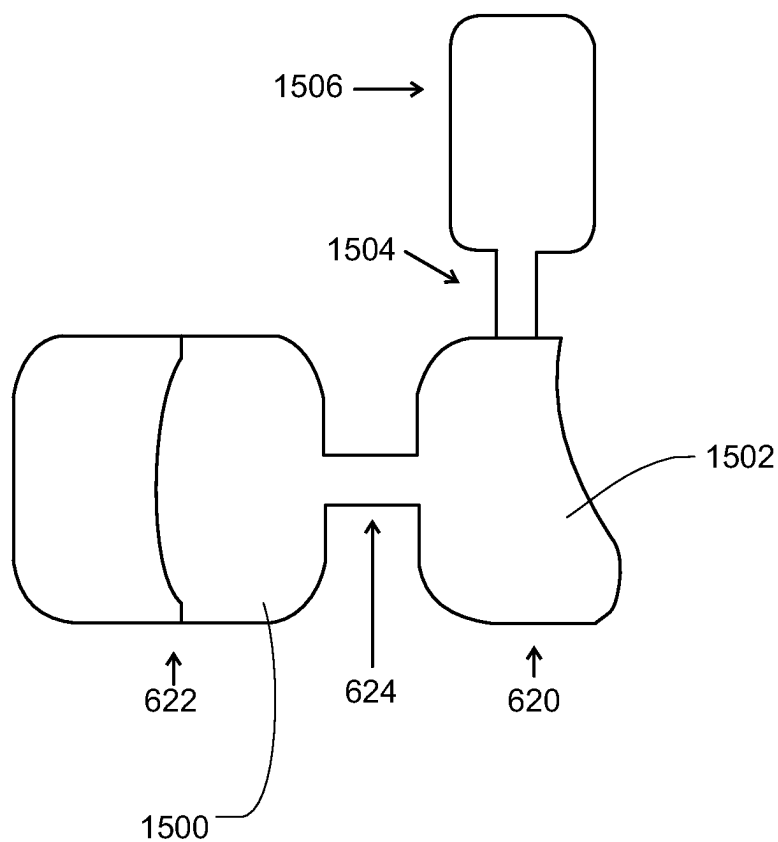


FIG. 100

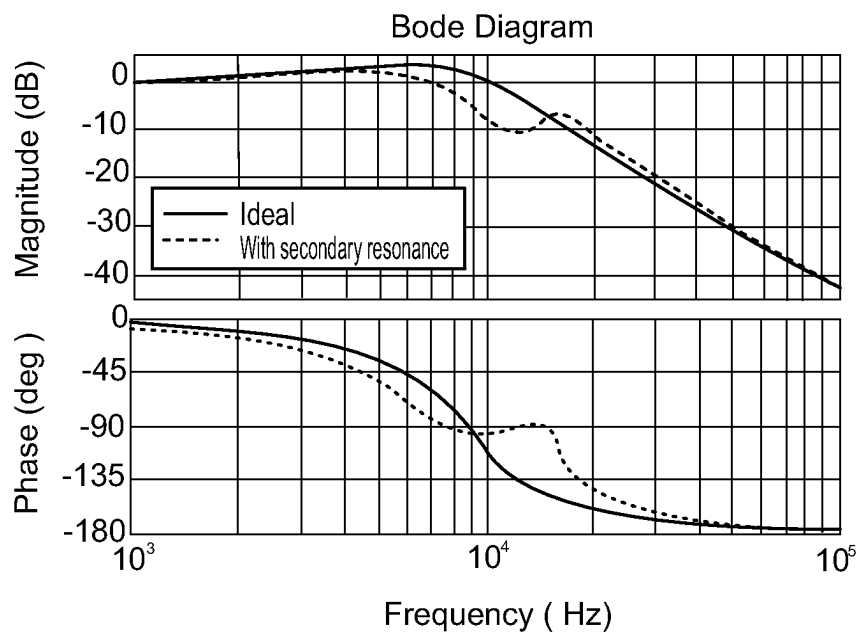


FIG. 101

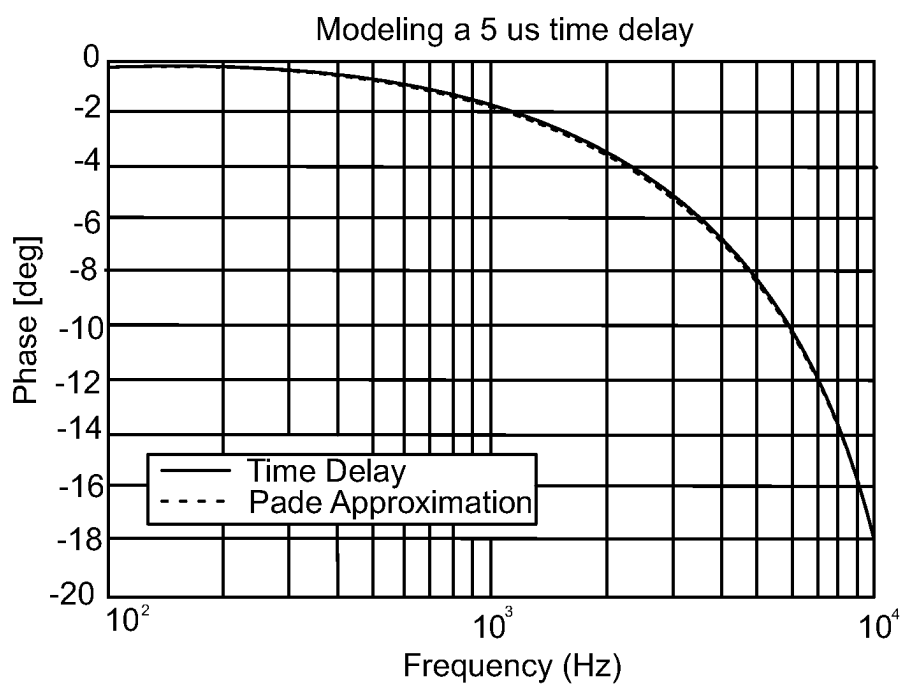


FIG. 102

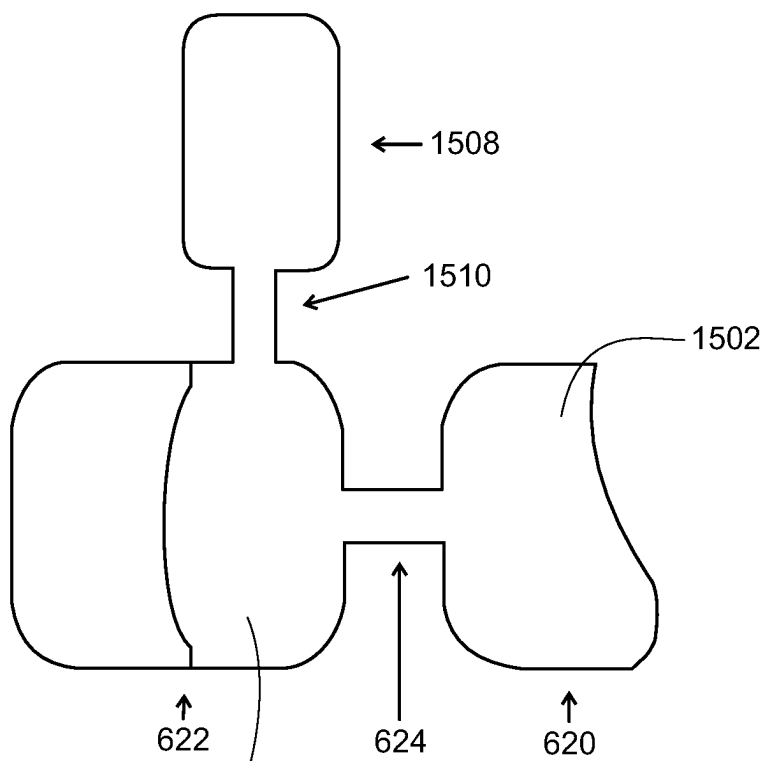


FIG.103

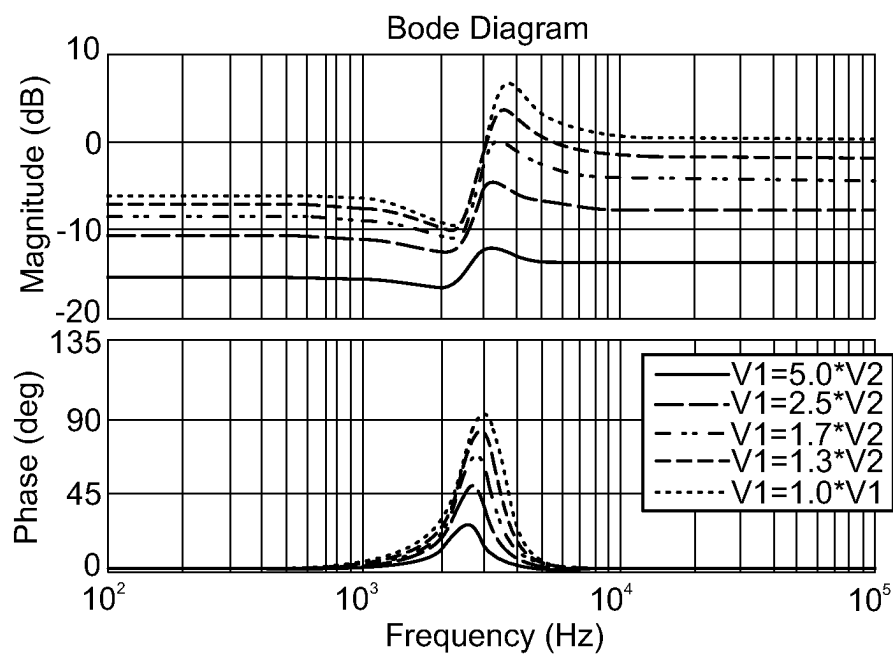


FIG. 104

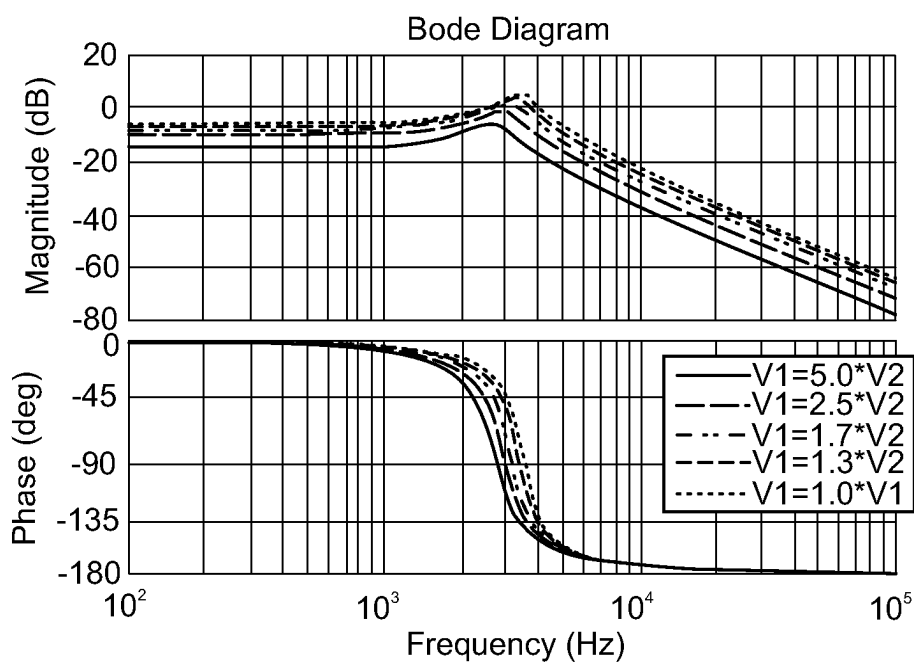


FIG. 105

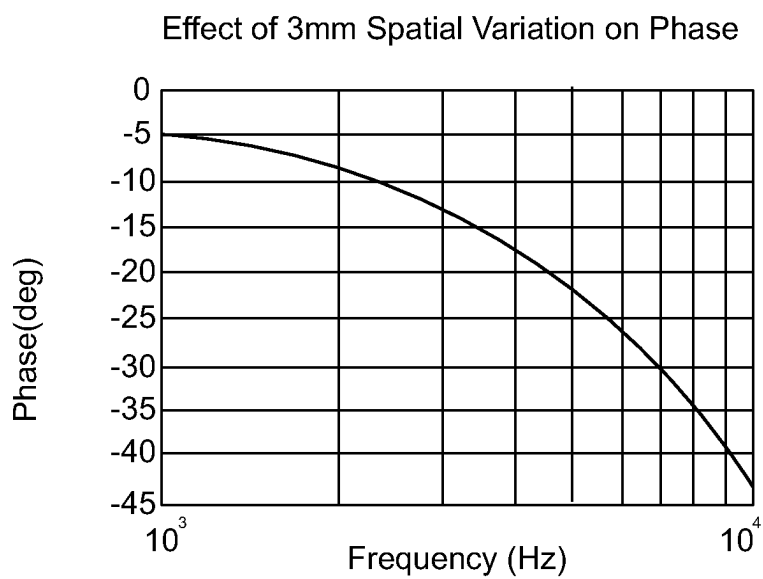


FIG. 106

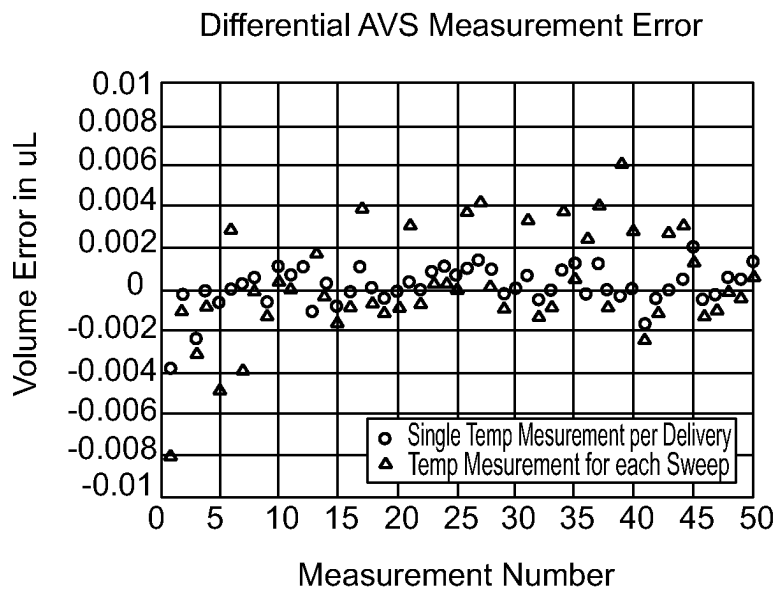


FIG. 107

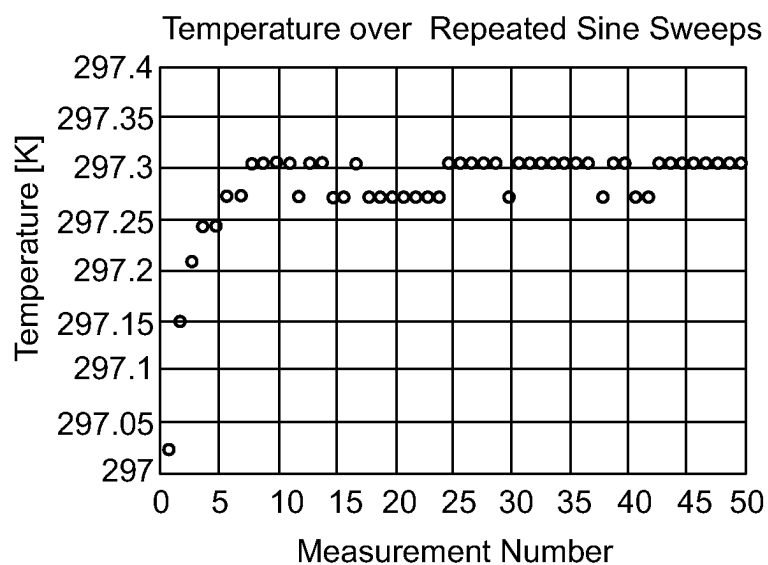


FIG. 108

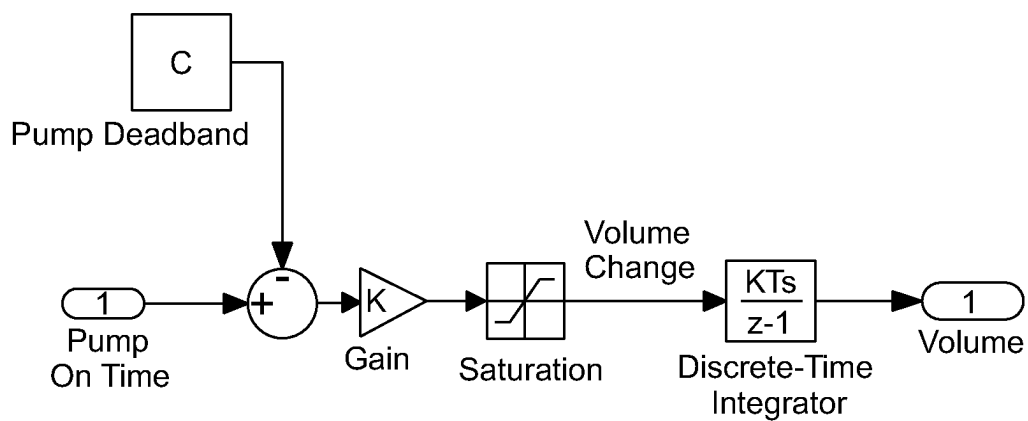


FIG. 109

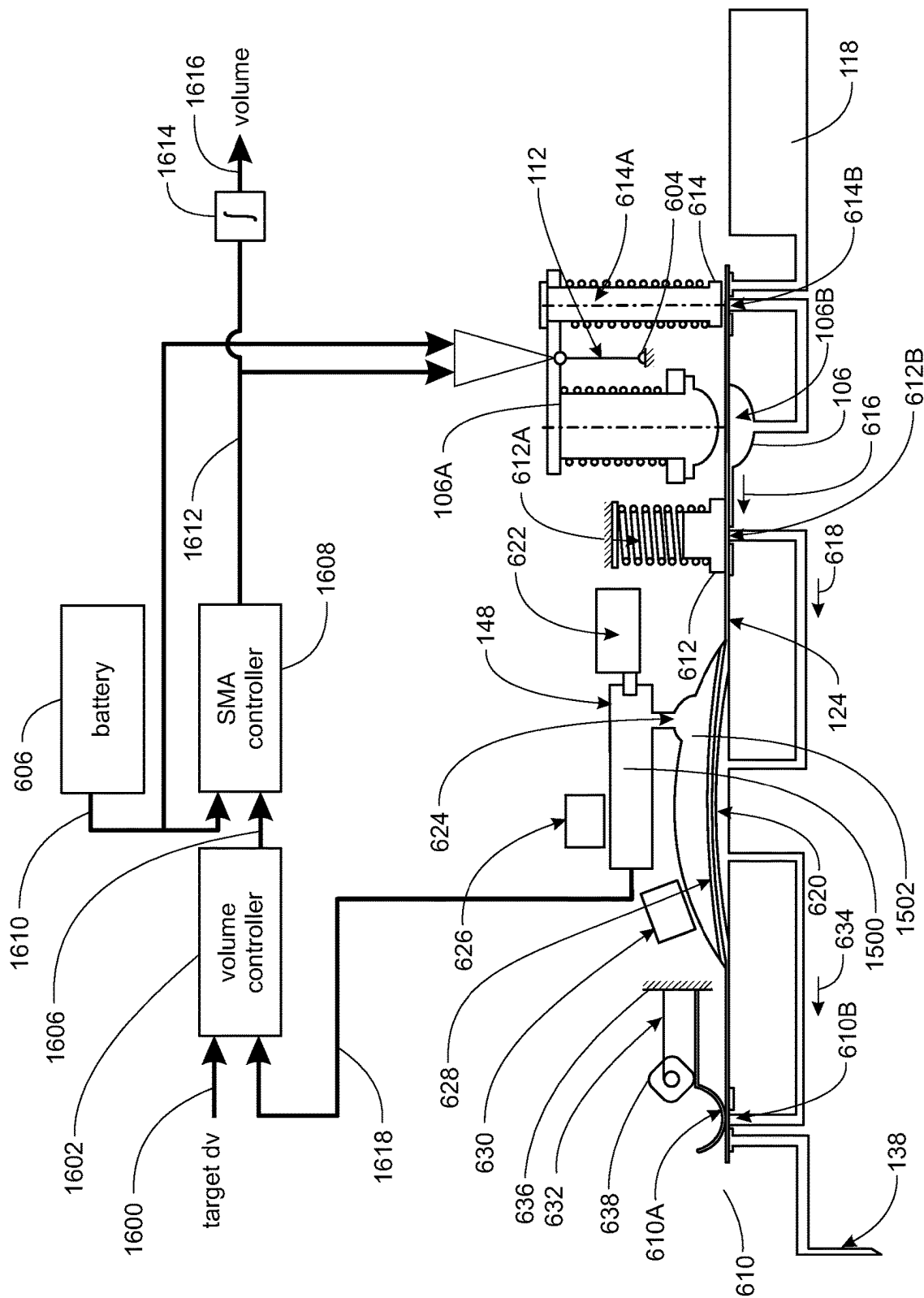


FIG. 110

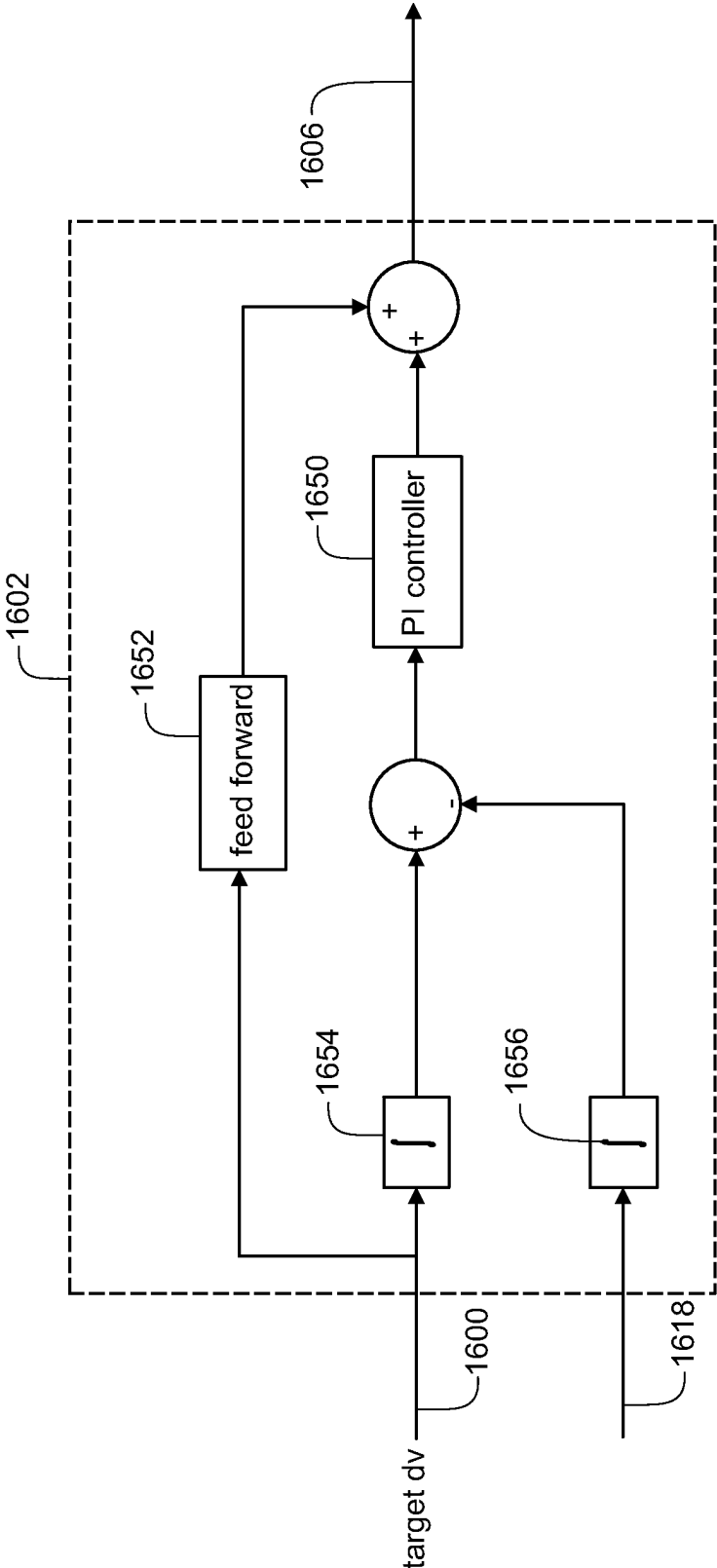


FIG.111

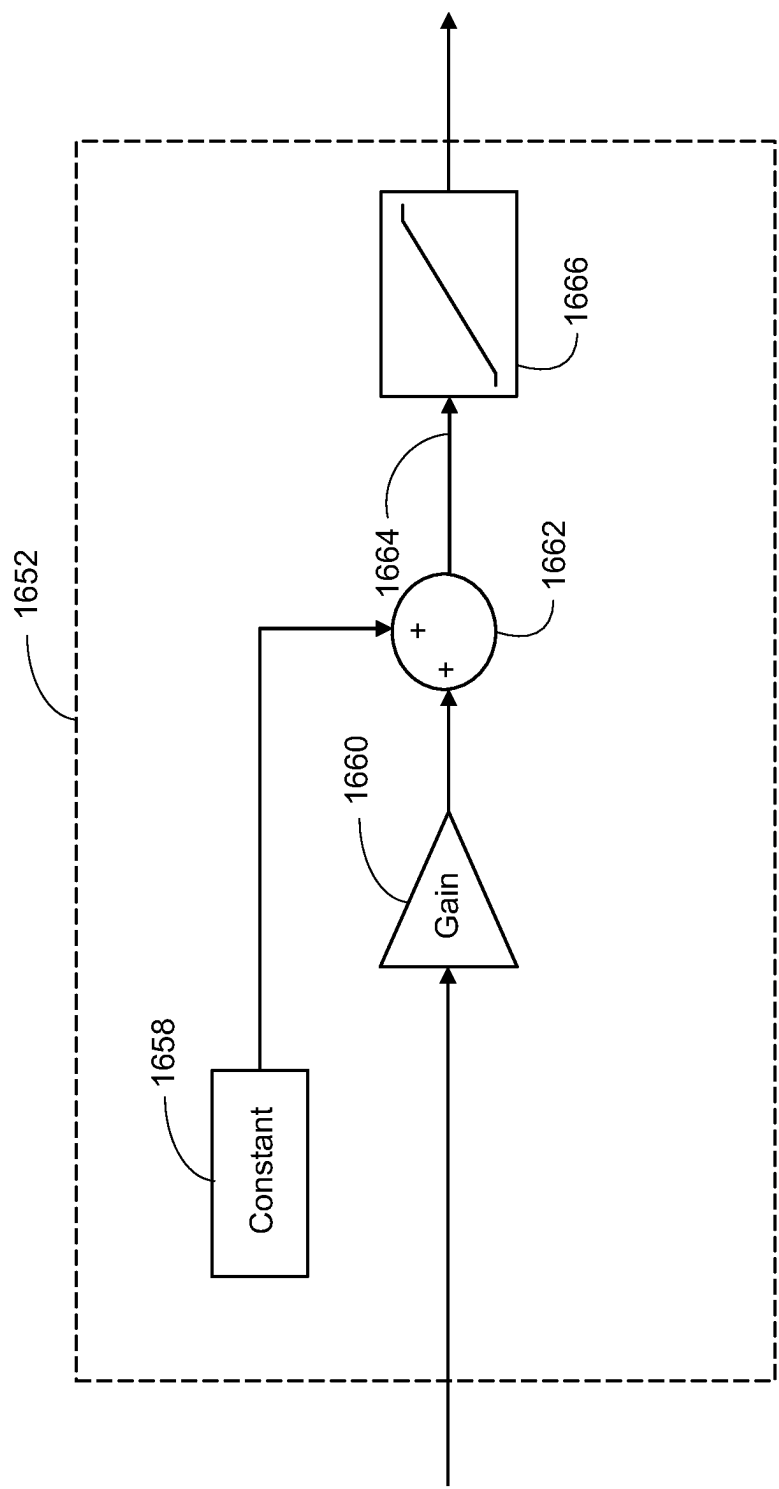


FIG. 112

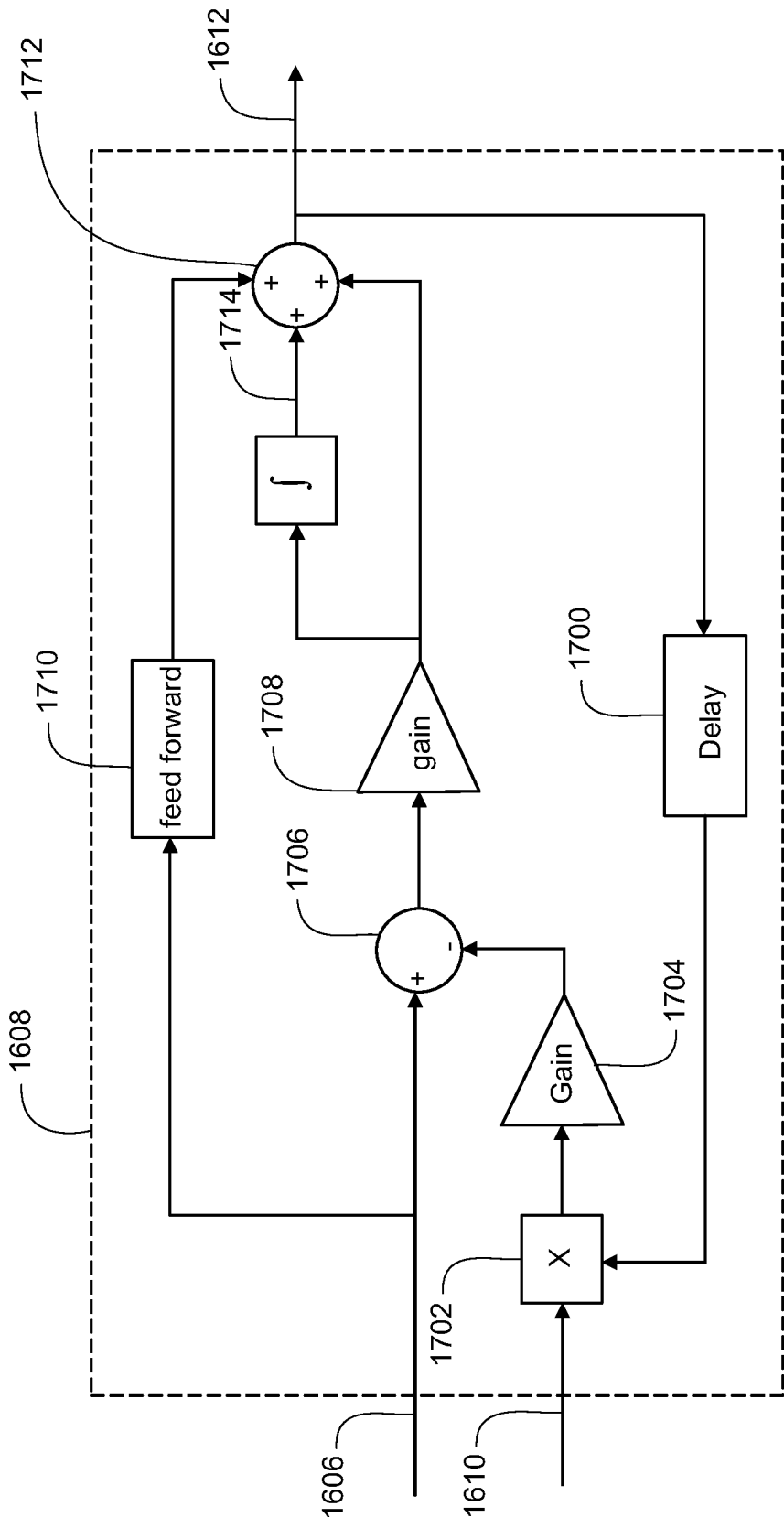


FIG. 113

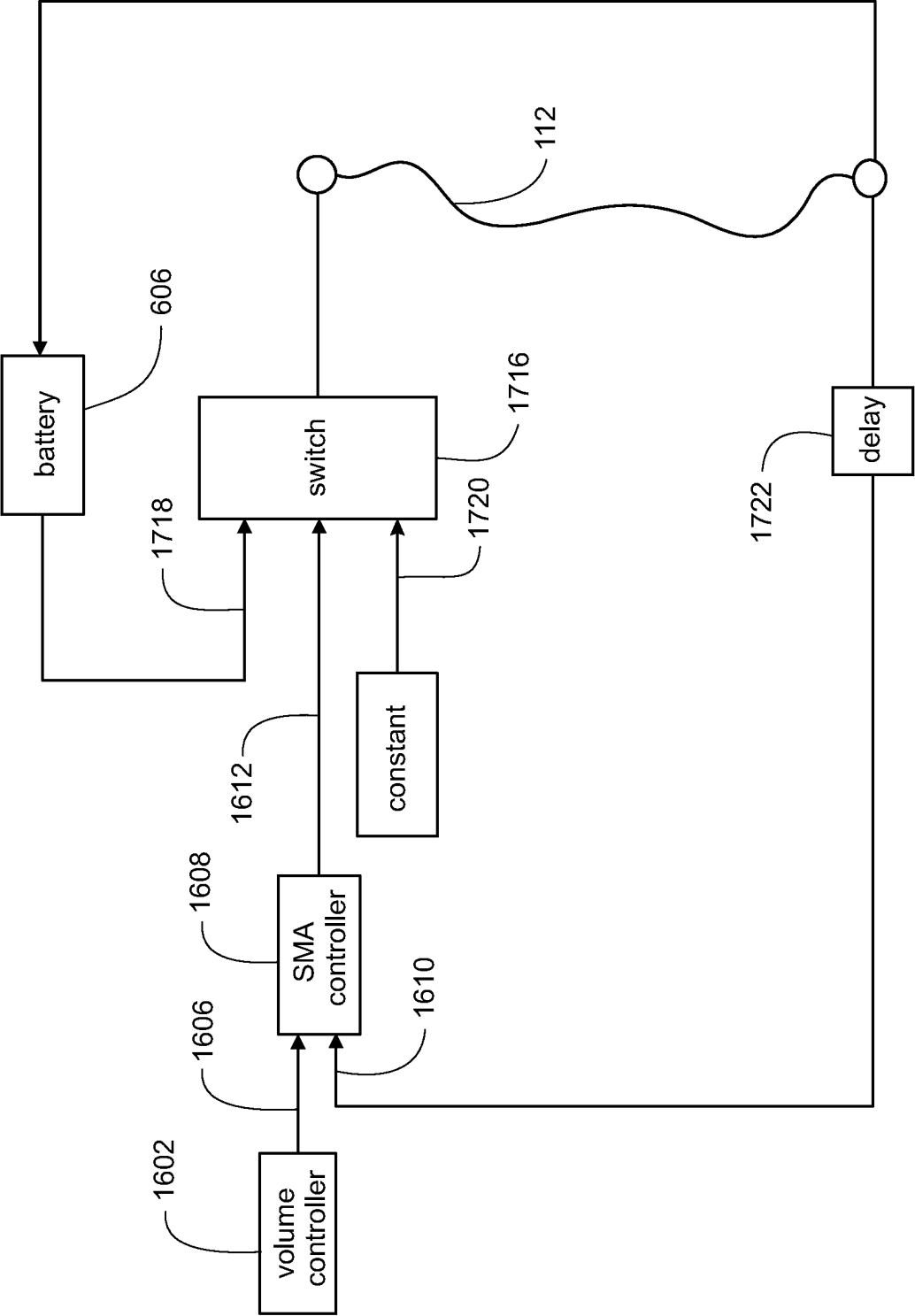


FIG. 114

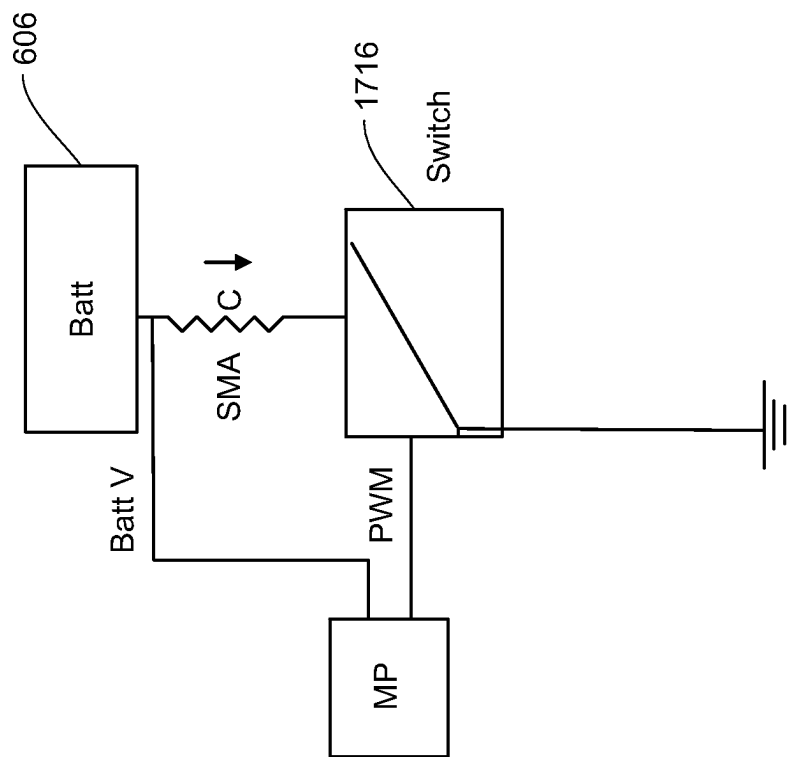


FIG. 114A

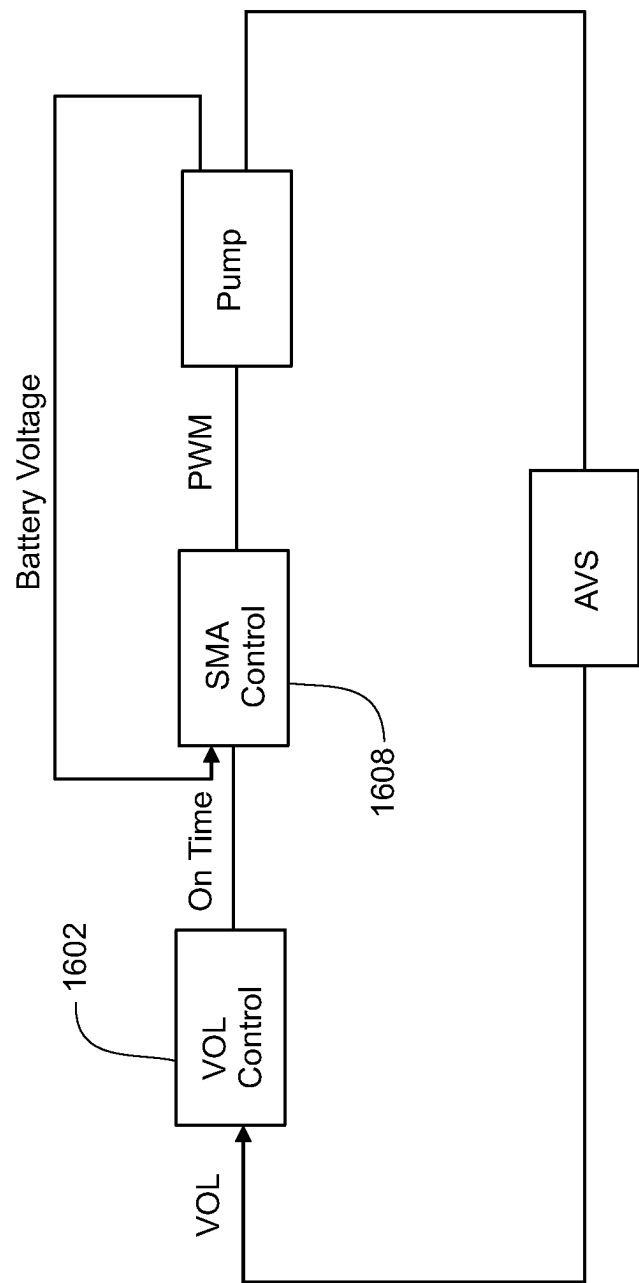


FIG.114B

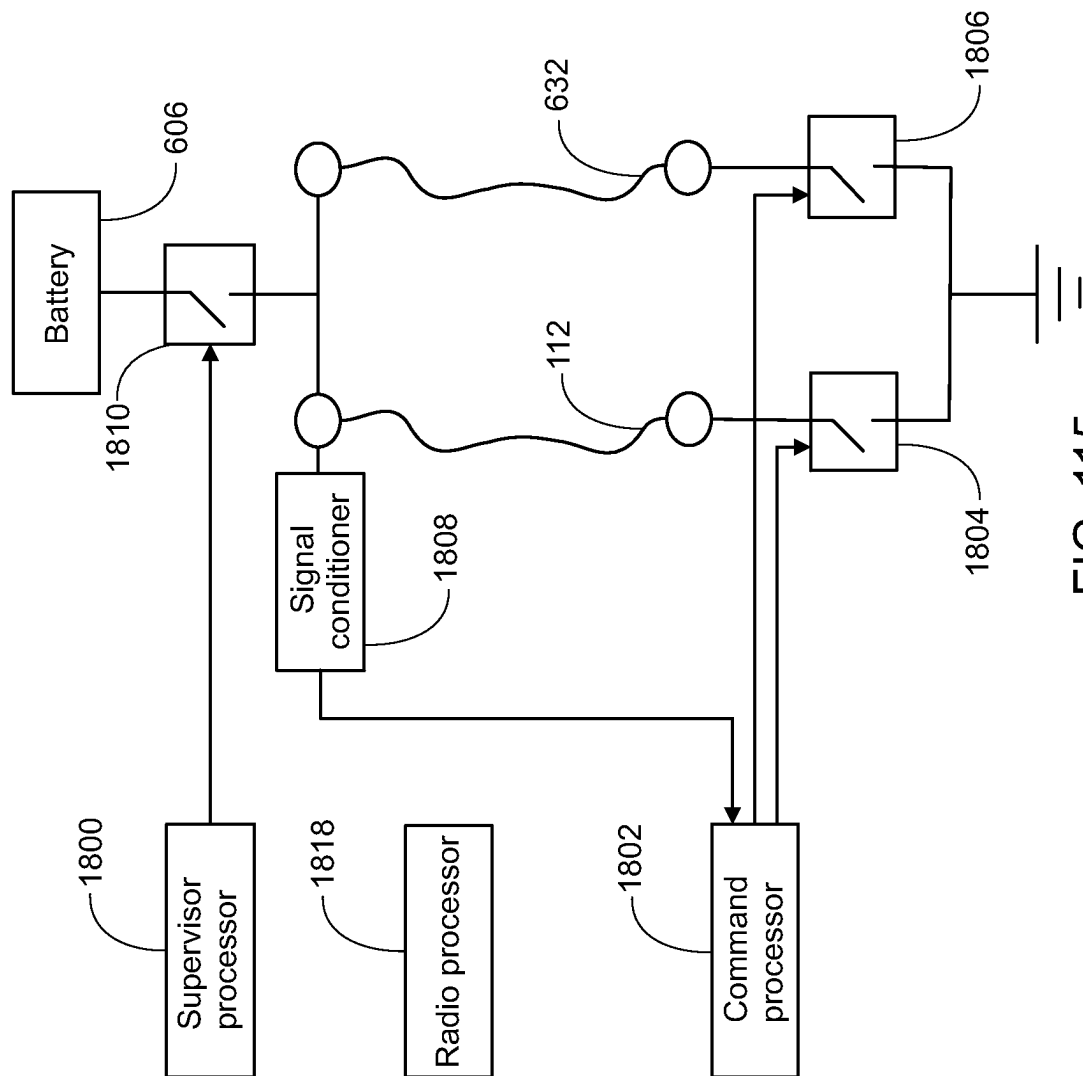


FIG. 115

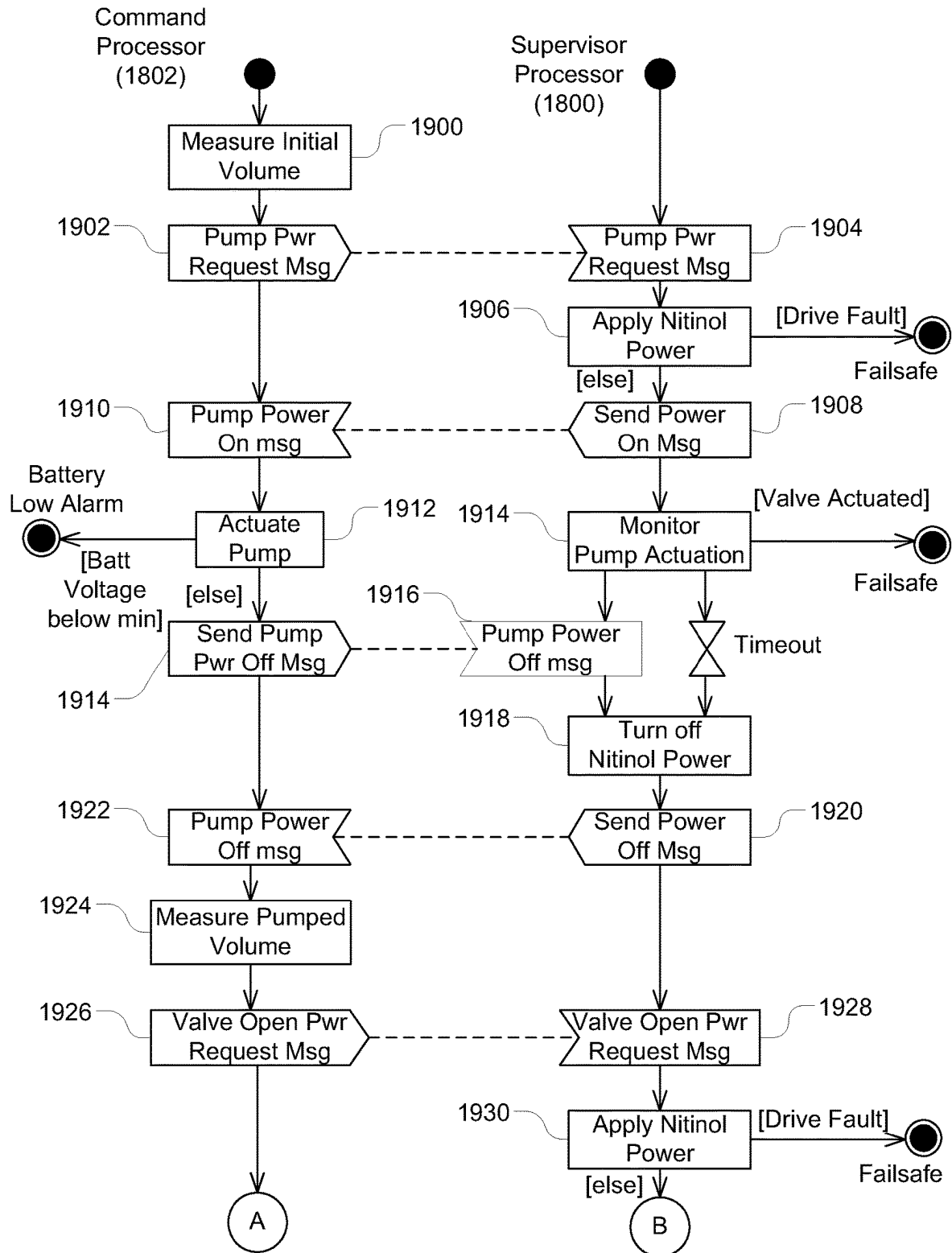
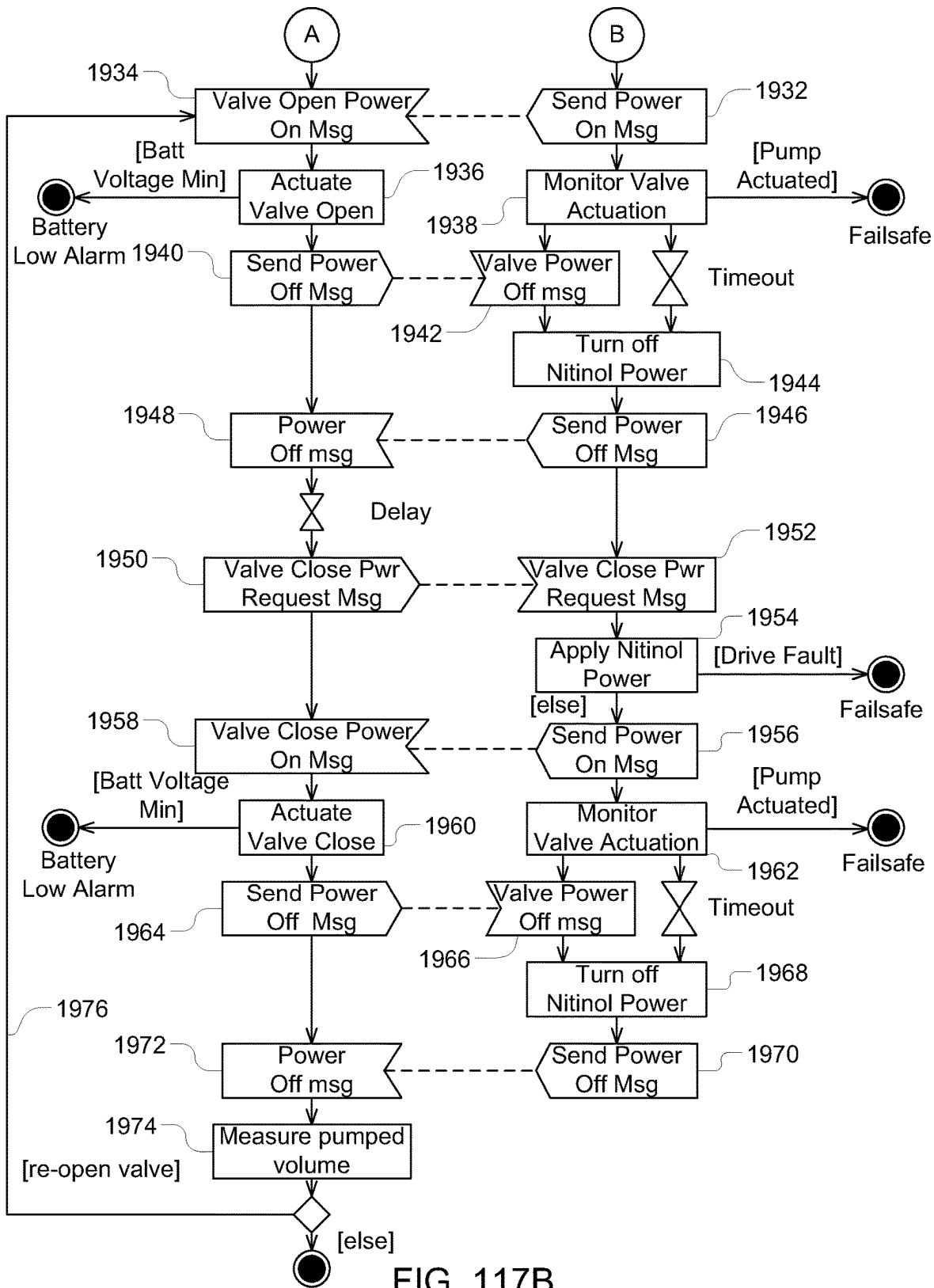
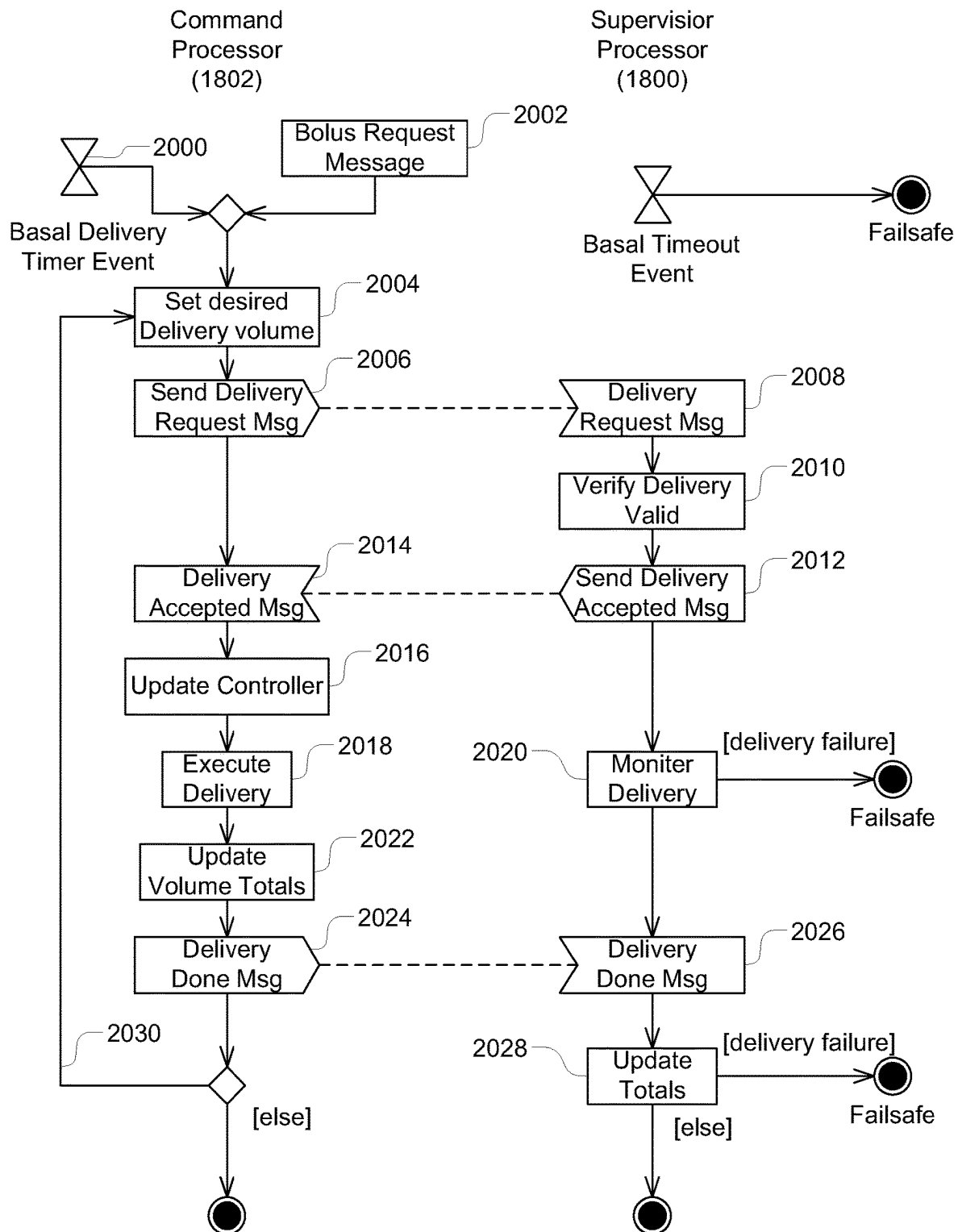


FIG. 117A





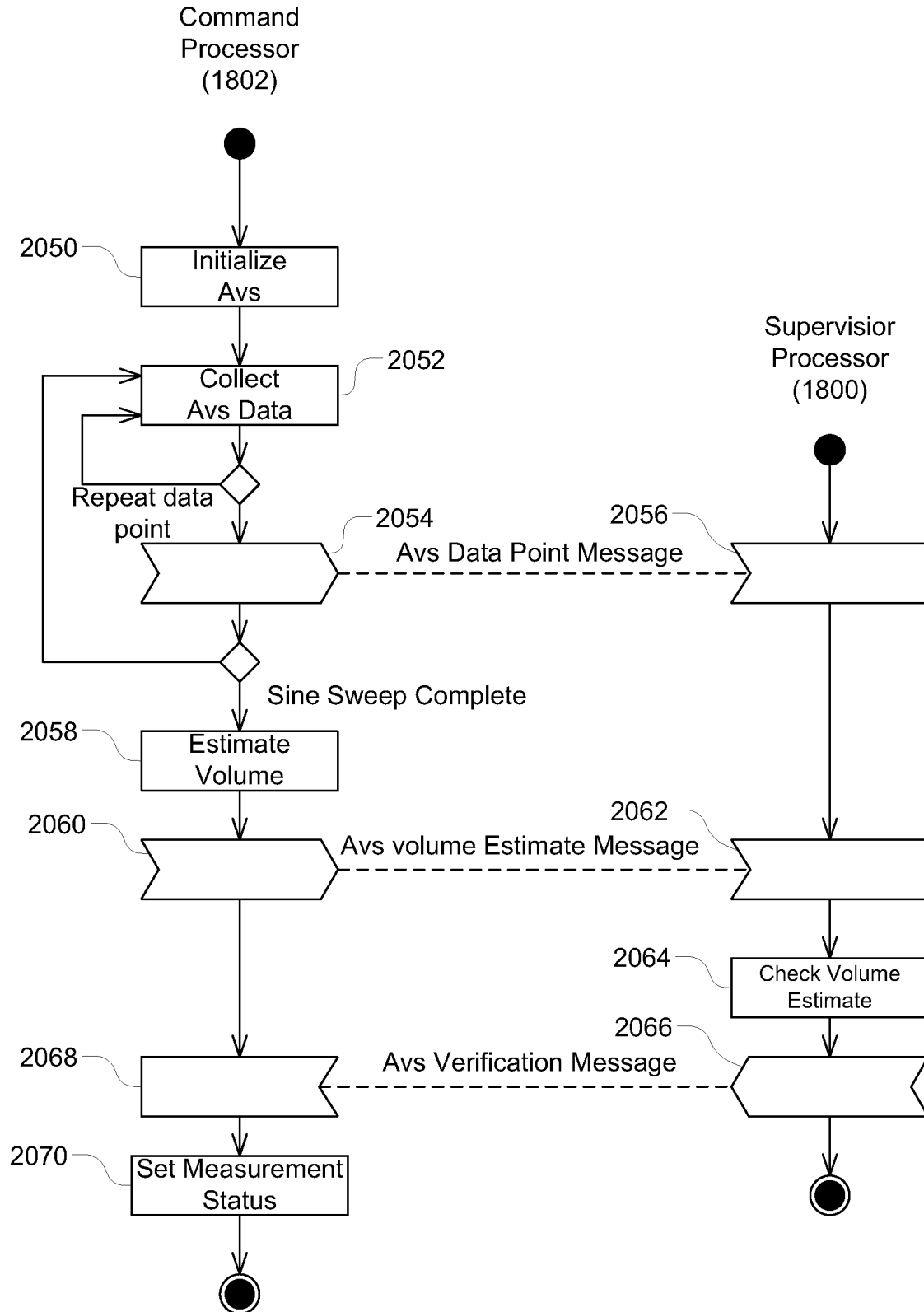


FIG. 119

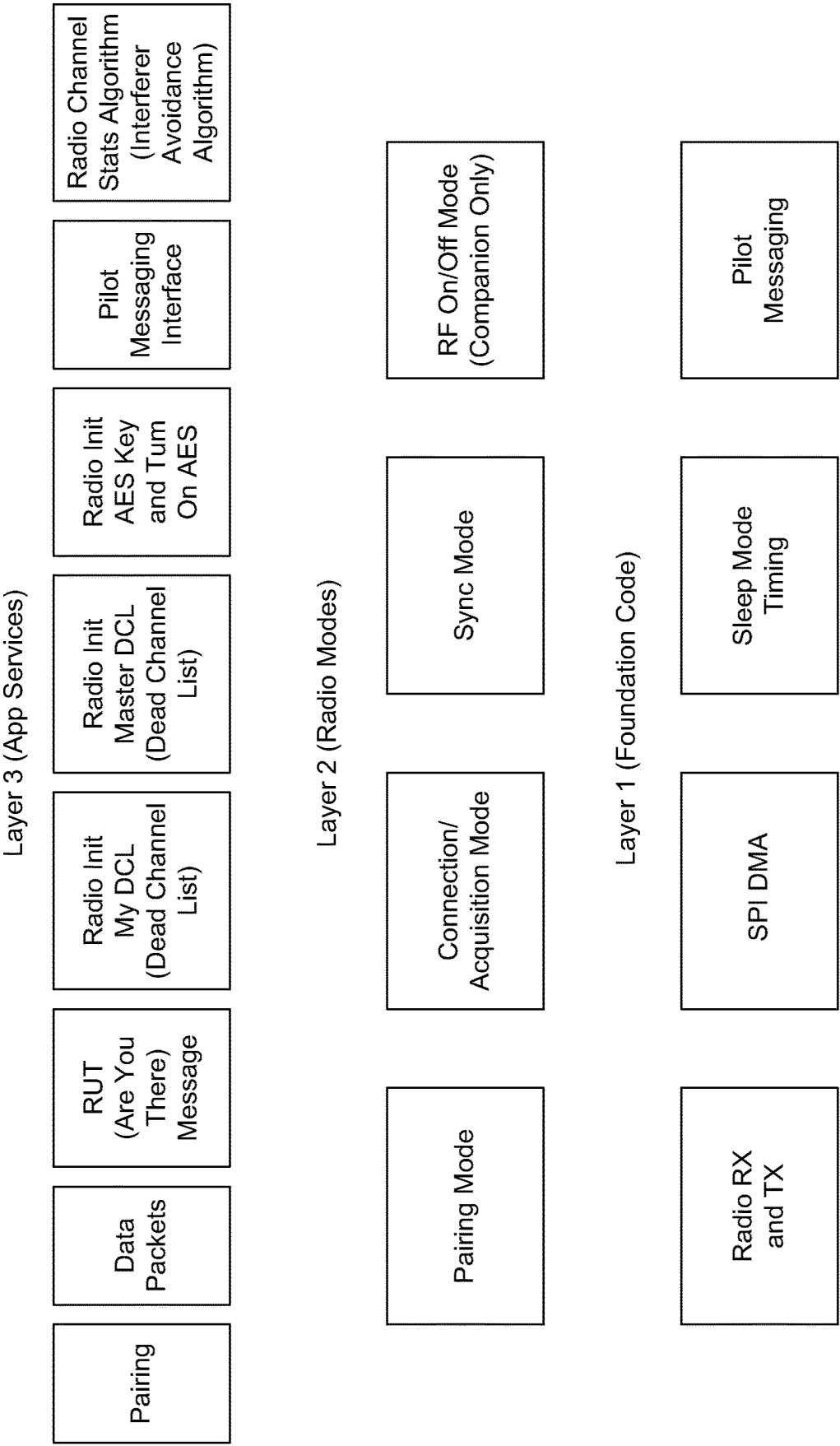


FIG. 120A

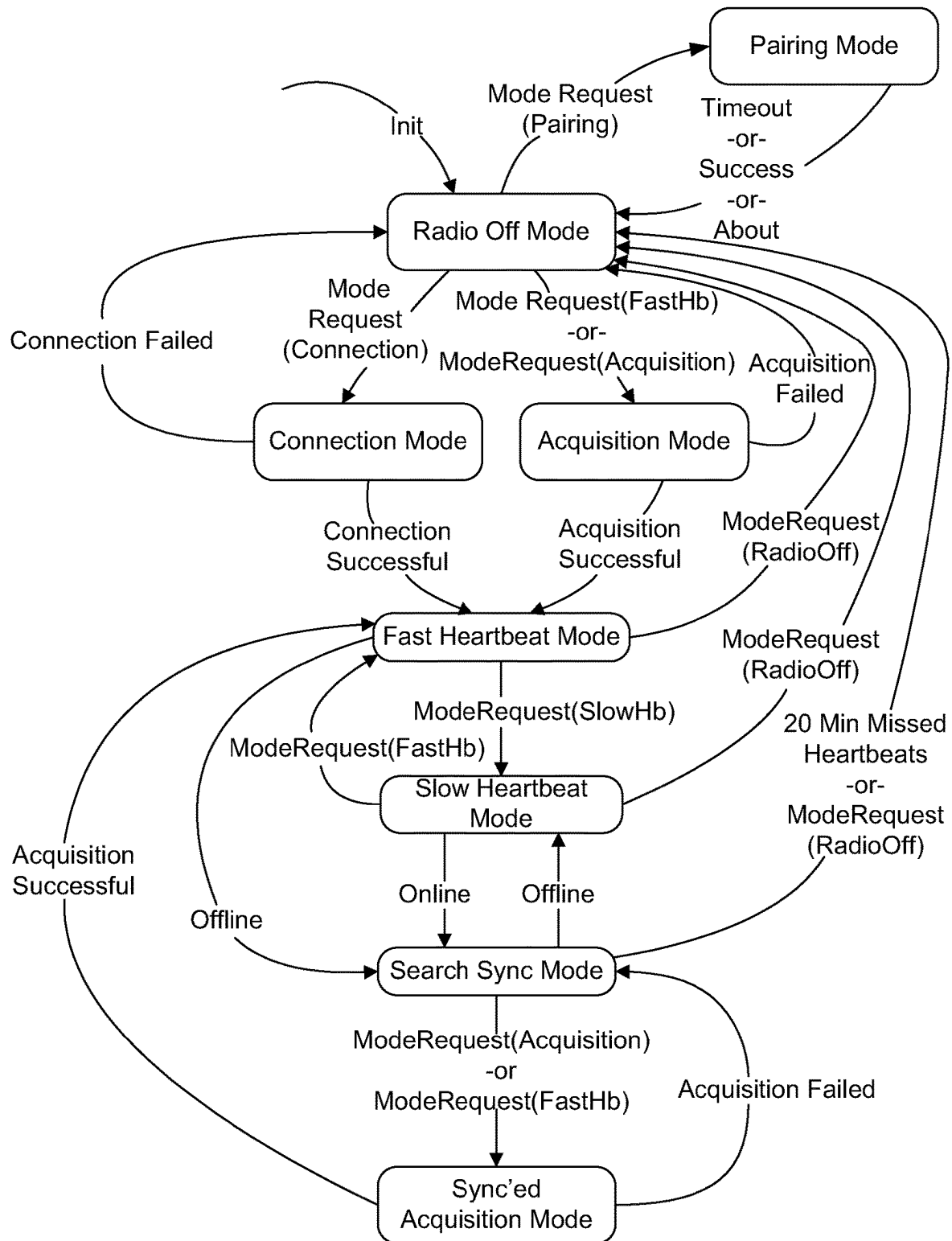


FIG. 120B

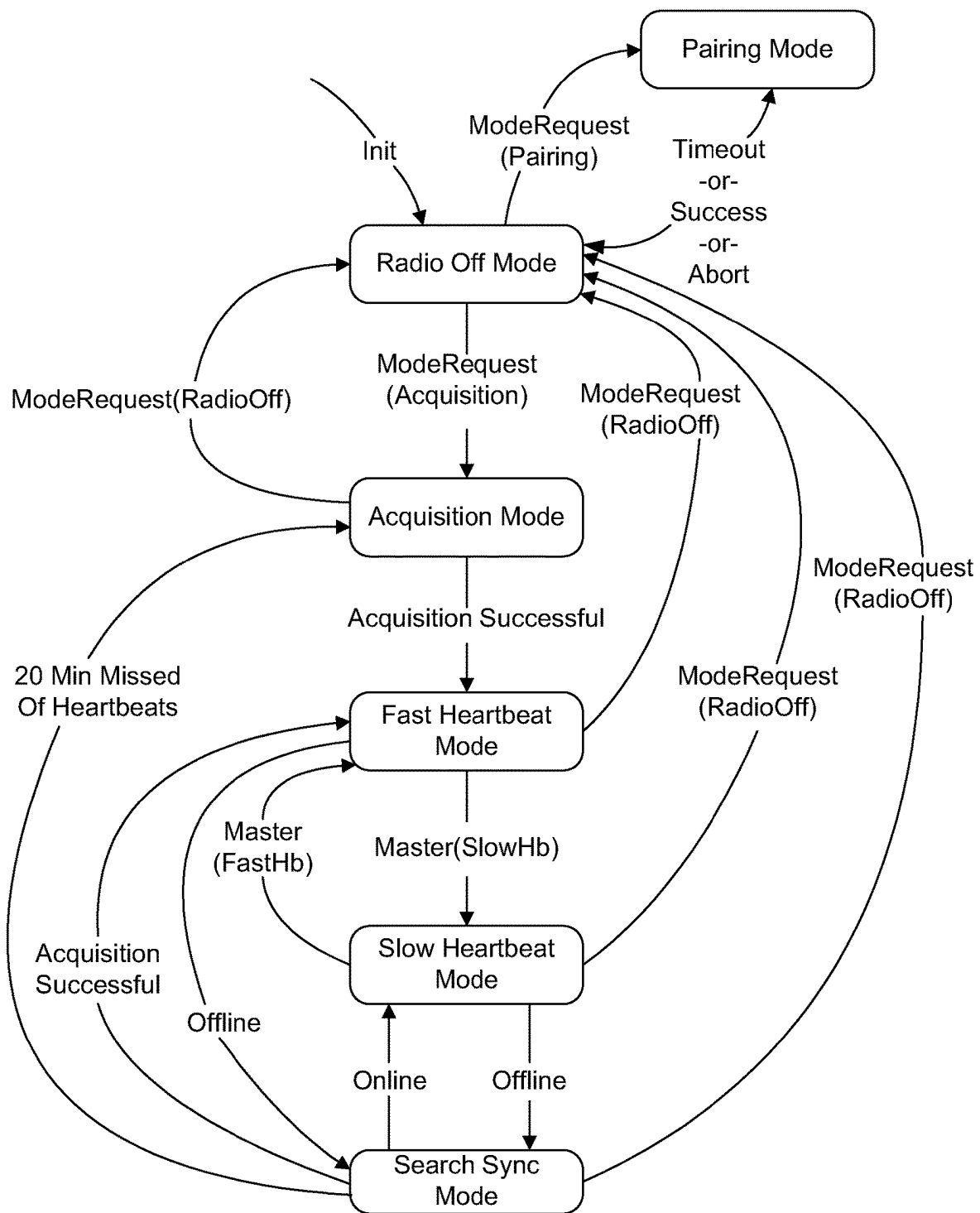


FIG. 120C

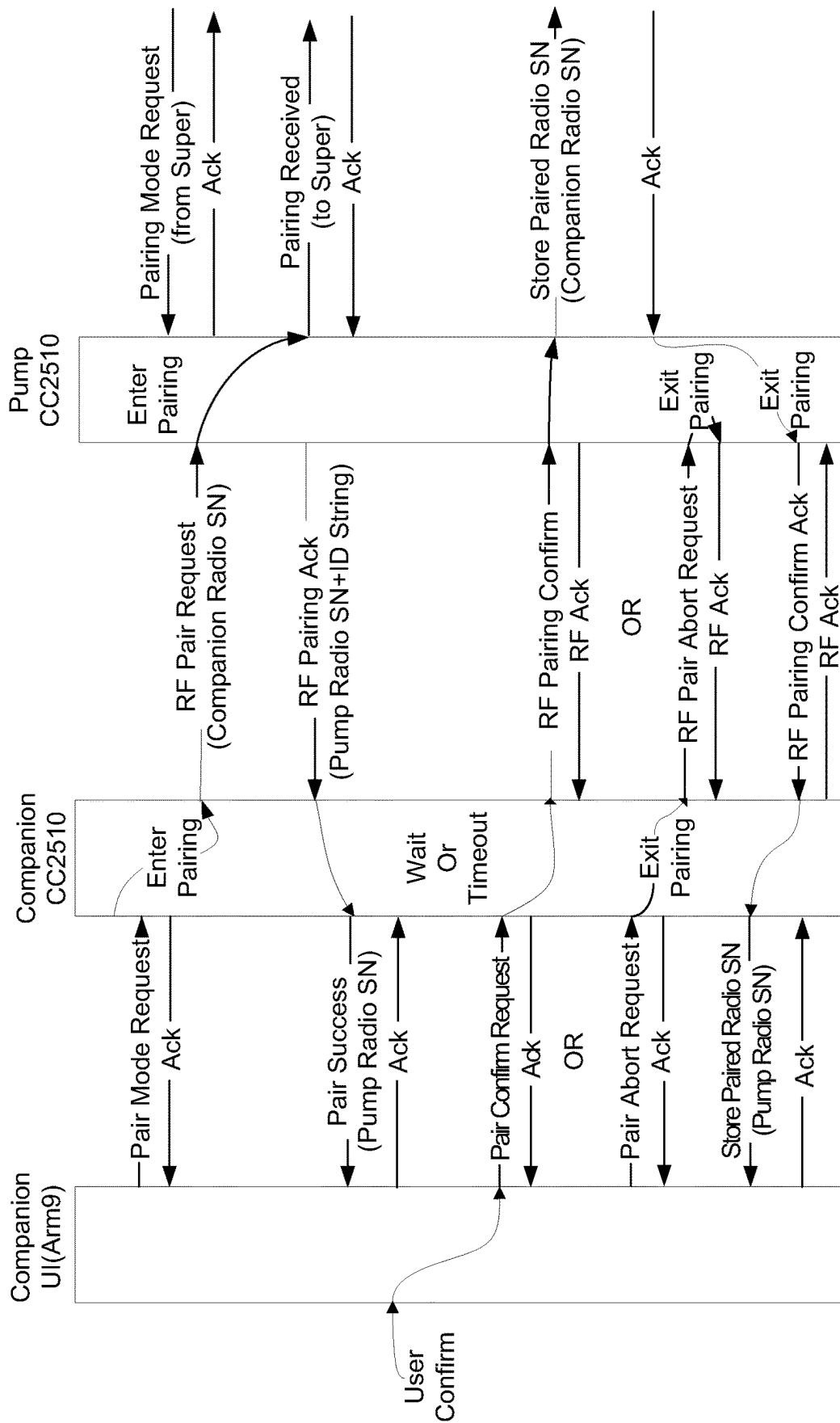


FIG. 120D

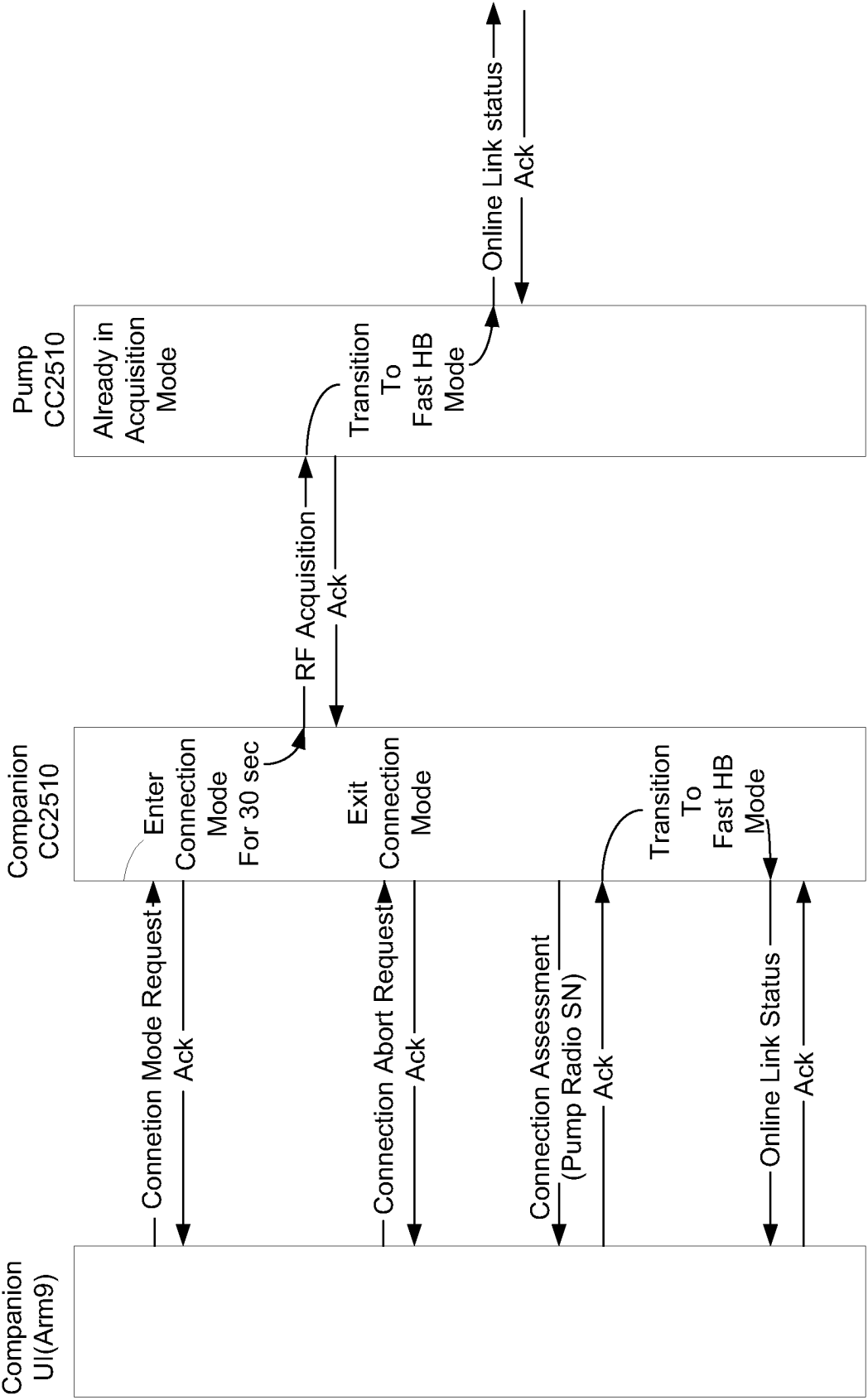


FIG. 120E

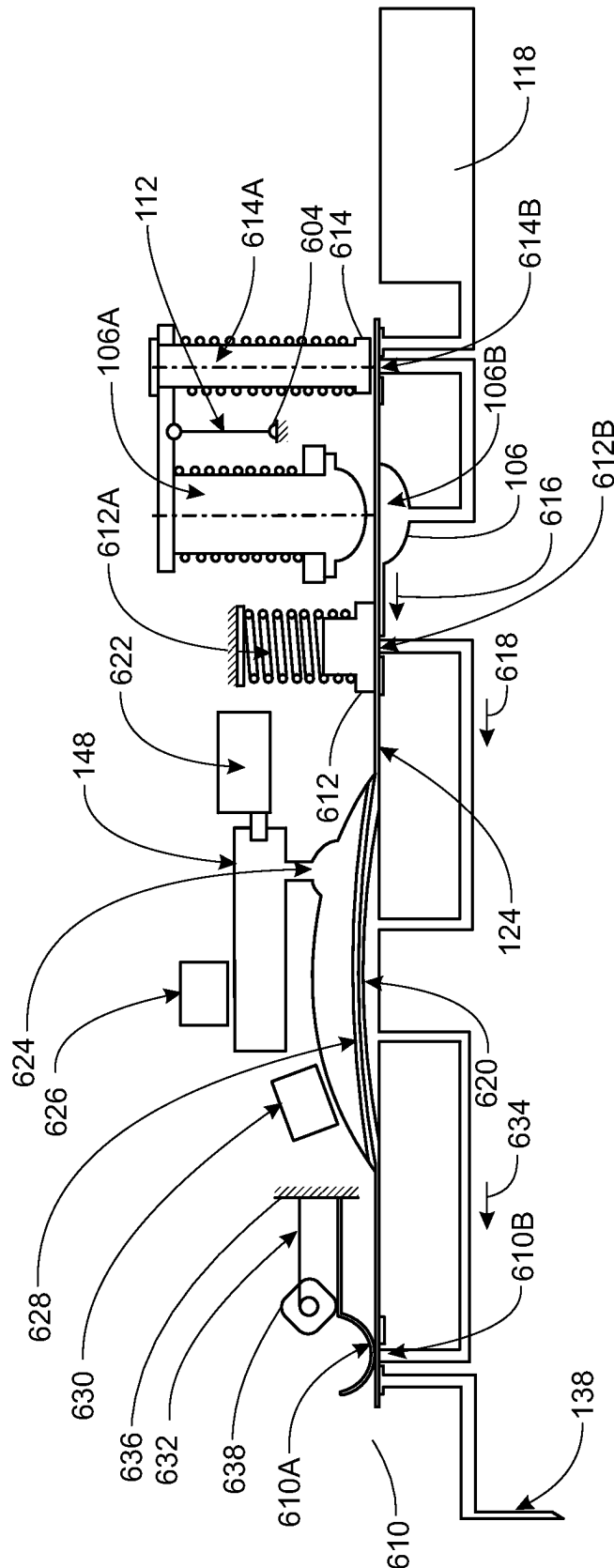


FIG. 121

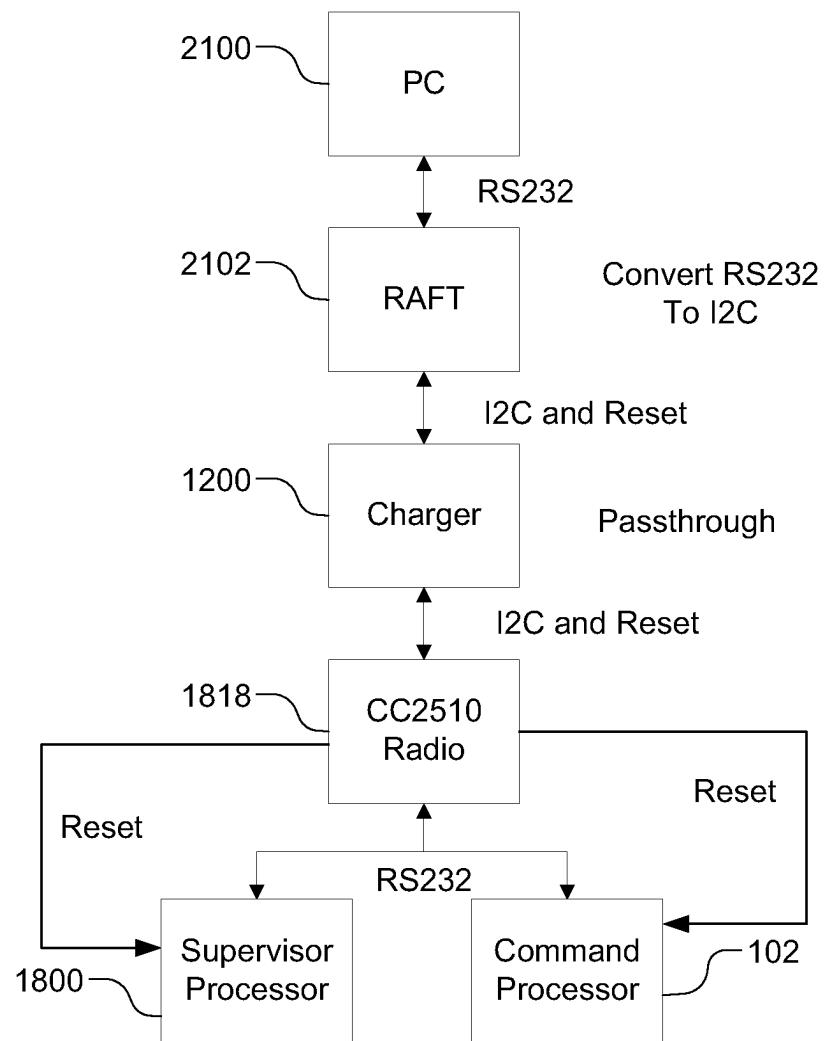


FIG. 122

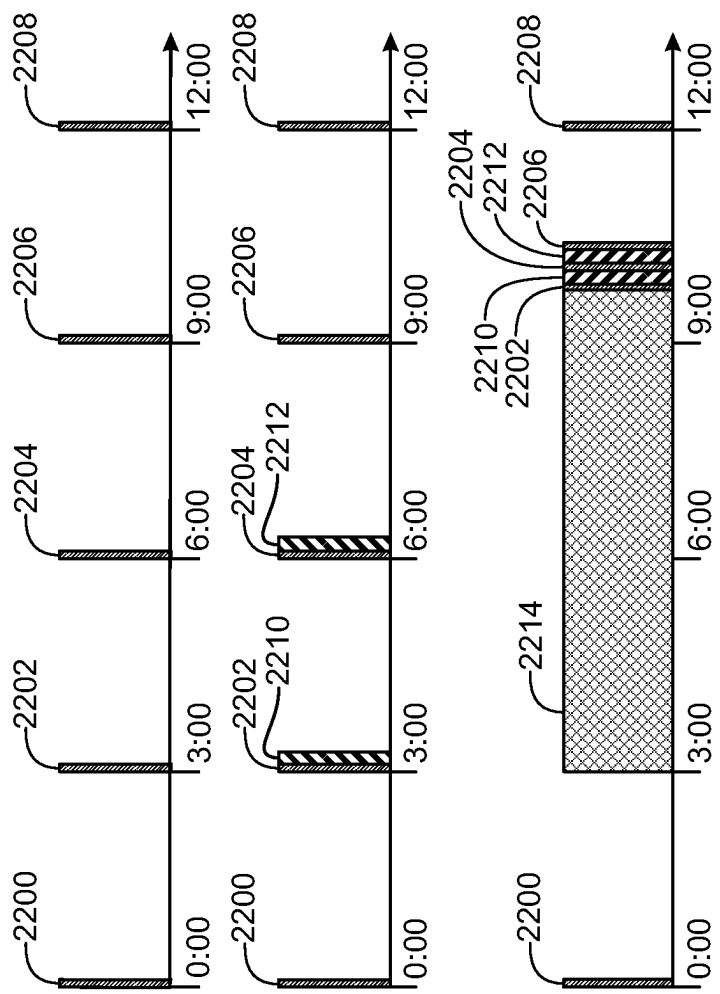


FIG. 123

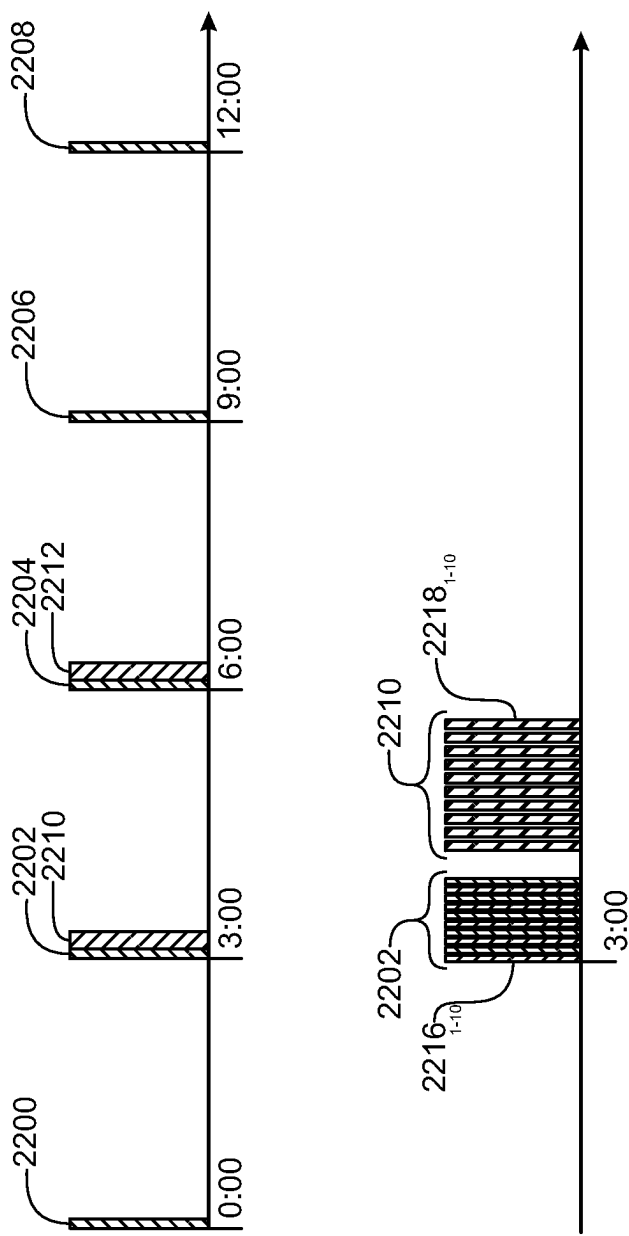


FIG. 124

2300

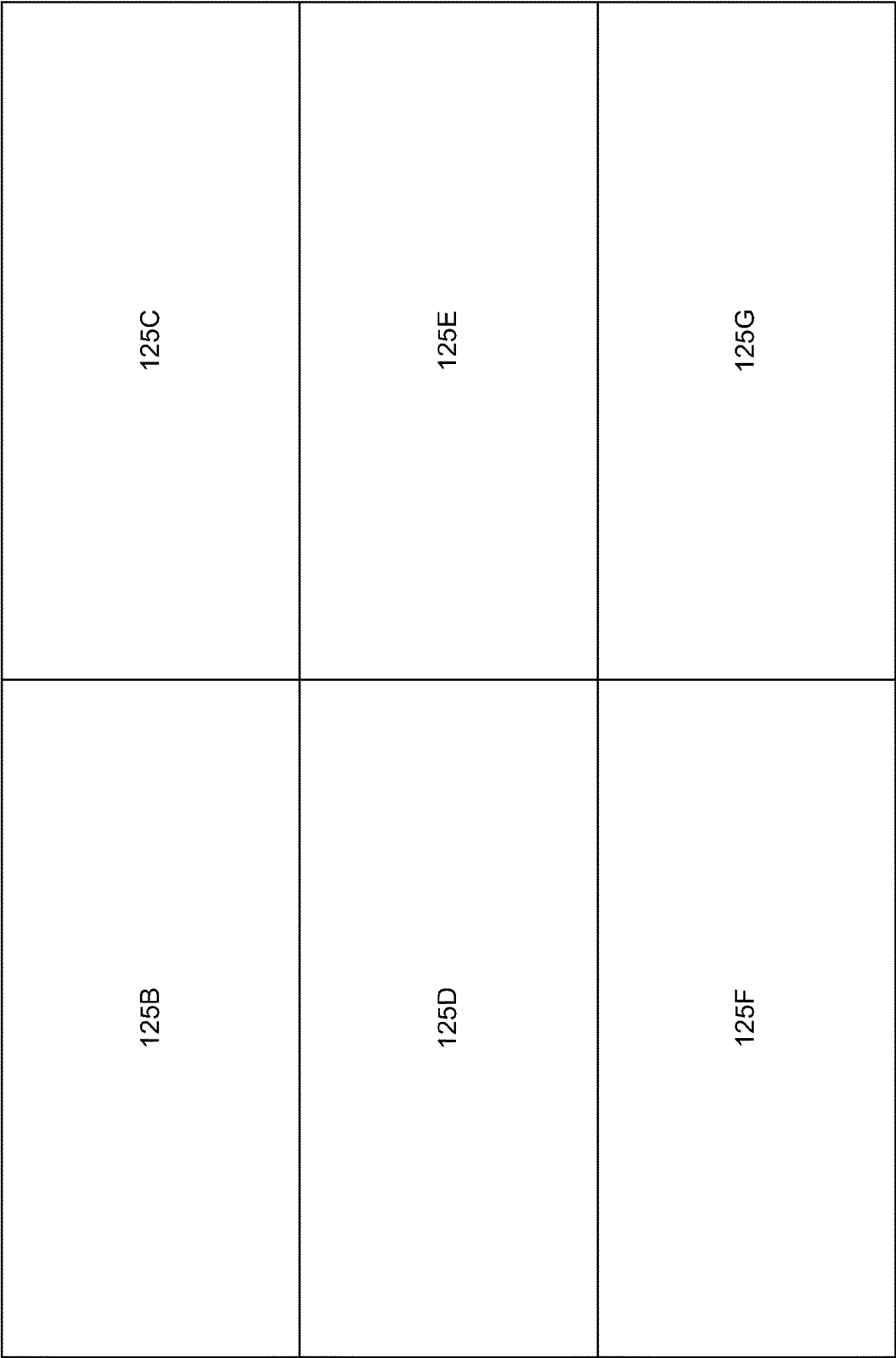
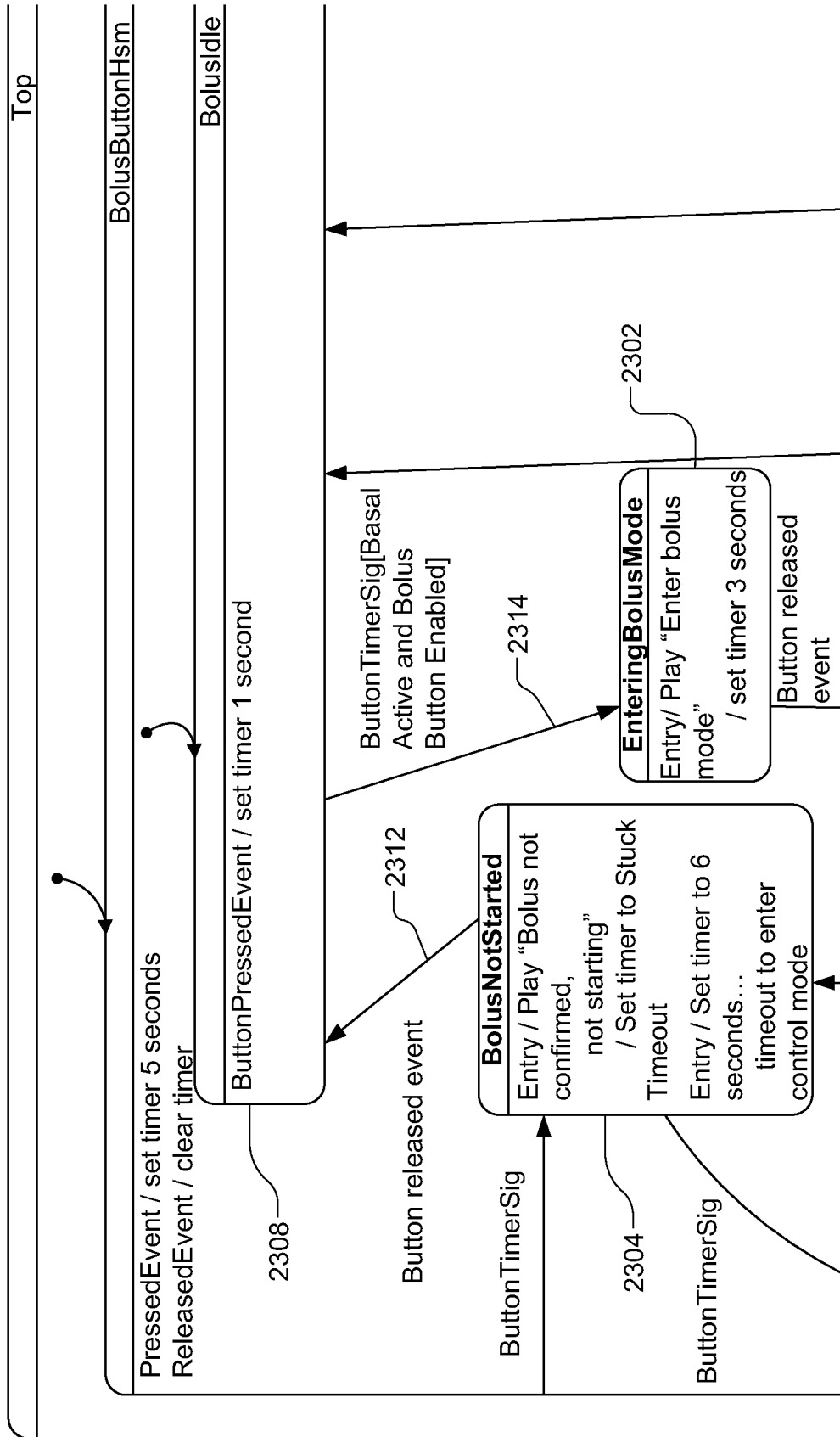
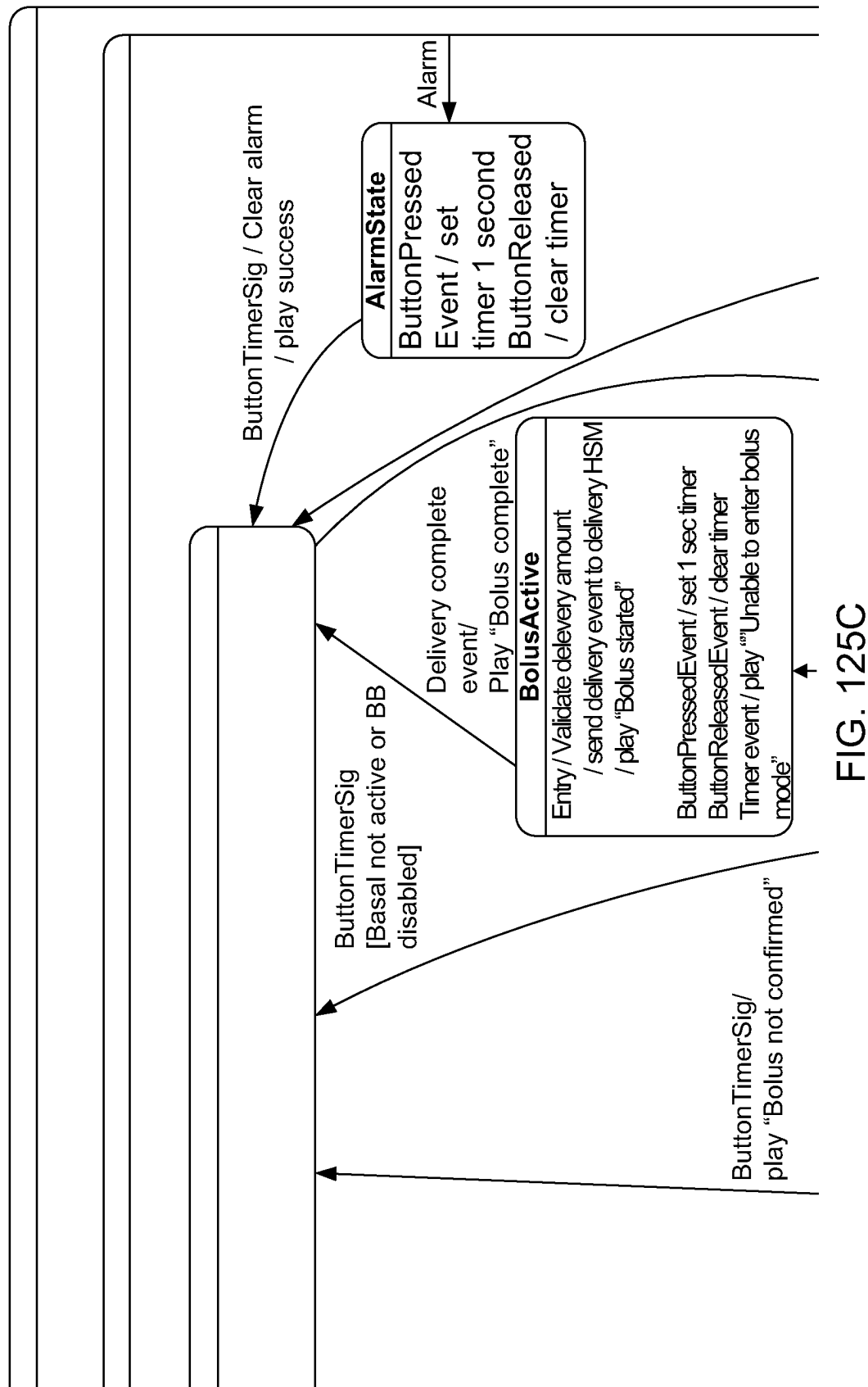


FIG. 125A





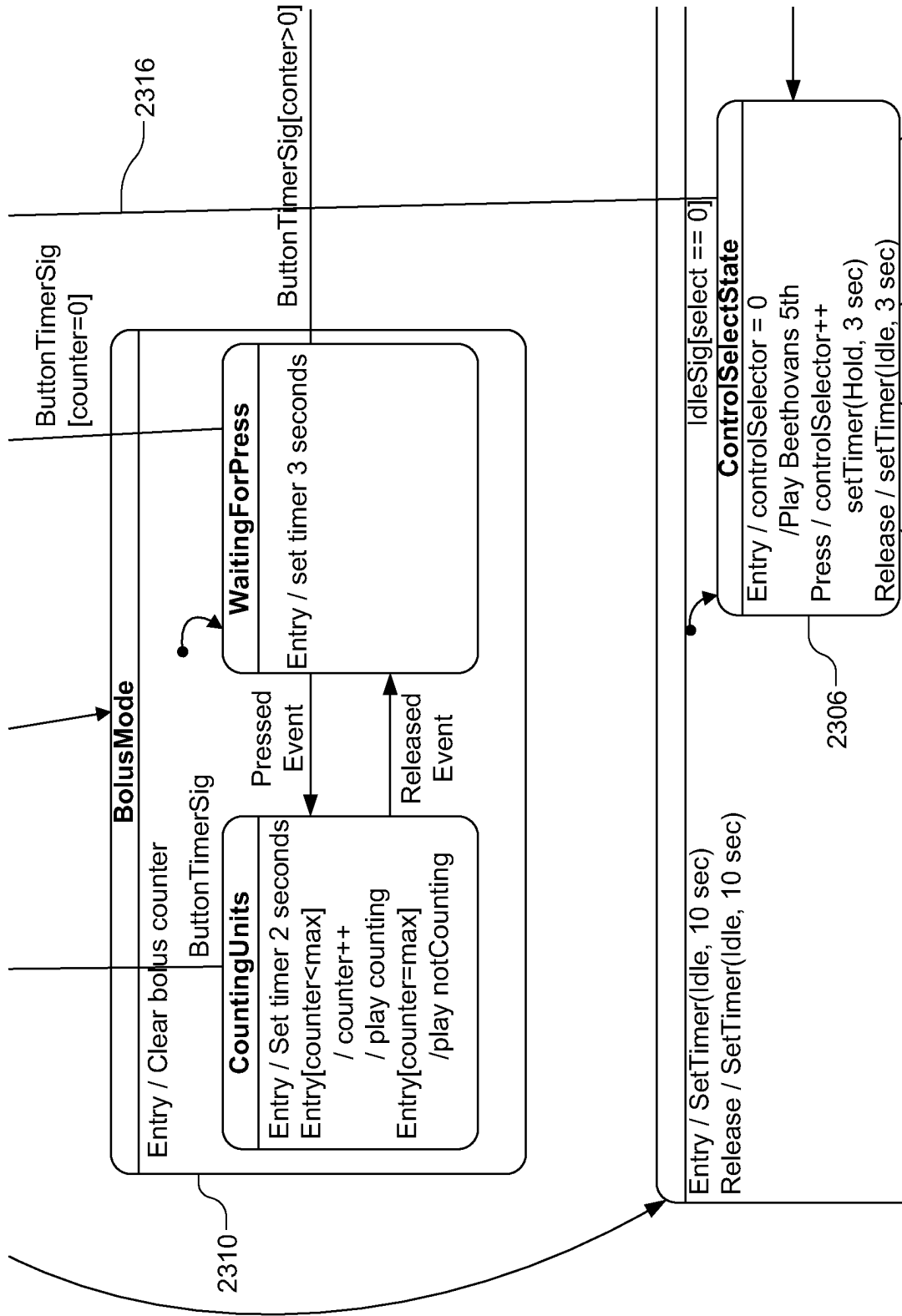


FIG. 125D

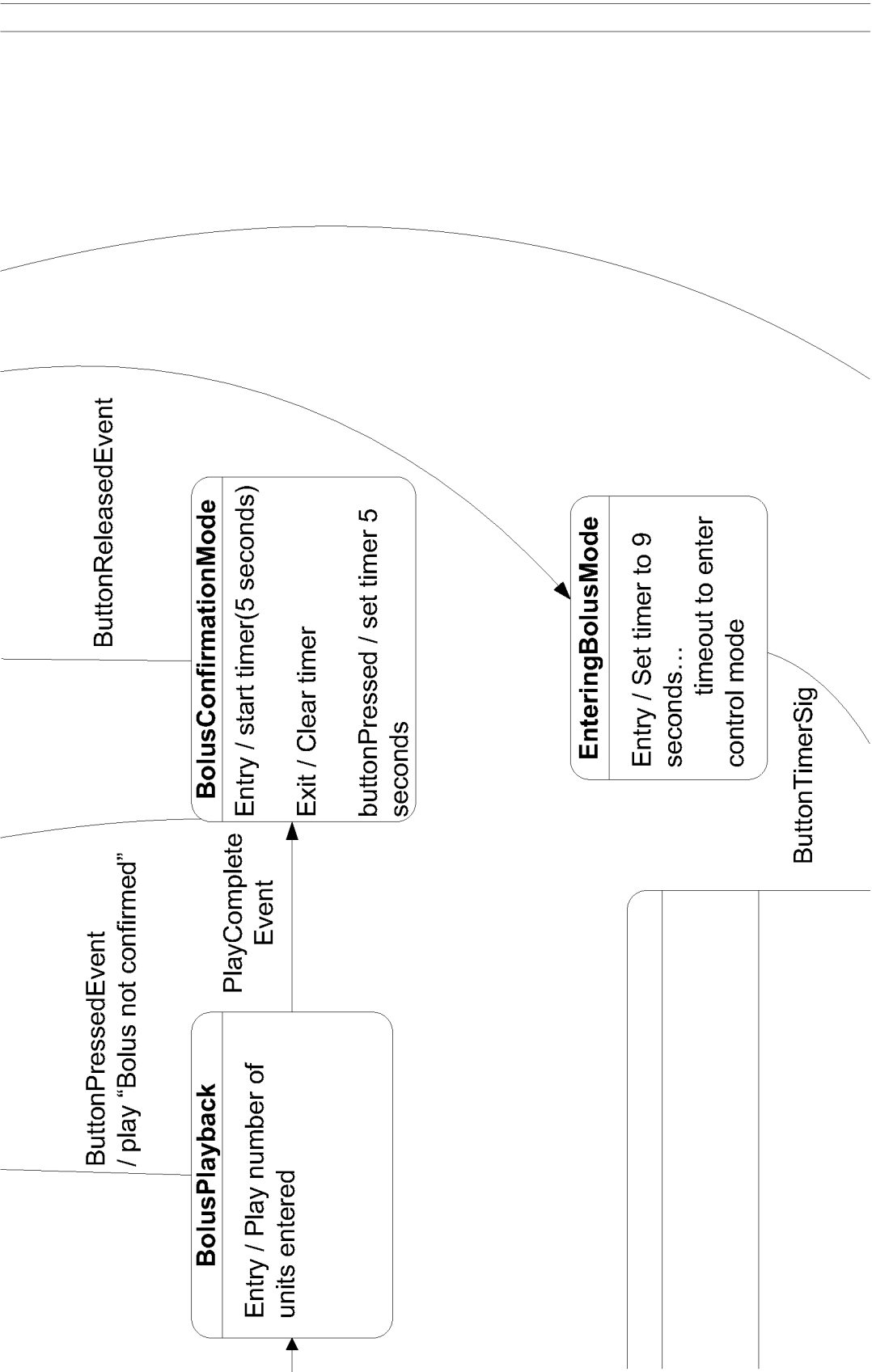


FIG. 125E

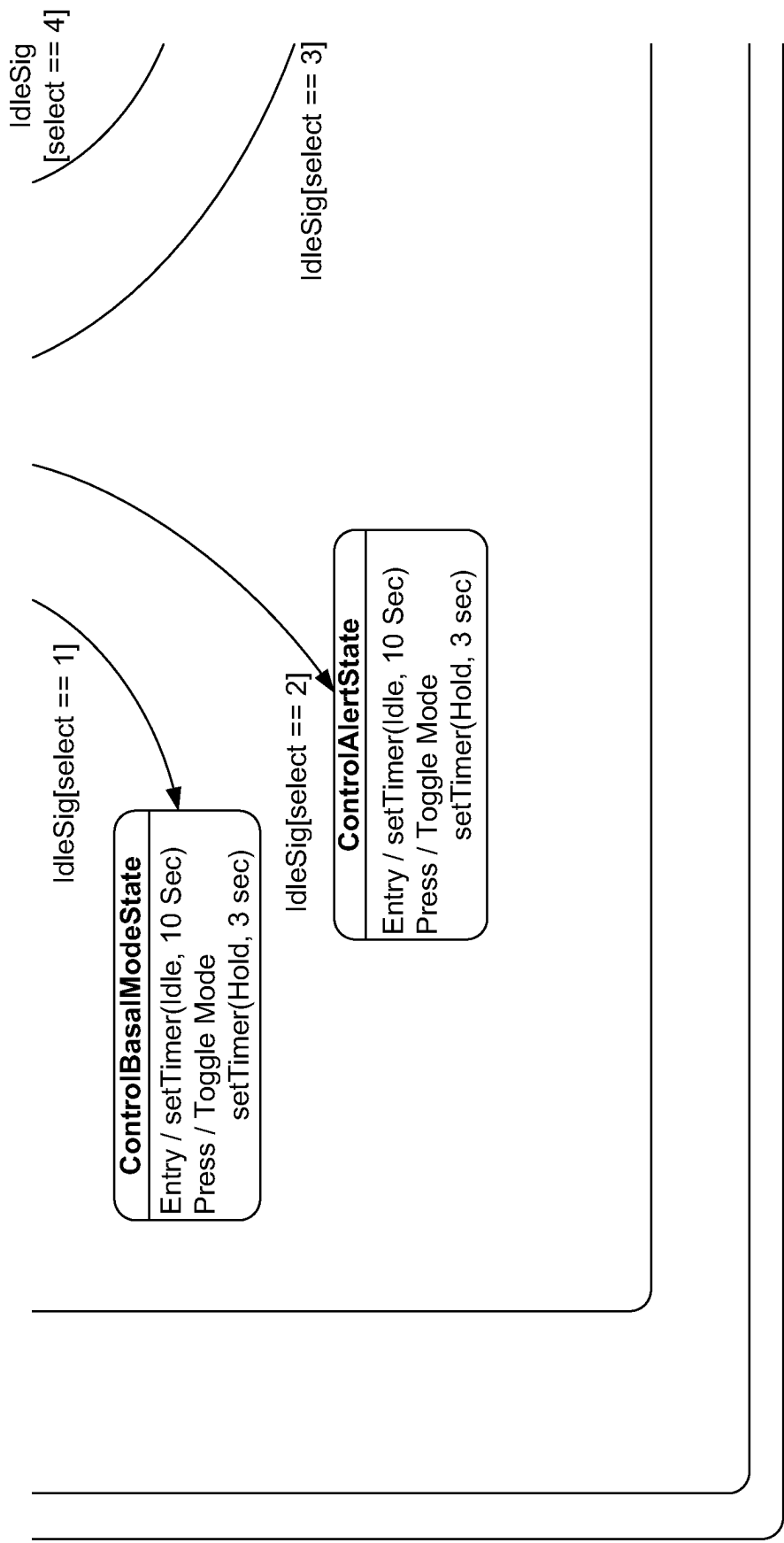


FIG. 125F

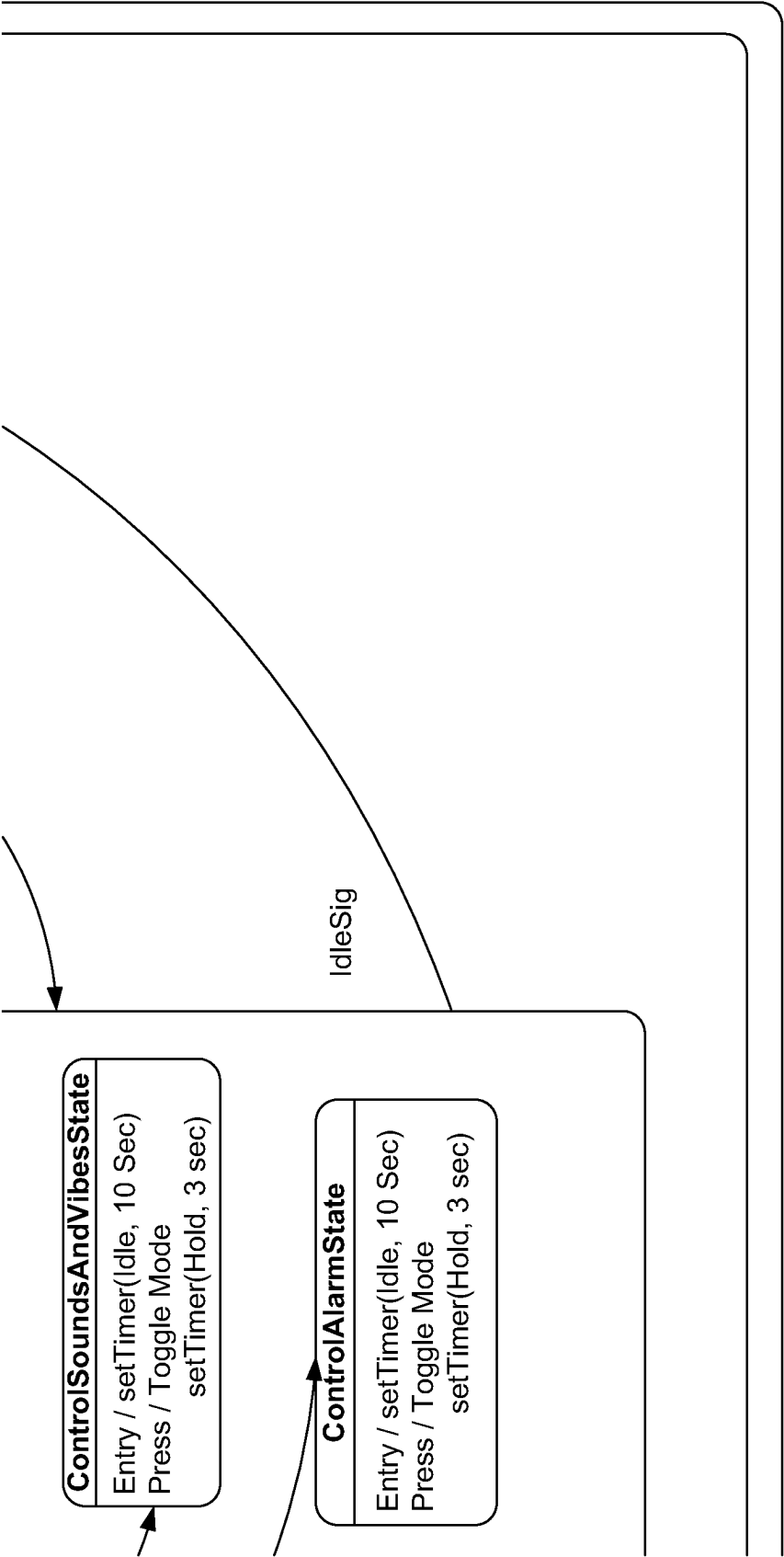


FIG. 125G

2400

126B	126C	126D
126E	126F	126G
126H	126I	126J
126K	126L	126M

FIG. 126A

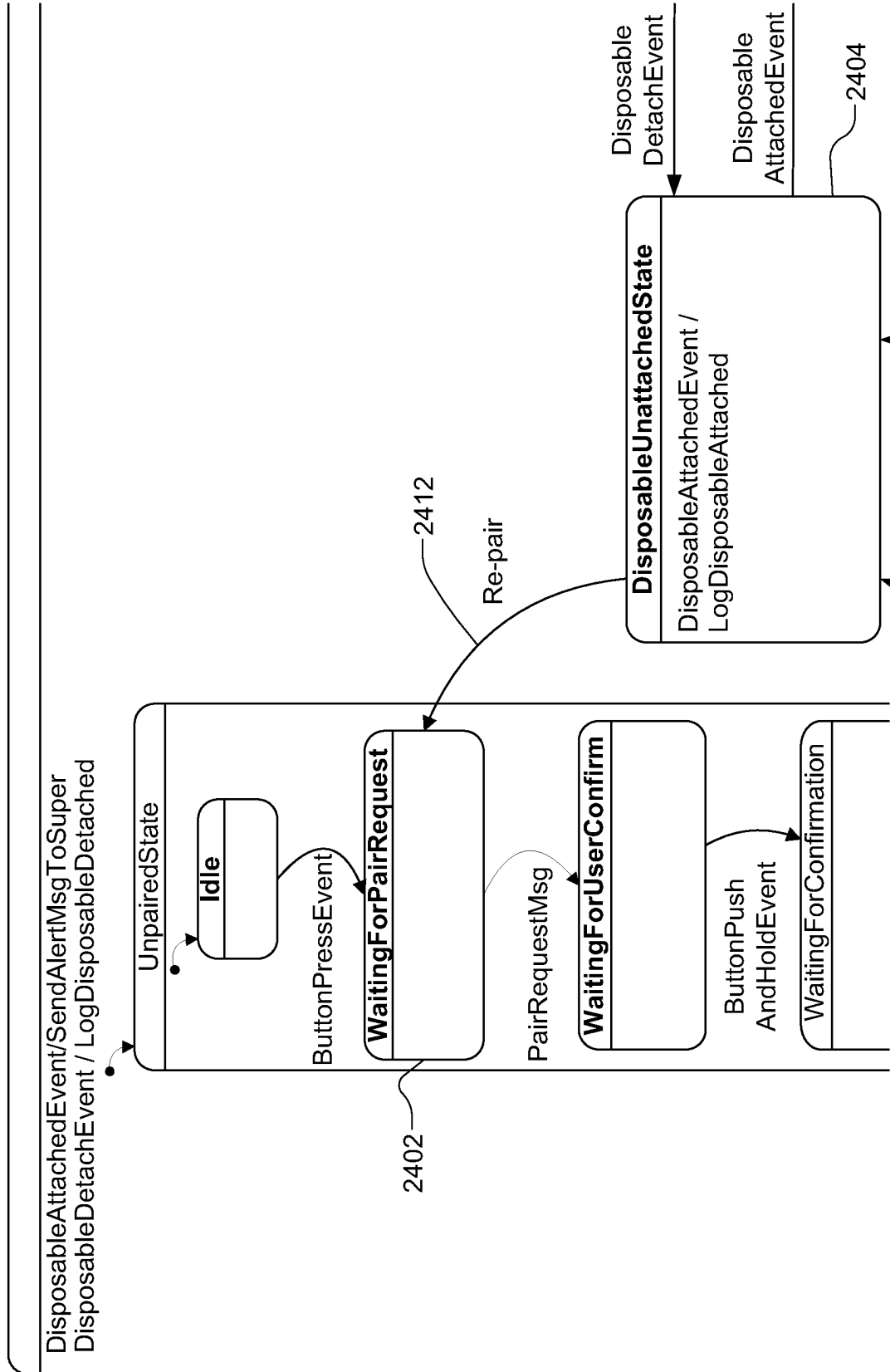


FIG. 126B

Top

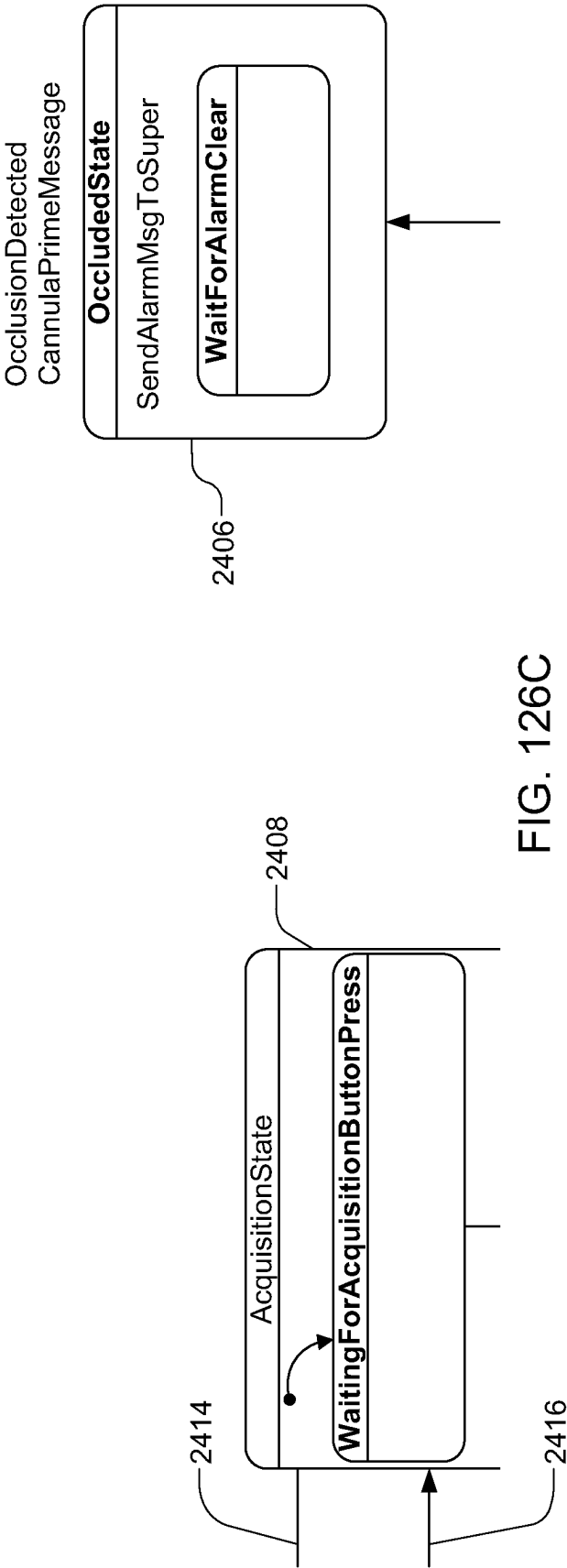


FIG. 126C

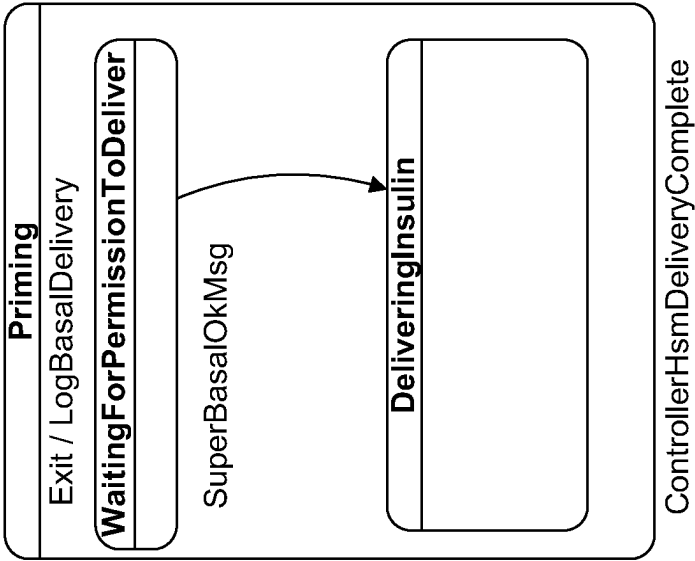


FIG. 126D

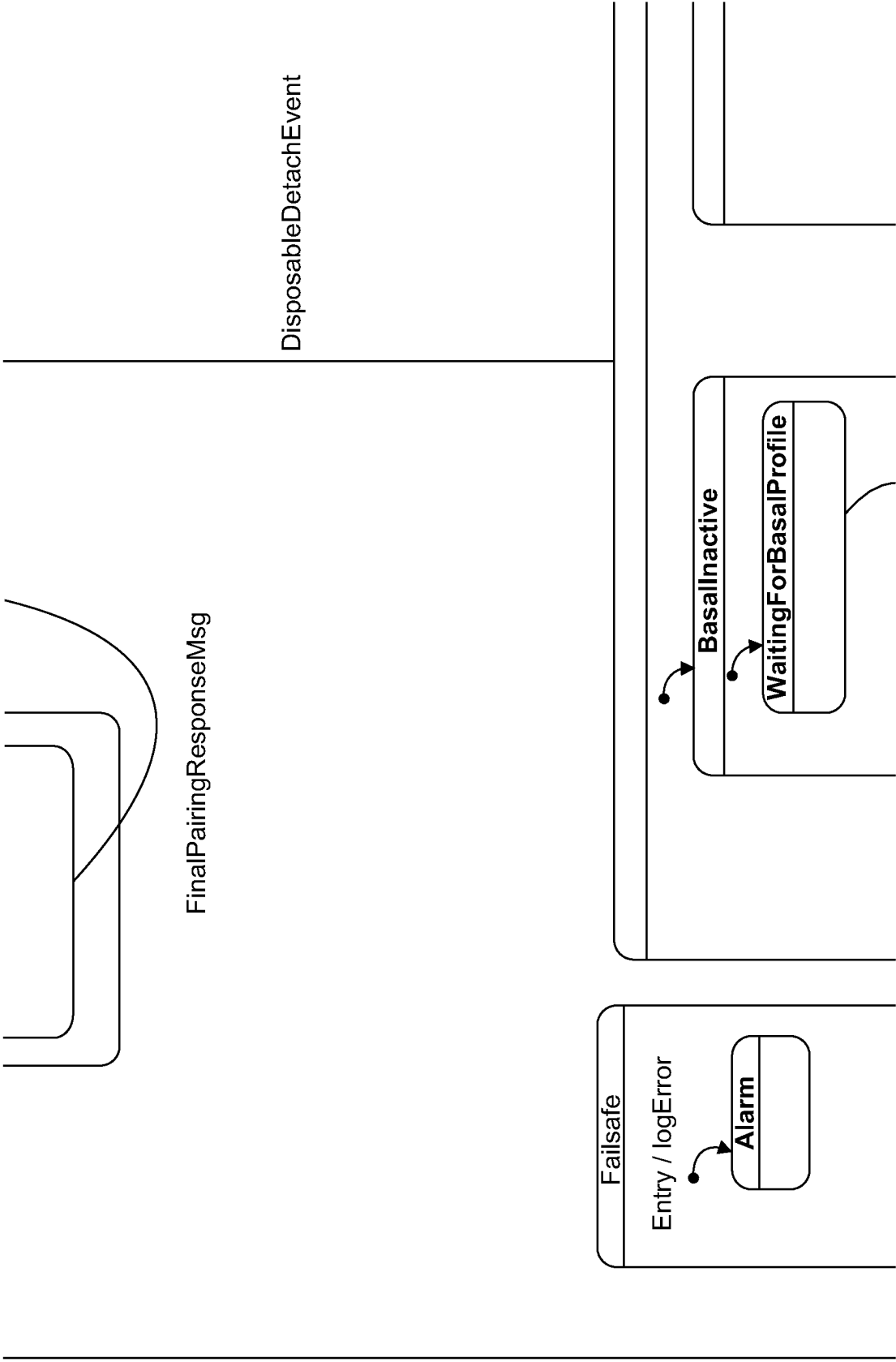


FIG. 126E

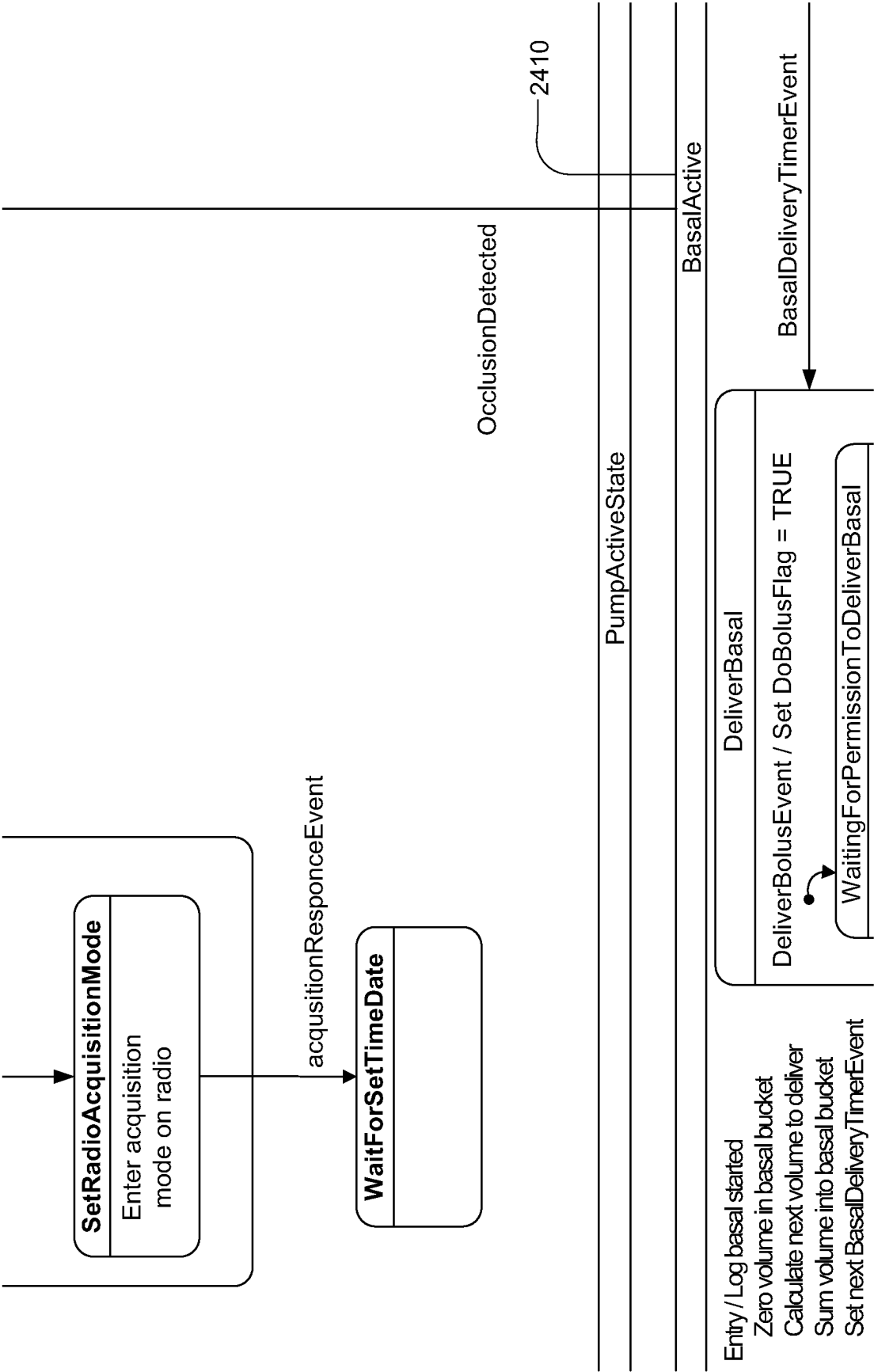


FIG. 126F

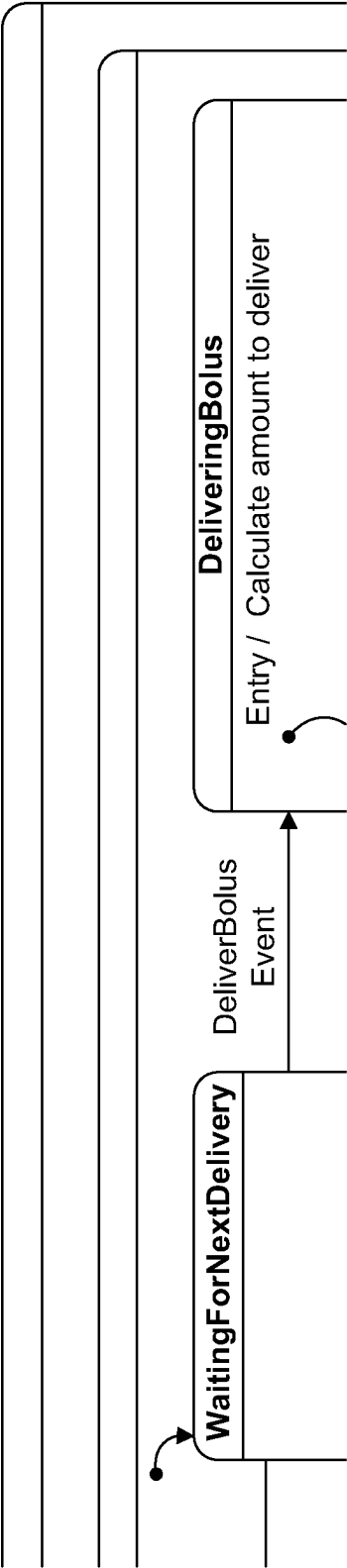


FIG. 126G

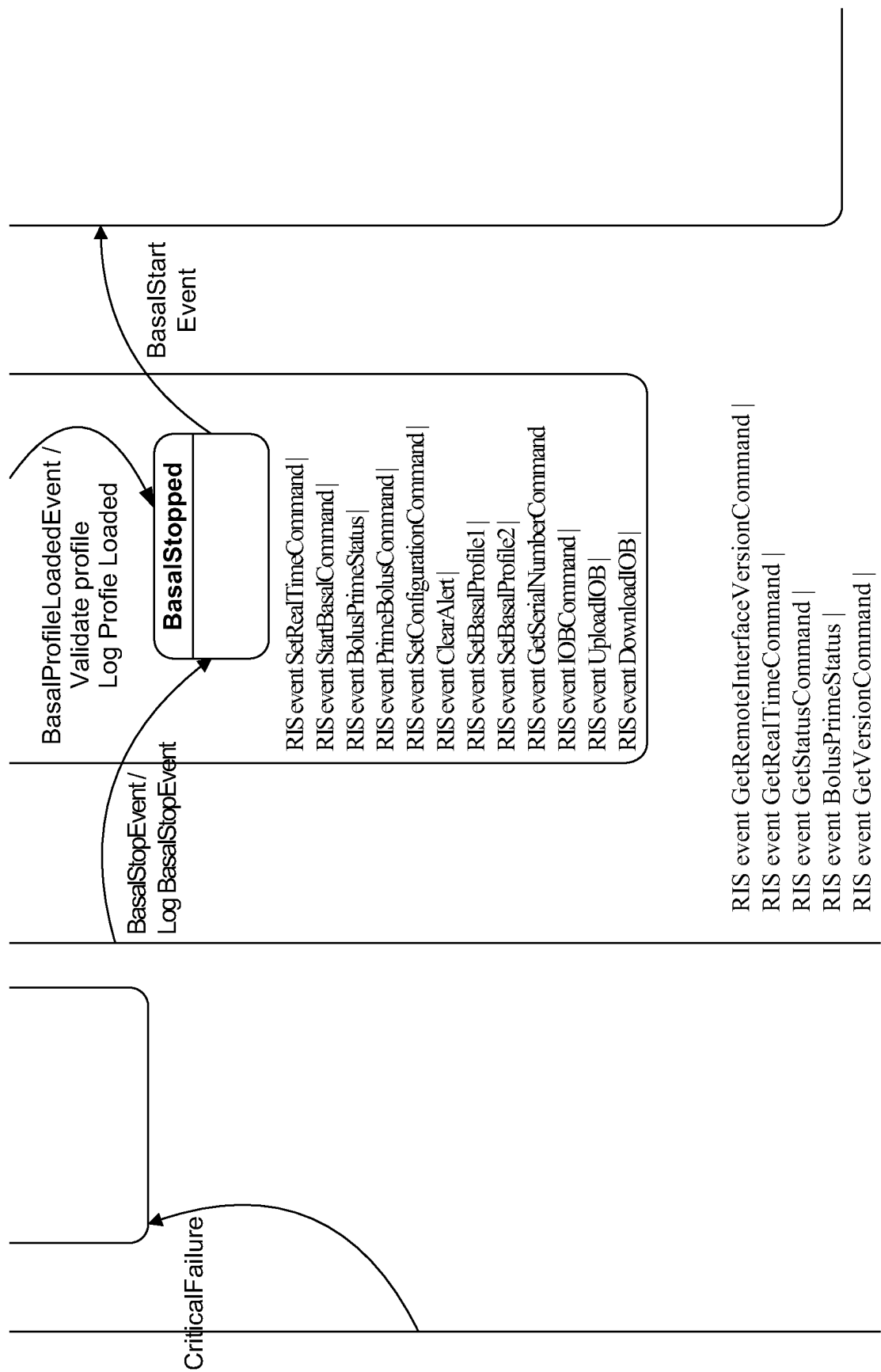


FIG. 126H

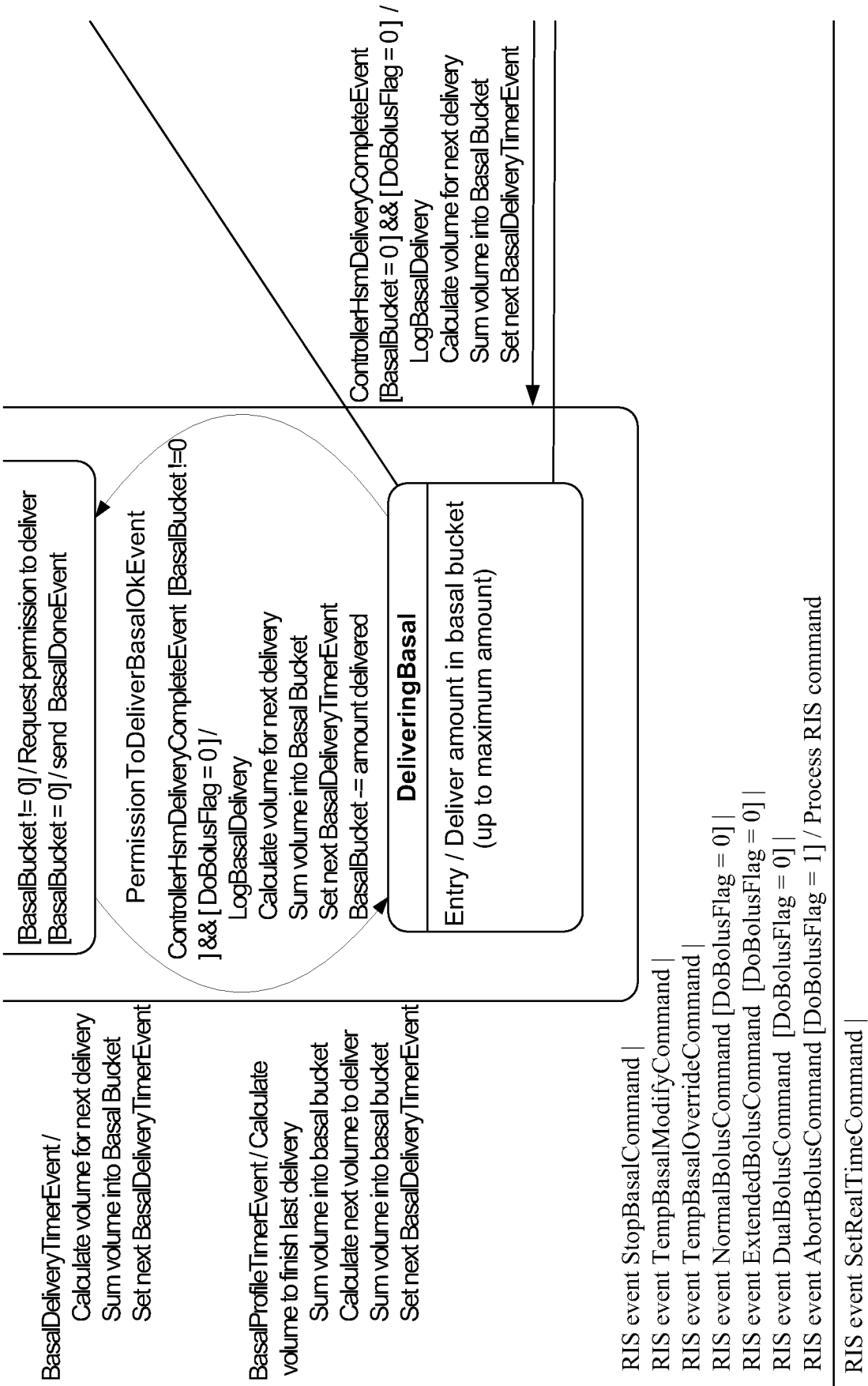


FIG. 126I

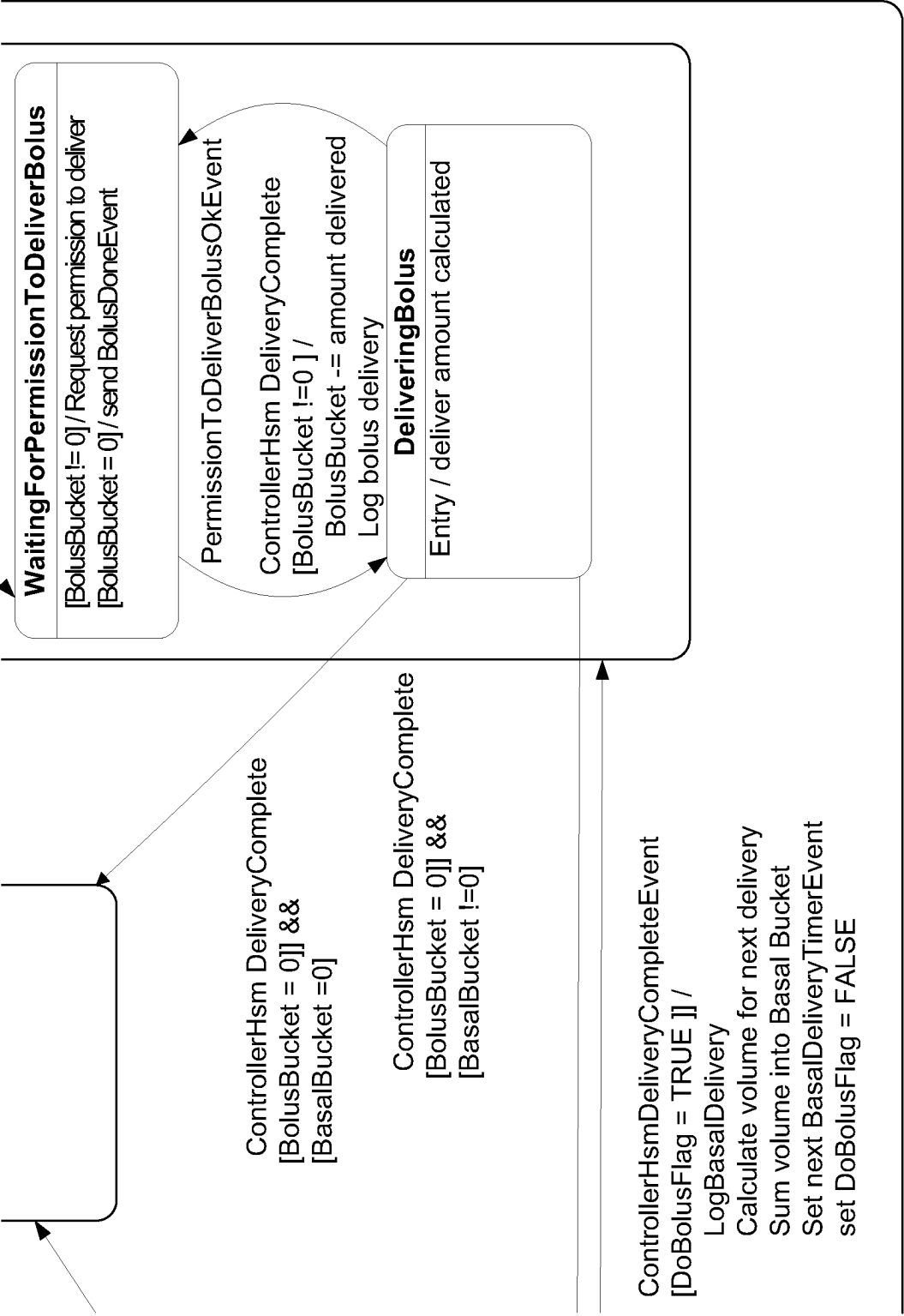


FIG. 126J

RIS event GetSingleVersionCommand |
RIS event GetAllVersionCommand |
RIS event ClearAlert |
RIS event LogCommand |
RIS event GetTotalRecordsLoggedCommand |
RIS event GetLogRecord |
RIS event ErasePumpLogs |
RIS event GetSerialNumberCommand
RIS event IOBCommand |
RIS event UploadIOB |
RIS event DownloadIOB |
RIS event Airplane Mode/ Process RIS command

FIG. 126K

RIS event StartBasalCommand |
RIS event StopBasalCommand |
RIS event TempBasalModifyCommand |
RIS event TempBasalOverrideCommand |
RIS event BolusPrimeStatus
RIS event StopBasalCommand |
RIS event TempBasalModifyCommand |
RIS event TempBasalOverrideCommand |
RIS event NormalBolusCommand |
RIS event ExtendedBolusCommand |
RIS event DualBolusCommand |
RIS event AbortBolusCommand / NACK RIS command

FIG. 126L

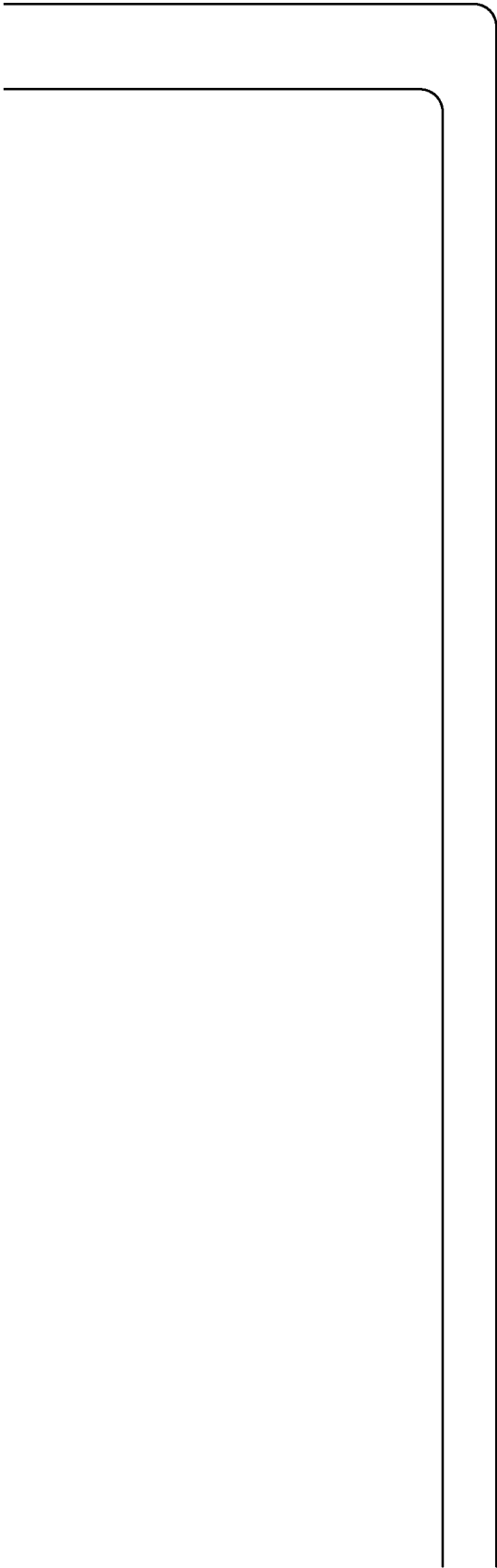


FIG. 126M

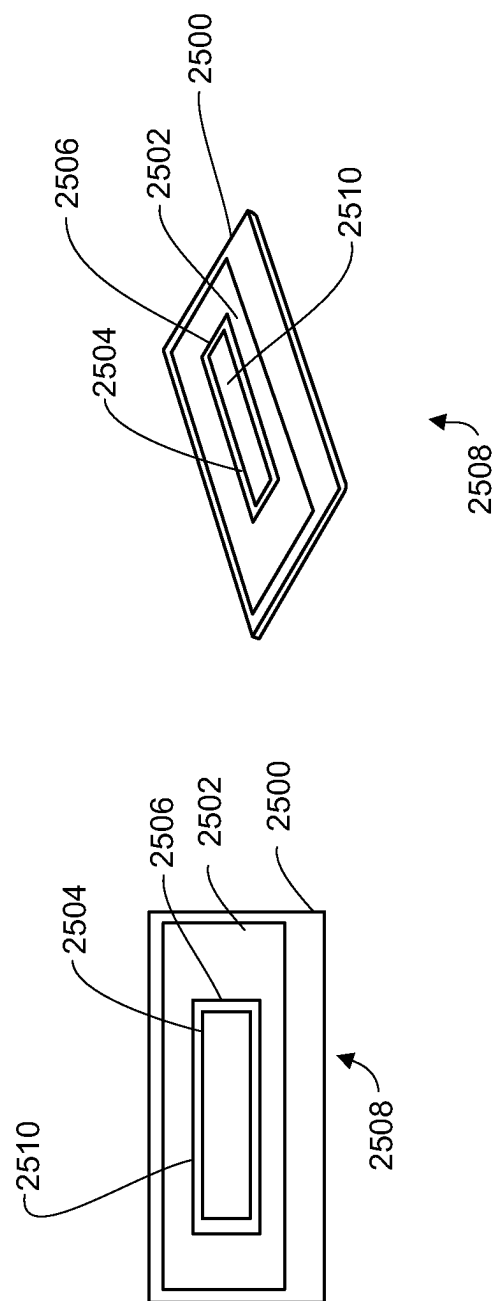


FIG. 127

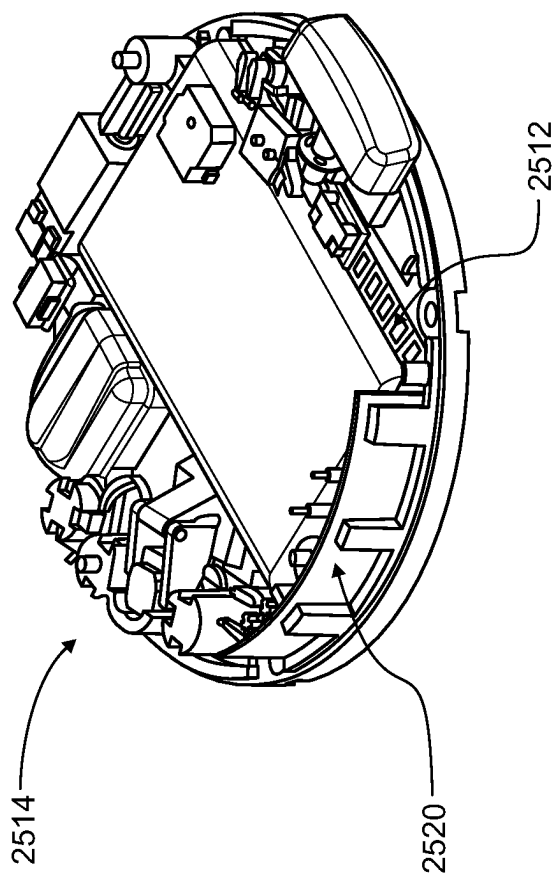


FIG.128

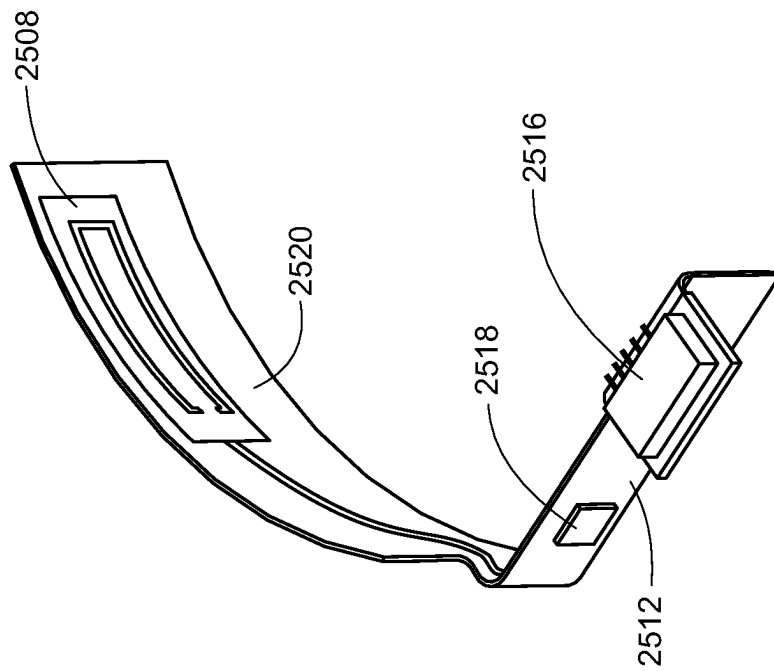


FIG. 129

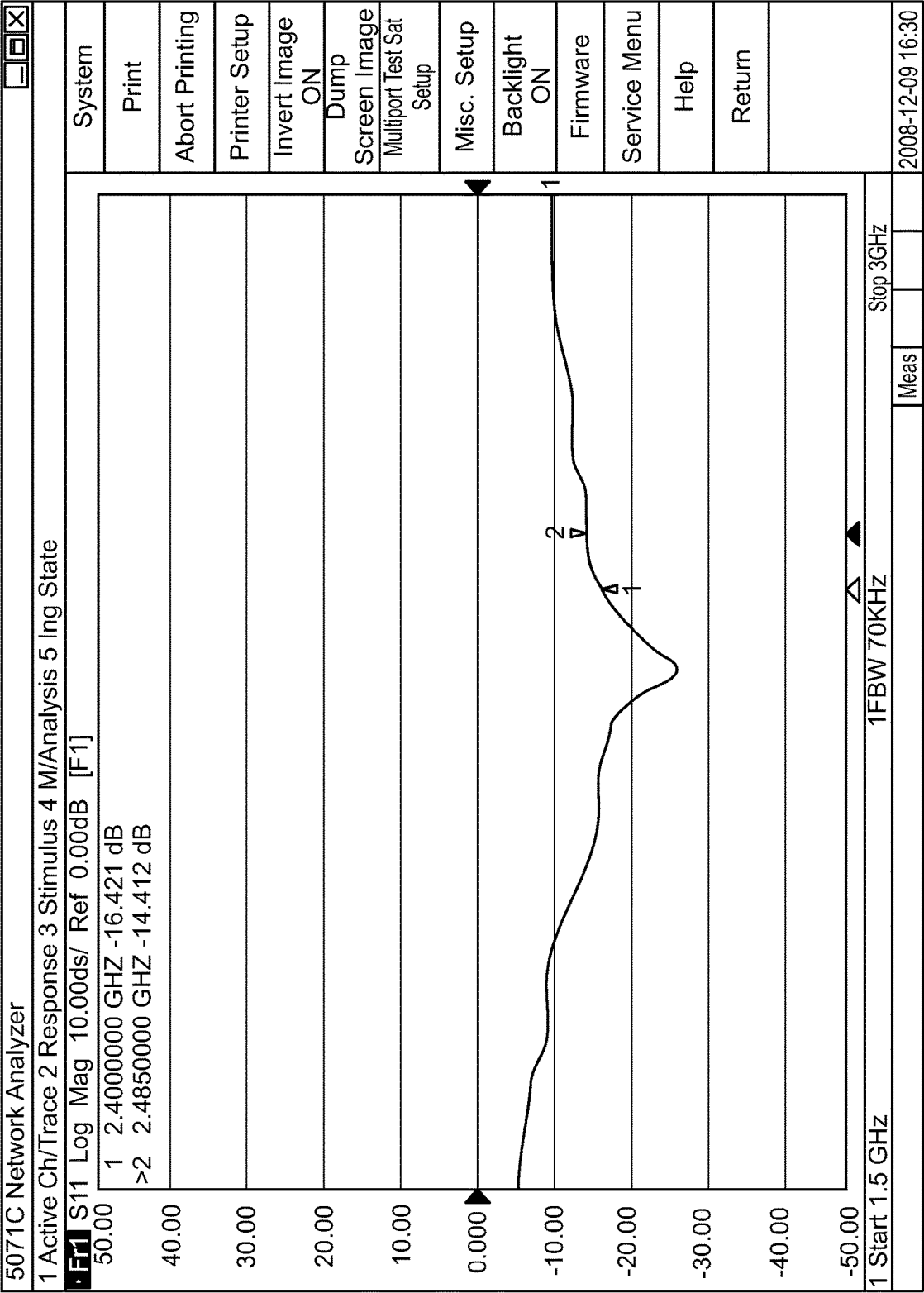


FIG. 130

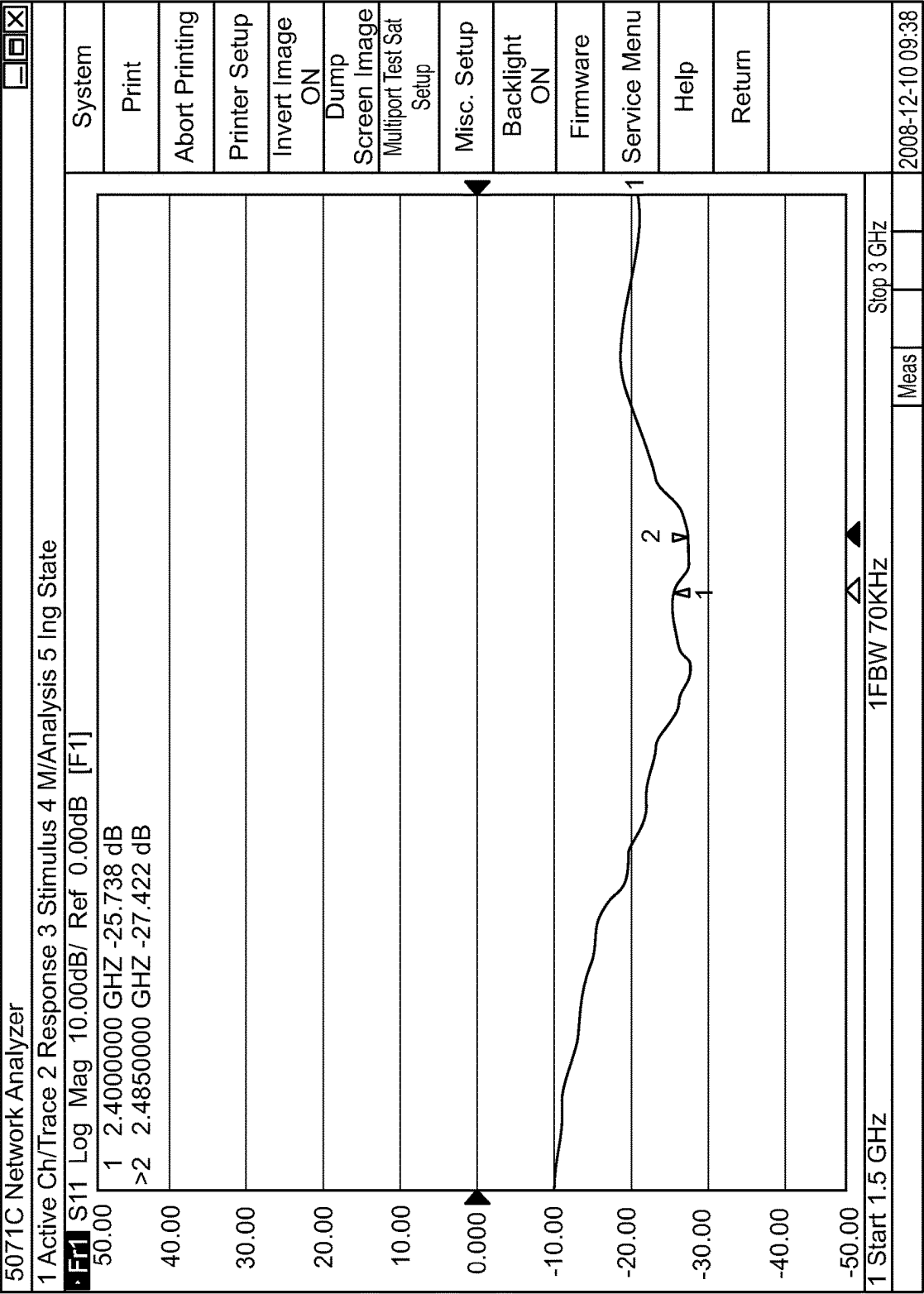


FIG. 130A

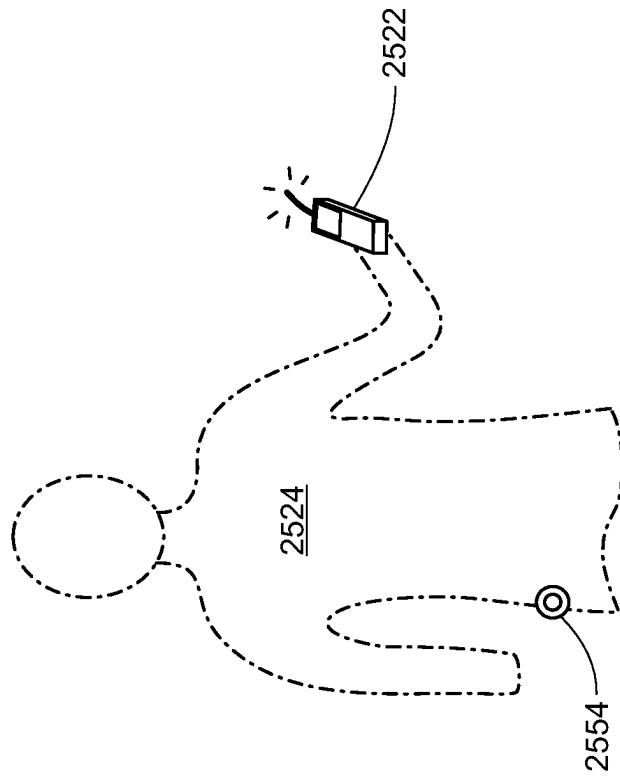


FIG. 131

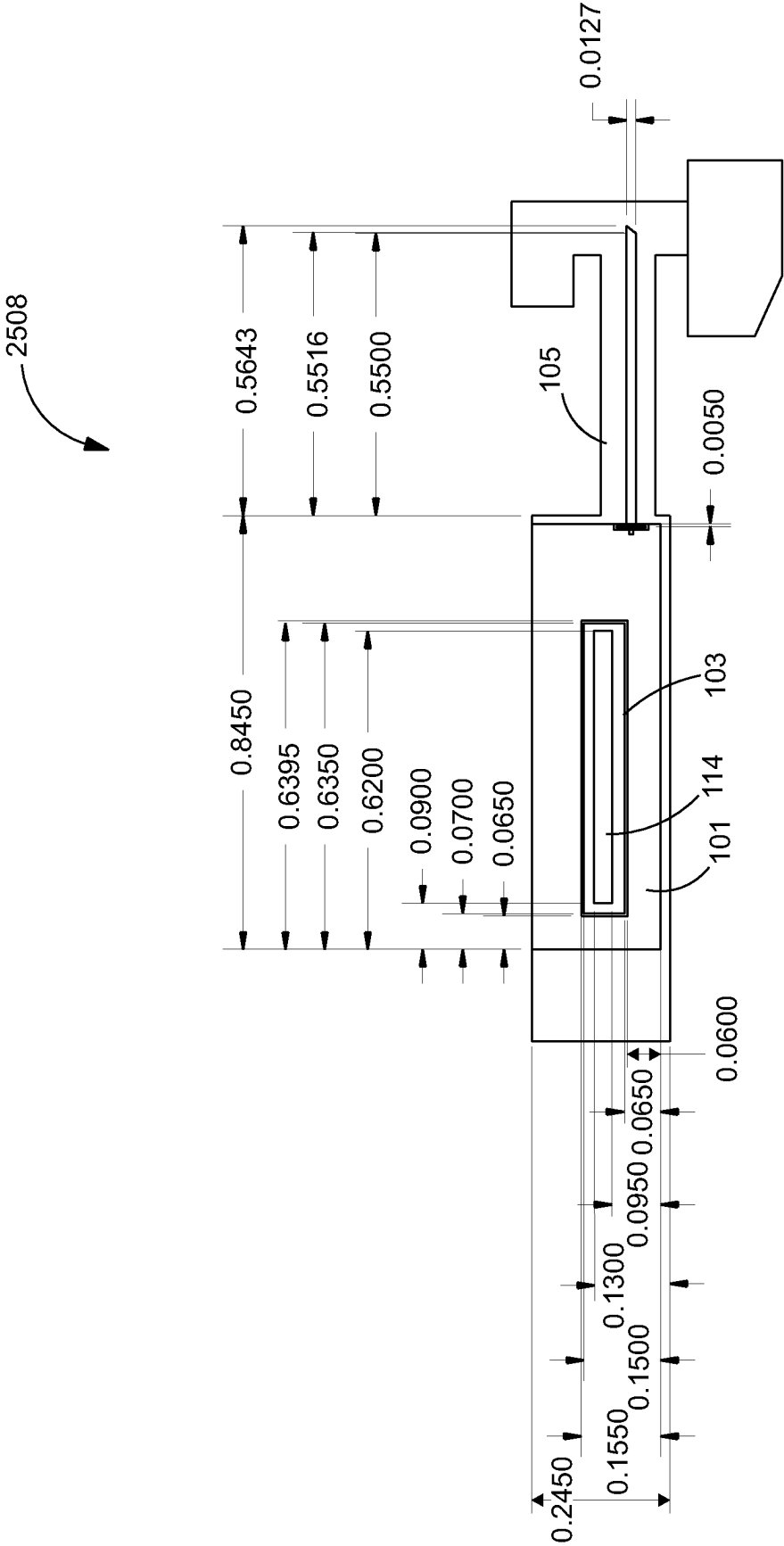


FIG.132

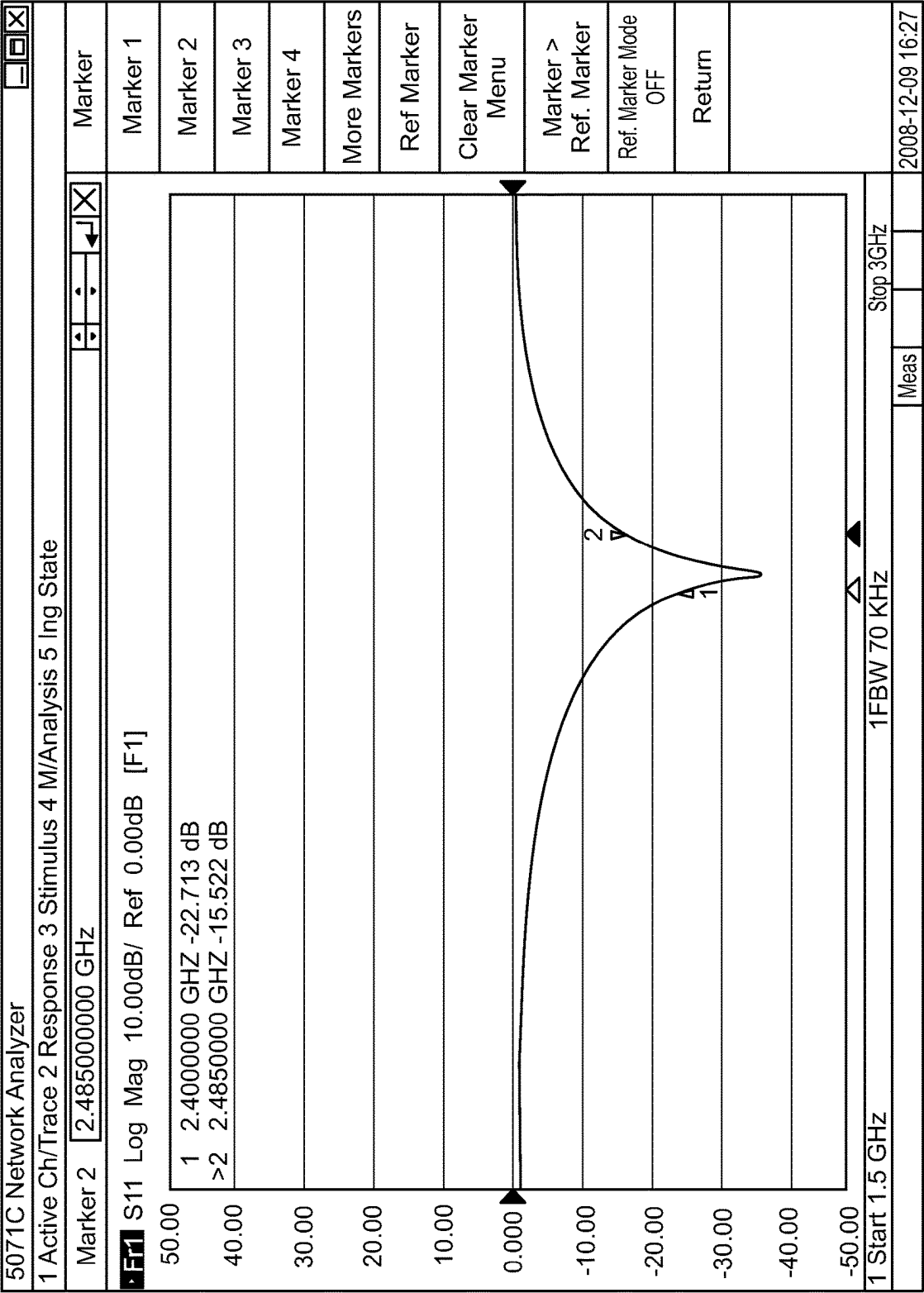


FIG. 133

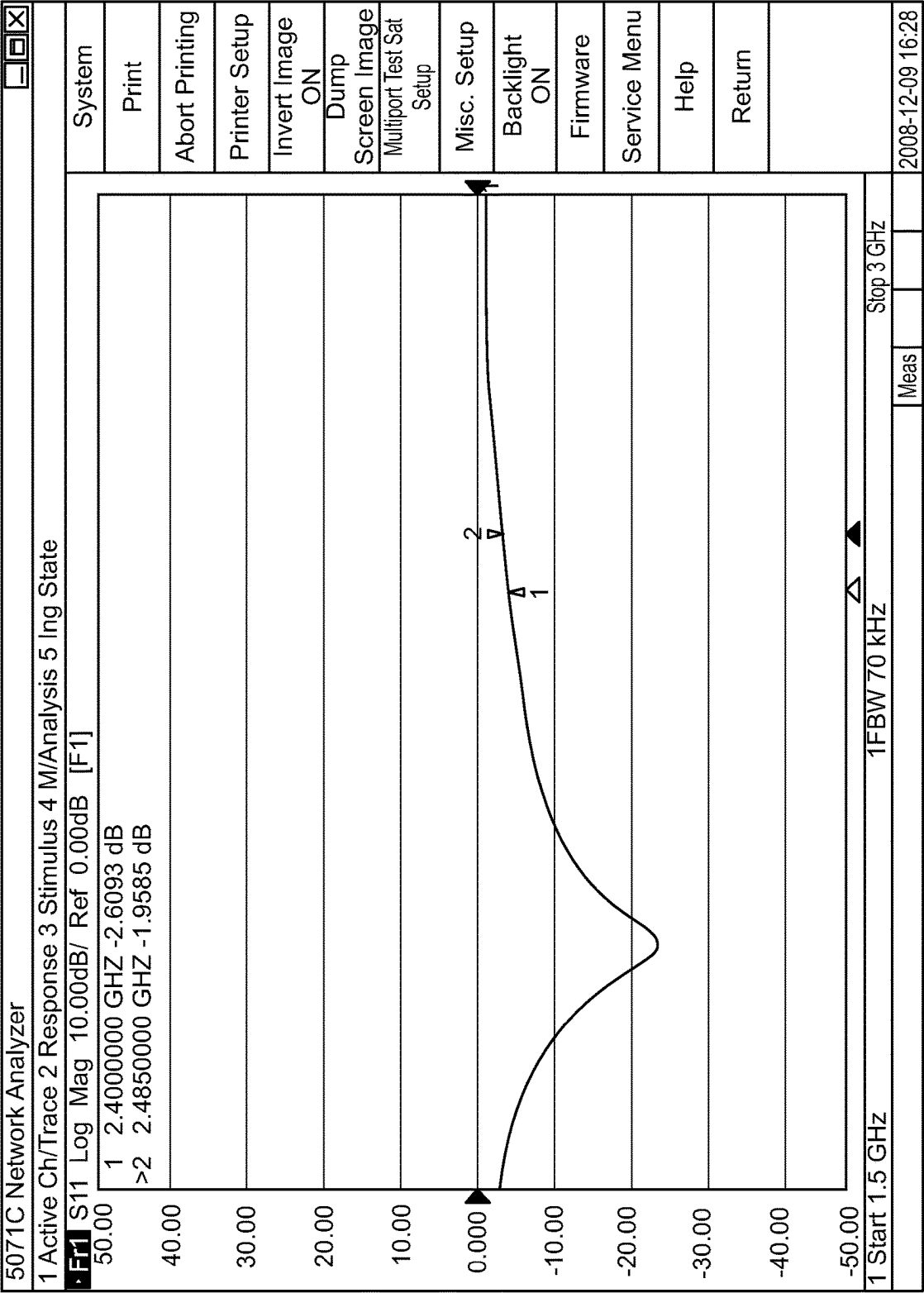


FIG. 133A

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WEARABLE PUMP ASSEMBLY**CROSS REFERENCE TO RELATED APPLICATIONS**

The present application is continuation of U.S. patent application Ser. No. 15/388,341 filed on Dec. 22, 2016, which is a continuation of U.S. patent application Ser. No. 12/347,982 filed on Dec. 31, 2008, which claims the benefit of the following provisional patent applications: U.S. Provisional Patent Application No. 61/018,054, filed Dec. 31, 2007; U.S. Provisional Patent Application No. 61/018,042, filed Dec. 31, 2007; U.S. Provisional Patent Application No. 61/017,989, filed Dec. 31, 2007; U.S. Provisional Patent Application No. 61/018,002, filed Dec. 31, 2007; U.S. Provisional Patent Application No. 61/018,339, filed Dec. 31, 2007; U.S. Provisional Patent Application No. 61/023,645, filed Jan. 25, 2008; U.S. Provisional Patent Application No. 61/101,053, filed Sep. 29, 2008; U.S. Provisional Patent Application No. 61/101,077, filed Sep. 29, 2008; U.S. Provisional Patent Application No. 61/101,105, filed Sep. 29, 2008 and U.S. Provisional Patent Application No. 61/101,115, filed Sep. 29, 2008. All of the above-identified applications are hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

This application relates generally to fluid delivery systems, and more particularly to infusion pump assemblies.

BACKGROUND

Many potentially valuable medicines or compounds, including biologicals, are not orally active due to poor absorption, hepatic metabolism or other pharmacokinetic factors. Additionally, some therapeutic compounds, although they can be orally absorbed, are sometimes required to be administered so often it is difficult for a patient to maintain the desired schedule. In these cases, parenteral delivery is often employed or could be employed.

Effective parenteral routes of drug delivery, as well as other fluids and compounds, such as subcutaneous injection, intramuscular injection, and intravenous (IV) administration include puncture of the skin with a needle or stylet. Insulin is an example of a therapeutic fluid that is self-injected by millions of diabetic patients. Users of parenterally delivered drugs may benefit from a wearable device that would automatically deliver needed drugs/compounds over a period of time.

To this end, there have been efforts to design portable and wearable devices for the controlled release of therapeutics. Such devices are known to have a reservoir such as a cartridge, syringe, or bag, and to be electronically controlled. These devices suffer from a number of drawbacks including the malfunction rate. Reducing the size, weight and cost of these devices is also an ongoing challenge. Additionally, these devices often apply to the skin and pose the challenge of frequent re-location for application.

SUMMARY OF THE INVENTION

According to a first implementation, a wearable infusion pump assembly includes a reusable housing assembly and a disposable housing assembly including a reservoir for receiving an infusible fluid. A releasable engagement assembly is configured to allow the reusable housing assembly to

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releasably engage the disposable housing assembly. The wearable infusion pump assembly further includes a detachable external infusion set configured to deliver the infusible fluid to a user.

One or more of the following features may be included. The reusable housing assembly may include a mechanical control assembly having a pump assembly and at least one valve assembly. The reusable housing assembly may further include an electrical control assembly configured to provide one or more control signals to the mechanical control assembly and to effectuate the delivery of the infusible fluid to the user via the detachable external infusion set. The mechanical control assembly may include at least one shape-memory actuator.

The detachable external infusion set may be a tethered infusion set. The detachable external infusion set may include a first disconnect assembly configured to releasably couple the detachable external infusion set to one of the reusable housing assembly and the disposable housing assembly. The detachable external infusion set may include a cannula assembly, a tubing assembly, and a second disconnect assembly configured to releasably couple the tubing assembly to the cannula assembly.

According to another implementation, a wearable infusion pump assembly includes a reusable housing assembly, and a disposable housing assembly including a reservoir for receiving an infusible fluid. The wearable infusion pump assembly further includes a releasable engagement assembly configured to allow the reusable housing assembly to rotationally engage the disposable housing assembly.

One or more of the following features may be included. The releasable engagement assembly may include a sealing assembly configured to provide a seal between the reusable housing assembly and the disposable housing assembly. The sealing assembly may include an o-ring assembly. The sealing assembly may include an over molded seal assembly. The sealing assembly may be a watertight seal assembly.

The wearable infusion pump assembly may further include a peripheral tubing storage assembly configured to store at least a portion of an external infusion set about a periphery of the wearable infusion pump assembly.

According to yet another implementation, a wearable infusion pump assembly includes a reusable housing assembly, and a disposable housing assembly including a reservoir for receiving an infusible fluid. A releasable engagement assembly is configured to allow the reusable housing assembly to releasably engage the disposable housing assembly. The wearable infusion pump assembly further includes a bolus switch assembly configured to effectuate a bolus infusion event.

One or more of the following features may be included. The wearable infusion pump may further include at least one processor, and a computer readable medium coupled to the at least one processor. The computer readable medium includes a plurality of instructions stored thereon which, when executed by the at least one processor, may cause the at least one processor to perform operations including receiving an initiation signal from the switch assembly indicative of a bolus infusion event. The computer readable medium may further include instructions for receiving a dose signal from the switch assembly indicative of at least a portion of a bolus quantity of the infusible fluid, and rendering an audible quantity signal in response to the dose signal. The computer readable medium may also include instructions for receiving an approval signal from the switch assembly indicative of a concurrence with the audible quan-

tity signal. The computer readable medium may further include instructions for administering the bolus quantity of the infusible fluid.

According to still another implementation, a wearable infusion pump assembly includes a fluidless reusable housing assembly, and a disposable housing assembly including a reservoir for receiving an infusible fluid. A releasable engagement assembly is configured to allow the reusable housing assembly to releasably engage the disposable housing assembly.

One or more of the following features may be included. The wearable infusion pump assembly may include an external infusion set that may be configured to deliver the infusible fluid to a user. The external infusion set may be a detachable external infusion set. The external infusion set may include a first disconnect assembly configured to releasably couple the external infusion set to the disposable housing assembly. The external infusion set may include a cannula assembly, a tubing assembly, and a second disconnect assembly configured to releasably couple the tubing assembly to the cannula assembly.

The details of one or more embodiments are set forth in the accompanying drawings and the description below. Other features and advantages will become apparent from the description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a side view of an infusion pump assembly;
 FIG. 2 is a perspective view of the infusion pump assembly of FIG. 1;
 FIG. 3 is an exploded view of various components of the infusion pump assembly of FIG. 1;
 FIG. 4 is a cross-sectional view of the disposable housing assembly of the infusion pump assembly of FIG. 1;
 FIGS. 5A-5C are cross-sectional views of an embodiment of a septum access assembly;
 FIGS. 6A-6B are cross-sectional views of another embodiment of a septum access assembly;
 FIGS. 7A-7B are partial top views of another embodiment of a septum access assembly;
 FIGS. 8A-8B are cross-sectional views of another embodiment of a septum access assembly;
 FIG. 9 is a perspective view of the infusion pump assembly of FIG. 1 showing an external infusion set;
 FIGS. 10A-10E depict a plurality of hook-and-loop fastener configurations;
 FIG. 11A is an isometric view of a remote control assembly and an alternative embodiment of the infusion pump assembly of FIG. 1;
 FIGS. 11B-11R depicts various views of high level schematics and flow charts of the infusion pump assembly of FIG. 1;
 FIGS. 12A-12F is a plurality of display screens rendered by the remote control assembly of FIG. 11A;
 FIG. 13 is an isometric view of an alternative embodiment of the infusion pump assembly of FIG. 1;
 FIG. 14 is an isometric view of the infusion pump assembly of FIG. 13;
 FIG. 15 is an isometric view of the infusion pump assembly of FIG. 13;
 FIG. 16 is an isometric view of an alternative embodiment of the infusion pump assembly of FIG. 1;
 FIG. 17 is an plan view of the infusion pump assembly of FIG. 16;
 FIG. 18 is a plan view of the infusion pump assembly of FIG. 16;

FIG. 19A is an exploded view of various components of the infusion pump assembly of FIG. 16;

FIG. 19B is an isometric view of a portion of the infusion pump assembly of FIG. 16;

FIG. 20 is a cross-sectional view of the disposable housing assembly of the infusion pump assembly of FIG. 16;

FIG. 21 is a diagrammatic view of a fluid path within the infusion pump assembly of FIG. 16;

FIGS. 22A-22C are diagrammatic views of a fluid path within the infusion pump assembly of FIG. 16;

FIG. 23 is an exploded view of various components of the infusion pump assembly of FIG. 16;

FIG. 24 is a cutaway isometric view of a pump assembly of the infusion pump assembly of FIG. 16;

FIGS. 25A-25D are other isometric views of the pump assembly of FIG. 24;

FIG. 26A-26B are isometric views of a measurement valve assembly of the infusion pump assembly of FIG. 16;

FIG. 27A-27B are side views of the measurement valve assembly of FIGS. 26A-26B;

FIGS. 28A-28D are views of a measurement valve assembly of the infusion pump assembly of FIG. 16;

FIG. 29 is an isometric view of an alternative embodiment of the infusion pump assembly of FIG. 1;

FIG. 30 is an isometric view of an alternative embodiment of the infusion pump assembly of FIG. 1;

FIG. 31 is another view of the alternative embodiment infusion pump assembly of FIG. 9;

FIG. 32 is an exploded view of another embodiment of an infusion pump assembly;

FIG. 33 is another exploded view of the infusion pump assembly of FIG. 32;

FIGS. 34A-34B depict another embodiment of an infusion pump assembly;

FIGS. 35A-35C are a top view, side view, and bottom view of a reusable housing assembly of the infusion pump assembly of FIG. 32;

FIG. 36 is an exploded view of the reusable housing assembly of FIGS. 35A-35C;

FIG. 37 is an exploded view of the reusable housing assembly of FIGS. 35A-35C;

FIG. 38A is an exploded view of the reusable housing assembly of FIGS. 35A-35C;

FIG. 38B-38D are top, side and bottom views of one embodiment of a dust cover;

FIGS. 39A-39C are a top view, side view, and bottom view of an electrical control assembly of the reusable housing assembly of FIGS. 35A-35C;

FIGS. 40A-40C are a top view, side view, and bottom view of a base plate of the reusable housing assembly of FIGS. 35A-35C;

FIGS. 41A-41B are a perspective top view and a perspective bottom view of the base plate of FIGS. 40A-40C;

FIGS. 42A-42C are a top view, side view, and bottom view of a base plate of the reusable housing assembly of FIGS. 35A-35C;

FIGS. 43A-43B depict a mechanical control assembly of the reusable housing assembly of FIGS. 35A-35C;

FIGS. 44A-44C depict the mechanical control assembly of the reusable housing assembly of FIGS. 35A-35C;

FIGS. 45A-45B depict the pump plunger and reservoir valve of the mechanical control assembly of the reusable housing assembly of FIGS. 35A-35C;

FIGS. 46A-46E depict various views of the plunger pump and reservoir valve of the mechanical control assembly of the reusable housing assembly of FIGS. 35A-35C;

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FIGS. 47A-47B depict the measurement valve of the mechanical control assembly of the reusable housing assembly of FIGS. 35A-35C;

FIG. 48 is an exploded view of the disposable housing assembly of the infusion pump assembly of FIG. 32;

FIG. 49A is a plan view of the disposable housing assembly of FIG. 48;

FIG. 49B is a sectional view of the disposable housing assembly of FIG. 49A taken along line B-B;

FIG. 49C is a sectional view of the disposable housing assembly of FIG. 49A taken along line C-C;

FIGS. 50A-50C depict the base portion of the disposable housing assembly of FIG. 48;

FIGS. 51A-51C depict the fluid pathway cover of the disposable housing assembly of FIG. 48;

FIGS. 52A-52C depict the membrane assembly of the disposable housing assembly of FIG. 48;

FIGS. 53A-53C depict the top portion of the disposable housing assembly of FIG. 48;

FIGS. 54A-54C depict the valve membrane insert of the disposable housing assembly of FIG. 48;

FIGS. 55A-55B depict the locking ring assembly of the infusion pump assembly of FIG. 32;

FIG. 56A-56C depict the locking ring assembly of the infusion pump assembly of FIG. 32;

FIGS. 57-58 is an isometric view of an infusion pump assembly and a fill adapter;

FIGS. 59-64 are various views of the fill adapter of FIG. 57;

FIG. 65 is an isometric view of another embodiment of a fill adapter;

FIGS. 66-67 depict an infusion pump assembly and another embodiment of a fill adapter;

FIGS. 68-74 are various views of the fill adapter of FIG. 66;

FIGS. 75-80 depict various views of an embodiment of a battery charger;

FIGS. 81-89B depict various embodiments of battery chargers/docking stations;

FIGS. 90A-90C are various views of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIGS. 91A-91I are various views of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIGS. 92A-92I are various views of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIGS. 93A-93I are various views of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIGS. 94A-94F are various views of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 95 is an exploded view of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 96 is a diagrammatic view of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 97 is a two-dimensional graph of a performance characteristic of the volume sensor assembly of FIG. 96;

FIG. 98 is a two-dimensional graph of a performance characteristic of the volume sensor assembly of FIG. 96;

FIG. 99 is a two-dimensional graph of a performance characteristic of the volume sensor assembly of FIG. 96;

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FIG. 100 is a diagrammatic view of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 101 is a two-dimensional graph of a performance characteristic of the volume sensor assembly of FIG. 100;

FIG. 102 is a two-dimensional graph of a performance characteristic of the volume sensor assembly of FIG. 100;

FIG. 103 is a diagrammatic view of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 104 is a two-dimensional graph of a performance characteristic of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 105 is a two-dimensional graph of a performance characteristic of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 106 is a two-dimensional graph of a performance characteristic of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 107 is a two-dimensional graph of a performance characteristic of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 108 is a two-dimensional graph of a performance characteristic of a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 109 is a diagrammatic view of a control model for a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 110 is a diagrammatic view of an electrical control assembly for the volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 111 is a diagrammatic view of a volume controller for the volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 112 is a diagrammatic view of a feed forward controller of the volume controller of FIG. 111;

FIGS. 113-114 diagrammatically depicts an implementation of an SMA controller of the volume controller of FIG. 111;

FIG. 114A-114B is an alternate implementation of an SMA controller;

FIG. 115 diagrammatically depicts a multi-processor control configuration that may be included within the infusion pump assembly of FIG. 1;

FIG. 116 is a diagrammatic view of a multi-processor control configuration that may be included within the infusion pump assembly of FIG. 1;

FIG. 117A-117B diagrammatically depicts multi-processor functionality;

FIG. 118 diagrammatically depicts multi-processor functionality;

FIG. 119 diagrammatically depicts multi-processor functionality;

FIGS. 120A-120E graphically depicts various software layers;

120B-120C depict various state diagrams;

120D graphically depicts device interaction;

120E graphically depicts device interaction;

FIG. 121 diagrammatically depicts a volume sensor assembly included within the infusion pump assembly of FIG. 1;

FIG. 122 diagrammatically depicts an inter-connection of the various systems of the infusion pump assembly of FIG. 1;

FIG. 123 diagrammatically depicts basal-bolus infusion events;

FIG. 124 diagrammatically depicts basal-bolus infusion events;

FIG. 125A-125G depicts a hierarchical state machine;

FIG. 126A-126M depicts a hierarchical state machine;

FIG. 127 is an exemplary diagram of a split ring resonator antenna;

FIG. 128 is an exemplary diagram of a medical device configured to utilize a split ring resonator antenna;

FIG. 129 is an exemplary diagram of a split ring resonator antenna and transmission line from a medical infusion device;

FIG. 130 is a graph of the return loss of a split ring resonator antenna prior to contact with human skin;

FIG. 130A is a graph of the return loss of a split ring resonator antenna during contact with human skin;

FIG. 131 is an exemplary diagram of a split ring resonator antenna integrated into a device which operates within close proximity to dielectric material;

FIG. 132 is a diagram of the dimensions of the inner and outer portion of the exemplary embodiment;

FIG. 133 is a graph of the return loss of a non-split ring resonator antenna prior to contact with human skin; and

FIG. 133A is a graph of the return loss of a non-split ring resonator antenna during contact with human skin.

Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Referring to FIGS. 1-3, an infusion pump assembly 100 may include a reusable housing assembly 102. Reusable housing assembly 102 may be constructed from any suitable material, such as a hard or rigid plastic, that will resist compression. For example, use of durable materials and parts may improve quality and reduce costs by providing a reusable portion that lasts longer and is more durable, providing greater protection to components disposed therein.

Reusable housing assembly 102 may include mechanical control assembly 104 having a pump assembly 106 and at least one valve assembly 108. Reusable housing assembly 102 may also include electrical control assembly 110 configured to provide one or more control signals to mechanical control assembly 104 and effectuate the basal and/or bolus delivery of an infusible fluid to a user. Disposable housing assembly 114 may include valve assembly 108 which may be configured to control the flow of the infusible fluid through a fluid path. Reusable housing assembly 102 may also include pump assembly 106 which may be configured to pump the infusible fluid from the fluid path to the user.

Electrical control assembly 110 may monitor and control the amount of infusible fluid that has been and/or is being pumped. For example, electrical control assembly 110 may receive signals from volume sensor assembly 148 and calculate the amount of infusible fluid that has just been dispensed and determine, based upon the dosage required by the user, whether enough infusible fluid has been dispensed. If enough infusible fluid has not been dispensed, electrical control assembly 110 may determine that more infusible fluid should be pumped. Electrical control assembly 110 may provide the appropriate signal to mechanical control assembly 104 so that any additional necessary dosage may be pumped or electrical control assembly 110 may provide the appropriate signal to mechanical control assembly 104 so that the additional dosage may be dispensed with the next dosage. Alternatively, if too much infusible fluid has been

dispensed, electrical control assembly 110 may provide the appropriate signal to mechanical control assembly 104 so that less infusible fluid may be dispensed in the next dosage.

Mechanical control assembly 104 may include at least one shape-memory actuator 112. Pump assembly 106 and/or valve assembly 108 of mechanical control assembly 104 may be actuated by at least one shape-memory actuator, e.g., shape-memory actuator 112, which may be a shape-memory wire in wire or spring configuration. Shape memory actuator 112 may be operably connected to and activated by electrical control assembly 110, which may control the timing and the amount of heat and/or electrical energy used to actuate mechanical control assembly 104. Shape memory actuator 112 may be, for example, a conductive shape-memory alloy wire that changes shape with temperature. The temperature of shape-memory actuator 112 may be changed with a heater, or more conveniently, by application of electrical energy. Shape memory actuator 112 may be a shape memory wire constructed of nickel/titanium alloy, such as NITI-NOL™ or FLEXINOL®.

Infusion pump assembly 100 may include a volume sensor assembly 148 configured to monitor the amount of fluid infused by infusion pump assembly 100. For example, volume sensor assembly 148 may employ, for example, acoustic volume sensing. Acoustic volume measurement technology is the subject of U.S. Pat. Nos. 5,575,310 and 5,755,683 assigned to DEKA Products Limited Partnership, as well as U.S. patent application Publication Nos. US 2007/0228071 A1, US 2007/0219496 A1, US 2007/0219480 A1, US 2007/0219597 A1, the entire disclosures of all of which are incorporated herein by reference. Other alternative techniques for measuring fluid flow may also be used; for example, Doppler-based methods; the use of Hall-effect sensors in combination with a vane or flapper valve; the use of a strain beam (for example, related to a flexible member over a fluid reservoir to sense deflection of the flexible member); the use of capacitive sensing with plates; or thermal time of flight methods. One such alternative technique is disclosed in U.S. patent application Ser. No. 11/704,899, entitled Fluid Delivery Systems and Methods, filed 9 Feb. 2007, the entire disclosure of which is incorporated herein by reference. Infusion pump assembly 100 may be configured so that the volume measurements produced by volume sensor assembly 148 may be used to control, through a feedback loop, the amount of infusible fluid that is infused into the user.

Infusion pump assembly 100 may further include a disposable housing assembly 114. For example, disposable housing assembly 114 may be configured for a single use or for use for a specified period of time, e.g., three days or any other amount of time. Disposable housing assembly 114 may be configured such that any components in infusion pump assembly 100 that come in contact with the infusible fluid are disposed on and/or within disposable housing assembly 114. For example, a fluid path or channel including a reservoir, may be positioned within disposable housing assembly 114 and may be configured for a single use or for a specified number of uses before disposal. The disposable nature of disposable housing assembly 114 may improve sanitation of infusion pump assembly 100.

Referring also to FIG. 4, disposable housing assembly 114 may be configured to releasably engage reusable housing assembly 102, and includes a cavity 116 that has a reservoir 118 for receiving an infusible fluid (not shown), e.g., insulin. Such releasable engagement may be accomplished by a screw-on, a twist-lock or a compression fit configuration, for example. Disposable housing assembly 114 and/or reusable

housing assembly 102 may include an alignment assembly configured to assist in aligning disposable housing assembly 114 and reusable housing assembly 102 for engagement in a specific orientation. Similarly, base nub 120 and top nub 122 may be used as indicators of alignment and complete engagement.

Cavity 116 may be at least partially formed by and integral to disposable housing assembly 114. Cavity 116 may include a membrane assembly 124 for at least partially defining reservoir 118. Reservoir 118 may be further defined by disposable housing assembly 114, e.g., by a recess 126 formed in base portion 128 of disposable housing assembly 114. For example, membrane assembly 124 may be disposed over recess 126 and attached to base portion 128, thereby forming reservoir 118. Membrane assembly 124 may be attached to base portion 128 by conventional means, such as gluing, heat sealing, and/or compression fitting, such that a seal 130 is formed between membrane assembly 124 and base portion 128. Membrane assembly 124 may be flexible and the space formed between membrane assembly 124 and recess 126 in base portion 128 may define reservoir 118. Reservoir 118 may be non-pressurized and in fluid communication with a fluid path (not shown). Membrane assembly 124 may be at least partially collapsible and cavity 116 may include a vent assembly, thereby advantageously preventing the buildup of a vacuum in reservoir 118 as the infusible fluid is delivered from reservoir 118 to the fluid path. In a preferred embodiment, membrane assembly 124 is fully collapsible, thus allowing for the complete delivery of the infusible fluid. Cavity 116 may be configured to provide sufficient space to ensure there is always some air space even when reservoir 118 is filled with infusible fluid.

The membranes and reservoirs described herein may be made from materials including but not limited to silicone, NITRILE, and any other material having desired resilience and properties for functioning as described herein. Additionally, other structures could serve the same purpose.

The use of a partially collapsible non pressurized reservoir may advantageously prevent the buildup of air in the reservoir as the fluid in the reservoir is depleted. Air buildup in a vented reservoir could prevent fluid egress from the reservoir, especially if the system is tilted so that an air pocket intervenes between the fluid contained in the reservoir and the septum of the reservoir. Tilting of the system is expected during normal operation as a wearable device.

Reservoir 118 may be conveniently sized to hold an insulin supply sufficient for delivery over one or more days. For example, reservoir 118 may hold about 1.00 to 3.00 ml of insulin. A 3.00 ml insulin reservoir may correspond to approximately a three day supply for about 90% of potential users. In other embodiments, reservoir 118 may be any size or shape and may be adapted to hold any amount of insulin or other infusible fluid. In some embodiments, the size and shape of cavity 116 and reservoir 118 is related to the type of infusible fluid that cavity 116 and reservoir 118 are adapted to hold.

Disposable housing assembly 114 may include a support member 132 (FIG. 3) configured to prevent accidental compression of reservoir 118. Compression of reservoir 118 may result in an unintentional dosage of infusible fluid being forced through the fluid path to the user. In a preferred embodiment, reusable housing assembly 102 and disposable housing assembly 114 may be constructed of a rigid material that is not easily compressible. However, as an added precaution, support member 132 may be included within disposable housing assembly 114 to prevent compression of infusion pump assembly 100 and cavity 116 therein. Support

member 132 may be a rigid projection from base portion 128. For example, support member 132 may be disposed within cavity 116 and may prevent compression of reservoir 118.

As discussed above, cavity 116 may be configured to provide sufficient space to ensure there is always some air space even when reservoir 118 is filled with infusible fluid. Accordingly, in the event that infusion pump assembly 100 is accidentally compressed, the infusible fluid may not be forced through cannula assembly 136 (e.g., shown in FIG. 9).

Cavity 116 may include a septum assembly 146 (FIG. 3) configured to allow reservoir 118 to be filled with the infusible fluid. Septum assembly 146 may be a conventional septum made from rubber or plastic and have a one-way fluid valve configured to allow a user to fill reservoir 118 from a syringe or other filling device. In some embodiments, septum 146 may be located on the top of membrane assembly 124. In these embodiments, cavity 116 may include a support structure (e.g., support member 132 in FIG. 3) for supporting the area about the back side of the septum so as to maintain the integrity of the septum seal when a needle is introducing infusible fluid into cavity 116. The support structure may be configured to support the septum while still allowing the introduction of the needle for introducing infusible fluid into cavity 116.

Infusion pump assembly 100 may include an overflow prevention assembly (not shown) that may e.g., protrude into cavity 116 and may e.g., prevent the overfilling of reservoir 118.

In some embodiments, reservoir 118 may be configured to be filled a plurality of times. For example, reservoir 118 may be refillable through septum assembly 146. As infusible fluid may be dispensed to a user, electronic control assembly 110 may monitor the fluid level of the infusible fluid in reservoir 118. When the fluid level reaches a low point, electronic control assembly 110 may provide a signal, such as a light or a vibration, to the user that reservoir 118 needs to be refilled. A syringe, or other filling device, may be used to fill reservoir 118 through septum 146.

Reservoir 118 may be configured to be filled a single time. For example, a refill prevention assembly (not shown) may be utilized to prevent the refilling of reservoir 118, such that disposable housing assembly 114 may only be used once. The refill prevention assembly (not shown) may be a mechanical device or an electro-mechanical device. For example, insertion of a syringe into septum assembly 146 for filling reservoir 118 may trigger a shutter to close over septum 146 after a single filling, thus preventing future access to septum 146. Similarly, a sensor may indicate to electronic control assembly 110 that reservoir 118 has been filled once and may trigger a shutter to close over septum 146 after a single filling, thus preventing future access to septum 146. Other means of preventing refilling may be utilized and are considered to be within the scope of this disclosure.

As discussed above, disposable housing assembly 114 may include septum assembly 146 that may be configured to allow reservoir 118 to be filled with the infusible fluid. Septum assembly 146 may be a conventional septum made from rubber or any other material that may function as a septum, or, in other embodiments, septum assembly 146 may be, but is not limited to, a plastic, or other material, one-way fluid valve. In various embodiments, including the exemplary embodiment, septum assembly 146 is configured to allow a user to fill reservoir 118 from a syringe or other filling device. Disposable housing assembly 114 may

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include a septum access assembly that may be configured to limit the number of times that the user may refill reservoir 118.

For example and referring also to FIGS. 5A-5C, septum access assembly 152 may include shutter assembly 154 that may be held in an "open" position by a tab assembly 156 that is configured to fit within a slot assembly 158. Upon penetrating septum 146 with filling syringe 160, shutter assembly 154 may be displaced downward, resulting in tab assembly 156 disengaging from slot assembly 158. Once disengaged, spring assembly 162 may displace shutter assembly 154 in the direction of arrow 164, resulting in septum 146 no longer being accessible to the user.

Referring also to FIG. 6A, an alternative-embodiment septum access assembly 166 is shown in the "open" position. In a fashion similar to that of septum access assembly 152, septum access assembly 166 includes shutter assembly 168 and spring assembly 170.

Referring also to FIG. 6B, an alternative-embodiment of septum access assembly 172 is shown in the "open" position where tab 178 may engage slot 180. In a fashion similar to that of septum access assembly 166, septum access assembly 172 may include shutter assembly 174 and spring assembly 176. Once shutter assembly 172 moves to the "closed" position (e.g., which may prevent further access of septum 146 by the user), tab 178 may at least partially engage slot 180a. Engagement between tab 178 and slot 180a may lock shutter assembly 172 in the "closed" position to inhibit tampering and reopening of shutter assembly 172. Spring tab 182 of shutter assembly 172 may bias tab 178 into engagement with slot 180a.

However, in various embodiments, septum access assemblies may not be actuated linearly. For example and referring also to FIGS. 7A-7B, there is shown alternative embodiment septum access assembly 184 that includes shutter assembly 186 that is configured to pivot about axis 188. When positioned in the open position (as shown in FIG. 7A), septum 146 may be accessible due to passage 190 (in shutter assembly 186) being aligned with passage 192 in e.g., a surface of disposable housing assembly 114. However, in a fashion similar to septum access assemblies 166, 172, upon penetrating septum 146 with filling syringe 160 (See FIG. 6B), shutter assembly 186 may be displaced in a clockwise fashion, resulting in passage 190 (in shutter assembly 186) no longer being aligned with passage 192 in e.g., a surface of disposable housing assembly 114, thus preventing access to septum 146.

Referring also to FIGS. 8A-8B, an alternative-embodiment septum access assembly 194 is shown. In a fashion similar to that of septum access assemblies 166, 172, septum access assembly 194 includes shutter assembly 196 and spring assembly 198 that is configured to bias shutter assembly 196 in the direction of arrow 200. Filling assembly 202 may be used to fill reservoir 118. Filling assembly 202 may include shutter displacement assembly 204 that may be configured to displace shutter assembly 196 in the direction of arrow 206, which in turn aligns passage 208 in shutter assembly 196 with septum 146 and passage 210 in septum access assembly 194, thus allowing filling syringe assembly 212 to penetrate septum 146 and fill reservoir 118.

Infusion pump assembly 100 may include a sealing assembly 150 (FIG. 3) configured to provide a seal between reusable housing assembly 102 and disposable housing assembly 114. For example, when reusable housing assembly 102 and disposable housing assembly 114 are engaged by e.g. rotational screw-on engagement, twist-lock engagement or compression engagement, reusable housing assem-

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bly 102 and disposable housing assembly 114 may fit together snugly, thus forming a seal. In some embodiments, it may be desirable for the seal to be more secure. Accordingly, sealing assembly 150 may include an o-ring assembly (not shown). Alternatively, sealing assembly 150 may include an over molded seal assembly (not shown). The use of an o-ring assembly or an over molded seal assembly may make the seal more secure by providing a compressible rubber or plastic layer between reusable housing assembly 102 and disposable housing assembly 114 when engaged thus preventing penetration by outside fluids. In some instances, the o-ring assembly may prevent inadvertent disengagement. For example, sealing assembly 150 may be a watertight seal assembly and, thus, enable a user to wear infusion pump assembly 100 while swimming, bathing or exercising.

Referring also to FIG. 9, infusion pump assembly 100 may include an external infusion set 134 configured to deliver the infusible fluid to a user. External infusion set 134 may be in fluid communication with cavity 118, e.g. by way of the fluid path. External infusion set 134 may be disposed adjacent to infusion pump assembly 100. Alternatively, external infusion set 134 may be configured for application remote from infusion pump assembly 100, as discussed in greater detail below. External infusion set 134 may include a cannula assembly 136, which may include a needle or a disposable cannula 138, and tubing assembly 140. Tubing assembly 140 may be in fluid communication with reservoir 118, for example, by way of the fluid path, and with cannula assembly 138 for example, either directly or by way of a cannula interface 142.

External infusion set 134 may be a tethered infusion set, as discussed above regarding application remote from infusion pump assembly 100. For example, external infusion set 134 may be in fluid communication with infusion pump assembly 100 through tubing assembly 140, which may be of any length desired by the user (e.g., 3-18 inches). Though infusion pump assembly 100 may be worn on the skin of a user with the use of adhesive patch 144, the length of tubing assembly 140 may enable the user to alternatively wear infusion pump assembly 100 in a pocket. This may be beneficial to users whose skin is easily irritated by application of adhesive patch 144. Similarly, wearing and/or securing infusion pump assembly 100 in a pocket may be preferable for users engaged in physical activity.

In addition to/as an alternative to adhesive patch 144, a hook and loop fastener system (e.g. such as hook and loop fastener systems offered by Velcro USA Inc. of Manchester, NH) may be utilized to allow for easy attachment/removal of an infusion pump assembly (e.g., infusion pump assembly 100) from the user. Accordingly, adhesive patch 144 may be attached to the skin of the user and may include an outward facing hook or loop surface. Additionally, the lower surface of disposable housing assembly 114 may include a complementary hook or loop surface. Depending upon the separation resistance of the particular type of hook and loop fastener system employed, it may be possible for the strength of the hook and loop connection to be stronger than the strength of the adhesive to skin connection. Accordingly, various hook and loop surface patterns may be utilized to regulate the strength of the hook and loop connection.

Referring also to FIGS. 10A-10E, five examples of such hook and loop surface patterns are shown. Assume for illustrative purposes that the entire lower surface of disposable housing assembly 114 is covered in a "loop" material. Accordingly, the strength of the hook and loop connection may be regulated by varying the pattern (i.e., amount) of the

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“hook” material present on the surface of adhesive patch 144. Examples of such patterns may include but are not limited to: a singular outer circle 220 of “hook” material (as shown in FIG. 10A); a plurality of concentric circles 222, 224 of “hook” material (as shown in FIG. 10B); a plurality of radial spokes 226 of “hook” material (as shown in FIG. 10C); a plurality of radial spokes 228 of “hook” material in combination with a single outer circle 230 of “hook” material (as shown in FIG. 10D); and a plurality of radial spokes 232 of “hook” material in combination with a plurality of concentric circles 234, 236 of “hook” material (as shown in FIG. 10E).

Additionally and referring also to FIG. 11A, in one exemplary embodiment of the above-described infusion pump assembly, infusion pump assembly 100' may be configured via a remote control assembly 300. In this particular embodiment, infusion pump assembly 100' may include telemetry circuitry (not shown) that allows for communication (e.g., wired or wireless) between infusion pump assembly 100' and e.g., remote control assembly 300, thus allowing remote control assembly 300 to remotely control infusion pump assembly 100'. Remote control assembly 300 (which may also include telemetry circuitry (not shown) and may be capable of communicating with infusion pump assembly 100') may include display assembly 302 and input assembly 304. Input assembly 304 may include slider assembly 306 and switch assemblies 308, 310. In other embodiments, the input assembly may include a jog wheel, a plurality of switch assemblies, or the like.

Remote control assembly 300 may include the ability to pre-program basal rates, bolus alarms, delivery limitations, and allow the user to view history and to establish user preferences. Remote control assembly 300 may also include a glucose strip reader.

During use, remote control assembly 300 may provide instructions to infusion pump assembly 100' via wireless communication channel 312 established between remote control assembly 300 and infusion pump assembly 100'. Accordingly, the user may use remote control assembly 300 to program/configure infusion pump assembly 100'. Some or all of the communication between remote control assembly 300 and infusion pump assembly 100' may be encrypted to provide an enhanced level of security.

Communication between remote control assembly 300 and infusion pump assembly 100' may be accomplished utilizing a standardized communication protocol. Further, communication between the various components included within infusion pump assembly 100, 100' may be accomplished using the same protocol. One example of such a communication protocol is the Packet Communication Gateway Protocol (PCGP) developed by DEKA Research & Development of Manchester, NH. As discussed above, infusion pump assembly 100, 100' may include electrical control assembly 110 that may include one or more electrical components. For example, electrical control assembly 110 may include a plurality of data processors (e.g., a supervisor processor and a command processor) and a radio processor for allowing infusion pump assembly 100, 100' to communicate with remote control assembly 300. Further, remote control assembly 300 may include one or more electrical components, examples of which may include but are not limited to a command processor and a radio processor for allowing remote control assembly 300 to communicate with infusion pump assembly 100, 100'. A high-level diagrammatic view of one example of such a system is shown in FIG. 11B.

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Each of these electrical components may be manufactured from a different component provider and, therefore, may utilize native (i.e. unique) communication commands. Accordingly, through the use of a standardized communication protocol, efficient communication between such disparate components may be accomplished.

PCGP may be a flexible extendable software module that may be used on the processors within infusion pump assembly 100, 100' and remote control assembly 300 to build and route packets. PCGP may abstract the various interfaces and may provide a unified application programming interface (API) to the various applications being executed on each processor. PCGP may also provide an adaptable interface to the various drivers. For illustrative purposes only, PCGP may have the conceptual structure illustrated in FIG. 11C for any given processor.

PCGP may ensure data integrity by utilizing cyclic redundancy checks (CRCs). PCGP may also provide guaranteed delivery status. For example, all new messages should have a reply. If such a reply isn't sent back in time, the message may time out and PCGP may generate a negative acknowledgment reply message for the application (i.e., a NACK). Accordingly, the message-reply protocol may let the application know whether the application should retry sending a message.

PCGP may also limit the number of messages in-flight from a given node, and may be coupled with a flow-control mechanism at the driver level to provide a deterministic approach to message delivery and may let individual nodes have different quantities of buffers without dropping packets. As a node runs out of buffers, drivers may provide back pressure to other nodes and prevent sending of new messages.

PCGP may use a shared buffer pool strategy to minimize data copies, and may avoid mutual exclusions, which may have a small affect on the API used to send/receive messages to the application, and a larger affect on the drivers. PCGP may use a “Bridge” base class that provides routing and buffer ownership. The main PCGP class may be sub-classed from the bridge base class. Drivers may either be derived from a bridge class, or talk to or own a derived bridge class.

PCGP may be designed to work in an embedded environment with or without an operating system by using a semaphore to protect shared data such that some calls can be re-entrant and run on a multiple threads. One illustrative example of such an implementation is shown in FIG. 11D. PCGP may operate the same way in both environments, but there may be versions of the call for specific processor types (e.g., the ARM 9/OS version). So while the functionality may be the same, there may be an operating system abstraction layer with slightly different calls tailored for e.g., the ARM 9 Nucleus OS environment.

Referring also to FIG. 11E, PCGP may:

- allow multiple Send/Reply calls to occur (on Pilot's ARM 9 on multiple tasks re-entrant);
- have multiple drivers running asynchronously for RX and TX on different interfaces; and
- provide packet ordering for send/receive, and deterministic timeout on message send.

Each software object may ask the buffer manager for the next buffer to use, and may then give that buffer to another object. Buffers may pass from one exclusive owner to another autonomically, and queues may occur automatically by ordering buffers by sequence number. When a buffer is no longer in use, the buffer may be recycled (e.g., object attempts to give the buffer to itself, or frees it for the buffer

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manager to re-allocate later). Accordingly, data generally doesn't need to be copied, and routing simply writes over the buffer ownership byte.

Such an implementation of PCGP may provide various benefits, examples of which may include but are not limited to:

dropping a message due to lack of buffers may be impossible, as once a message is put into a buffer, the message may live there until it is transferred or received by the application;

data may not need to be copied, as offsets are used to access driver, PCGP and payload sections of a buffer; drivers may exchange ownership of message data by writing over one byte (i.e., the buffer ownership byte);

there may be no need for multiple exclusions except for re-entrant calls, as a mutual exclusion may be needed only when a single buffer owner could simultaneously want to use a buffer or get a new sequence number;

there may be fewer rules for application writers to follow to implement a reliable system;

drivers may use ISR/push/pull and polled data models, as there are a set of calls provided to push/pull data out of the buffer management system from the drivers;

drivers may not do much work beyond TX and RX, as drivers may not copy, CRC or check anything but the destination byte and CRC and other checks may be done off of the ISR hot path later;

as the buffer manager may order access by sequence number, queue ordering may automatically occur; and a small code/variable foot print may be utilized; hot path code may be small and overhead may be low.

As shown in FIG. 11F, when a message needs to be sent, the PCGP may build the packet quickly and may insert it into the buffer management system. Once in the buffer management system, a call to "packetProcessor" may apply protocol rules and may give the messages to the drivers/application.

To send a new message or send a reply, PCGP may:

check the call arguments to e.g., make sure the packet length is legal, destination is ok, etc.;

avoid trying to send a message across a link that is down unless the down link is the radio node, which may allow PCGP to be used by the radio processors to establish a link, pair, etc. and may notify the application when PCGP is trying to talk across a link that is not functional (instead of timing out);

obtain a sequence number for a new message or utilize an existing sequence number for an existing message;

build the packet, copy the payload data and write in the CRC, wherein (from this point forward) the packet integrity may be protected by the CRC; and

either give the message to the buffer manager as a reply or as a new message, and check to see if putting this buffer into the buffer manager would exceed the maximum number of en-queued send messages.

Referring also to FIGS. 11G-11H, PCGP may work by doing all of the main work on one thread to avoid mutual exclusions, and to avoid doing considerable work on the send/reply or driver calls. The "packetProcessor" call may have to apply protocol rules to replies, new sent messages, and received messages. Reply messages may simply get routed, but new messages and received messages may have rules for routing the messages. In each case, the software may loop while a message of the right type is available to apply protocol rules until it cannot process the packets.

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Sending a new message may conform to the following rules:

only two messages may be allowed "in-flight" on the network; and

enough data about an in-flight message may be stored to match the response and handle timeout.

Receiving a message may conform to the following rules: responses that match may clear out the "in-flight" information slot so a new packet can be sent;

responses that do not match may be dropped;

new messages may be for the protocol (e.g., getting/clearing network statistics for this node);

to receive a message, the buffer may be given up to the application and may use a call back; and

the buffer may be freed or left owned by the application. Accordingly, PCGP may be configured such that:

the call back function may copy the payload data out or may use it completely before returning;

the call back function owns the buffer and may reference the buffer and the buffer's payload by the payload address, wherein the message may be processed later; applications may poll the PCGP system for received messages; and

applications may use the call back to set an event and then poll for received messages.

The communication system may have a limited number of buffers. When PCGP runs out of buffers, drivers may stop receiving new packets and the application may be told that the application cannot send new packets. To avoid this and maintain optimal performance, the application may try to perform one or more procedures, examples of which may include but are not limited to:

a) The application should keep PCGP up to date with radio status: Specifically, if the link goes down and PCGP doesn't know, PCGP may accept and queue new messages to send (or not timeout messages optimally), which may jam the send queue and delay the application from using the link optimally.

b) The application should call "decrement timeouts" regularly: Optimally, every 20-100 milliseconds unless the processor is asleep. In general, a message moves fast (milliseconds) slow (seconds) or not at all. Timeouts are an attempt to remove "in-flight" messages that should be dropped to free up buffers and bandwidth. Doing this less often may delay when a new message gets sent, or when the application can queue a new message.

c) The application should ask PCGP if it has work to do that is pending before going to sleep: If PCGP has nothing to do, driver activity may wake up the system and thus PCGP, and then PCGP won't need a call to "packetProcessor" or "decrement timeouts" until new packets enter the system. Failure to do this may cause messages that could have been sent/forwarded/received successfully to be dropped due to a timeout condition.

d) The application should not hold onto received messages indefinitely: The message system relies on prompt replies. If the application is sharing PCGP buffers, then holding onto a message means holding onto a PCGP buffer. The receiving node doesn't know if the sending node has timeout configured for slow or fast radio. This means when a node receives a message it should assume the network's fast timeout speed.

e) The application should call the "packetProcessor" often: The call may cause new messages queued by the application to get sent and may handle receipt of new messages. The call may also cause buffers to re-allocate and calling it infrequently may delay message traffic.

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As shown in FIG. 11I, at some point the RX driver may be asked to receive a message from the other side of the interface. To ensure a message does not get dropped, the RX driver may ask the buffer manager if there is an available buffer for storing a new message. The driver may then ask for a buffer pointer and may start filling the buffer with received data. When a complete message is received, the RX driver may call a function to route the packet. The route function may examine the destination byte in the packet header and may change the owner to either the other driver, or the application, or may detect that the packet is bad and may drop the packet by freeing the buffer.

PCGP RX overhead may consist of asking for the next available buffer and calling the route function. An example of code that performs such a function is as follows:

```
@ Receive request
uint8 i=0, *p;
if (Bridge::canReceiveFlowControl( ))
{
    p = Bridge::nextBufferRX( );
    while (not done) { p[i] = the next byte; }
    Bridge::route(p);
}
```

A driver may perform a TX by asking the buffer manager for the pointer to the next buffer to send. The TX driver may then ask the other side of the interface if it can accept a packet. If the other side denies the packet, the TX driver may do nothing to the buffer, as its status has not changed. Otherwise, the driver may send the packet and may recycle/free the buffer. An example of code that performs such a function is as follows:

```
uint8 *p = Bridge::nextBufferTX( );
if (p != (uint8 *)0)
{
    send the buffer p;
    Bridge::recycle(p);
}
```

To avoid forwarding packets that are past the maximum message system timeout time, asking for the nextBuffer may call the BufferManager::first(uint8 owner) function that may scan for buffers to free. Accordingly, full TX buffers with no hope of making a timeout may be freed on the thread that owns the buffer. Abridge that is doing TX (i.e., while looking for the next TX buffer) may free all of the TX buffers that are expired before receiving the next TX buffer for processing.

As shown in FIG. 11J-11L, during the buffer allocation process, buffers marked free may be transferred to the drivers to receive new packets, or to PCGP to receive new payloads for TX. Allocation from “free” may be done by the “packetProcessor” function. The number of sends and receives between “packetProcessor” calls may dictate how many LT_Driver_RX, GT_Driver_RX and PCGP_Free buffers need to be allocated. LT_Driver may represent drivers that handle addresses that are less than the node address. GT_Driver may represent drivers that handle addresses that are greater than the node address.

When a driver receives a packet, the driver may put the data into an RX buffer that gets handed to the router. The router may then reassign the buffer to PCGP_Receive or to the other driver’s TX (not shown). If the buffer contains obviously invalid data, the buffer may transition to free.

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After a router marks a buffer for TX, the driver may discover the buffer is TX and may send the message. After sending the message, the buffer may immediately become an RX buffer if the driver was low in RX buffers, or the buffer may be freed for re-allocation.

During the “packetProcessor” call, PCGP may process all buffers that the router marked as PCGP_Receive. At this point, data may be acted upon, so the CRC and other data items may be checked. If the data is corrupted, a statistic may be incremented and the buffer may be freed. Otherwise, the buffer may be marked as owned by the application. Buffers marked as owned by the application may be either recycled for the use of PCGP or freed for reallocation by the buffer manager.

When the application wants to send a new message, it may be done in a re-entrant friendly/mutual exclusion manner. If the buffer may be allocated, PCGP may mark the buffer as busy. Once marked busy, no other thread calling the send or reply functions may grab this buffer, as it is owned by this function call’s invocation. The remainder of the process of error checking and building the message may be done outside the isolated race condition mutual exclusion guarded code. The buffer may either transition to free or may become a valid filled CRC-checked buffer and passed to the router. These buffers may not be routed immediately and may be queued so that messages can be sent later (assuming that protocol rules allow). Reply messages may be marked differently than new send messages because reply messages may be routed with a higher priority than regular send messages and reply messages may have no rules limiting how many/when they can be sent.

PCGP was designed to work with flow control, and flow control may negotiate the transfer of messages from one node to another node so that a buffer is never dropped because the other side of an interface lacks a buffer (which may cause back pressure on the sending node).

Flow control may be apart of the shared buffer format. The first two bytes may be reserved for the driver so that the driver never needs to shift the packet bytes. Two bytes may be used so that one byte is the DMA length-1, and the second byte is to control the flow of messages. These same two bytes may be synchronizing bytes if a PCGP message is transmitted over RS232.

When a packet is “in-flight”, the packet may be in the process of being sent by a driver on the way to its destination, being processed by the destination, or being sent back as a response.

Typical delays are as follows:

Interface/ Delay cause	Delay (seconds)	Notes
SPI	<3	Roughly 400 kbps
I2C	<1	
Waking a CC2510	<6?	Clock calibration, min. sleep time
Flow control	<0.2	
RF link	20 to 2000	Minutes, never
Interference/ separation	Minutes, never	

Accordingly, messages tend to complete the round trip either: quickly (e.g., <50 ms); slowly (e.g., one or more seconds); or not at all.

PCGP may use two different times (set at initialization) for all timeouts, one for when the RF link is in fast heartbeat mode, and another for when the RF link is in slow mode. If

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a message is in-flight and the link status changes from fast to slow, the timeout may be adjusted and the difference between fast and slow may be added to the time-to-live counter for the packet. No additional transitions back and forth may affect the time-to-live time for the message.

There is a second timeout that may be twice as long as the slow timeout that is used to monitor buffer allocation inside PCGP. Accordingly, if a message is “stuck” inside a driver and hasn’t been sent due to e.g., flow control or hardware damage, the buffer may be freed by the buffer manager, resulting in the buffer being dropped. For a “new” message, this may mean that the packet already timed out and the application was already given a reply saying the message wasn’t delivered, resulting in the buffer being freed. Since the driver polls the buffer manager for buffers that need to be sent, the buffer is freed up so that a message that could be sent is handed to the driver the next time that it unblocks. For a reply message, the reply may simply get dropped and the sending node may time out.

The PCGP messaging system may pass messages that contain header information and payload. Outside of PCGP, the header may be a set of data items in a call signature. However, internal to PCGP, there may be a consistent, driver friendly byte layout. Drivers may insert bytes either into the PCGP packet or before the PCGP packet such:

DE, CA: Synch bytes for use with RS232, nominal value of 0xDE, 0xCA or 0x5A, 0xA5.

LD: Driver DMA length byte, equals amount driver is pushing in this DMA transfer, which is the total size, not including the size byte or synch bytes.

Cmd: Driver command and control byte used for flow control.

LP: PCGP packet length, always the total header+payload size in bytes+CRC size. LD=LP+1.

Dst: Destination address.

Src: Source address

Cmd: Command byte

Scd: Sub command byte

AT: Application Tag is defined by the application and has no significance to PCGP. It allows the application to attach more information to a message e.g., the thread from which the message originated.

SeqNum: thirty-two bit sequence number is incremented by PCGP for a new message sent, guarantees the number will not wrap, acts as a token, endianness isn’t relevant.

CRC16: A sixteen bit CRC of the PCGP header and payload.

An example of a message with no payload, cmd=1, subcmd=2 is as follows:

0xDE, 0xCA, 0xC, 0x5, 0x14, 1, 2, 0, 0, 0, 0, 0x1, crchigh, crclow.

0x0D, cmd, 0xC, 0x5, 0x14, 1, 2, 0, 0, 0, 0, 0x1, crchigh, crclow.

There may be several advantages to this methodology, examples of which may include but are not limited to:

Most of our hardware DMA engines may use the first byte to define how many additional bytes to move, so in this methodology, drivers and PCGP may share buffers.

A byte may be provided right after the DMA length to pass flow control information between drivers.

Driver length and “Cmd” byte may be outside the CRC region so they may be altered by the driver, may be owned by the driver transport mechanism, and the driver may guard for invalid lengths.

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There may be a separate PCGP packet length byte that is CRC protected. Accordingly, the application may trust the that payload length is correct.

The endianness of the sequence number may not be relevant, as it is just a byte pattern that may be matched that happens to also be a thirty-two bit integer.

The sequence number may be four bytes aligned to the edge of the shared buffer pool length.

There may be optional RS232 synchronizing bytes so that users may move cables around while debugging a message stream and both sides of the interface may resynchronize.

The application, driver and PCGP may share buffers and may release them by pointer.

PCGP may not be an event driven software design, but may be used in event driven architectures by how the sub-classes are written. Data may be exchanged between the classes conceptually (as shown in FIG. 11M-11N).

Some event model in the driver may wake the driver, may receive a message and may pass the message through the bridge into the buffer manager that routes the message to new owner of the new message (through a bridge to either a driver or PCGP).

The following summarizes some exemplary events:

Event:	Possible use:	Where this occurs:
When a new send or reply is queued, or decTimeouts generates a timeout reply.	Decide to run packetProcessor.	Inside PCGP: :sendInternal
When a messages is received for PCGP.	Decide to run packetProcessor.	BufferManager: :give
When a driver has something new to send.	Wake driver for TX.	BufferManager: :give
When a Driver RX buffer becomes available.	Turn off flow control.	BufferManager: :give

The following illustrative example shows how the PCGP event model may work with Nucleus to wakeup the PCGP task after every message send, reply, or decTimeout that generated a NACK:

```

class PcpOS : public Pcp
{
    virtual void schedulePacketProcessor(void)
    {
        OS_EventGrp_Set(g_RCVevGrps[EVG_RF_TASK].pEvgHandle,
            RfRadioTxEvent, OS_EV_OR_NO_CLEAR);
    }
}

```

The following is a pseudo code driver that is event based, illustrating how driver events work. The Driver subclasses Bridge and overrides hasMessagesToSend and flowControlTurnedOff to schedule the TX and RX functions to run if they aren’t already running.

```

class SPI_Driver : public Bridge
{
    virtual void hasMessagesToSend( )
    {
        Trigger_ISR(TX_ISR, this);
    }
    virtual void flowControlTurnedOff( )
    {
        Trigger_ISR(RX_ISR, this);
    }
    static void TX_RetryTimer( )

```

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-continued

```

{
    Trigger_ISR(TX_ISR, this);
}
static void TX_ISR(Bridge *b)
{
    DisableISRs( );
    do
    {
        uint8 *p = b->nextBufferTX( );
        if (p == null) break;
        if (b->_bufferManager->bufferTimedOut(p)==false)
        {
            if (OtherSideSPI_FlowControl( ) == false)
            {
                Trigger TX_RetryTimer in 20 msec.
                break;
            }
            send(p);
        }
        free(p)
    } while (true) ;
    EnableISRs( );
}
static void RX_ISR(Bridge *b)
{
    DisableISRs( );
    do
    {
        uint8* p = b->nextBufferRX( );
        if (p == null) break;
        uint i;
        while (not done receiving)
            p[i++] = getChar( );
        b->route(p);
    } while (true) ;
    EnableISRs( );
}
}

```

The following statistics may be supported by PCGP:

Number of packets sent;
 Number of packets received;
 CRC errors;
 Timeouts; and
 Buffer unavailable (ran out of buffers)

PCGP may be designed to run in multiple processing environments. Most parameters may be run time configured because it facilitates testing, and any run time fine tuning for performance. Other parameters may be compile time e.g., anything that alters memory allocation must be done statically at compile time.

The following may be compile time configuration #defines that may vary where PCGP is implemented:

driver bytes: may be two bytes reserved in the common buffer scheme for the driver, but this may be a compile time option to accommodate other drivers such as RF protocol.

#RX driver buffers: may be tuned to how many buffers would be good for that processor/traffic flow, etc.

#PCGP RX buffers: may be tuned to how many buffers would be good for that processor/traffic flow, etc.

Total # of buffers: may be tuned to how many buffers should be at that processor.

The CRC may be used to ensure data integrity. If a CRC is invalid, it may not be delivered to the application and the CRC error may be tracked. The message may eventually timeout and may be retried by the originator.

Likewise, if the messaging system informs the application that a message was delivered when it was not, this may be a hazard to the system. The Stop Bolus Command is an example of such a command. This may be mitigated by the Request/Action sequence of messages which may be

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required by the application to change therapy. The Controller may receive a matching command from the Pump application to consider the message delivered.

DEKA may provide a reference way of interfacing PCGP into the Nucleus OS system on the ARM 9 (as shown in FIG. 110).

As shown in FIG. 11P, the pcgpOS.cpp file may instantiate a PCGP node instance (Pcgp, a Bridge, etc.) and may provide through pcgpOS.h a 'C' linkable set of function calls that provide a 'C' language interface to the C++ code. This may simplify the 'C' code as the objects acted upon are implicit.

The following general rules may be applied:

PCGP may run on all nodes: any driver may support a generic driver interface.

Race conditions may not be permitted.

May support half duplex on the SPI port between slave processor and master processor.

Data transfer may not be attempted; as it either succeeds or returns fail/false.

May require low overhead (time, processing, bandwidth wasted).

May support CC2510 operating at DMA (fast) SPI clock rates.

SPI flow control may prevent data from being sent if the receiving side does not currently have an empty buffer to place the packet. This may be accomplished by asking for permission to send and waiting for a response indicating that you have been cleared to do so. There may also be a way to tell the other side that there are currently no free buffers and the transfer should be attempted at a later time.

All transmission may begin with a length byte that indicates the number of bytes to be sent, not including the length byte itself. Following the length may be a single byte indicating the command being sent.

The actual transmission of a packet may be the length of packet plus one for the command byte, followed by the command byte for a message appended and finally the packet itself.

In addition to the command bytes that will be sent, an additional hardware line called the FlowControl line may be added to the traditional four SPI signals. The purpose of this line is to allow the protocol to run as quickly as possible without a need for preset delays. It also allows the slave processor to tell the master processor that it has a packet waiting to be sent, thus eliminating the need for the master processor to poll the slave processor for status.

The following exemplary command values may be used: Commands to be Sent by the Master Processor:

Command	Value	Description
M_RTS	0xC1	Master is requesting to send a packet
M_MSG_APPENDED	0xC2	Master is sending a packet
M_CTS	0xC3	Master is tell slave it is Cleared to Send
M_ERROR	0xC4	An Error condition has been encountered

Commands to be Sent by the Slave Processor:

Command	Value	Description
S_PREPARING_FOR_RX	0xA1	Slave is prepare the dma to receive a packet

-continued

Command	Value	Description
S_RX_BUFF_FULL	0xA2	Slave is currently out of RX buffers, retry later
S_MSG_APPENDED	0xA3	Slave is sending a packet
S_ERROR	0xA4	An Error condition has been encountered

As illustrated in FIG. 11Q, when the slave processor has a packet to send to the master processor, the slave processor may notify the master processor (by asserting the FlowControl line) that it has a pending packet that is waiting to be sent. Doing so may result in an IRQ on the master processor at which time the master processor may decide when to go retrieve the message from the slave processor. Retrieving the packet may be delayed at the discretion of the master processor, and the master processor may even decide to attempt to send a packet to the slave processor before retrieving from the slave processor.

The master processor may begin the retrieval by sending the slave processor M_CTS commands; this shall be repeated until the slave processor responds by sending the S_MSG_APPENDED command along with the packet itself. The FlowControl line may be cleared after the packet has been sent. If a M_CTS command is received by the slave processor when one is not expected, the M_CTS command may be ignored.

As illustrated in FIG. 11R, when the master processor has a packet to send to the slave processor, the master processor may initiate the transfer by sending a M_RTS command. Upon receiving the M_RTS command, if the slave processor currently has a send packet pending, the slave processor will lower the FlowControl line so that it may be re-used as a Cleared To Send signal. The slave processor may then tell the master processor that it is in the process of preparing the SPI DMA to receive the packet, during which time the master processor may stop clocking bytes onto the bus and may allow the slave processor to finish preparing for the receive.

The slave processor may then indicate it is ready to receive the full packet by raising the FlowControl line (which is now used as the CTS signal). Upon receiving the CTS signal, the master processor may proceed to send the M_MSG_APPENDED command along with the packet itself.

After the completion of the transfer, the slave processor may lower the FlowControl line. If a packet was pending at the start of the transfer, or a send occurred on the slave processor when the packet was being received, the slave processor may reassert the FlowControl line now indicating that it has a pending packet.

Referring again to FIG. 11A, infusion pump assembly 100, 100' may include switch assembly 318 coupled to electrical control assembly 110 (FIG. 3) that may allow a user (not shown) to perform at least one, and in some embodiments, a plurality of tasks. One illustrative example of such a task is the administration of a bolus dose of the infusible fluid (e.g., insulin) without the use of a display assembly. Remote control assembly 300 may allow the user to enable/disable/configure infusion pump assembly 100, 100' to administer the bolus dose of insulin.

Referring also to FIG. 12A, slider assembly 306 may be configured, at least in part, to enable the user to manipulate the menu-based information rendered on display assembly 302. An example of slider assembly 306 may include a

capacitive slider assembly, which may be implemented using a CY8C21434-24LFXI PSOC offered by Cypress Semiconductor of San Jose, California, the design an operation of which are described within the "CSD User Module" published by Cypress Semiconductor. For example, via slider assembly 306, the user may slide their finger in the direction of arrow 314, resulting in the highlighted portion of the information included within main menu 350 (shown in FIG. 12A) rendered on display assembly 302 scrolling upward. Alternatively, the user may slide their finger in the direction of arrow 316, resulting in the highlighted portion of the information included within main menu 350 rendered on display assembly 302 scrolling downward.

Slider assembly 306 may be configured so that the rate at which e.g. the highlighted portion of main menu 350 scrolls "upward" or "downward" varies depending upon the displacement of the finger of the user with respect to point of origin 320. Therefore, if the user wishes to quickly scroll "upward", the user may position their finger near the top of slider assembly 306. Likewise, if the user wishes to quickly scroll "downward", the user may position their finger near the bottom of slider assembly 306. Additionally, if the user wishes to slowly scroll "upward", the user may position their finger slightly "upward" with respect to point of origin 320. Further, if the user wishes to slowly scroll "downward", the user may position their finger slightly "downward" with respect to point of origin 320. Once the appropriate menu item is highlighted, the user may select the highlighted menu item via one or more switch assemblies 308, 310.

Referring also to FIGS. 12B-12F, assume for illustrative purposes that infusion pump assembly 100, 100' is an insulin pump and the user wishes to configure infusion pump assembly 100, 100' so that when switch assembly 318 is depressed by the user, a 0.20 unit bolus dose of insulin is administered. Accordingly, the user may use slider assembly 306 to highlight "Bolus" within main menu 350 rendered on display assembly 302. The user may then use switch assembly 308 to select "Bolus". Once selected, processing logic (not shown) within remote control assembly 300 may then render submenu 352 on display assembly 302 (as shown in FIG. 12B).

The user may then use slider assembly 306 to highlight "Manual Bolus" within submenu 352, which may be selected using switch assembly 308. Processing logic (not shown) within remote control assembly 300 may then render submenu 354 on display assembly 302 (as shown in FIG. 12C).

The user may then use slider assembly 306 to highlight "Bolus: 0.0 Units" within submenu 354, which may be selected using switch assembly 308. Processing logic (not shown) within remote control assembly 300 may then render submenu 356 on display assembly 302 (as shown in FIG. 12D).

The user may then use slider assembly 306 to adjust the "Bolus" insulin amount to "0.20 units", which may be selected using switch assembly 308. Processing logic (not shown) within remote control assembly 300 may then render submenu 358 on display assembly 302 (as shown in FIG. 12E).

The user 14 may then use slider assembly 306 to highlight "Confirm", which may be selected using switch assembly 308. Processing logic (not shown) within remote control assembly 300 may then generate the appropriate signals that may be sent to the above-described telemetry circuitry (not shown) included within remote control assembly 300. The telemetry circuitry (not shown) included within the remote control assembly may then transmit, via wireless commu-

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nication channel 312 established between remote control assembly 300 and infusion pump assembly 100', the appropriate configuration commands to configure infusion pump assembly 100' so that whenever switch assembly 318 is depressed by the user, a 0.20 unit bolus dose of insulin is administered.

Once the appropriate commands are successfully transmitted, processing logic (not shown) within remote control assembly 300 may once again render submenu 350 on display assembly 302 (as shown in FIG. 12F).

Specifically and once programmed via remote control assembly 300, the user may depress switch assembly 318 of infusion pump assembly 100' to administer the above-described 0.20 unit bolus dose of insulin. Via the above-described menuing system included within remote control assembly 300, the user may define a quantity of insulin to be administered each time that the user depresses switch assembly 318. While this particular example specifies that a single depression of switch assembly 318 is equivalent to 0.20 units of insulin, this is for illustrative purposes only and is not intended to be a limitation of this disclosure, as other values (e.g. 1.00 units of insulin per depression) are equally applicable.

Assume for illustrative purposes that the user wishes to administer a 2.00 unit bolus dose of insulin. To activate the above-described bolus dose administration system, the user may be required to press and hold switch assembly 318 for a defined period of time (e.g. five seconds), at which point infusion pump assembly 100, 100' may generate an audible signal indicating to the user that infusion pump assembly 100, 100' is ready to administer a bolus dose of insulin via switch assembly 318. Accordingly, the user may depress switch assembly 318 ten times (i.e., 2.00 units is ten 0.20 unit doses). After each time that switch assembly 318 is depressed, infusion pump assembly 100, 100' may provide on audible response to the user via an internal speaker/sound generation device (not shown). Accordingly, the user may depress switch assembly 318 the first time and infusion pump assembly 100, 100' may generate a confirmation beep in response, thus indicating to the user that infusion pump assembly 100, 100' received the command for (in this particular example) 0.20 units of insulin. As the desired bolus dose is 2.00 units of insulin, the user may repeat this procedure nine more times in order to effectuate a bolus dose of 2.00 units, wherein infusion pump assembly 100, 100' generates a confirmation beep after each depression of switch assembly 318.

While in this particular example, infusion pump assemblies 100, 100' are described as providing one beep after each time the user depresses switch assembly 318, this is for illustrative purposes only and is not intended to be a limitation of this disclosure. Specifically, infusion pump assembly 100, 100' may be configured to provide a single beep for each defined quantity of insulin. As discussed above, a single depression of switch assembly 318 may be equivalent to 0.20 units of insulin. Accordingly, infusion pump assembly 100, 100' may be configured to provide a single beep for each 0.10 units of insulin. Accordingly, if infusion pump assembly 100, 100' is configured such that a single depression of switch assembly 318 is equivalent to 0.20 units of insulin, each time switch assembly 318 is depressed, infusion pump assembly 100, 100' may provide the user with two beeps (i.e. one for each 0.10 units of insulin).

Once the user has depressed switch assembly 318 on infusion pump assembly 100' a total of ten times, the user may simply wait for infusion pump assembly 100, 100' to acknowledge receipt of the instructions to administer a 2.00

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unit bolus dose of insulin (as opposed to the confirmation beep received at each depression of switch assembly 318). Once a defined period of time (e.g., two seconds) passes, infusion pump assembly 100, 100' may provide an audible confirmation to the user concerning the quantity of units to be administered via the bolus insulin dose that the user just requested. For example, as (in this example) infusion pump assembly 100, 100' was programmed by the user so that a single depression of switch assembly 318 is equivalent to 0.20 units of insulin, infusion pump assembly 100, 100' may beep ten times (i.e., 2.00 units is ten 0.20 unit doses).

When providing feedback to the user concerning the quantity of units to be administered via the bolus insulin dose, infusion pump assembly 100, 100' may provide a multifrequency audible confirmation. For example and continuing with the above-stated example in which ten beeps are to be provided to the user, infusion pump assembly 100, 100' may group the beeps into groups of five (to facilitate easier counting by the user) and the beeps within each group of five may be rendered by infusion pump assembly 100, 100' so that each subsequent beep has a higher frequency than the preceding beep (in a manner similar to a musical scale). Accordingly and continuing with the above-stated example, infusion pump assembly 100, 100' may render a 1,000 Hz beep, followed by an 1,100 Hz beep, followed by a 1,200 Hz beep, followed by a 1,300 Hz beep, followed by a 1,400 Hz beep (thus completing a group of five beeps), followed by a short pause, and then a 1,000 Hz beep, followed by an 1,100 Hz beep, followed by a 1,200 Hz beep, followed by a 1,300 Hz beep, followed by a 1,400 Hz beep (thus completing the second group of five beeps). According to various additional/alternative embodiments the multifrequency audible confirmation may utilize various numbers of tones incrementing in frequency. For example, an embodiment may utilize twenty different tones incrementing in frequency. However, the number of tones should not be construed as a limitation of the present disclosure as number of tones may vary according to design criteria and user need.

Once infusion pump assembly 100, 100' completes the rendering of the multifrequency audible confirmation (i.e. the ten beeps described above), the user may, within a defined period of time (e.g. two seconds), depress switch assembly 318 to provide a confirmation signal to infusion pump assembly 100, 100', indicating that the multifrequency audible confirmation was accurate and indicative of the size of the bolus dose of insulin to be administered (i.e. 2.00 units). Upon receiving this confirmation signal, infusion pump assembly 100, 100' may render a "confirmation received" audible tone and effectuate the delivery of (in this particular example) the 2.00 unit bolus dose of insulin. In the event that infusion pump assembly 100, 100' fails to receive the above-described confirmation signal, infusion pump assembly 100, 100' may render a "confirmation failed" audible tone and will not effectuate the delivery of the bolus dose of insulin. Accordingly, if the multifrequency audible confirmation was not accurate/indicative of the size of the bolus dose of insulin to be administered, the user may simply not provide the above-described confirmation signal, thereby canceling the delivery of the bolus dose of insulin.

As discussed above, in one exemplary embodiment of the above-described infusion pump assembly, infusion pump assembly 100' may be used to communicate with a remote control assembly 300. When such a remote control assembly 300 is utilized, infusion pump assembly 100' and remote control assembly 300 may routinely contact each other to ensure that the two devices are still in communication with each other. For example, infusion pump assembly 100' may

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“ping” remote control assembly 300 to ensure that remote control assembly 300 is present and active. Further, remote control assembly 300 may “ping” infusion pump assembly 100' to ensure that infusion pump assembly 100' is still present and active. In the event that one of infusion pump assembly 100' and remote control assembly 300 fails to establish communication with the other assembly, the assembly that is unable to establish communication may sound a “separation” alarm. For example, assume that remote control assembly 300 is left in the car of the user, while infusion pump assembly 100' is in the pocket of the user. Accordingly and after a defined period of time, infusion pump assembly 100' may begin sounding the “separation” alarm, indicating that communication with remote control assembly 300 cannot be established. Using switch assembly 318, the user may acknowledge/silence this “separation” alarm.

As the user may define and administer a bolus insulin dose via switch assembly 318 of infusion pump assembly 100' while remote control assembly 300 is not in communication with infusion pump assembly 100', infusion pump assembly 100' may store information concerning the administered bolus insulin dose within a log file (not shown) stored within infusion pump assembly 100'. This log file (not shown) may be stored within nonvolatile memory (not shown) included within infusion pump assembly 100'. Upon communication being reestablished between infusion pump assembly 100' and remote control assembly 300, infusion pump assembly 100' may provide the information concerning the administered bolus insulin dose stored within the log file (not shown) of infusion pump assembly 100' to remote control assembly 300.

Further, if the user anticipates separating remote control assembly 300 from infusion pump assembly 100', the user (via the above-described menuing system) may configure infusion pump assembly 100' and remote control assembly 300 to be in “separation” mode, thus eliminating the occurrence of the above-described “separation” alarms. However, the devices may continue to “ping” each other so that when they come back into communication with each other, infusion pump assembly 100' and remote control assembly 300 may automatically exit “separation” mode.

Further, if the user anticipates traveling in an airplane, the user (via the above-described menuing system of remote control assembly 300) may configure infusion pump assembly 100' and remote control assembly 300 to be in “airplane” mode, in which each of infusion pump assembly 100' and remote control assembly 300 suspend any and all data transmissions. While in “airplane” mode, infusion pump assembly 100' and remote control assembly 300 may or may not continue to receive data.

Switch assembly 318 may be used to perform additional functions, such as: checking the battery life of reusable housing assembly 102; pairing reusable housing assembly 102 with remote control assembly 300; and aborting the administration of a bolus dose of infusible fluid.

Checking Battery Life: Reusable housing assembly 102 may include a rechargeable battery assembly that may be capable of powering infusion pump assembly 100, 100' for approximately three days (when fully charged). Such a rechargeable battery assembly may have a usable life of a predetermined number of usable hours, for example, or years, or other predetermined length of usage. However, the predetermined life may depend on many factors, including but not limited to, one or more of the following: climate, daily usage, and number of recharges. Whenever reusable housing assembly 102 is disconnected from disposable

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housing assembly 114, infusion pump assembly 100, 100' may perform a battery check on the above-described rechargeable battery assembly whenever switch assembly 318 is depressed for a defined period of time (e.g. in excess of two seconds). In the event that the above-described rechargeable battery assembly is determined to be charged above a desired threshold, infusion pump assembly 100, 100' may render a “battery pass” tone. Alternatively, in the event that the above-described rechargeable battery assembly is determined to be charged below a desired threshold, infusion pump assembly 100, 100' may render a “battery fail” tone. Infusion pump assembly 100, 100' may include components and/or circuitry to determine whether reusable housing assembly 102 is disconnected from disposable housing assembly 114.

Pairing: As discussed above and in one exemplary embodiment of the above-described infusion pump assembly, infusion pump assembly 100' may be used to communicate with remote control assembly 300. In order to effectuate communication between infusion pump assembly 100' and remote control assembly 300, a pairing process may be performed. During such a pairing process, one or more infusion pump assemblies (e.g. infusion pump assembly 100') may be configured to communicate with remote control assembly 300 and (conversely) remote control assembly 300 may be configured to communicate with one or more infusion pump assemblies (e.g. infusion pump assembly 100'). Specifically, the serial numbers of the infusion pump assemblies (e.g. infusion pump assembly 100') may be recorded within a pairing file (not shown) included within remote control assembly 300 and the serial number of remote control assembly 300 may be recorded within a pairing file (not shown) included within the infusion pump assemblies (e.g. infusion pump assembly 100').

According to an embodiment, in order to effectuate such a pairing procedure, the user may simultaneously hold down one or more switch assemblies on both remote control assembly 300 and infusion pump assembly 100'. For example, the user may simultaneously hold down switch assembly 310 included within remote control assembly 300 and switch assembly 318 included within infusion pump assembly 100' for a defined period exceeding e.g. five seconds. Once this defined period is reached, one or more of remote control assembly 300 and infusion pump assembly 100' may generate an audible signal indicating that the above-described pairing procedure has been effectuated.

According to another embodiment, prior to performing the pairing process, the user may uncouple reusable housing assembly 102 from disposable housing assembly 114. By requiring this initial step, further assurance is provided that an infusion pump assembly being worn by a user may not be surreptitiously paired with a remote control assembly.

Once uncoupled, the user may enter pairing mode via input assembly 304 of remote control assembly 300. For example, the user may enter pairing mode on remote control assembly 300 via the above-described menuing system in combination with e.g., switch assembly 310. The user may be prompted on display assembly 302 of remote control assembly 300 to depress and hold switch assembly 318 on infusion pump assembly 100'. Additionally, remote control assembly 304 may switch to a low power mode to e.g., avoid trying to pair with distant infusion pump assemblies. The user may then depress and hold switch assembly 318 on infusion pump assembly 100' so that infusion pump assembly 100' enters a receive mode and waits for a pairing command from remote control assembly 300.

Remote control assembly 300 may then transmit a pairing request to infusion pump assembly 100', which may be acknowledged by infusion pump assembly 100'. Infusion pump assembly 100' may perform a security check on the pairing request received from remote control assembly 300 and (if the security check passes) infusion pump assembly 100' may activate a pump pairing signal (i.e., enter active pairing mode). Remote control assembly 300 may perform a security check on the acknowledgment received from infusion pump assembly 100'.

The acknowledgment received from infusion pump assembly 100' may define the serial number of infusion pump assembly 100' and remote control assembly 300 may display that serial number on display assembly 302 of remote control assembly 300. The user may be asked if they wish to pair with the pump found. If the user declines, the pairing process may be aborted. If the user agrees to the pairing process, remote control assembly 300 may prompt the user (via display assembly 302) to depress and hold switch assembly 318 on infusion pump assembly 100'.

The user may then depress and hold switch assembly 318 on infusion pump assembly 100' and depress and hold e.g. switch assembly 310 on remote control assembly 300.

Remote control assembly 300 may confirm that remote switch assembly 310 was held (which may be reported to infusion pump assembly 100'). Infusion pump assembly 100' may perform a security check on the confirmation received from remote control assembly 300 to confirm the integrity of same. If the integrity of the confirmation received is not verified, the pairing process is aborted. If the integrity of the confirmation received is verified, any existing remote pair configuration file is overwritten to reflect newly-paired remote control assembly 300, the pump pairing completed signal is activated, and the pairing process is completed.

Additionally, infusion pump assembly 100' may confirm that switch assembly 318 was held (which may be reported to remote control assembly 300). Remote control assembly 300 may perform a security check on the confirmation received from infusion pump assembly 100' to confirm the integrity of same. If the integrity of the confirmation received is not verified, the pairing process is aborted. If the integrity of the confirmation received is verified, a pair list file within remote control assembly 300 may be modified to add infusion pump assembly 100'. Typically, remote control assembly 300 may be capable of pairing with multiple infusion pump assemblies, while infusion pump assembly 100' may be capable of only pairing with a single remote control assembly. The pairing completed signal may be activated and the pairing process may be completed.

When the pairing process is completed, one or more of remote control assembly 300 and infusion pump assembly 100' may generate an audible signal indicating that the above-described pairing procedure has been successfully effectuated.

Aborting Bolus Dose: in the event that the user wishes to cancel a bolus dose of e.g. insulin being administered by infusion pump assembly 100', the user may depress switch assembly 318 (e.g., shown in FIGS. 1 & 2) for a defined period exceeding e.g. five seconds. Once this defined period is reached, infusion pump assembly 100' may render an audible signal indicating that the above-described cancellation procedure has been effectuated.

While switch assembly 318 is shown as being positioned on the top of infusion pump assembly 100, 100', this is for illustrative purposes only and is not intended to be a limitation of this disclosure, as other configurations are possible.

For example, switch assembly 318 may be positioned about the periphery of infusion pump assembly 100, 100'.

Referring also to FIGS. 13-15, there is shown an alternative-embodiment infusion pump assembly 400. As with pump assembly 100, 100', infusion pump assembly 400 may include reusable housing assembly 402 and disposable housing assembly 404.

In a fashion similar to reusable housing assembly 102, reusable housing assembly 402 may include a mechanical control assembly (that includes at least one pump assembly and at least one valve assembly). Reusable housing assembly 402 may also include an electrical control assembly that is configured to provide control signals to the mechanical control assembly and effectuate the delivery of an infusible fluid to a user. The valve assembly may be configured to control the flow of the infusible fluid through a fluid path and the pump assembly may be configured to pump the infusible fluid from the fluid path to the user.

In a fashion similar to disposable housing assembly 114, disposable housing assembly 404 may be configured for a single use or for use for a specified period of time, e.g., three days or any other amount of time. Disposable housing assembly 404 may be configured such that any components in infusion pump assembly 400 that come in contact with the infusible fluid are disposed on and/or within disposable housing assembly 404.

In this particular embodiment of the infusion pump assembly, infusion pump assembly 400 may include switch assembly 406 positioned about the periphery of infusion pump assembly 400. For example, switch assembly 406 may be positioned along a radial edge of infusion pump assembly 400, which may allow for easier use by a user. Switch assembly 406 may be covered with a waterproof membrane configured to prevent the infiltration of water into infusion pump assembly 400. Reusable housing assembly 402 may include main body portion 408 (housing the above-described mechanical and electrical control assemblies) and locking ring assembly 410 that may be configured to rotate about main body portion 408 (in the direction of arrow 412).

In a fashion similar to reusable housing assembly 102 and disposable housing assembly 114, reusable housing assembly 402 may be configured to releasably engage disposable housing assembly 404. Such releasable engagement may be accomplished by a screw-on, a twist-lock or a compression fit configuration, for example. In an embodiment in which a twist-lock configuration is utilized, the user of infusion pump assembly 400 may first properly position reusable housing assembly 402 with respect to disposable housing assembly 404 and may then rotate locking ring assembly 410 (in the direction of arrow 412) to releasably engage reusable housing assembly 402 with disposable housing assembly 404.

Through the use of locking ring assembly 410, reusable housing assembly 402 may be properly positioned with respect to disposable housing assembly 404 and then releasably engaged by rotating locking ring assembly 410, thus eliminating the need to rotate reusable housing assembly 402 with respect to disposable housing assembly 404. Accordingly, reusable housing assembly 402 may be properly aligned with disposable housing assembly 404 prior to engagement, and such alignment may not be disturbed during the engagement process. Locking ring assembly 410 may include a latching mechanism (not shown) that may prevent the rotation of locking ring assembly 410 until reusable housing assembly 402 and disposable housing assembly 404 are properly positioned with respect to each other.

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Referring also to FIGS. 16-18, there is shown an alternative-embodiment infusion pump assembly 500. As with pump assembly 100, 100', infusion pump assembly 500 may include reusable housing assembly 502 and disposable housing assembly 504.

In a fashion similar to reusable housing assembly 402, reusable housing assembly 502 may include a mechanical control assembly (that includes at least one pump assembly and at least one valve assembly). Reusable housing assembly 502 may also include an electrical control assembly that is configured to provide control signals to the mechanical control assembly and effectuate the delivery of an infusible fluid to a user. The valve assembly may be configured to control the flow of the infusible fluid through a fluid path and the pump assembly may be configured to pump the infusible fluid from the fluid path to the user.

In a fashion similar to disposable housing assembly 404, disposable housing assembly 504 may be configured for a single use or for use for a specified period of time, e.g., e.g., three days or any other amount of time. Disposable housing assembly 504 may be configured such that any components in infusion pump assembly 500 that come in contact with the infusible fluid are disposed on and/or within disposable housing assembly 504.

In this particular embodiment of the infusion pump assembly, infusion pump assembly 500 may include switch assembly 506 positioned about the periphery of infusion pump assembly 500. For example, switch assembly 506 may be positioned along a radial edge of infusion pump assembly 500, which may allow for easier use by a user. Switch assembly 506 may be covered with a waterproof membrane and/or an o-ring or other sealing mechanism may be included on the stem 507 of the switch assembly 506 configured to prevent the infiltration of water into infusion pump assembly 500. However, in some embodiments, switch assembly 506 may include an overmolded rubber button, thus providing functionality as a waterproof seal without the use of a waterproof membrane or an o-ring. However, in still other embodiments, the overmolded rubber button may additionally be covered by a waterproof membrane and/or include an o-ring. Reusable housing assembly 502 may include main body portion 508 (housing the above-described mechanical and electrical control assemblies) and locking ring assembly 510 that may be configured to rotate about main body portion 508 (in the direction of arrow 512).

In a fashion similar to reusable housing assembly 402 and disposable housing assembly 404, reusable housing assembly 502 may be configured to releasably engage disposable housing assembly 504. Such releasable engagement may be accomplished by a screw-on, a twist-lock or a compression fit configuration, for example. In an embodiment in which a twist-lock configuration is utilized, the user of infusion pump assembly 500 may first properly position reusable housing assembly 502 with respect to disposable housing assembly 504 and may then rotate locking ring assembly 510 (in the direction of arrow 512) to releasably engage reusable housing assembly 502 with disposable housing assembly 404.

As locking ring assembly 510 included within infusion pump assembly 500 may be taller (i.e., as indicated by arrow 514) than locking ring assembly 410, locking ring assembly 510 may include a passage 516 through which button 506 may pass. Accordingly, when assembling reusable housing assembly 502, locking ring assembly 510 may be installed onto main body portion 508 (in the direction of arrow 518). Once locking ring assembly 510 is installed onto main body

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portion 508, one or more locking tabs (not shown) may prevent locking ring assembly 510 from being removed from main body portion 508. The portion of switch assembly 506 that protrudes through passage 516 may then be pressed into main body portion 508 (in the direction of arrow 520), thus completing the installation of switch assembly 506.

Although button 506 is shown in various locations on infusion pump assembly 500, button 506, in other embodiments, may be located anywhere desirable on infusion pump assembly 500.

Through the use of locking ring assembly 510, reusable housing assembly 502 may be properly positioned with respect to disposable housing assembly 504 and then releasably engaged by rotating locking ring assembly 510, thus eliminating the need to rotate reusable housing assembly 502 with respect to disposable housing assembly 504. Accordingly, reusable housing assembly 502 may be properly aligned with disposable housing assembly 504 prior to engagement, and such alignment may not be disturbed during the engagement process. Locking ring assembly 510 may include a latching mechanism (not shown) that prevents the rotation of locking ring assembly 510 until reusable housing assembly 502 and disposable housing assembly 504 are properly positioned with respect to each other. Passage 516 may be elongated to allow for the movement of locking ring 510 about switch assembly 506.

Referring also to FIGS. 19A-19B & 20-21, there are shown various views of infusion pump assembly 500, which is shown to include reusable housing assembly 502, switch assembly 506, and main body portion 508. As discussed above, main body portion 508 may include a plurality of components, examples of which may include but are not limited to volume sensor assembly 148, printed circuit board 600, vibration motor assembly 602, shape memory actuator anchor 604, switch assembly 506, battery 606, antenna assembly 608, pump assembly 106, measurement valve assembly 610, volume sensor valve assembly 612 and reservoir valve assembly 614. To enhance clarity, printed circuit board 600 has been removed from FIG. 19B to allow for viewing of the various components positioned beneath printed circuit board 600.

The various electrical components that may be electrically coupled with printed circuit board 600 may utilize spring-biased terminals that allow for electrical coupling without the need for soldering the connections. For example, vibration motor assembly 602 may utilize a pair of spring-biased terminals (one positive terminal and one negative terminal) that are configured to press against corresponding conductive pads on printed circuit board 600 when vibration motor assembly 602 is positioned on printed circuit board 600. However, in the exemplary embodiment, vibration motor assembly 602 is soldered directly to the printed circuit board.

As discussed above, volume sensor assembly 148 may be configured to monitor the amount of fluid infused by infusion pump assembly 500. For example, volume sensor assembly 148 may employ acoustic volume sensing, which is the subject of U.S. Pat. Nos. 5,575,310 and 5,755,683 assigned to DEKA Products Limited Partnership, as well as the U.S. patent application Publication Nos. US 2007/0228071 A1, US 2007/0219496 A1, US 2007/0219480 A1, US 2007/0219597 A1, the entire disclosures of all of which are incorporated herein by reference.

Vibration motor assembly 602 may be configured to provide a vibration-based signal to the user of infusion pump assembly 500. For example, in the event that the voltage of battery 606 (which powers infusion pump assembly 500) is

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below the minimum acceptable voltage, vibration motor assembly 602 may vibrate infusion pump assembly 500 to provide a vibration-based signal to the user of infusion pump assembly 500. Shape memory actuator anchor 604 may provide a mounting point for the above-described shape memory actuator (e.g. shape memory actuator 112). As discussed above, shape memory actuator 112 may be, for example, a conductive shape-memory alloy wire that changes shape with temperature. The temperature of shape-memory actuator 112 may be changed with a heater, or more conveniently, by application of electrical energy. Accordingly, one end of shape memory actuator 112 may be rigidly affixed (i.e., anchored) to shape memory actuator anchor 604 and the other end of shape memory actuator 112 may be applied to e.g. a valve assembly and/or a pump actuator. Therefore, by applying electrical energy to shape memory actuator 112, the length of shape memory actuator 112 may be controlled and, therefore, the valve assembly and/or the pump actuator to which it is attached may be manipulated.

Antenna assembly 608 may be configured to allow for wireless communication between e.g. infusion pump assembly 500 and remote control assembly 300 (FIG. 11). As discussed above, remote control assembly 300 may allow the user to program infusion pump assembly 500 and e.g. configure bolus infusion events. As discussed above, infusion pump assembly 500 may include one or more valve assemblies configured to control the flow of the infusible fluid through a fluid path (within infusion pump assembly 500) and pump assembly 106 may be configured to pump the infusible fluid from the fluid path to the user. In this particular embodiment of infusion pump assembly 500, infusion pump assembly 500 is shown to include three valve assemblies, namely measurement valve assembly 610, volume sensor valve assembly 612, and reservoir valve assembly 614.

As discussed above and referring also to FIG. 21, the infusible fluid may be stored within reservoir 118. In order to effectuate the delivery of the infusible fluid to the user, the processing logic (not shown) included within infusion pump assembly 500 may energize shape memory actuator 112, which may be anchored on one end using shape memory actuator anchor 604. Referring also to FIG. 22A, shape memory actuator 112 may result in the activation of pump assembly 106 and reservoir valve assembly 614. Reservoir valve assembly 614 may include reservoir valve actuator 614A and reservoir valve 614B, and the activation of reservoir valve assembly 614 may result in the downward displacement of reservoir valve actuator 614A and the closing of reservoir valve 614B, resulting in the effective isolation of reservoir 118. Further, pump assembly 106 may include pump plunger 106A and pump chamber 106B and the activation of pump assembly 106 may result in pump plunger 106A being displaced in a downward fashion into pump chamber 106B and the displacement of the infusible fluid (in the direction of arrow 616).

Volume sensor valve assembly 612 may include volume sensor valve actuator 612A and volume sensor valve 612B. Referring also to FIG. 22B, volume sensor valve actuator 612A may be closed via a spring assembly that provides mechanical force to seal volume sensor valve 612B. However, when pump assembly 106 is activated, if the displaced infusible fluid is of sufficient pressure to overcome the mechanical sealing force of volume sensor valve assembly 612, the displacement of the infusible fluid occurs in the direction of arrow 618. This may result in the filling of volume sensor chamber 620 included within volume sensor assembly 148. Through the use of speaker assembly 622,

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port assembly 624, reference microphone 626, spring diaphragm 628, invariable volume microphone 630, volume sensor assembly 148 may determine the volume of infusible fluid included within volume sensor chamber 620.

Referring also to FIG. 22C, once the volume of infusible fluid included within volume sensor chamber 620 is calculated, shape memory actuator 632 may be energized, resulting in the activation of measurement valve assembly 610, which may include measurement valve actuator 610A and measurement valve 610B. Once activated and due to the mechanical energy asserted on the infusible fluid within volume sensor chamber 620 by spring diaphragm 628, the infusible fluid within volume sensor chamber 620 may be displaced (in the direction of arrow 634) through disposable cannula 138 and into the body of the user.

Referring also to FIG. 23, there is shown an exploded view of infusion pump assembly 500. Shape memory actuator 632 may be anchored (on a first end) to shape memory actuator anchor 636. Additionally, the other end of shape memory actuator 632 may be used to provide mechanical energy to valve assembly 638, which may activate measurement valve assembly 610. Volume sensor assembly spring retainer 642 may properly position volume sensor assembly 148 with respect to the various other components of infusion pump assembly 500. Valve assembly 638 may be used in conjunction with shape memory actuator 112 to activate pump plunger 106A. Measurement valve 610B, volume sensor valve 612B and/or reservoir valve 614B may be self-contained valves that are configured to allow for installation during assembly of infusion pump assembly 500 by pressing the valves upward into the lower surface of main body portion 508.

Referring also to FIG. 24 & FIGS. 25A-25D, there is shown a more-detailed view of pump assembly 106. Pump actuator assembly 644 may include pump actuator support structure 646, bias spring 648, and lever assembly 650.

Referring also to FIGS. 26A-26B & FIGS. 27A-27B, there is shown a more-detailed view of measurement valve assembly 610. As discussed above, valve assembly 638 may activate measurement valve assembly 610.

Referring also to FIGS. 28A-28D, infusion pump assembly 500 may include measurement valve assembly 610. As discussed above, valve assembly 638 may be activated via shape memory actuator 632 and actuator assembly 640. Accordingly, to infuse the quantity of infusible fluid stored within volume sensor chamber 620, shape memory actuator 632 may need to activate valve assembly 638 for a considerable period of time (e.g. one minute or more). As this would consume a considerable amount of power from battery 606, measurement valve assembly 610 may allow for the temporary activation of valve assembly 638, at which point measurement valve latch 656 may prevent valve assembly 638 from returning to its non-activated position. Shape memory actuator 652 may be anchored on a first end using electrical contact 654. The other end of shape memory actuator 652 may be connected to a valve latch 656. When shape memory actuator 652 is activated, shape memory actuator 652 may pull valve latch 656 forward and release valve assembly 638. As such, measurement valve assembly 610 may be activated via shape memory actuator 632. Once measurement valve assembly 610 has been activated, valve latch 656 may automatically latch valve assembly 638 in the activated position. Actuating shape memory actuator 652 may pull valve latch 656 forward and release valve assembly 638. Assuming shape memory actuator 632 is no longer activated, measurement valve assembly 610 may move to a de-activated state once valve latch 656 has released valve

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assembly 638. Accordingly, through the use of measurement valve assembly 610, shape memory actuator 632 does not need to be activated during the entire time that it takes to infuse the quantity of infusible fluid stored within volume sensor chamber 620.

As discussed above, the above-described infusion pump assemblies (e.g., infusion pumps assemblies 100, 100', 400, 500) may include an external infusion set 134 configured to deliver the infusible fluid to a user. External infusion set 134 may include a cannula assembly 136, which may include a needle or a disposable cannula 138, and tubing assembly 140. Tubing assembly 140 may be in fluid communication with reservoir 118, for example, by way of the fluid path, and with cannula assembly 138 for example, either directly or by way of a cannula interface 142.

Referring also to FIG. 29, there is shown an alternative embodiment infusion pump assembly 700 that is configured to store a portion of tubing assembly 140. Specifically, infusion pump assembly 700 may include peripheral tubing storage assembly 702 that is configured to allow the user to wind a portion of tubing assembly 140 about the periphery of infusion pump assembly 700 (in a manner similar to that of a yoyo). Peripheral tubing storage assembly 702 may be positioned about the periphery of infusion pump assembly 700. Peripheral tubing storage assembly 702 may be configured as an open trough into which a portion of tubing assembly 140 may be wound. Alternatively, peripheral tubing storage assembly 702 may include one or more divider portions 704, 706 that form a plurality of narrower troughs that may be sized to generate an interference fit between the walls of the narrower trough and the exterior surface of the portion of tubing 140. When peripheral tubing storage assembly 705 includes plurality of divider portions 704, 706, the resulting narrower troughs may be wound in a spiral fashion about the periphery of infusion pump assembly 700 (in a manner similar to the thread of a screw).

Referring also to FIGS. 30-31, there is shown an alternative embodiment infusion pump assembly 750 that is configured to store a portion of tubing assembly 140. Specifically, infusion pump assembly 750 may include peripheral tubing storage assembly 752 that is configured to allow the user to wind a portion of tubing assembly 140 about the periphery of infusion pump assembly 750 (again, in a manner similar to that of a yoyo). Peripheral tubing storage assembly 752 may be positioned about the periphery of infusion pump assembly 750. Peripheral tubing storage assembly 752 may be configured as an open trough into which a portion of tubing assembly 140 is wound. Alternatively, peripheral tubing storage assembly 752 may include one or more divider portions 754, 756 that form a plurality of narrower troughs that may be sized to generate an interference fit between the walls of the narrower trough and the exterior surface of the portion of tubing 140. When peripheral tubing storage assembly 752 includes plurality of divider portions 754, 756, the resulting narrower trough may be wound in a spiral fashion about the periphery of infusion pump assembly 750 (again, in a manner similar to the thread of a screw).

Infusion pump assembly 750 may include tubing retainer assembly 758. Tubing retainer assembly 758 may be configured to releasably secure tubing assembly 140 so as to prevent tubing assembly 140 from unraveling from around infusion pump assembly 750. In one embodiment of tubing retainer assembly 758, tubing retainer assembly 758 may include downward facing pin assembly 760 positioned above upward facing pin assembly 762. The combination of pin assemblies 760, 762 may define a "pinch point" through

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which tubing assembly 140 may be pushed. Accordingly, the user may wrap tubing assembly 140 around the periphery of infusion pump assembly 750, wherein each loop of tubing assembly 140 is secured within peripheral tubing storage assembly 752 via tubing retainer assembly 758. In the event that the user wishes to lengthen the unsecured portion of tubing assembly 140, the user may release one loop of tubing assembly 140 from tubing retainer assembly 758. Conversely, in the event that the user wishes to shorten the unsecured portion of tubing assembly 140, the user may secure one additional loop of tubing assembly 140 within tubing retainer assembly 758.

Referring also to FIGS. 32-33, there is shown an exemplary embodiment of infusion pump assembly 800. As with infusion pump assemblies 100, 100', 400, and 500, infusion pump assembly 800 may include reusable housing assembly 802 and disposable housing assembly 804.

With reference also to FIGS. 34A-34B, in a fashion similar to infusion pump assembly 100, reusable housing assembly 802 may be configured to releasably engage disposable housing assembly 804. Such releasable engagement may be effectuated by a screw-on, twist-lock, or compression fit configuration, for example. Infusion pump assembly 800 may include locking ring assembly 806. For example, reusable housing assembly 802 may be properly positioned relative to disposable housing assembly, and locking ring assembly 806 may be rotated to releasably engage reusable housing assembly 802 and disposable housing assembly 804.

Locking ring assembly 806 may include nub 808 that may facilitate rotation of locking ring assembly 806. Additionally, the position of nub 808, e.g., relative to tab 810 of disposable housing assembly 804, may provide verification that reusable housing assembly 802 is fully engaged with disposable housing assembly 804. For example, as shown in FIG. 34A, when reusable housing assembly 802 is properly aligned with disposable housing assembly 804, nub 808 may be aligned in a first position relative to tab 810. Upon achieving a fully engaged condition, by rotation locking ring assembly 806, nub 808 may be aligned in a second position relative to tab 810, as shown in FIG. 34B.

Referring also to FIGS. 35A-35C and FIGS. 36-38A, in a fashion similar to reusable housing assembly 102, reusable housing assembly 802 may include mechanical control assembly 812 (e.g., which may include valve assembly 814, shown in FIG. 36, including one or more valves and one or more pumps for pumping and controlling the flow of the infusible fluid). Reusable housing assembly 802 may also include an electrical control assembly 816 that may be configured to provide control signals to the mechanical control assembly 812 to effectuate the delivery of an infusible fluid to the user. Valve assembly 814 may be configured to control the flow of the infusible fluid through a fluid path and the pump assembly may be configured to pump the infusible fluid from the fluid path to the user.

Mechanical control assembly 812 and electrical control assembly 816 may be contained within a housing defined by base plate 818, body 820. In some embodiments one or more of base plate 818 and body 820 may provide electromagnetic shielding. In such an embodiment, the electromagnetic shielding may prevent and/or reduce electromagnetic interference received by electrical control assembly 816 and/or created by electrical control assembly 816. Additionally/alternatively, EMI shield 822 may be included, as shown in FIG. 36 and FIG. 37. EMI shield 822 may provide shielding against generated and/or received electromagnetic interference.

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Reusable housing assembly **802** may include a switch assembly that may be configured to receive user commands (e.g., for bolus delivery, pairing with a remote control assembly, or the like). The switch assembly may include button **824** that may be disposed in opening **826** of body **820**. As shown, e.g., in FIG. **35B**, locking ring assembly **806** may include radial slot **828** that may be configured to allow locking ring assembly **806** to be rotated relative to body **820** while still providing facile access to button **824**.

Referring also to FIGS. **39A-39C**, electrical control assembly **816** may include printed circuit board **830** as well as battery **832**. Printed circuit board **830** may include the various control electronics for monitoring and controlling the amount of infusible fluid that has been and/or is being pumped. For example, electrical control assembly **816** may measure the amount of infusible fluid that has just been dispensed, and determine, based upon the dosage required by the user, whether enough infusible fluid has been dispensed. If not enough infusible fluid has been dispensed, electrical control assembly **816** may determine that more infusible fluid should be pumped. Electrical control assembly **816** may provide the appropriate signal to mechanical control assembly **812** so that any additional necessary dosage may be pumped or electrical control assembly **816** may provide the appropriate signal to mechanical control assembly **812** so that the additional dosage may be dispensed with the next dosage. Alternatively, if too much infusible fluid has been dispensed, electrical control assembly **816** may provide the appropriate signal to mechanical control assembly **812** so that less infusible fluid may be dispensed in the next dosage. Electrical control assembly **816** may include one or more microprocessors. In an exemplary embodiment, electrical control assembly **816** may include three microprocessors. One processor (e.g., which may include, but is not limited to a CC2510 microcontroller/RF transceiver, available from Chipcon AS, of Oslo, Norway) may be dedicated to radio communication, e.g., for communicating with a remote control assembly. Two additional microprocessors (example of which may include, but is not limited to an MSP430 microcontroller, available from Texas Instruments Inc. of Dallas, Texas) may be dedicated to issuing and carrying out commands (e.g., to dispense a dosage of infusible fluid, process feedback signals from a volume measurement device, and the like).

As shown in FIG. **35C**, base plate **818** may provide access to electrical contacts **834**, e.g., which may be electrically coupled to electrical control assembly **816** for recharging battery **832**. Base plate **818** may include one or more features (e.g., openings **836**, **838**) which may be configured to facilitate proper alignment with disposable housing assembly **804** by way of cooperating features (e.g., tabs) of disposable housing assembly **804**. Additionally, as shown in FIGS. **40A-40C**, **41A-41B**, and **42A-42C**, base plate **818** may include various features for mounting valve assembly **814** and electrical control assembly **816**, as well as providing access to disposable housing assembly **804** by valve assembly **814**.

Locking ring assembly **806** may include grip inserts **840**, **842**, e.g., which may include an elastomeric or textured material that may facilitate gripping and twisting locking ring assembly **806**, e.g., for engaging/disengaging reusable housing assembly **802** and disposable housing assembly **804**. Additionally, locking ring assembly **806** may include a sensing component (e.g., magnet **844**) that may interact with a component of reusable housing assembly **802** (e.g., a Hall Effect sensor), e.g., to provide an indication of the nature of a mating component (e.g., which in some embodiments may

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include, but is not limited to, one or more of disposable housing assembly **804**, a charging station, or a filling station) and/or of whether reusable housing assembly **802** is properly engaged with the mating component. In the exemplary embodiment, a Hall Effect sensor (not shown) may be located on the pump printed circuit board. The Hall Effect sensor may detect when the locking ring has been rotated to a closed position. Thus, the Hall Effect sensor together with magnet **844** may provide a system for determining whether the locking ring has been rotated to a closed position.

The sensing component (magnet) **844** together with the reusable housing assembly components, i.e., in the exemplary embodiment, the Hall Effect sensor, may work to provide for a determination of whether the reusable housing assembly is properly attached to the intended component or device. Locking ring assembly **806** may not turn without being attached to a component, i.e., disposable housing assembly **804**, a dust cover or a charger. Thus, the sensing component together with the reusable housing assembly component may function to provide many advantageous safety features to the infusion pump system. These features may include, but are not limited to, one or more of the following. Where the system does not detect being attached to a disposable assembly, a dust cover or a charger, the system may notify, alert or alarm the user as the reusable portion, e.g., the valves and pumping components, may be vulnerable to contamination or destruction which may compromise the integrity of the reusable assembly. Thus, the system may provide for an integrity alarm to alert the user of potential reusable integrity threats. Also, where the system senses the reusable assembly is attached to a dust cover, the system may power off or reduce power to conserve power. This may provide for more efficient use of power where the reusable assembly is not connecting to a component in which it needs to interact.

Reusable housing assembly **802** may attach to a number of different components, including but not limited to, a disposable housing assembly, a dust cover or a battery charger/battery charging station. In each case, the Hall Effect sensor may detect that the locking ring is in the closed position, and therefore, that reusable housing assembly **802** is releasably engaged to a disposable housing assembly, a dust cover, or a battery charger/battery charging station (or, another component). The infusion pump system may determine the component to which it is attached by using the AVS system described in more detail below or by an electronic contact. Referring now also to FIGS. **38B-38D**, one embodiment of a dust cover (e.g., dust cover **839**) is shown. In the exemplary embodiment, dust cover **839** may include features **841**, **843**, **845**, **847** such that the locking ring of reusable housing assembly **802** may releasably engage dust cover **839**. In addition, dust cover **839** may further include recess region **849** for accommodating the valving and pumping features of reusable housing assembly **804**. For example, with respect to the dust cover, the AVS system may determine that a dust cover, and not a disposable housing assembly, is connected to the reusable housing assembly. The AVS system may distinguish using a look-up table or other comparative data and comparing the measurement data with characteristic dust cover or empty disposable housing assembly data. With respect to the battery charger, the battery charger, in the exemplary embodiments, may include electric contacts. When the reusable housing assembly is attached to the battery charger, the infusion pump assembly electronic system may sense that the contacts have been made, and will thus indicate that the reusable housing assembly is attached to a battery charger.

Referring also to FIGS. 43A-45B and FIGS. 44A-44C an embodiment of valve assembly **814**, which may include one or more valves and one or more pumps, is shown. As with infusion pump assemblies **100**, **100'**, **400**, and **500**, valve assembly **814** may generally include reservoir valve **850**, plunger pump **852**, volume sensor valve **854**, and measurement valve **856**. Similar to the previous description, reservoir valve **850** and plunger pump **852** may be actuated by shape memory actuator **858**, which may be anchored (on a first end) to shape memory actuator anchor **860**. Additionally, measurement valve **856** may be actuated, via valve actuator **862**, by shape memory actuator **864**, which may be anchored (on a first end) to shape memory actuator anchor **866**. In a similar manner as discussed above, measurement valve may be maintained in an open position via measurement valve latch assembly **868**. Measurement valve **856** may be released via actuation of shape memory actuator **870**, which may be anchored (on a first end) by shape memory actuator anchor **872**. In some embodiments, shape memory actuator anchor **860** may be potted onto the reusable housing assembly. Using this process during manufacture ensures shape memory length actuator **858** is installed and maintains the desired length and tension/strain.

Referring also to FIGS. 45A-45B and FIGS. 46A-46E, shape memory actuator **858** (e.g., which may include one or more shape memory wires) may actuate plunger pump **852** via actuator assembly **874**. Actuator assembly **874** may include bias spring **876** and lever assembly **878**. Actuator assembly **874** may actuate both plunger pump **852** and measurement valve **850**.

Referring also to FIGS. 47A-47B, measurement valve **856** may be actuated by shape memory actuator **864**, via valve actuator **862** and lever assembly **878**. Once actuated, measurement valve latch assembly **868** may maintain measurement valve **856** in an open position. Measurement valve latch assembly **868** actuated by shape memory actuator **870** to release measurement valve **856**, allowing it to return to a closed position.

Disposable housing assembly **804** may be configured for a single use or for use for a specified period of time, e.g., e.g., three days or any other amount of time. Disposable housing assembly **804** may be configured such that any of the component of infusion pump assembly **800** that come in contact with the infusible fluid may be disposed on and/or within disposable housing assembly **804**. As such, the risk of contaminating the infusible fluid may be reduced.

Referring also to FIG. 48 and FIGS. 49A-49C, disposable housing assembly **804** may include base portion **900**, membrane assembly **902**, and top portion **904**. Base portion **900** may include recess **906** that together with membrane assembly **902** defines reservoir **908** for receiving an infusible fluid (not shown), e.g., insulin. Referring also to FIGS. 50A-50C, recess **906** may be at least partially formed by and integral with base portion **900**. Membrane assembly **902** may be sealingly engaged with base portion **900**, e.g., by being compressively pinched between base portion **900** and top portion **904**. Top portion **904** may be attached to base portion **900** by conventional means, such as gluing, heat sealing, ultrasonic welding, and compression fitting. Additionally/alternatively, membrane assembly **902** may be attached to base portion **900**, e.g., via gluing, ultrasonic welding, heat sealing, and the like, to provide a seal between membrane assembly **902** and base portion **900**.

Still referring to FIGS. 48 and 50A, recess **906**, in the exemplary embodiment, includes raised portion **901** which includes area **903** about fluid openings **905** leading to the fluid line. Raised portion **901**, in the exemplary embodi-

ment, extends about the perimeter of recess **906**. However, in other embodiments, raised portion **901** may not extend the entire perimeter, but may be partially about the perimeter. Area **903** about fluid openings **905** may be shaped as shown in the exemplary embodiment, including an angled portion, which in some embodiments, includes 45 degree angles, however in other embodiments, the angle may be greater or lesser. In some embodiments, the pump may not generate a sufficient enough vacuum to collapse the reservoir so as to eliminate the entire volume of fluid that may be stored in the reservoir. Raised portion **901** may act to minimize wasted fluid.

Fluid openings **905**, which, in the exemplary embodiment, may include three openings, however, in other embodiments may include more openings or fewer openings, may be surrounded by area **903** of the raised portion. In the exemplary embodiment, fluid openings **905** may be narrow in the center, thus creating a surface tension that may prevent the air from being drawn into the opening. In the exemplary embodiment, this area may be designed to encourage any air that is present in the reservoir to be drawn above one of fluid openings **905** rather than be pulled through fluid openings **905** and into the fluid line. Additionally, because there may be more than one fluid opening **905**, where an air bubble is caught above one, the air may not prevent fluid from flowing through the other two openings.

Referring also to FIGS. 51A-51C, disposable housing assembly **804** may also include fluid pathway cover **910**. Fluid pathway cover **910** may be received in cavity **912** formed on/within base portion **900**. Fluid pathway cover **910** may, in some embodiments, include at least a portion of one or more channels (e.g., channel **914**). The channels included in fluid pathway cover **910** may fluidly couple one or more volcano valve features (e.g. volcano valves **916**) included on base portion **900**. Volcano valves **916** may include a protrusion having an opening extending through it. Additionally, fluid pathway cover **910** and base portion **900** may each define a portion of recess (e.g., recess portions **918**, **920** included in base portion **900** and fluid pathway cover **910** respectively) for fluidly coupling to an infusion set (e.g., including cannula **922**). Cannula **922** may be coupled to disposable housing assembly **804** by conventional means (e.g., gluing, heat sealing, compression fit, or the like). The fluid pathways defined by fluid pathway cover **910** and the volcano valves (e.g., volcano valves **916**) of base portion **900** may define a fluid pathway between reservoir **908** and cannula **922** for the delivery of the infusible fluid to the user via the infusion set. However, in some embodiments, fluid path cover **910** may include at least a portion of the fluid path, and in some embodiments, fluid path cover **910** may not include at least a portion of the fluid path. In the exemplary embodiment, fluid pathway cover **910** may be laser welded to base portion **900**. However, in other embodiments, fluid pathway cover **910** may also be connected to base portion **900** by conventional means (e.g., gluing, heat sealing, ultrasonic welding, compression fit, or the like) to achieve a generally fluid tight seal between fluid pathway cover **910** and base portion **900**.

With reference also to FIGS. 54A-54C, disposable housing assembly **804** may further include valve membrane cover **924**. Valve membrane cover **924** may be at least partially disposed over the volcano valves (e.g., volcano valve **916**) and pumping recess **926** included on/within base portion **900**. Valve membrane cover **924** may include a flexible material, e.g., which may be selectively engaged against the volcano valves by reservoir valve **850**, volume sensor valve **854**, and measurement valve **856** of reusable

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housing assembly 802, e.g., for controlling the flow of the infusible fluid. Additionally, valve membrane cover 924 may be resiliently deformed into pumping recess 926 by plunger pump 852 to effectuate pumping of the infusible fluid. Valve membrane cover 924 may be engaged between base portion 900 and top portion 904 of disposable housing assembly 804 to form seal 928 between valve membrane cover 924 and base portion 900. For example, in the exemplary embodiment, valve membrane cover 924 may be overmolded onto base portion 900. In other embodiment, valve membrane cover 924 may be compressively pinched between base portion 900 and top portion 904 to form seal 928. Additionally/alternatively, valve membrane insert may be connected to one or more of base portion 900 and top portion 904, e.g., by gluing, heat sealing, or the like.

Referring also to FIGS. 53A-C, top portion 904 may include alignment tabs 930, 932 that may be configured to be at least partially received in openings 836, 838 of base plate 818 of reusable housing assembly 802 to ensure proper alignment between reusable housing assembly 802 and disposable housing assembly 804. Additionally, top portion 904 may include one or more radial tabs 934, 936, 938, 940 configured to be engaged by cooperating tabs 942, 944, 946, 948 of locking ring assembly 806. The one or more radial tabs (e.g., radial tab 940) may include stops (e.g., alignment tab stop 950, which may be used for welding, it's the tab that fits in the recess to locate and ultrasonically weld), e.g., which may prevent further rotation of locking ring assembly 806 once reusable housing assembly 802 and disposable housing assembly 804 are fully engaged.

As discussed above, valve membrane insert 924 may allow for pumping and flow of the infusible fluid by reservoir valve 850, plunger pump 852, volume sensor valve 854, and measurement valve 856. Accordingly, top portion 904 may include one or more openings (e.g., openings 952, 954, 956) that may expose at least a portion of valve membrane insert 924 for actuation by reservoir valve 850, plunger pump 852, volume sensor valve 854, and measurement valve 856. Additionally, top portion 904 may include one or more openings 958, 960, 962 which may be configured to allow the fill volume to be controlled during filling of reservoir 908, as will be discussed in greater detail below. Reservoir assembly 902 may include ribs 964, 966, 968 (e.g., as shown in FIG. 52A), which may be at least partially received in respective openings 958, 960, 962. As will be described in greater detail below, a force may be applied to one or more of ribs 964, 966, 968 to, at least temporarily, reduce the volume of reservoir 908.

In some embodiments, it may be desirable to provide a seal between reusable housing assembly 802 and disposable housing assembly 804. Accordingly, disposable housing assembly 804 may include sealing assembly 970. Sealing assembly 970 may include, for example, an elastomeric member that may provide a compressible rubber or plastic layer between reusable housing assembly 802 and disposable housing assembly 804 when engaged, thus preventing inadvertent disengagement and penetration by outside fluids. For example, sealing assembly 970 may be a watertight seal assembly and, thus, enable a user to wear infusion pump assembly 800 while swimming, bathing or exercising.

In a fashion similar to, e.g., disposable housing assembly 114, disposable housing assembly 802 may, in some embodiments, be configured to have reservoir 908 filled a plurality of times. However, in some embodiments, disposable housing assembly 114 may be configured such that reservoir 908 may not be refilled. Referring also to FIGS. 57-64, fill adapter 1000 may be configured to be coupled to

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disposable housing assembly 804 for refilling reservoir 908 using a syringe (not shown). Fill adapter 1000 may include locking tabs 1002, 1004, 1006, 1008 that may be configured to engage radial tabs 934, 936, 938, 940 of disposable housing assembly 804 in a manner generally similar to tabs 942, 944, 946, 948 of locking ring assembly 806. Accordingly, fill adapter 1000 may be releasably engaged with disposable housing assembly 804 by aligning fill adapter 1000 with disposable housing assembly 804 and rotating fill adapter 1000 and disposable housing assembly 804 relative to one another to releasably engage locking tabs 1002, 1004, 1006, 1008 with radial tabs 934, 936, 938, 940.

Fill adapter 1000 may further include filling aid 1010, which may include guide passage 1012, e.g., which may be configured to guide a needle of a syringe (not shown) to a septum of disposable housing assembly 804 to allow reservoir 908 of disposable housing assembly 804 to be filled by the syringe. In some embodiments, guide passage 1012 may be an angled bevel or other gradual angled bevel to further guide a syringe to a septum. Fill adapter 1000 may facilitate filling reservoir 908 by providing a relatively large insertion area, e.g., at the distal opening of guide passage 1012. Guide passage 1012 may generally taper to a smaller proximal opening that may be properly aligned with the septum of disposable housing assembly 804, when fill adapter 1000 is engaged with disposable housing assembly 804. Accordingly, fill adapter 1000 may reduce the dexterity and aim necessary to properly insert a needle through the septum of disposable housing assembly 804 for the purpose of filling reservoir 908.

As discussed above, disposable housing assembly 804 may be configured to facilitate controlling the quantity of infusible fluid delivered to reservoir 908 during filling. For example, membrane assembly 902 of disposable housing assembly 804 may include ribs 964, 966, 968 that may be depressed and at least partially displaced into reservoir 908, thereby reducing the volume of reservoir 908. Accordingly, when infusible fluid is delivered to reservoir 908, the volume of fluid that may be accommodated by reservoir 908 may be correspondingly reduced. Ribs 964, 966, 968 may be accessible via openings 958, 960, 962 in top portion 904 of disposable housing assembly 804.

Fill adapter 1000 may include one or more button assemblies (e.g., button assemblies 1014, 1016, 1018) corresponding to ribs 964, 966, 968. That is, when fill adapter 1000 is releasably engaged with disposable housing assembly 804, buttons 1014, 1016, 1018 may be aligned with ribs 964, 966, 968. Button assemblies 1014, 1016, 1018 may be, for example, cantilever members capable of being depressed. When fill adapter 1000 is releasably engaged with disposable housing assembly 804, one or more of button assemblies 1014, 1016, 1018 may be depressed, and may correspondingly displace a respective one of ribs 964, 966, 968 into reservoir 908, causing an attendant reduction in the volume of reservoir 908.

For example, assume for illustrative purposes that reservoir 908 has a maximum capacity of 3.00 mL. Further, assume that button assembly 1014 is configured to displace rib 964 into disposable housing assembly 804, resulting in a 0.5 mL reduction in the 3.00 mL capacity of disposable housing assembly 804. Further, assume that button assembly 1016 is configured to displace rib 966 into disposable housing assembly 804, also resulting in a 0.5 mL reduction in the 3.00 mL capacity of disposable housing assembly 804. Further, assume that button assembly 1018 is configured to displace slot assembly 968 into disposable housing assembly 804, also resulting in a 0.5 mL reduction in the 3.00 mL

capacity of disposable housing assembly **804**. Therefore, if the user wishes to fill reservoir **908** within disposable housing assembly **804** with 2.00 mL of infusible fluid, in some embodiments, the user may first fill the reservoir to the 3.00 mL capacity and then depresses button assemblies **1016** and **1014** (resulting in the displacement of rib **966** into disposable housing assembly **804**), effectively reducing the 3.00 mL capacity of reservoir **908** within disposable housing assembly **804** to 2.00 mL. In some embodiments, the user may first depress a respective number of button assemblies, effectively reducing the capacity of reservoir **908**, and then fill reservoir **908**. Although a particular number of button assemblies are shown, representing the exemplary embodiment, in other embodiments, the number of button assemblies may vary from a minimum of 1 to as many as is desired. Additionally, although for descriptive purposes, and in the exemplary embodiment, each button assembly may displace 0.5 mL, in other embodiments, the volume of displacement per button may vary. Additionally, the reservoir may be, in various embodiments, include a larger or smaller volume than described in the exemplary embodiment.

According to the above-described configuration, the button assemblies (e.g., button assemblies **1014**, **1016**, **1018**) may employed, at least in part, to control the fill volume of reservoir **908**. By not depressing any of the button assemblies, the greatest fill volume of reservoir **908** may be achieved. Depressing one button assembly (e.g., button assembly **1014**) may allow the second greatest fill volume to be achieved. Depressing two button assemblies (e.g., button assemblies **1014**, **1016**) may achieve the third greatest fill volume. Depressing all three button assemblies (e.g., button assemblies **1014**, **1016**, **1018**) may allow the smallest fill volume to be achieved.

Further, in an embodiment button assemblies **1014**, **1016**, **1018** may be utilized, at least in part, to facilitate filling of reservoir **908**. For example, once a filling needle (e.g., which may be fluidly coupled to a vial of infusible fluid) has been inserted into reservoir **908**, button assemblies **1014**, **1016**, **1018** may be depressed to pump at least a portion of any air that may be contained within reservoir into the vial of infusible fluid. Button assemblies **1014**, **1016**, **1018** may subsequently be released to allow infusible fluid to flow from the vial into reservoir **908**. Once reservoir **908** has been filled with the infusible fluid, one or more button assemblies (e.g., one or more of button assemblies **1014**, **1016**, **1018**) may be depressed, thereby squeezing at least a portion of the infusible fluid from reservoir **908** (e.g., via a needle used to fill reservoir **908** and back into the vial of infusible fluid). As discussed above, the volume of infusible fluid contained within reservoir **908** may be controlled, e.g., depending upon how many button assemblies are depressed (e.g., which may control how much infusible fluid is squeezed back into the vial of infusible fluid).

With particular reference to FIGS. **62-64**, filling aid **1010** may be pivotally coupled to fill adapter base plate **1020**. For example, filling aid **1010** may include pivot members **1022**, **1024** that may be configured to be received in pivot supports **1026**, **1028**, thereby allowing filling aid to pivot between an open position (e.g., as shown in FIGS. **57-61**) and a closed position (e.g., as shown in FIGS. **63-64**). The closed position may be suitable, e.g., for packaging fill adapter **1000**, storage of fill adapter **1000**, or the like. In order to ensure that filling aid **1010** is properly oriented for filling reservoir **908**, fill adapter **1000** may include support member **1030**. To prop-

erly orient filling aid **1010**, a user may pivot filling aid **1010** to a fully open position, wherein filling aid **1010** may contact support member **1030**.

According to an alternative embodiment, and referring also to FIG. **65**, fill adapter **1050** may be configured to releasably engage disposable housing assembly **804** via a plurality of locking tabs (e.g., locking tabs **1052**, **1054**). Additionally, fill adapter **1050** may include a plurality of button assemblies (e.g., button assemblies **1056**, **1058**, **1060**) that may interact with ribs **964**, **966**, **968** of disposable housing assembly **804** to adjust a fill volume of reservoir **908**. Fill adapter **1050** may further include filling aid **1062**, having guide passage **1064** configured to align a needle of a syringe with the septum of disposable housing **804**, e.g., for accessing reservoir **908** for the purpose of filling reservoir **908** with an infusible fluid. Filling aid **1062** may be connected to base plate **1066**, e.g., as an integral component therewith, by gluing, heat sealing, compression fit, or the like.

Referring also to FIGS. **66-74**, vial fill adapter **1100** may be configured to facilitate filling reservoir **908** of disposable housing assembly **804** directly from a vial. Similar to fill adapter **1000**, vial fill adapter **1100** may include locking tabs **1102**, **1104**, **1106**, **1108** that may be configured to engage radial tabs **934**, **936**, **938**, **940** of disposable housing assembly in a manner generally similar to tabs **942**, **944**, **946**, **948** of locking ring assembly **806**. Accordingly, vial fill adapter **1100** may be releasably engaged with disposable housing assembly **804** by aligning vial fill adapter **1100** with disposable housing assembly **804** and rotating vial fill adapter **1100** and disposable housing assembly **804** relative to one another to releasably engage locking tabs **1102**, **1104**, **1106**, **1108** with radial tabs **934**, **936**, **938**, **940**.

As discussed above, disposable housing assembly **804** may be configured to facilitate controlling the quantity of infusible fluid delivered to reservoir **908** during filling. For example, membrane assembly **902** of disposable housing assembly **804** may include ribs **964**, **966**, **968** that may be depressed and at least partially displaced into reservoir **908**, thereby reducing the volume of reservoir **908**. Accordingly, when infusible fluid is delivered to reservoir **908**, the volume of fluid that may be accommodated by reservoir **908** may be correspondingly reduced. Ribs **964**, **966**, **968** may be accessible via openings **958**, **960**, **962** in top portion **904** of disposable housing assembly **804**.

Vial fill adapter **1100** may include one or more button assemblies (e.g., button assemblies **1110**, **1112**, **1114**) corresponding to ribs **964**, **966**, **968** (e.g., shown in FIG. **52A**). That is, when vial fill adapter **1100** is releasably engaged with disposable housing assembly **804**, buttons **1110**, **1112**, **1114** may be aligned with ribs **964**, **966**, **968**. Button assemblies **1110**, **1112**, **1114** may be, for example, cantilever members capable of being depressed. When vial fill adapter **1100** is releasably engaged with disposable housing assembly **804**, one or more of button assemblies **1110**, **1112**, **1114** may be depressed, and may correspondingly displace a respective one of ribs **964**, **966**, **968** into reservoir **908**, thereby reducing the volume of reservoir **908**.

For example, assume for illustrative purposes that reservoir **908** has a maximum capacity of 3.00 mL. Further, assume that button assembly **1110** is configured to displace rib **964** into disposable housing assembly **804**, resulting in a 0.5 mL reduction in the 3.00 mL capacity of disposable housing assembly **804**. Further, assume that button assembly **1112** is configured to displace rib **966** into disposable housing assembly **804**, also resulting in a 0.5 mL reduction in the 3.00 mL capacity of disposable housing assembly **804**.

Further, assume that button assembly **1114** is configured to displace rib **968** into disposable housing assembly **804**, also resulting in a 0.50 mL reduction in the 3.00 mL capacity of disposable housing assembly **804**. Therefore, if the user wishes to fill reservoir **908** within disposable housing assembly **804** with 2.00 mL of infusible fluid, the user may depress button assemblies **1112** and **1114** (resulting in the displacement of ribs **966** and **968** into disposable housing assembly **804**), effectively reducing the 3.00 mL capacity of reservoir **908** within disposable housing assembly **804** to 2.0 mL.

Vial fill adapter **1100** may further include vial filling aid assembly **1116** that may be configured to fluidly couple a vial of infusible fluid to reservoir **908** of disposable housing assembly **804** via a septum. With particular reference to FIG. **71**, vial filling aid assembly may include double ended needle assembly **1118**. Double ended needle assembly **1118** may include first needle end **1120** configured to penetrate the septum of a vial (not shown) and second needle end **1122** configured to penetrate the septum of disposable housing assembly **804**. As such, the vial and reservoir **908** may be fluidly coupled allowing infusible fluid to be transferred from the vial to reservoir **908**. Double ended needle assembly **1118** may include vial engagement portion **1124** adjacent first end **1120**. Vial engagement arms **1124**, **1126** may be configured to releasably engage, e.g., a vial cap, to assist in maintaining the fluid connection between double ended needle assembly **1118** and the vial. Additionally, double ended needle assembly **1118** may include body **1128** that may be slidably received in opening **1130** of vial filling aid body **1132**. Vial filling aid body **1132** may include stabilizer arms **1134**, **1136**, e.g., which may be configured to stabilize the vial during filling of disposable housing assembly **804**. In one embodiment, the vial may be engaged with double ended needle assembly **1118** e.g., such that first end **1120** may penetrate the septum of the vial and the cap of the vial may be engaged by engagement arms **1124**, **1126**. Body **1128** may be slidably inserted into opening **1130** such that second end **1122** of double ended needle assembly **1118** may penetrate the septum of disposable body assembly **804**.

Similar to fill adapter **1000**, vial filling aid assembly **1116** may be configured to be pivotally coupled to vial fill adapter base plate **1138**. For example, vial filling aid **1116** may include pivot members **1140**, **1142** that may be configured to be received in pivot supports **1144**, **1146** (e.g., shown in FIG. **71**), thereby allowing vial filling aid **1116** to pivot between an open position (e.g., as shown in FIGS. **66-70**) and a closed position (e.g., as shown in FIGS. **72-74**). The closed position may be suitable, e.g., for packaging vial fill adapter **1100**, storage of vial fill adapter **1100**, or the like. In order to ensure that vial filling aid **1116** is properly oriented for filling reservoir **908**, vial fill adapter **1100** may include support member **1148**. To properly orient vial filling aid **1116**, a user may pivot vial filling aid **1116** to a fully open position, wherein vial filling aid **1116** may contact support member **1148**. Additionally, vial fill adapter base plate **1138** may include one or more locking features (e.g., locking tabs **1150**, **1152**) that may engage vial filling aid **1116**, and may maintain vial filling aid **1116** in the closed position. Vial fill adapter base plate **1138** may also include features (e.g., tabs **1154**, **1156**) that may be configured to assist in retaining double ended needle assembly **1118**, e.g., by preventing slidable separation of double ended needle assembly **1118** from vial filling aid body **1132**.

As shown in FIGS. **72-74**, filling aid assembly **1116** is in a closed position. In this configuration, support member **1148** may additionally function as a needle guard. When removing filling aid assembly **1116** from disposable housing

assembly **804**, support member **1148** may function to safely allow a user to squeeze the ends and rotate filling aid assembly **1116** for removal. As shown in FIG. **70**, in the open position, support member **1148** may function as a stop to maintain proper orientation.

Referring again to FIGS. **57-73**, the exemplary embodiments of the fill adapter include a grip feature (e.g., **1166** in FIG. **72**). Grip feature **1166** may provide a grip interface for removal of the fill adapter from disposable housing assembly **804**. Although shown in one configuration in these figures, in other embodiments, the configuration may vary. In still other embodiments, a grip feature may not be included.

According to one embodiment, fill adapter base plate **1020** and vial fill adapter base plate **1138** may be interchangeable components. Accordingly, a single base plate (e.g., either fill adapter base plate **1020** or vial fill adapter base plate **1138** may be used with either filling aid **1010** or vial filling aid **1116**. Accordingly, the number of distinct components that are required for both filling adapters may be reduced, and a user may have the ability to select the filling adapter that may be the most suitable for a given filling scenario.

The various embodiments of the fill adapters may provide many safely benefits, including but not limited to: providing a system for filling the reservoir without handling a needle; protecting the reservoir from unintentional contact with the needle, i.e., destruction of the integrity of the reservoir through unintentional puncture; designed to be ambidextrous; in some embodiments, may provide a system for maintaining air in the reservoir.

As discussed above, reusable housing assembly **802** may include battery **832**, e.g., which may include a rechargeable battery. Referring also to FIGS. **75-80**, battery charger **1200** may be configured to recharge battery **832**. Battery charger **1200** may include housing **1202** having top plate **1204**. Top plate **1204** may include one or more electrical contacts **1206**, generally, configured to be electrically coupled to electrical contacts **834** of reusable housing assembly **802**. Electrical contacts **1206** may include, but are not limited to, electrical contact pads, spring biased electrical contact members, or the like. Additionally, top plate **1204** may include alignment tabs **1208**, **1210**, which may be configured to mate with openings **836**, **838** in base plate **818** of reusable housing assembly **802** (e.g., as shown in FIG. **35C**). The cooperation of alignment tabs **1208**, **1210** and openings **836**, **838** may ensure that reusable housing assembly **802** is aligned with battery charger **1200** such that electrical contacts **1206** of battery charger **1200** may electrically couple with electrical contacts **834** of reusable housing assembly **802**.

With reference also to FIGS. **77** and **78**, battery charger **1200** may be configured to releasably engage reusable housing assembly **802**. For example, in a similar manner as disposable housing assembly **804**, battery charger **1200** may include one or more locking tabs (e.g., locking tabs **1212**, **1214** shown in FIG. **76**). The locking tabs (e.g., locking tabs **1212**, **1214**) may be engaged by tabs **942**, **944**, **946**, **948** of locking ring assembly **806**. As such, reusable housing assembly **802** may be aligned with battery charger **1200** (via alignment tabs **1208**, **1210**) with locking ring **806** in a first, unlocked position, as shown in FIG. **77**. Locking ring **806** may be rotated relative to battery charger **1200** in the direction of arrow **1216** to releasably engage tabs **942**, **944**, **946**, **948** of locking ring **806** with the locking tabs (e.g., locking tabs **1212**, **1214**) of battery charger **1200**, as shown in FIG. **78**.

In an embodiment, battery charger **1200** may include recessed region **1218**, e.g., which may, in the exemplary embodiments, provide clearance to accommodate reusable housing assembly **802** pumping and valving components. Referring also to FIGS. **79** & **80**, battery charger **1200** may provide electrical current to electrical contacts **1206** (and thereby to reusable housing assembly **802** via electrical contacts **834**) for recharging battery **832** of reusable housing assembly **802**. In some embodiments, when a signal indicative of a fully engaged reusable housing is not provided, current may not be provided to electrical contacts **1206**. According to such an embodiment, the risk associated with an electrical short circuit (e.g., resulting from foreign objects contacting electrical contacts **1206**) and damage to reusable housing assembly **802** (e.g., resulting from improper initial alignment between electrical contacts **1206** and electrical contacts **834**) may be reduced. Additionally, battery charger **1200** may not unnecessarily draw current when battery charger is not charging reusable housing assembly **802**.

Still referring to FIGS. **79** and **80**, battery charger **1200** may include a lower housing portion **1224** and top plate **1204**. Printed circuit board **1222** (e.g., which may include electrical contacts **1206**) may be disposed within a cavity included between top plate **1204** and lower housing portion **1224**.

Referring also to FIGS. **81-89B**, various embodiments of battery charger/docking stations are shown. FIGS. **81** and **82** depicts desktop charger **1250** including recess **1252** configured to mate with and recharge a reusable housing assembly (e.g., reusable housing assembly **802**). The reusable housing assembly may rest in recess **1252** and or may be releasably engaged in recess **1252**, in a similar manner as discussed above. Additionally, desktop charger **1250** may include recess **1254** configured to mate with a remote control assembly (e.g., remote control assembly **300**). Recess **1254** may include a USB plug **1256**, e.g., which may be configured to couple with the remote control assembly when the remote control assembly is disposed within recess **1254**. USB plug **1256** may allow for data transfer to/from the remote control assembly, as well as charging of remote control assembly. Desktop charger **1250** may also include USB port **1258** (e.g., which may include a mini-USB port), allowing desktop charger to receive power (e.g., for charging the reusable housing assembly and/or the remote control assembly). Additionally/alternatively USB port **1258** may be configured for data transfer to/from remote control assembly and/or reusable housing assembly, e.g., by connection to a computer (not shown).

Referring to FIGS. **83A-83B**, similar to the previous embodiment, desktop charger **1260** may include recess **1262** for mating with a reusable housing assembly (e.g., reusable housing assembly **1264**). Desktop charger may also include recess **1266** configured to receive a remote control assembly (e.g., remote control assembly **1268**). One or more of recess **1262**, **1266** may include electrical and/or data connections configured to charge and/or transfer data to/from reusable housing assembly **1262** and/or remote control assembly **1268**, respectively.

Referring to FIGS. **84A-84B**, another embodiment of a desktop charger is shown. Similar to desktop charger **1260**, desktop charger **1270** may include recesses (not shown) for respectively mating with reusable housing assembly **1272** and remote control assembly **1274**. As shown, desktop charger **1270** may hold reusable housing assembly **1272** and remote control assembly **1274** in a side-by-side configuration. Desktop charger **1270** may include various electrical and data connection configured to charge and/or transfer

data to/from reusable housing assembly **1272** and/or remote control assembly **1274**, as described in various embodiments above.

Referring to FIG. **85A-85D**, collapsible charger **1280** may include recess **1282** for receiving reusable housing assembly **1284** and remote control assembly **1286**. Collapsible charger **1280** may include various electrical and data connection configured to charge and/or transfer data to/from reusable housing assembly **1284** and/or remote control assembly **1286**, as described in various embodiments above. Additionally, as shown in FIGS. **85B-85D**, collapsible charger **1280** may include pivotable cover **1288**. Pivotable cover **1288** may be configured to pivot between an open position (e.g., as shown in FIG. **85B**), in which reusable housing assembly **1284** and remote control assembly **1286** may be docked in collapsible charger **1280**, and a closed position (e.g., as shown in FIG. **85D**), in which recess **1282** may be covered by pivotable cover **1288**. In the closed position, recess **1282**, as well as any electrical and/or data connections disposed therein, may be protected from damage.

Referring to FIG. **86**, wall charger **1290** may include recess **1292** configured to receive reusable housing assembly **1294**. Additionally, wall charger **1290** may include recess **1296** configured to receive remote control assembly **1298**. Reusable housing assembly **1294** and remote control assembly **1298** may be positioned in a stacked configuration, e.g., thereby providing a relatively slim profile. A rear portion of wall charger **1290** may include an electrical plug, configured to allow wall charger to be plugged into an electrical receptacle. As such, wall charger **1290**, while plugged into the electrical receptacle, may achieve a wall mounted configuration. Additionally, while plugged into the electrical receptacle, wall charger **1290** may be provided with power for charging reusable housing assembly **1294** and/or remote control assembly **1298**.

Referring to FIG. **87**, wall charger **1300** may include recess **1302** configured to receive remote control assembly **1304**. Additionally, wall charger may include a recess (not shown) configured to receive reusable housing assembly **1306**. Wall charger **1300** may be configured to position remote control assembly **1304** and reusable housing assembly **1306** in a back-to-back configuration, which may provide a relatively thin profile. Additionally, wall charger **1300** may include an electrical plug **1308** configured to be plugged into an electrical receptacle. Electrical plug **1308** may include a stowable configuration, in which electrical plug **1308** may be pivotable between a deployed position (e.g., as shown), and a stowed position. In the deployed position, electrical plug **1308** may be oriented to be plugged into an electrical receptacle. In the stowed position electrical plug **1308** may be disposed within recess **1310**, which may protect electrical plug **1308** from damage and/or from damaging other items.

Referring to FIG. **88**, charger **1320** may include recess **1322** configured to receive reusable housing assembly **1324**. Charger **1320** may additionally include a recess (not shown) configured to receive remote control assembly **1326**. Charger **1320** may additionally include cover **1328**. Cover **1328** may be configured to pivot between an open position (as shown) and a closed position. When cover **1328** is in the open position, reusable housing assembly **1324** and remote control assembly **1326** may be accessible (e.g., allowing a user to remove/install reusable housing assembly **1324** and/or remote control assembly **1326** from/into charger **1320**). When cover **1324** is in the closed position, cover **1328** and charger body **1330** may substantially enclose reusable housing assembly **1324** and/or remote control assembly **1326**.

and/or recess 1322 and the recess configured to receive remote control assembly 1326, thereby providing damage and/or tamper protection for reusable housing assembly 1324, remote control assembly 1326 and/or any electrical and/or data connection associated with charger 1320.

Referring to FIGS. 89A-89B, wall charger 1350 may include recess 1352 configured to receive remote control assembly 1354. Wall charger 1350 may also include recess 1356 configured to receive reusable housing assembly 1358. Wall charger 1350 may be configured to position remote control assembly 1354 and reusable housing assembly 1358 in a generally side-by-side configuration, thereby providing a relatively slim profile. Charger 1350 may additionally include electrical plug 1360, e.g., which may be configured to be plugged into an electrical receptacle. Electrical plug 1360 may include a stowable configuration, in which electrical plug 1360 may be pivotable between a deployed position (e.g., as shown), and a stowed position. In the deployed position, electrical plug 1360 may be oriented to be plugged into an electrical receptacle. In the stowed position electrical plug 1360 may be disposed within recess 1362, which may protect electrical plug 1308 from damage and/or from damaging other items.

Infusion pump therapy may include volume and time specifications. The amount of fluid dispensed together with the dispense timing may be two critical factors of infusion pump therapy. As discussed in detail below, the infusion pump apparatus and systems described herein may provide for a method of dispensing fluid together with a device, system and method for measuring the amount of fluid dispensed. However, in a circumstance where the calibration and precision of the measurement device calibration is critical, there may be advantages to determining any compromise in the precision of the measurement device as soon as possible. Thus, there are advantages to off-board verification of volume and pumping.

As discussed above, infusion pump assembly 100 may include volume sensor assembly 148 configured to monitor the amount of fluid infused by infusion pump assembly 100. Further and as discussed above, infusion pump assembly 100 may be configured so that the volume measurements produced by volume sensor assembly 148 may be used to control, through a feedback loop, the amount of infusible fluid that is infused into the user.

Referring also to FIGS. 90A-90C, there is shown one diagrammatic view and two cross-sectional views of volume sensor assembly 148. Referring also to FIGS. 91A-91I, there is shown various isometric and diagrammatic views of volume sensor assembly 148 (which is shown to include upper housing 1400). Referring also to FIGS. 92A-92I, there is shown various isometric and diagrammatic views of volume sensor assembly 148 (with upper housing 1400 removed), exposing speaker assembly 622, reference microphone 626, and printed circuit board assembly 830. Referring also to FIGS. 93A-93I, there is shown various isometric and diagrammatic views of volume sensor assembly 148 (with printed circuit board assembly 830 removed), exposing port assembly 624. Referring also to FIGS. 94A-94F, there is shown various isometric and diagrammatic cross-sectional views of volume sensor assembly 148 (with printed circuit board assembly 830 removed), exposing port assembly 624. Referring also to FIG. 95, there are shown an exploded view of volume sensor assembly 148, exposing upper housing 1400, speaker assembly 622, reference microphone 626, seal assembly 1404, lower housing 1402, port assembly 624, spring diaphragm 628, and retaining ring assembly 1406.

The following discussion concerns the design and operation of volume sensor assembly 148 (which is shown in a simplified form in FIG. 96). For the following discussion, the following nomenclature may be used:

Symbols	
P	Pressure
p	Pressure Perturbation
V	Volume
v	Volume Perturbation
γ	Specific Heat Ratio
R	Gas Constant
ρ	Density
Z	Impedance
f	Flow friction
A	Cross sectional Area
L	Length
ω	Frequency
ξ	Damping ratio
α	Volume Ratio
Subscripts	
0	Speaker Volume
1	Reference Volume
2	Variable Volume
k	Speaker
r	Resonant Port
z	Zero
p	Pole

Derivation of the Equations for Volume Sensor Assembly 148:

Modeling the Acoustic Volumes

The pressure and volume of an ideal adiabatic gas may be related by:

$$PV^\gamma = K \quad [\text{EQ \#1}]$$

where K is a constant defined by the initial conditions of the system.

EQ #1 may be written in terms of a mean pressure, P, and volume, V, and a small time-dependent perturbation on top of those pressures, p(t), v(t) as follows:

$$(P+p(t))(V+v(t))^\gamma = K \quad [\text{EQ \#2}]$$

Differentiating this equation may result in:

$$\dot{p}(t)(V+v(t))^\gamma + \gamma(V+v(t))^{\gamma-1}(P+p(t))\dot{v}(t) = 0 \quad [\text{EQ \#3}]$$

which may simplify to:

$$\dot{p}(t) + \gamma \frac{P+p(t)}{V+v(t)} \dot{v}(t) = 0 \quad [\text{EQ\#4}]$$

If the acoustic pressure levels are much less than the ambient pressure, the equation may be further simplified to:

$$\dot{p}(t) + \frac{\gamma P}{V} \dot{v}(t) = 0 \quad [\text{EQ\#5}]$$

How good is this assumption? Using the adiabatic relation it may be shown that:

$$\frac{P}{V} = \left(\frac{P+p(t)}{V+v(t)} \right) \left(\frac{P+p(t)}{P} \right)^{\frac{\gamma+1}{\gamma}} \quad [\text{EQ\#6}]$$

Accordingly, the error in the assumption would be:

$$\text{error} = 1 - \left(\frac{P + p(t)}{P} \right)^{\frac{\gamma+1}{\gamma}} \quad [\text{EQ\#7}]$$

A very loud acoustic signal (120 dB) may correspond to pressure sine wave with amplitude of roughly 20 Pascal. Assuming air at atmospheric conditions ($\gamma=1.4$, $P=101325$ Pa), the resulting error is 0.03%. The conversion from dB to Pa is as follows:

$$\lambda = 20 \log_{10} \left(\frac{p_{rms}}{p_{ref}} \right) \text{ or } p_{rms} = p_{ref} 10^{\frac{\lambda}{20}} \quad [\text{EQ\#8}]$$

where $p_{ref}=20 \cdot \mu\text{Pa}$.

Applying the ideal gas law, $P=\rho RT$, and substituting in for pressure may result in the following:

$$\dot{p}(t) + \frac{\gamma RT \rho}{V} \dot{v}(t) = 0 \quad [\text{EQ\#9}]$$

EQ #9 may be written in terms of the speed of sound, $a=\sqrt{\gamma RT}$ as follows:

$$\dot{p}(t) + \frac{\rho a^2}{V} \dot{v}(t) = 0 \quad [\text{EQ\#10}]$$

Acoustic impedance for a volume may be defined as follows:

$$Z_v = \frac{p(t)}{\dot{v}(t)} = - \frac{1}{\left(\frac{V}{\rho a^2} \right) s} \quad [\text{EQ\#11}]$$

Modeling the Acoustic Port

The acoustic port may be modeled assuming that all of the fluid in the port essentially moves as a rigid cylinder reciprocating in the axial direction. All of the fluid in the channel is assumed to travel at the same velocity, the channel is assumed to be of constant cross section, and the “end effects” resulting from the fluid entering and leaving the channel are neglected.

If we assume laminar flow friction of the form $\Delta p=f\dot{p}\dot{v}$, the friction force acting on the mass of fluid in the channel may be written as follows:

$$F=f\dot{p}A^2\dot{x} \quad [\text{EQ \#12}]$$

A second order differential equation may then be written for the dynamics of the fluid in the channel:

$$\mu LA\ddot{x}=\Delta pA-f\dot{p}A^2\dot{x} \quad [\text{EQ \#13}]$$

or, in terms of volume flow rate:

$$\dot{v} = - \frac{fA}{L} \dot{v} + \Delta p \frac{A}{\rho L} \quad [\text{EQ\#14}]$$

The acoustic impedance of the channel may then be written as follows:

$$Z_p = \frac{\Delta p}{\dot{v}} = \frac{\rho L}{A} \left(s + \frac{fA}{L} \right) \quad [\text{EQ\#15}]$$

System Transfer Functions

Using the volume and port dynamics defined above, volume sensor assembly **148** may be described by the following system of equations: (k=speaker, r=resonator)

$$\dot{p}_0 - \frac{\rho a^2}{V_0} \dot{v}_k = 0 \quad [\text{EQ\#16}]$$

$$\dot{p}_1 + \frac{\rho a^2}{V_1} (\dot{v}_k - \dot{v}_r) = 0 \quad [\text{EQ\#17}]$$

$$\dot{p}_2 + \frac{\rho a^2}{V_2} \dot{v}_r = 0 \quad [\text{EQ\#18}]$$

$$\dot{v}_r = - \frac{fA}{L} \dot{v}_r + \frac{A}{\rho L} (p_2 - p_1) \quad [\text{EQ\#19}]$$

One equation may be eliminated if p_0 is treated as the input substituting in

$$\dot{v}_k = \frac{V_0}{\rho a^2} \dot{p}_0.$$

$$\dot{p}_1 + \frac{V_0}{V_1} \dot{p}_0 - \frac{\rho a^2}{V_1} \dot{v}_r = 0 \quad [\text{EQ\#20}]$$

$$\dot{p}_2 + \frac{\rho a^2}{V_2} \dot{v}_r = 0 \quad [\text{EQ\#21}]$$

$$\ddot{v}_r = - \frac{fA}{L} \dot{v}_r + \frac{A}{\rho L} p_2 - \frac{A}{\rho L} p_1 \quad [\text{EQ\#22}]$$

Cross System Transfer Function

The relationship between the speaker volume and the variable volume may be referred to as the Cross System transfer function. This transfer function may be derived from the above equations and is as follows:

$$\frac{p_2}{p_0} = - \frac{V_0}{V_1} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \alpha\omega_n^2} \quad [\text{EQ\#23}]$$

where

$$\omega_n^2 = \frac{a^2 A}{L} \frac{1}{V_2}, \zeta = \frac{fA}{2L\omega_n} \text{ and } \alpha = \left(1 + \frac{V_2}{V_1} \right) \quad [\text{EQ\#24}]$$

Referring also to FIG. 97, a bode plot of EQ #23 is shown.

The difficulty of this relationship is that the complex poles depend on both the variable volume, V_2 , and the reference volume, V_1 . Any change in the mean position of the speaker may result in an error in the estimated volume.

Cross Port Transfer Function

The relationship between the two volumes on each side of the acoustic port may be referred to as the Cross Port transfer function. This relationship is as follows:

$$\frac{p_2}{p_1} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad [\text{EQ\#25}]$$

which is shown graphically in FIG. 98.

This relationship has the advantage that the poles are only dependent on the variable volume and not on the reference volume. It does, however, have the difficulty that the resonant peak is actually due to the inversion of the zero in the response of the reference volume pressure. Accordingly, the pressure measurement in the reference chamber will have a low amplitude in the vicinity of the resonance, potentially increasing the noise in the measurement.

Cross Speaker Transfer Function

The pressures may also be measured on each side of the speaker. This is referred to as the cross speaker transfer function:

$$\frac{p_1}{p_0} = -\frac{V_0}{V_1} \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \alpha\omega_n^2} \quad [\text{EQ\#26}]$$

which is shown graphically in FIG. 99.

This transfer function has a set of complex zeros in addition to the set of complex poles.

Looking at the limits of this transfer function: as

$$s \rightarrow 0, \frac{p_1}{p_0} \rightarrow -\frac{V_0}{V_1 + V_2};$$

and as

$$s \rightarrow \infty, \frac{p_1}{p_0} \rightarrow -\frac{V_0}{V_1}.$$

Resonance Q Factor and Peak Response

The quality of the resonance is the ratio of the energy stored to the power loss multiplied by the resonant frequency. For a pure second-order system, the quality factor may be expressed as a function of the damping ratio:

$$Q = \frac{1}{2\zeta} \quad [\text{EQ\#27}]$$

The ratio of the peak response to the low-frequency response may also be written as a function of the damping ratio:

$$|G|_{\omega_d} = \frac{1}{\zeta\sqrt{5-4\zeta^2}} \quad [\text{EQ\#28}]$$

This may occur at the damped natural frequency:

$$\omega_d = \omega_n \sqrt{1-\zeta^2} \quad [\text{EQ \#29}]$$

Volume Estimation

Volume Estimation Using Cross-Port Phase

The variable volume (i.e., within volume sensor chamber 620) may also be estimated using the cross-port phase. The transfer function for the pressure ratio across the resonant port may be as follows:

$$\frac{p_2}{p_1} = \frac{\omega_n^2}{s^2 + bs + \omega_n^2} \quad [\text{EQ\#30}]$$

At the 90° phase point, $\omega = \omega_n$; where

$$\omega_n^2 = \frac{1}{V_2} \frac{a^2 A}{L}$$

The resonant frequency may be found on the physical system using a number of methods. A phase-lock loop may be employed to find the 90° phase point—this frequency may correspond to the natural frequency of the system. Alternatively, the resonant frequency may be calculated using the phase at any two frequencies:

The phase, ϕ , at any given frequency will satisfy the following relation:

$$\tan\phi = \frac{b\omega}{\omega^2 - \omega_n^2} \quad [\text{EQ\#31}]$$

$$\text{where } b = \frac{fA}{L}.$$

Solving for V_2 results in:

$$V_2 = \frac{a^2 A}{\omega - f\omega \cot\phi} \quad [\text{EQ\#32}]$$

Accordingly, the ratio of the phases at two different frequencies ω_1 and ω_2 can be used to compute the natural frequency of the system:

$$\alpha\omega_n^2 = \omega_1\omega_2 \frac{\left(\omega_1 \frac{\tan\phi_1}{\tan\phi_2} - \omega_2\right)}{\left(\omega_2 \frac{\tan\phi_1}{\tan\phi_2} - \omega_1\right)} \quad [\text{EQ\#33}]$$

For computational efficiency, the actual phase does not need to be calculated. All that is needed is the ratio of the real and imaginary parts of the response ($\tan\phi$).

Re-writing EQ #33 in terms of the variable volume results in:

$$\frac{1}{V_2} = \frac{1}{a^2} \frac{L}{A} \omega_1\omega_2 \frac{\left(\omega_1 \frac{\tan\phi_1}{\tan\phi_2} - \omega_2\right)}{\left(\omega_2 \frac{\tan\phi_1}{\tan\phi_2} - \omega_1\right)} \quad [\text{EQ\#34}]$$

Volume Estimation Using Swept Sine

The resonant frequency of the system may be estimated using swept-sine system identification. In this method, the response of the system to a sinusoidal pressure variation may be found at a number of different frequencies. This frequency response data may then be used to estimate the system transfer function using linear regression.

The transfer function for the system may be expressed as a rational function of s . The general case is expressed below

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for a transfer function with an n^{th} order numerator and an m^{th} order denominator. N and D are the coefficients for the numerator and denominator respectively. The equation has been normalized such that the leading coefficient in the denominator is 1.

$$G(s) = \frac{N_n s^n + N_{n-1} s^{n-1} + \dots + N_0}{s^m + D_{m-1} s^{m-1} + D_{m-2} s^{m-2} + \dots + D_0} \quad [\text{EQ\#35}]$$

$$G(s) = \frac{\sum_{k=0}^n N_k s^k}{s^m + \sum_{k=0}^{m-1} D_k s^k} \quad [\text{EQ\#36}]$$

This equation may be re-written as follows:

$$G s^m = \sum_{k=0}^n N_k s^k - G \sum_{k=0}^{m-1} D_k s^k \quad [\text{EQ\#37}]$$

Representing this summation in matrix notation resulting in the following:

$$\begin{bmatrix} G_1 s_1^m \\ \vdots \\ G_k s_k^m \end{bmatrix} = \begin{bmatrix} s_1^n & \dots & s_1^0 & -G_1 s_1^{m-1} & \dots & -G_1 s_1^0 \\ \vdots & & \vdots & \vdots & & \vdots \\ s_k^n & \dots & s_k^0 & -G_k s_k^{m-1} & \dots & -G_k s_k^0 \end{bmatrix} \begin{bmatrix} N_n \\ \vdots \\ N_0 \\ D_{m-1} \\ \vdots \\ D_0 \end{bmatrix} \quad [\text{EQ\#38}]$$

where k is the number of data points collected in the swept sine. To simplify the notation, this equation may be summarized using the vectors:

$$y = Xc \quad [\text{EQ \#39}]$$

where y is k by 1, x is k by (m+n-1) and c is (m+n-1) by 1. The coefficients may then be found using a least square approach. The error function may be written as follows:

$$e = y - Xc \quad [\text{EQ \#40}]$$

The function to be minimized is the weighted square of the error function; W is a k×k diagonal matrix.

$$e^T W e = (y - Xc)^T W (y - Xc) \quad [\text{EQ \#41}]$$

$$e^T W e = y^T W y - (y^T W X c) - (y^T W X c)^T - y^T W X c + c^T X^T W X c \quad [\text{EQ \#42}]$$

As the center two terms are scalars, the transpose may be neglected.

$$e^T W e = y^T W y - 2 y^T W X c + c^T X^T W X c \quad [\text{EQ\#43}]$$

$$\frac{\partial e^T W e}{\partial c} = -2 X^T W y + 2 X^T W X c = 0 \quad [\text{EQ\#44}]$$

$$c = (X^T W X)^{-1} X^T W y \quad [\text{EQ\#45}]$$

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It may be necessary to use the complex transpose in all of these cases. This approach may result in complex coefficients, but the process may be modified to ensure that all the coefficients are real. The least-square minimization may be modified to give only real coefficients if the error function is changed to be

$$e^T W e = \text{Re}(y - Xc)^T W \text{Re}(y - Xc) + \text{Im}(y - Xc)^T W \text{Im}(y - Xc) \quad [\text{EQ \#46}]$$

Accordingly, the coefficients may be found with the relation:

$$c = (\text{Re}(X)^T W \text{Re}(X) + \text{Im}(X)^T W \text{Im}(X)) (\text{Re}(X)^T W \text{Re}(y) + \text{Im}(X)^T W \text{Im}(y)) \quad [\text{EQ \#47}]$$

Solution for a 2nd Order System

For a system with a 0th order numerator and a second order denominator as shown in the transfer function:

$$G(s) = \frac{N_0}{s^2 + D_1 s + D_0} \quad [\text{EQ\#48}]$$

The coefficients in this transfer function may be found based on the expression found in the previous section:

$$c = (\text{Re}(X)^T W \text{Re}(X) + \text{Im}(X)^T W \text{Im}(X)) (\text{Re}(X)^T W \text{Re}(y) + \text{Im}(X)^T W \text{Im}(y)) \quad [\text{EQ \#49}]$$

where:

$$y = \begin{bmatrix} G_1 s_1^2 \\ \vdots \\ G_k s_k^2 \end{bmatrix}, X = \begin{bmatrix} 1 & -G_1 s_1 & -G_1 \\ \vdots & \vdots & \vdots \\ 1 & -G_k s_k & -G_k \end{bmatrix}, \text{ and } c = \begin{bmatrix} N_0 \\ D_1 \\ D_0 \end{bmatrix} \quad [\text{EQ\#50}]$$

To simplify the algorithm, we may combine some of terms:

$$c = D^{-1} b \quad [\text{EQ \#51}]$$

where:

$$D = \text{Re}(X)^T W \text{Re}(X) + \text{Im}(X)^T W \text{Im}(X) \quad [\text{EQ \#52}]$$

$$b = \text{Re}(X)^T W \text{Re}(y) + \text{Im}(X)^T W \text{Im}(y) \quad [\text{EQ \#53}]$$

To find an expression for D in terms of the complex response vector G and the natural frequency $s = j\omega$, X may be split into its real and imaginary parts:

$$\text{Re}(X) = \begin{bmatrix} 1 & \omega_k \text{Im}(G_1) & -\text{Re}(G_1) \\ \vdots & \vdots & \vdots \\ 1 & \omega_k \text{Im}(G_k) & -\text{Re}(G_k) \end{bmatrix} \quad [\text{EQ\#54}]$$

$$\text{Im}(X) = \begin{bmatrix} 0 & -\omega_k \text{Re}(G_1) & -\text{Im}(G_1) \\ \vdots & \vdots & \vdots \\ 0 & -\omega_k \text{Re}(G_k) & -\text{Im}(G_k) \end{bmatrix}$$

The real and imaginary portions of the expression for D above may then become

$$\text{Re}(X)^T W \text{Re}(X) = \begin{bmatrix} \sum_{i=1}^k w_i & \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i & -\sum_{i=1}^k w_i \text{Re}(G_i) \\ \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i & \sum_{i=1}^k w_i \text{Im}(G_i)^2 \omega_i^2 & -\sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i \\ -\sum_{i=1}^k w_i \text{Re}(G_i) & -\sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i & \sum_{i=1}^k w_i \text{Re}(G_i)^2 \end{bmatrix} \quad [\text{EQ}\#55]$$

$$\text{Im}(X)^T W \text{Im}(X) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \sum_{i=1}^k w_i \text{Re}(G_i)^2 \omega_i^2 & \sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i \\ 0 & \sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i & \sum_{i=1}^k w_i \text{Im}(G_i)^2 \end{bmatrix} \quad [\text{EQ}\#56]$$

Combining these terms results in the final expression for the D matrix, which may contain only real values.

$$D = \begin{bmatrix} \sum_{i=1}^k w_i & \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i & -\sum_{i=1}^k w_i \text{Re}(G_i) \\ \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i & \sum_{i=1}^k w_i (\text{Re}(G_i)^2 + \text{Im}(G_i)^2) \omega_i^2 & 0 \\ -\sum_{i=1}^k w_i \text{Re}(G_i) & 0 & \sum_{i=1}^k w_i (\text{Re}(G_i)^2 + \text{Im}(G_i)^2) \end{bmatrix} \quad [\text{EQ}\#57]$$

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The same approach may be taken to find an expression for the b vector in terms of G and co. The real and imaginary parts of y are as follows:

$$D^{-1} = \frac{1}{\det(D)} \text{adj}(D) \quad [\text{EQ}\#60]$$

$$\text{Re}(y) = \begin{bmatrix} -\text{Re}(G_1) \omega_1^2 \\ \vdots \\ -\text{Re}(G_k) \omega_k^2 \end{bmatrix}, \quad \text{Im}(y) = \begin{bmatrix} -\text{Im}(G_1) \omega_1^2 \\ \vdots \\ -\text{Im}(G_k) \omega_k^2 \end{bmatrix} \quad [\text{EQ}\#58]$$

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If D is expressed as follows:

$$D = \begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{12} & d_{22} & 0 \\ d_{13} & 0 & d_{33} \end{bmatrix} \quad [\text{EQ}\#61]$$

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Combining the real and imaginary parts results in the expression for the b vector as follows:

$$b = \text{Re}(X)^T W \text{Re}(y) + \text{Im}(X)^T W \text{Im}(y) = \quad [\text{EQ}\#59]$$

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then the adjugate matrix may be written as follows:

$$\begin{bmatrix} -\sum_{i=1}^k w_i \text{Re}(G_i) \omega_i^2 \\ 0 \\ \sum_{i=1}^k w_i (\text{Re}(G_i)^2 + \text{Im}(G_i)^2) \omega_i^2 \end{bmatrix}$$

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$$\text{adj}(D) = \begin{bmatrix} \begin{vmatrix} d_{22} & 0 \\ 0 & d_{33} \end{vmatrix} & -\begin{vmatrix} d_{12} & 0 \\ d_{13} & d_{33} \end{vmatrix} & \begin{vmatrix} d_{12} & d_{22} \\ d_{13} & 0 \end{vmatrix} \\ -\begin{vmatrix} d_{12} & d_{13} \\ 0 & d_{33} \end{vmatrix} & \begin{vmatrix} d_{11} & d_{13} \\ d_{13} & d_{33} \end{vmatrix} & -\begin{vmatrix} d_{11} & d_{12} \\ d_{13} & 0 \end{vmatrix} \\ \begin{vmatrix} d_{12} & d_{13} \\ d_{22} & 0 \end{vmatrix} & -\begin{vmatrix} d_{11} & d_{13} \\ d_{12} & 0 \end{vmatrix} & \begin{vmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{vmatrix} \end{bmatrix} = \quad [\text{EQ}\#62]$$

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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{32} & a_{33} \end{bmatrix}$$

The next step is to invert the D matrix. The matrix is symmetric and positive-definite so the number of computations needed to find the inverse will be reduced from the general 3x3 case. The general expression for a matrix inverse is:

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Due to symmetry, only the upper diagonal matrix may need to be calculated.

59

The Determinant may then be computed in terms of the adjugate matrix values, taking advantage of the zero elements in the original array:

$$\det(D)=a_{12}d_{12}+a_{22}d_{22} \quad [\text{EQ \#63}] \quad 5$$

Finally, the inverse of D may be written as follows:

$$D^{-1} = \frac{1}{\det(D)} \text{adj}(D) \quad [\text{EQ\#64}] \quad 10$$

Since we are trying to solve:

$$c = D^{-1}b = \frac{1}{\det(D)} \text{adj}(D)b \quad [\text{EQ\#65}]$$

then :

$$c = \quad [\text{EQ\#66}] \quad 20$$

$$\frac{1}{\det(D)} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_1 \\ 0 \\ b_3 \end{bmatrix} = \frac{1}{\det(D)} \begin{bmatrix} a_{11}b_1 + a_{13}b_3 \\ a_{12}b_1 + a_{23}b_3 \\ a_{13}b_1 + a_{33}b_3 \end{bmatrix}$$

25

The final step is to get a quantitative assessment of how well the data fits the model. Accordingly, the original expression for the error is as follows:

$$e^T W e = \text{Re}(y-Xc)^T W \text{Re}(y-Xc) + \text{Im}(y-Xc)^T W \text{Im}(y-Xc) \quad [\text{EQ \#67}] \quad 30$$

This may be expressed in terms of the D matrix and the b and c vectors as follows:

$$e^T W e = h - 2c^T b + c^T D c \quad [\text{EQ\#68}] \quad 35$$

where

$$h = \text{Re}(y^T) W \text{Re}(y) + \text{Im}(y^T) W \text{Im}(y) \quad [\text{EQ\#69}] \quad 40$$

$$h = \sum_{i=1}^k w_i (\text{Re}(G_i)^2 + \text{Im}(G_i)^2) \omega_i^4 \quad [\text{EQ\#70}]$$

The model fit error may also be used to detect sensor failures.

Alternate Solution for a 2nd Order System

$$G(s) = \frac{N_n s^n + N_{n-1} s^{n-1} + \dots + N_0}{s^m + D_{m-1} s^{m-1} + D_{m-2} s^{m-2} + \dots + D_0} \quad [\text{EQ\#71}] \quad 50$$

or

$$G(s) = \frac{\sum_{k=0}^n N_k s^k}{s^m + \sum_{k=0}^{m-1} D_k s^k} \quad [\text{EQ\#72}] \quad 55$$

This equation may be re-written as follows:

$$G = \sum_{k=0}^n N_k s^{k-m} - G \sum_{k=0}^{m-1} D_k s^{k-m} \quad [\text{EQ\#73}] \quad 60$$

60

Putting this summation into matrix notation results in the following:

$$\begin{bmatrix} G_1 \\ \vdots \\ G_k \end{bmatrix} = \begin{bmatrix} s_1^n & \dots & s_1^{-m} & -G_1 s_1^{-1} & \dots & -G_1 s_1^{-m} \\ \vdots & & \vdots & \vdots & & \vdots \\ s_k^{n-m} & \dots & s_k^{-m} & -G_k s_k^{-1} & \dots & -G_k s_k^{-m} \end{bmatrix} \begin{bmatrix} N_n \\ \vdots \\ N_0 \\ D_{m-1} \\ \vdots \\ D_0 \end{bmatrix} \quad [\text{EQ\#74}]$$

For a system with a 0th order numerator and a second order denominator as shown in the transfer function:

$$G(s) = \frac{N_0}{s^2 + D_1 s + D_0} \quad [\text{EQ\#75}]$$

The coefficients in this transfer function may be found based on the expression found in the previous section:

$$c = (\text{Re}(X)^T W \text{Re}(X) + \text{Im}(X)^T W \text{Im}(X))^{-1} (\text{Re}(X)^T W \text{Re}(y) + \text{Im}(X)^T W \text{Im}(y)) \quad [\text{EQ\#76}]$$

where

$$y = \begin{bmatrix} G_1 \\ \vdots \\ G_k \end{bmatrix}, X = \begin{bmatrix} s_1^{-2} & -G_1 s_1^{-1} & -G_1 s_1^{-2} \\ \vdots & \vdots & \vdots \\ s_k^{-2} & -G_k s_k^{-1} & -G_k s_k^{-2} \end{bmatrix}, \text{ and} \quad [\text{EQ\#77}]$$

$$c = \begin{bmatrix} N_0 \\ D_1 \\ D_0 \end{bmatrix}$$

To simplify the algorithm, some terms may be combined:

$$c = D^{-1}b \quad [\text{EQ \#78}]$$

where:

$$D = \text{Re}(X)^T W \text{Re}(X) + \text{Im}(X)^T W \text{Im}(X) \quad [\text{EQ \#79}]$$

$$b = \text{Re}(X)^T W \text{Re}(y) + \text{Im}(X)^T W \text{Im}(y) \quad [\text{EQ \#80}]$$

To find an expression for D in terms of the complex response vector G and the natural frequency s=jω, split X may be split into its real and imaginary parts:

$$\text{Re}(X) = \begin{bmatrix} -\omega_1^{-2} & -\omega_1^{-1} \text{Im}(G_1) & \omega_1^{-2} \text{Re}(G_1) \\ \vdots & \vdots & \vdots \\ -\omega_k^{-2} & -\omega_k^{-1} \text{Im}(G_k) & \omega_k^{-2} \text{Re}(G_k) \end{bmatrix} \quad [\text{EQ\#81}]$$

$$\text{Im}(X) = \begin{bmatrix} 0 & -\omega_1^{-1} \text{Re}(G_1) & \omega_1^{-2} \text{Im}(G_1) \\ \vdots & \vdots & \vdots \\ 0 & -\omega_k^{-1} \text{Re}(G_k) & \omega_k^{-2} \text{Im}(G_k) \end{bmatrix} \quad [\text{EQ\#82}]$$

65

The real and imaginary portions of the expression for D above may then become:

$$\text{Re}(X)^T W \text{Re}(X) = \begin{bmatrix} \sum_{i=1}^k w_i \omega_i^{-4} & \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i^{-3} & -\sum_{i=1}^k w_i \text{Re}(G_i) \omega_i^{-4} \\ \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i^{-3} & \sum_{i=1}^k w_i \text{Im}(G_i)^2 \omega_i^{-2} & -\sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i^{-3} \\ -\sum_{i=1}^k w_i \text{Re}(G_i) \omega_i^{-4} & -\sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i^{-3} & \sum_{i=1}^k w_i \text{Re}(G_i)^2 \omega_i^{-4} \end{bmatrix} \quad [\text{EQ}\#83]$$

$$\text{Im}(X)^T W \text{Im}(X) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \sum_{i=1}^k w_i (G_i)^2 \omega_i^{-4} & -\sum_{j=1}^k w_j \text{Im}(G_i) \text{Re}(G_j) \omega_i^{-3} \\ 0 & -\sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i^{-3} & \sum_{i=1}^k w_i \text{Im}(G_i)^2 \omega_i^{-4} \end{bmatrix} \quad [\text{EQ}\#84]$$

Combining these terms results in the final expression for the D matrix, which may contain only real values.

$$D = \begin{bmatrix} \sum_{i=1}^k w_i \omega_i^{-4} & \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i^{-3} & -\sum_{i=1}^k w_i \text{Re}(G_i) \omega_i^{-4} \\ \sum_{i=1}^k w_i \text{Im}(G_i) \omega_i^{-3} & \sum_{i=1}^k w_i (\text{Re}(G_i)^2 + \text{Im}(G_i)^2) \omega_i^{-2} & -2 \sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i^{-3} \\ -\sum_{i=1}^k w_i \text{Re}(G_i) \omega_i^{-4} & -2 \sum_{i=1}^k w_i \text{Im}(G_i) \text{Re}(G_i) \omega_i^{-3} & \sum_{i=1}^k w_i (\text{Re}(G_i)^2 + \text{Im}(G_i)^2) \omega_i^{-4} \end{bmatrix} \quad [\text{EQ}\#85]$$

The same approach may be taken to find an expression for the b vector in terms of G and ω . The real and imaginary parts of y areas follows:

$$\text{Re}(y) = \begin{bmatrix} -\text{Re}(G_1) \\ \vdots \\ -\text{Re}(G_k) \end{bmatrix}, \quad \text{Im}(y) = \begin{bmatrix} -\text{Im}(G_1) \\ \vdots \\ -\text{Im}(G_k) \end{bmatrix} \quad [\text{EQ}\#86]$$

Combining the real and imaginary parts results in the expression for the b vector as follows:

$$b = \text{Re}(X)^T W \text{Re}(y) + \text{Im}(X)^T W \text{Im}(y) = \begin{bmatrix} -\sum_{i=1}^k w_i \text{Re}(G_i) \omega_i^{-2} \\ -\sum_{i=1}^k w_i (\text{Im}(G_i) + \text{Re}(G_i) \omega_i^{-1}) \\ \sum_{i=1}^k w_i (\text{Re}(G_i)^2 + \text{Im}(G_i)^2) \omega_i^{-2} \end{bmatrix} \quad [\text{EQ}\#87]$$

Implementing Acoustic Volume Sensing
Collecting the Frequency Response Data and Computing the Complex Response

To implement volume sensor assembly 148, volume sensor assembly 148 should determine the relative response of reference microphone 626 and invariable volume micro-

phone 630 to the acoustic wave set up by speaker assembly 622. This may be accomplished by driving speaker assembly 622 with a sinusoidal output at a known frequency; the complex response of microphones 626, 630 may then be found at that driving frequency. Finally, the relative response of microphones 626, 630 may be found and corrected for alternating sampling by e.g., an analog-to-digital convertor (i.e., ADC).

Additionally, the total signal variance may be computed and compared to the variance of pure tone extracted using the discrete Fourier transform (i.e., DFT). This may result in a measure of how much of the signal power comes from noise sources or distortion. This value may then be used to reject and repeat bad measurements.

Computing the Discrete Fourier Transform

The signal from the microphone may be sampled synchronously with the output to speaker assembly 622 such that a fixed number of points, N, are taken per wavelength. The measured signal at each point in the wavelength may be summed over an integer number of wavelengths, M, and stored in an array x by the ISR for processing after all the data for that frequency has been collected.

A DFT may be performed on the data at the integer value corresponding to the driven frequency of the speaker. The general expression for the first harmonic of a DFT is as follows:

$$x_k = \frac{2}{MN} \sum_{n=0}^{N-1} x_n e^{-\frac{2\pi i}{N} kn} \quad [\text{EQ}\#88]$$

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The product MN may be the total number of points and the factor of two may be added such that the resulting real and imaginary portions of the answer match the amplitude of the sine wave:

$$x_n = \text{re}(x_k) \cos\left(\frac{2\pi}{N}kn\right) + \text{im}(x_k) \sin\left(\frac{2\pi}{N}kn\right) \quad [\text{EQ\#89}]$$

This real part of this expression may be as follows:

$$\text{re}(x) = \frac{2}{MN} \sum_{n=0}^{N-1} x_n \cos\left(\frac{2\pi}{N}n\right) \quad [\text{EQ\#90}]$$

We may take advantage of the symmetry of the cosine function to reduce the number of computations needed to compute the DFT. The expression above may be equivalent to:

$$\text{re}(x) = \frac{2}{MN} \left[\left(x_0 - x_{\frac{1}{2}N} \right) + \sum_{n=1}^{\frac{1}{4}N-1} \sin\left(\frac{\pi}{2} - \frac{2\pi}{N}n\right) \left[\left(x_n - x_{\frac{1}{2}N+n} \right) - \left(x_{\frac{1}{2}N-n} - x_{N-n} \right) \right] \right] \quad [\text{EQ\#91}]$$

Similarly, for the imaginary portion of the equation:

$$\text{im}(x) = -\frac{2}{MN} \sum_{n=0}^{N-1} x_n \sin\left(\frac{2\pi}{N}n\right) \quad [\text{EQ\#92}]$$

which may be expressed as follows:

$$\text{im}(x) = -\frac{2}{MN} \left[\left(x_{\frac{1}{4}N} - x_{\frac{3}{4}N} \right) + \sum_{n=1}^{\frac{1}{4}N-1} \sin\left(\frac{2\pi}{N}n\right) \left[\left(x_n - x_{\frac{1}{2}N+n} \right) + \left(x_{\frac{1}{2}N-n} - x_{N-n} \right) \right] \right] \quad [\text{EQ\#93}]$$

The variance of this signal may be calculated as follows:

$$\sigma^2 = \frac{1}{2}(\text{re}(x)^2 + \text{im}(x)^2) \quad [\text{EQ \#94}]$$

The maximum possible value of the real and imaginary portions of x may be 2^{11} ; which corresponds to half the AD range. The maximum value of the tone variance may be 2^{21} ; half the square of the AD range.

Computing the Signal Variance

The pseudo-variance of the signal may be calculated using the following relation:

$$\sigma^2 = \frac{1}{NM^2} \sum_{n=0}^{N-1} x_n^2 - \frac{1}{N^2 M^2} \left(\sum_{n=0}^{N-1} x_n \right)^2 \quad [\text{EQ\#95}]$$

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The result may be in the units of AD counts squared. It may only be the “pseudo-variance” because the signal has been averaged over M periods before the variance is calculated over the N samples in the “averaged” period. This may be a useful metric, however, for finding if the “averaged” signal looks like a sinusoid at the expected frequency. This may be done by comparing the total signal variance to that of the sinusoid found in the discrete Fourier transform.

The summation may be on the order of

$$\sum_{n=0}^{N-1} x_n^2 = O(NM^2 2^{24})$$

for a 12-bit ADC. If $N < 2^7 = 128$ and $M < 2^6 = 64$, then the summation will be less than 2^{43} and may be stored in a 64-bit integer. The maximum possible value of the variance may result if the ADC oscillated between a value of 0 and 2^{12} on each consecutive sample. This may result in a peak variance of $\frac{1}{4}(2^{12})^2 = 2^{22}$ so the result may be stored at a maximum of a $\frac{1}{2}^9$ resolution in a signed 32-bit integer.

Computing the Relative Microphone Response

The relative response (G) of microphones **626**, **630** may be computed from the complex response of the individual microphones:

$$G = \frac{X_{var}}{X_{ref}} = \frac{X_{var}}{X_{ref}} \frac{X_{ref}^*}{X_{ref}^*} \quad [\text{EQ\#96}]$$

$$\text{Re}(G) = \frac{\text{Re}(x_{var})\text{Re}(x_{ref}) + \text{Im}(x_{var})\text{Im}(x_{ref})}{\text{Re}(x_{ref})^2 + \text{Im}(x_{ref})^2} \quad [\text{EQ\#97}]$$

$$\text{Im}(G) = \frac{\text{Re}(x_{ref})\text{Im}(x_{var}) - \text{Re}(x_{var})\text{Im}(x_{ref})}{\text{Re}(x_{ref})^2 + \text{Im}(x_{ref})^2} \quad [\text{EQ\#98}]$$

The denominator of either expression may be expressed in terms of the reference tone variance computed in the previous section as follows:

$$\text{Re}(x_{ref})^2 + \text{Im}(x_{ref})^2 = 2\sigma_{ref}^2 \quad [\text{EQ \#99}]$$

Correcting for A/D Skew

The signals from microphones **626**, **630** may not be sampled simultaneously; the A/D ISR alternates between microphones **626**, **630**, taking a total of N samples per wavelength for each of microphones **626**, **630**. The result may be a phase offset between two microphones **626**, **630** of π/N . To correct for this phase offset, a complex rotation may be applied to the relative frequency response computed in the previous section:

$$G_{rotated} = G \cdot \left(\cos\left(\frac{\pi}{N}\right) + i \sin\left(\frac{\pi}{N}\right) \right) \quad [\text{EQ\#100}]$$

Reference Models

Second and Higher Order Models

Leakage through the seals (e.g., seal assembly **1404**) of volume sensor chamber **620** may be modeled as a second resonant port (e.g., port **1504**, FIG. **100**) connected to an external volume (e.g., external volume **1506**, FIG. **100**).

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The system of equations describing the three-chamber configuration may be as follows:

$$\dot{p}_1 + \frac{\rho a^2}{V_1} (\dot{v}_k - \dot{v}_{r12}) = 0 \quad [\text{EQ\#101}]$$

5

$$\dot{p}_2 + \frac{\rho a^2}{V_2} (\dot{v}_{r12} - \dot{v}_{r23}) = 0 \quad [\text{EQ\#102}]$$

$$v_{r12} = -\frac{f_{12} A_{12}}{L_{12}} \dot{v}_{r12} + \frac{A_{12}}{\rho L_{12}} (p_2 - p_1) \quad [\text{EQ\#103}]$$

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$$\dot{p}_3 + \frac{\rho a^2}{V_3} \dot{v}_{r23} = 0 \quad [\text{EQ\#104}]$$

$$v_{r23} = -\frac{f_{23} A_{23}}{L_{23}} \dot{v}_{r23} + \frac{A_{23}}{\rho L_{23}} (p_3 - p_2) \quad [\text{EQ\#105}]$$

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Putting these equations into state-space results in the following:

$$\begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \\ \dot{p}_3 \\ \dot{v}_{12} \\ \dot{v}_{23} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \frac{\rho a^2}{V_1} & 0 \\ 0 & 0 & 0 & -\frac{\rho a^2}{V_2} & \frac{\rho a^2}{V_2} \\ 0 & 0 & 0 & 0 & -\frac{\rho a^2}{V_3} \\ -\frac{A_{12}}{\rho L_{12}} & \frac{A_{23}}{\rho L_{23}} & 0 & -b_{12} & 0 \\ 0 & -\frac{A_{23}}{\rho L_{23}} & \frac{A_{23}}{\rho L_{23}} & 0 & -b_{23} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ v_{12} \\ v_{23} \end{bmatrix} + \begin{bmatrix} -\frac{\rho a^2}{V_1} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \dot{v}_k \quad [\text{EQ\#106}]$$

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Second Order with Time Delay

The volume sensor assembly **148** equations derived above assume that the pressure is the same everywhere in the acoustic volume. This is only an approximation, as there are time delays associated with the propagation of the sound waves through the volume. This situation may look like a time delay or a time advance based on the relative position of the microphone and speakers.

A time delay may be expressed in the Laplace domain as:

$$G(s) = e^{-\Delta T s} \quad [\text{EQ \#112}]$$

which makes for a non-linear set of equations. However, a first-order Pade approximation of the time delay may be used as follows:

$$G(s) = -\frac{s + \frac{2}{\Delta T}}{s - \frac{2}{\Delta T}} \quad [\text{EQ\#113}]$$

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the frequency response of which may be represented graphically in the Bode diagram shown in FIG. **101** and which may also be written in transfer function form:

$$\frac{p_2}{p_1} = \frac{\omega_{12}^2 (s^2 + b_{23}s + \omega_{23}^2)}{(s^2 + b_{12}s + \omega_{12}^2)(s^2 + b_{23}s + \omega_{23}^2) + \frac{V_3}{V_2} \omega_{23}^2 (s + b_{12})s} \quad [\text{EQ\#107}]$$

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which is shown graphically in FIG. **102**.

Three Chamber Volume Estimation

Volume sensor assembly **148** may also be configured using a third reference volume (e.g., reference volume **1508**; FIG. **103**) connected with a separate resonant port (e.g., port **1510**; FIG. **103**). This configuration may allow for temperature-independent volume estimation.

The system of equations describing the three-chamber configuration are as follows:

$$\frac{p_2}{p_1} = \frac{\omega_{12}^2 (s^2 + b_{23}s + \omega_{23}^2)}{s^4 + (b_{12} + b_{23})s^3 + (b_{12}b_{23} + \omega_{12}^2 + \omega_{23}^2 \left(1 + \frac{V_3}{V_2}\right))s^2 + (b_{23}\omega_{12}^2 + b_{12}\omega_{23}^2 \left(1 + \frac{V_3}{V_2}\right))s + \omega_{12}^2 \omega_{23}^2} \quad [\text{EQ\#108}]$$

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$$\dot{p}_1 + \frac{\rho a^2}{V_1} (\dot{v}_k - \dot{v}_{r12} - \dot{v}_{r13}) = 0 \quad [\text{EQ\#114}]$$

$$\dot{p}_2 + \frac{\rho a^2}{V_2} \dot{v}_{r12} = 0 \quad [\text{EQ\#115}]$$

$$\dot{v}_{r12} = -\frac{f_{12} A_{12}}{L_{12}} \dot{v}_{r12} + \frac{A_{12}}{\rho L_{12}} (p_2 - p_1) \quad [\text{EQ\#116}]$$

$$\dot{p}_3 + \frac{\rho a^2}{V_3} \dot{v}_{r13} = 0 \quad [\text{EQ\#117}]$$

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A bubble underneath the diaphragm material in the variable volume will follow the same dynamic equations as a leakage path. In this case, the diaphragm material may act as the resonant mass rather than the leakage port. Accordingly, the equation may be as follows:

$$m\ddot{x} = \Delta p A - b_m \dot{x} \quad [\text{EQ \#109}]$$

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wherein m is the mass of the diaphragm, A is the cross sectional area of the diaphragm that can resonate, and b_m is the mechanical damping. EQ #106 may be written in terms of the volume flow rate:

$$\dot{v} = -\frac{b}{m} \dot{v} + \Delta p \frac{A^2}{m} \quad [\text{EQ\#110}]$$

wherein the volume of the air bubble is V_3 . If the bubble volume is substantially smaller than the acoustic volume $V_3 \ll V_2$ than the transfer function may be simplified to:

$$\frac{p_2}{p_1} = \frac{\omega_{12}^2 (s^2 + b_{23}s + \omega_{23}^2)}{(s^2 + b_{12}s + \omega_{12}^2)(s^2 + b_{23}s + \omega_{23}^2 \left(1 + \frac{V_3}{V_2}\right))} \quad [\text{EQ\#111}]$$

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$$\ddot{v}_{r13} = -\frac{f_{13}A_{13}}{L_{13}}\dot{v}_{r13} + \frac{A_{13}}{\rho L_{13}}(p_2 - p_1) \quad [\text{EQ\#118}]$$

Using these equations and solving for the transfer function across each of the resonant ports results in the following:

$$\frac{P_2}{P_1} = \frac{\omega_{n12}^2}{s^2 + 2\zeta_{12}\omega_{n12}s + \omega_{n12}^2} \quad [\text{EQ\#119}]$$

where

$$\omega_{n12} = \frac{1}{V_2} \frac{a^2 A_{12}}{L_{12}} \quad [\text{EQ\#120}]$$

and

$$\zeta = \frac{f_{12}A_{12}}{2L_{12}\omega_{n12}} \quad [\text{EQ\#121}]$$

$$\frac{P_3}{P_1} = \frac{\omega_{n13}^2}{s^2 + 2\zeta_{13}\omega_{n13}s + \omega_{n13}^2} \quad [\text{EQ\#122}]$$

where

$$\omega_{n13} = \frac{1}{V_2} \frac{a^2 A_{13}}{L_{13}} \quad [\text{EQ\#123}]$$

and

$$\zeta = \frac{f_{13}A_{13}}{2L_{13}\omega_{n13}} \quad [\text{EQ\#124}]$$

The volume of volume sensor chamber **620** may be estimated using the ratio of the natural frequency of the two resonant ports as follows:

$$\frac{\omega_{n13}^2}{\omega_{n12}^2} = \frac{V_2}{V_3} \frac{A_{13}}{A_{12}} \frac{L_{12}}{L_{13}} \quad [\text{EQ\#125}]$$

EQ #120 illustrates that the volume of volume sensor chamber **620** may be proportional to reference volume **1508**. The ratio of these two volumes (in the ideal model) may only depend on the geometry of the resonant port (e.g., port **1510**; FIG. **103**) and has no dependence upon temperature.

Exponential Volume Model

Assume the flow out through the flow resistance has the following form:

$$\dot{V}_{out} = \frac{V_{avs}}{\tau} \quad [\text{EQ\#126}]$$

Assuming a fixed input flow rate from the pump chamber, the volume of volume sensor chamber **620** is based upon the following differential equation:

$$\dot{V}_{avs} = \dot{V}_{in} - \dot{V}_{out} = \dot{V}_{in} - \frac{V_{avs}}{\tau} \quad [\text{EQ\#127}]$$

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which gives the following solution assuming a zero initial volume:

$$V_{avs} = \dot{V}_{in}\tau\left(1 - e^{-\frac{t}{\tau}}\right) \quad [\text{EQ\#128}]$$

Accordingly, the output flow rate flows:

$$\dot{V}_{out} = \dot{V}_{in}\left(1 - e^{-\frac{t}{\tau}}\right) \quad [\text{EQ\#129}]$$

The volume delivered during the pump phase may be written:

$$V_{out} = \dot{V}_{in}\left[t - \tau\left(1 - e^{-\frac{t}{\tau}}\right)\right] \quad [\text{EQ\#130}]$$

Device Calibration

The model fit allows the resonant frequency of the port to be extracted from the sine sweep data. The next step is to relate this value to the delivered volume. The ideal relationship between the resonant frequency and the delivered volume to be expressed as follows:

$$\omega_n^2 = \frac{a^2 A}{L} \frac{1}{V_2} \quad [\text{EQ\#131}]$$

The speed of sound will vary with temperature, so it may be useful to split out the temperature effects.

$$\omega_n^2 = \frac{\gamma R A}{L} \frac{T}{V_2} \quad [\text{EQ\#132}]$$

The volume may then be expressed as a function of the measured resonant frequency and the temperature:

$$V_2 = C \frac{T}{\omega_n^2} \quad [\text{EQ\#133}]$$

Where c is the calibration constant

$$C = \frac{\gamma R A}{L}$$

Implementation Details

End Effects

The air resonating in the port (e.g., port assembly **624**) may extend out into the acoustic volumes at the end of each oscillation. The distance the air extends may be estimated based on the fundamental volume sensor assembly equations. For any given acoustic volume, the distance the air extends into the volume may be expressed as a function of the pressure and port cross-sectional area:

$$x = \frac{V}{\rho a^2 A} p$$

[EQ#132]

If we assume the following values:

$$V = 28.8 \times 10^{-6} L$$

[EQ#133]

$$\rho = 1.292 \frac{\text{kg}}{\text{m}^3}$$

[EQ#134]

$$a = 340 \frac{\text{m}}{\text{s}}$$

[EQ#135]

$$d = 0.5 \cdot \text{mm}$$

[EQ#136]

$$p = 1 \cdot \text{Pa (Approximately 100 dB)}$$

[EQ#137]

Accordingly, the air will extend roughly 1.9 mm in to the acoustic chamber.

Sizing V_1 (i.e., the Fixed Volume) Relative to V_2 (i.e., the Variable Volume)

Sizing V_1 (e.g., fixed volume **1500**) may require trading off acoustic volume with the relative position of the poles and zeros in the transfer function. The transfer function for both V_1 and V_2 (e.g., variable volume **1502**) are shown below relative to the volume displacement of speaker assembly **622**.

$$\frac{P_2}{v_k} = -\frac{\rho a^2}{V_1} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \alpha\omega_n^2}$$

[EQ#138]

$$\frac{P_1}{v_k} = -\frac{\rho a^2}{V_1} \frac{s^2 + 2\zeta\omega_n s + \alpha\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

[EQ#139]

where

$$\omega_n^2 = \frac{a^2 A}{L} \frac{1}{V_2},$$

[EQ#140]

$$\zeta = \frac{fA}{2L\omega_n}$$

and

$$\alpha = \left(1 + \frac{V_2}{V_1}\right)$$

As V_1 is increased the gain may decrease and the speaker may be driven at a higher amplitude to get the same sound pressure level. However, increasing V_1 may also have the benefit of moving the complex zeros in the p_1 transfer function toward the complex poles. In the limiting case where $V_1 \rightarrow \infty$, $\alpha \rightarrow 1$ and you have pole-zero cancellation and a flat response. Increasing V_1 , therefore, may have the benefit of reducing both the resonance and the notch in the p_1 transfer function, and moving the p_2 poles toward ω_n ; resulting in a lower sensitivity to measurement error when calculating the p_2/p_1 transfer function.

FIG. **104** is a graphical representation of:

$$\frac{P_1}{v_k}$$

[EQ#141]

FIG. **105** is a graphical representation of

$$\frac{P_2}{v_k}$$

[EQ#141]

Aliasing

Higher frequencies may alias down to the frequency of interest, wherein the aliased frequency may be expressed as follows:

$$f = |f_n - nf_s|$$

[EQ #143]

where f_s is the sampling frequency, f_n is the frequency of the noise source, n is a positive integer, and f is the aliased frequency of the noise source.

The demodulation routine may effectively filter out noise except at the specific frequency of the demodulation. If the sample frequency is set dynamically to be a fixed multiple of the demodulation frequency, then the frequency of the noise that can alias down to the demodulation frequency may be a fixed set of harmonics of that fundamental frequency.

For example, if the sampling frequency is eight times the demodulation frequency, then the noise frequencies that can alias down to that frequency are as follows:

$$\frac{f_n}{f} = \left\{ \frac{1}{n\beta + 1}, \frac{1}{n\beta - 1} \right\} = \left\{ \frac{1}{7}, \frac{1}{9}, \frac{1}{15}, \frac{1}{17}, \frac{1}{23}, \frac{1}{25}, \dots \right\}$$

[EQ#144]

where

$$\beta = \frac{f_s}{f} = 8.$$

For $\beta=16$, the following series would result:

$$\frac{f_n}{f} = \left\{ \frac{1}{15}, \frac{1}{17}, \frac{1}{31}, \frac{1}{33}, \dots \right\}$$

[EQ#145]

Performance

Sensitivity to Temperature

The sensitivity to temperature may be split into a gain change and a noise change. If the temperature is off by a factor of dT , the resulting gain error may be:

$$V_2 = c \left(\frac{T_2}{\omega_2^2} - \frac{T_1}{\omega_1^2} \right)$$

[EQ#147]

Accordingly, if the same temperature is used for both sine sweeps, any error in the temperature measurement may look like a gain change to the system.

$$e_{\text{gain}} = 1 - \frac{T_{\text{measured}}}{T_{\text{actual}}}$$

[EQ#148]

Therefore, for a 1° K temperature error, the resulting volume error may be 0.3% at 298° K. This error may include both the error in the temperature sensor and the difference between the sensor temperature and the temperature of the air within volume sensor assembly **148**.

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The measurement, however, may be more susceptible to noise in the temperature measurement. A temperature change during the differential sine sweeps may result in an error that looks more like an offset rather than a gain change:

$$V_{error} = \frac{c}{\omega^2} \Delta T \quad [\text{EQ\#149}]$$

Accordingly, if the measurement varies by 0.1 K during the two measurement sine sweeps, the difference may be 0.012 uL. Therefore, it may be better to use a consistent temperature estimate for each delivery rather than taking a separate temperature measurement for each sine sweep (as shown in FIG. 107).

The LM73 temperature sensor has a published accuracy of $\pm 1^\circ \text{C}$. and a resolution of 0.03 C. Further, the LM73 temperature sensor seems to consistently have a startup transient of about 0.3°C . that takes about five sine sweeps to level out (as shown in FIG. 108).

Since the above-described infusion pump assemblies (e.g., infusion pump assembly 100, 100', 400, 500) provides discrete deliveries of infusible fluid, the above-described infusion pump assemblies may be modeled entirely in the discrete domain (in the manner shown in FIG. 109), which may be reduced to the following:

$$G_p(z) = \frac{Kz}{z-1} \quad [\text{EQ\#150}]$$

A discrete-time PI regulator may perform according to the following:

$$G_c(z) = K_p \left(1 + \frac{T_s}{T_i} \frac{Z}{z-1} \right) \quad [\text{EQ\#151}]$$

The AVS system described above works by comparing the acoustic response in fixed volume 1500 and variable volume 1502 to a speaker driven input and extracting the volume of the variable volume 1502. As such, there is a microphone in contact with each of these separate volumes (e.g., microphones 626, 630). The response of variable volume microphone 630 may also be used in a more gross manner to detect the presence or absence of disposable housing assembly 114. Specifically, if disposable housing assembly 114 is not attached to (i.e., positioned proximate) variable volume 1502, essentially no acoustic response to the speaker driven input should be sensed. The response of fixed volume 1500, however, should remain tied to the speaker input. Thus, the microphone data may be used to determine whether disposable housing assembly 114 by simply ensuring that both microphones exhibit an acoustic response. In the event that microphone 626 (i.e., the microphone positioned proximate fixed volume 1500) exhibits an acoustic response and microphone 630 (i.e., the microphone positioned proximate variable volume 1502) does not exhibit an acoustic response, it may be reasonably concluded that disposable housing assembly 114 is not attached to reusable housing assembly 102. It should be noted that a failure of variable volume microphone 630 may also appear to be indicative of disposable housing assembly 114 not being attached, as the failure of variable volume microphone 630 may result in a mid-range reading that is nearly indistinguishable from the

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microphone response expected when disposable housing assembly 114 is not attached.

For the following discussion, the following nomenclature may be used:

Symbols	
$\alpha_{max}(f)$	maximum read at a given frequency
$\alpha_{min}(f)$	minimum read at a given frequency
δ	difference between max and min sums
f	individual frequency
F	set of sine sweep frequencies
N	number of frequencies in each sine sweep, F
Φ	boolean disposable attached flag
σ_{max}	sum of maximum ADC reads
σ_{min}	sum of minimum ADC reads
T	max/min ADC difference threshold
Subscripts	
i	sweep number
ref	reference volume
var	variable volume

As part of the demodulation routine employed in each frequency response calculation, the minimum and maximum readings of both fixed volume microphone 626 and variable volume microphone 630 may be calculated. The sum of these maximum and minimum values may be calculated over the entire sine-sweep (as discussed above) for both microphone 626 and microphone 630 as follows.

$$\sigma_{max} = \sum_{f \in F} \alpha_{max}(f) \quad [\text{EQ\#152}]$$

$$\sigma_{min} = \sum_{f \in F} \alpha_{min}(f) \quad [\text{EQ\#153}]$$

and the difference between these two summations may be simplified as follows:

$$\delta = \sigma_{max} - \sigma_{min} \quad [\text{EQ \#154}]$$

While δ may be divided by the number of sine sweeps to get the average minimum/maximum difference for the sine sweep (which is then compared to a threshold), the threshold may equivalently be multiplied by N for computational efficiency. Accordingly, the basic disposable detection algorithm may be defined as follows:

$$\phi_i = \begin{cases} 1 & \text{if } \delta_{var} > N * T \\ 0 & \text{if } \delta_{var} < N * T \text{ \& } \delta_{ref} > N * T \end{cases} \quad [\text{EQ\#155}]$$

The additional condition that the maximum/minimum difference be greater than the threshold is a check performed to ensure that a failed speaker is not the cause of the acoustic response received. This algorithm may be repeated for any sine-sweep, thus allowing a detachment of disposable housing assembly 114 to be sensed within e.g., at most two consecutive sweeps (i.e., in the worst case scenario in which disposable housing assembly 114 is removed during the second half of an in-progress sine sweep).

Thresholding for the above-described algorithm may be based entirely on numerical evidence. For example, examination of typical minimum/maximum response differences may show that no individual difference is ever less than five hundred ADC counts. Accordingly, all data examined while disposable housing assembly 114 is detached from reusable

housing assembly **102** may show that all minimum/maximum response differences as being well under five hundred ADC counts. Thus, the threshold for δ may be set at $T=500$.

While volume sensor assembly **148** is described above as being utilized within an infusion pump assembly (e.g., infusion pump assembly **100**), this is for illustrative purposes only and is not intended to be a limitation of this disclosure, as other configurations are possible and are considered to be within the scope of this disclosure. For example, volume sensor assembly **148** may be used within a process control environment for e.g., controlling the quantity of chemicals mixed together. Alternatively, volume sensor assembly **148** may be used within a beverage dispensing system to control e.g., the quantity of ingredients mixed together.

While volume sensor assembly **148** is described above as utilizing a port (e.g., port assembly **624**) as a resonator, this is for illustrative purposes only, as other configurations are possible and are considered to be within the scope of this disclosure. For example, a solid mass (not shown) may be suspended within port assembly **624** and may function as a resonator for volume sensor assembly **148**. Specifically, the mass (not shown) for the resonator may be suspended on a diaphragm (not shown) spanning port assembly **624**. Alternatively, the diaphragm itself (not shown) may act as the mass for the resonator. The natural frequency of volume sensor assembly **148** may be a function of the volume of variable volume **1502**. Accordingly, if the natural frequency of volume sensor assembly **148** can be measured, the volume of variable volume **1502** may be calculated.

The natural frequency of volume sensor assembly **148** may be measured in a number of different ways. For example, a time-varying force may be applied to the diaphragm (not shown) and the relationship between that force and the motion of the diaphragm (not shown) may be used to estimate the natural frequency of volume sensor assembly **148**. Alternately the mass (not shown) may be perturbed and then allowed to oscillate. The unforced motion of the mass (not shown) may then be used to calculate the natural frequency of volume sensor assembly **148**.

The force applied to the resonant mass (not shown) may be accomplished in various ways, examples of which may include but are not limited to:

- speaker assembly **622** may create a time-varying pressure within fixed volume **1500**;
- the resonant mass (not shown) may be a piezoelectric material responding to a time-varying voltage/current; and
- the resonant mass (not shown) may be a voice coil responding to a time-varying voltage/current

The force applied to the resonant mass may be measured in various ways, examples of which may include but are not limited to:

- measuring the pressure in the fixed volume;
- the resonant mass (not shown) may be a piezoelectric material; and
- a strain gauge may be connected to the diaphragm (not shown) or other structural member supporting the resonant mass (not shown).

Similarly, the displacement of the resonant mass (not shown) may be estimated by measuring the pressure in the variable volume, or measured directly in various ways, examples of which may include but are not limited to:

- via piezoelectric sensor;
- via capacitive sensor;
- via optical sensor;
- via Hall-effect sensor;

- via a potentiometer (time varying impedance) sensor;
- via an inductive type sensor; and
- via a linear variable differential transformer (LVDT)

Further, the resonant mass (not shown) may be integral to either the force or displacement type sensor (i.e. the resonant mass (not shown) may be made of piezoelectric material).

The application of force and measurement of displacement may be accomplished by a single device. For example, a piezoelectric material may be used for the resonant mass (not shown) and a time-varying voltage/current may be applied to the piezoelectric material to create a time-varying force. The resulting voltage/current applied to the piezoelectric material may be measured and the transfer function between the two used to estimate the natural frequency of volume sensor assembly **148**.

As discussed above, the resonant frequency of volume sensor assembly **148** may be estimated using swept-sine system identification. Specifically, the above-described model fit may allow the resonant frequency of the port assembly to be extracted from the sine sweep data, which may then be used to determine the delivered volume. The ideal relationship between the resonant frequency and the delivered volume may be expressed as follows:

$$\omega_n^2 = \frac{a^2 A}{L} \frac{1}{V_2} \quad [\text{EQ\#126}]$$

The speed of sound will vary with temperature, so it may be useful to split out the temperature effects.

$$\omega_n^2 = \frac{\gamma R A}{L} \frac{T}{V_2} \quad [\text{EQ\#126}]$$

The volume may then be expressed as a function of the measured resonant frequency and the temperature:

$$V_2 = C \frac{T}{\omega_n^2} \quad [\text{EQ\#127}]$$

Where c is the calibration constant

$$C = \frac{\gamma R A}{L}.$$

Infusion pump assembly **100** may then compare this calculated volume V_2 (i.e., representative of the actual volume of infusible fluid delivered to the user) to the target volume (i.e., representative of the quantity of fluid that was supposed to be delivered to the user). For example, assume that infusion pump assembly **100** was to deliver a 0.100 unit basal dose of infusible fluid to the user every thirty minutes. Further, assume that upon effectuating such a delivery, volume sensor assembly **148** indicates a calculated volume V_2 (i.e., representative of the actual volume of infusible fluid delivered to the user) of 0.095 units of infusible fluid.

When calculating volume V_2 , infusion pump assembly **100** may first determine the volume of fluid within volume sensor chamber **620** prior to the administration of the dose of infusible fluid and may subsequently determine the volume of fluid within volume sensor chamber **620** after the

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administration of the dose of infusible fluid, wherein the difference of those two measurements is indicative of V_2 (i.e., the actual volume of infusible fluid delivered to the user). Accordingly, V_2 is a differential measurement.

V_2 may be the total air space over the diaphragm in the variable volume chamber. The actual fluid delivery to the patient may be the difference in V_2 from when the chamber was full to after the measurement valve was opened and the chamber was emptied. V_2 may not directly be the delivered volume. For example, the air volume may be measured and a series of differential measurements may be taken. For occlusion, an empty measurement may be taken, the chamber may be filled, a full measurement may be taken, and then a final measurement may be taken after the exit valve is open. Accordingly, the difference between the first and second measurement may be the amount pumped and the difference between the second and third is the amount delivered to the patient.

Accordingly, electrical control assembly 110 may determine that the infusible fluid delivered is 0.005 units under what was called for. In response to this determination, electrical control assembly 110 may provide the appropriate signal to mechanical control assembly 104 so that any additional necessary dosage may be pumped. Alternatively, electrical control assembly 110 may provide the appropriate signal to mechanical control assembly 104 so that the additional dosage may be dispensed with the next dosage. Accordingly, during administration of the next 0.100 unit dose of the infusible fluid, the output command for the pump may be modified based on the difference between the target and amount delivered.

Referring also to FIG. 110, there is shown one particular implementation of a control system for controlling the quantity of infusible fluid currently being infused based, at least in part, on the quantity of infusible fluid previously administered. Specifically and continuing with the above-stated example, assume for illustrative purposes that electrical control assembly 110 calls for the delivery of a 0.100 unit dose of the infusible fluid to the user. Accordingly, electrical control assembly 110 may provide a target differential volume signal 1600 (which identifies a partial basal dose of 0.010 units of infusible fluid per cycle of shape memory actuator 112) to volume controller 1602. Accordingly and in this particular example, shape memory actuator 112 may need to be cycled ten times in order to achieve the desired basal dose of 0.100 units of infusible fluid (i.e., 10 cycles $\times$ 0.010 units per cycle = 0.100 units). Volume controller 1602 in turn may provide "on-time" signal 1606 to SMA (i.e., shape memory actuator) controller 1608. Also provided to SMA controller 1608 is battery voltage signal 1610.

Specifically, shape-memory actuator 112 may be controlled by varying the amount of thermal energy (e.g., joules) applied to shape-memory actuator 112. Accordingly, if the voltage level of battery 606 is reduced, the quantity of joules applied to shape-memory actuator 112 may also be reduced for a defined period of time. Conversely, if the voltage level of battery 606 is increased, the quantity of joules applied to shape memory actuator 112 may also be increased for a defined period of time. Therefore, by monitoring the voltage level of battery 606 (via battery voltage signal 1610), the type of signal applied to shape-memory actuator 112 may be varied to ensure that the appropriate quantity of thermal energy is applied to shape-memory actuator 112 regardless of the battery voltage level.

SMA controller 1608 may process "on-time" signal 1606 and battery voltage signal 1610 to determine the appropriate SMA drive signal 1612 to apply to shape-memory actuator

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112. One example of SMA drive signal 1612 may be a series of binary pulses in which the amplitude of SMA drive signal 1612 essentially controls the stroke length of shape-memory actuator 112 (and therefore pump assembly 106) and the duty cycle of SMA drive signal 1612 essentially controls the stroke rate of shape-memory actuator 112 (and therefore pump assembly 106). Further, since SMA drive signal 1612 is indicative of a differential volume (i.e., the volume infused during each cycle of shape memory actuator 112), SMA drive signal 1612 may be integrated by discrete time integrator 1614 to generate volume signal 1616 which may be indicative of the total quantity of infusible fluid infused during a plurality of cycles of shape memory actuator 112. For example, since (as discussed above) it may take ten cycles of shape memory actuator 112 (at 0.010 units per cycle) to infuse 0.100 units of infusible fluid, discrete time integrator 1614 may integrate SMA drive signal 1612 over these ten cycles to determine the total quantity infused of infusible fluid (as represented by volume signal 1616).

SMA drive signal 1612 may actuate pump assembly 106 for e.g. one cycle, resulting in the filling of volume sensor chamber 620 included within volume sensor assembly 148. Infusion pump assembly 100 may then make a first measurement of the quantity of fluid included within volume sensor chamber 620 (as discussed above). Further and as discussed above, measurement valve assembly 610 may be subsequently energized, resulting in all or a portion of the fluid within volume sensor chamber 620 being delivered to the user. Infusion pump assembly 100 may then make a measurement of the quantity of fluid included within volume sensor chamber 620 (as described above) and use those two measurements to determine V_2 (i.e., the actual volume of infusible fluid delivered to the user during the current cycle of shape memory actuator 112). Once determined, V_2 (i.e., as represented by signal 1618) may be provided (i.e., fed back) to volume controller 1602 for comparison to the earlier-received target differential volume.

Continuing with the above-stated example in which the differential target volume was 0.010 units of infusible fluid, assume that V_2 (i.e., as represented by signal 1618) identifies 0.009 units of infusible fluid as having been delivered to the user. Accordingly, infusion pump assembly 100 may increase the next differential target volume to 0.011 units to offset the earlier 0.001 unit shortage. Accordingly and as discussed above, the amplitude and/or duty cycle of SMA drive signal 1612 may be increased when delivering the next basal dose of the infusible fluid to the user. This process may be repeated for the remaining nine cycles of shape memory actuator 112 (as discussed above) and discrete time integrator 1614 may continue to integrate SMA drive signal 1612 (to generate volume signal 1616) which may define the total quantity of infusible fluid delivered to the user.

Referring also to FIG. 111, there is shown one possible embodiment of volume controller 1602. In this particular implementation, volume controller 1602 may include PI (proportional-integrator) controller 1650. Volume controller 1602 may include feed forward controller 1652 for setting an initial "guess" concerning "on-time" signal 1606. For example, for the situation described above in which target differential volume signal 1600 identifies a partial basal dose of 0.010 units of infusible fluid per cycle of shape memory actuator 112, feed forward controller 1652 may define an initial "on-time" of e.g., one millisecond. Feed forward controller 1652 may include e.g., a lookup table that define an initial "on-time" that is based, at least in part, upon target differential volume signal 1600. Volume controller 1602 may further include discrete time integrator 1654 for inte-

grating target differential volume signal **1600** and discrete time integrator **1656** for integrating V_2 (i.e., as represented by signal **1618**).

Referring also to FIG. **112**, there is shown one possible embodiment of feed forward controller **1652**. In this particular implementation, feed forward controller **1652** may define a constant value signal **1658** and may include amplifier **1660** (e.g., a unity gain amplifier), the output of which may be summed with constant value signal **1658** at summing node **1662**. The resulting summed signal (i.e., signal **1664**) may be provided to as an input signal to e.g., lookup table **1666**, which may be processed to generate the output signal of feed forward controller **1652**.

As discussed above, pump assembly **106** may be controlled by shape memory actuator **112**. Further and as discussed above, SMA controller **1608** may process “on-time” signal **1606** and battery voltage signal **1610** to determine the appropriate SMA drive signal **1612** to apply to shape-memory actuator **112**.

Referring also to FIGS. **113-114**, there is shown one particular implementation of SMA controller **1608**. As discussed above, SMA controller **1608** may be responsive to “on-time” signal **1606** and battery voltage signal **1610** and may provide SMA drive signal **1612** to shape-memory actuator **112**. SMA controller **1608** may include a feedback loop (including unit delay **1700**), the output of which may be multiplied with battery voltage signal **1610** at multiplier **1702**. The output of multiplier **1702** may be amplified with e.g., unity gain amplifier **1704**. The output of amplifier **1704** may be applied to the negative input of summing node **1706** (to which “on-time” signal **1606** is applied). The output of summing node **1706** may be amplified (via e.g., unity gain amplifier **1708**). SMA controller may also include feed forward controller **1710** to provide an initial value for SMA drive signal **1612** (in a fashion similar to feed forward controller **1652** of volume controller **1602**; See FIG. **112**). The output of feed forward controller **1710** may be summed at summing node **1712** with the output of amplifier **1708** and an integrated representation (i.e., signal **1714**) of the output of amplifier **1708** to form SMA drive signal **1612**.

SMA drive signal **1612** may be provided to control circuitry that effectuates the application of power to shape-memory actuator **112**. For example, SMA drive signal **1612** may be applied to switching assembly **1716** that may selectively apply current signal **1718** (supplied from battery **606**) and/or fixed signal **1720** to shape-memory actuator. For example, SMA drive signal **1612** may effectuate the application of energy (supplied from battery **606** via current signal **1718**) via switching assembly **1716** in a manner that achieves the duty cycle defined by SMA drive signal **1612**. Unit delay **1722** may generate a delayed version of the signal applied to shape-memory actuator **112** to form battery voltage signal **1610** (which may be applied to SMA controller **1608**).

When applying power to shape-memory actuator **112**, voltage may be applied for a fixed amount of time and: a) at a fixed duty cycle with an unregulated voltage; b) at a fixed duty cycle with a regulated voltage; c) at a variable duty cycle based upon a measured current value; d) at a variable duty cycle based upon a measured voltage value; and e) at a variable duty cycle based upon the square of a measured voltage value. Alternatively, voltage may be applied to shape-memory actuator **112** for a variable amount of time based upon a measured impedance.

When applying an unregulated voltage for a fixed amount of time at a fixed duty cycle, inner loop feedback may not

be used and shape memory actuator may be driven at a fixed duty cycle and with an on-time determined by the outer volume loop.

When applying a regulated voltage for a fixed amount of time at a fixed duty cycle, inner loop feedback may not be used and shape memory actuator **112** may be driven at a fixed duty cycle and with an on-time determined by the outer volume loop.

When applying an unregulated voltage at a variable duty cycle based upon a measured current value, the actual current applied to shape-memory actuator **112** may be measured and the duty cycle may be adjusted during the actuation of shape-memory actuator **112** to maintain the correct mean current.

When applying an unregulated voltage at a variable duty cycle based upon a measured voltage value, the actual voltage applied to shape-memory actuator **112** may be measured and the duty cycle may be adjusted during the actuation of shape-memory actuator **112** to maintain the correct mean voltage.

When applying an unregulated voltage at a variable duty cycle based upon the square of a measured voltage value, the actual voltage applied to shape-memory actuator **112** may be measured and the duty cycle may be adjusted during the actuation of shape-memory actuator **112** to maintain the square of the voltage at a level required to provide the desired level of power to shape-memory actuator **112** (based upon the impedance of shape-memory actuator **112**).

Referring also to FIG. **114A-114B**, there is shown other implementations of SMA controller **1608**. Specifically, FIG. **114A** is an electrical schematic that includes a microprocessor and various control loops that may be configured to provide a PWM signal that may open and close the switch assembly. The switch assembly may control the current that is allowed to flow through the shape memory actuator. The battery may provide the current to the shape memory actuator. Further, **114B** discloses a volume controller and an inner shape memory actuator controller. The shape memory actuator controller may provide a PWM signal to the pump, which may be modified based on the battery voltage. This may occur for a fixed ontime, the result being a volume that may be measured by volume sensor assembly **148** and fed back into the volume controller.

In our preferred embodiment, we vary the duty cycle based on the measured battery voltage to give you approximately consistent power. We adjust the duty cycle to compensate for a lower battery voltage. Battery voltage may change for two reasons: 1) as batteries are discharged, the voltage slowly decreases; and 2) when you apply a load to a battery it has an internal impedance so its voltage dips. This is something that happens in any type of system, and we compensate for that by adjusting the duty cycle, thus mitigating the lower or varying battery voltage. Battery voltage may be measured by the microprocessor. In other systems: 1) voltage may be regulated (put a regulator to maintain the voltage at a steady voltage); 2) feedback based on something else (i.e., speed or position of a motor, not necessarily measuring the battery voltage).

Other configurations may be utilized to control the shape memory actuator. For example: A) the shape memory actuator may be controlled at fixed duty cycle with unregulated voltage. As voltage varies, the repeatability of heating the shape memory actuator is reduced. B) a fixed duty cycle, regulated voltage may be utilized which compensate for changes in battery voltage. However, regulate the voltage down is less efficient due to energy of energy. C) the duty cycle may be varied based on changes in current (which may

required more complicated measurement circuitry. D) The duty cycle may be varied based on measured voltage. E) The duty cycle may be varied based upon the square of the current, or the square of the voltage divided by resistance. F) the voltage may be applied for a variable amount of time based on the measured impedance (e.g., may measure impedance using Wheatstone gauge (not shown)). The impedance of the shape memory actuator may be correlated to strain (i.e., may correlate how much the SMA moves based on its impedance).

Referring also to FIG. 115 and as discussed above, to enhance the safety of infusion pump assembly 100, electrical control assembly 110 may include two separate and distinct microprocessors, namely supervisor processor 1800 and command processor 1802. Specifically, command processor 1802 may perform the functions discussed above (e.g., generating SMA drive signal 1612) and may control relay/switch assemblies 1804, 1806 that control the functionality of (in this example) shape memory actuators 112, 632 (respectively). Command processor 1802 may receive feedback from signal conditioner 1808 concerning the condition (e.g., voltage level) of the voltage signal applied to shape memory actuators 112, 632. Command processor 1800 may control relay/switch assembly 1810 independently of relay/switch assemblies 1804, 1806. Accordingly, when an infusion event is desired, both of supervisor processor 1800 and command processor 1802 must agree that the infusion event is proper and must both actuate their respective relays/switches. In the event that either of supervisor processor 1800 and command processor 1802 fails to actuate their respective relays/switches, the infusion event will not occur. Accordingly through the use of supervisor processor 1800 and command processor 1802 and the cooperation and concurrence that must occur, the safety of infusion pump assembly 100 is enhanced.

The supervisor processor may prevent the command processor from delivering when it is not supposed and also may alarm if the command processor does not deliver when it should be delivering. The supervisor processor may deactivate the relay/switch assembly if the command processor actuates the wrong switch, or if the command processor it tries to apply power for too long.

The supervisor processor may redundantly doing calculations for how much insulin should be delivered (i.e., double checking the calculations of the command processor). Command processor may decide the delivery schedule, and the supervisor processor may redundantly check those calculations.

Supervisor also redundantly holds the profiles (delivery profiles) in RAM, so the command processor may be doing the correct calculations, but if it has bad RAM, would cause the command to come up with the wrong result. The Supervisor uses its local copy of the basal profile, etc., to double check.

Supervisor can double check AVS measurements, looks at the AVS calculations and applies safety checks. Every time AVS measurement is taken, it double checks.

Referring also to FIG. 116, one or more of supervisor processor 1800 and command processor 1802 may perform diagnostics on various portions of infusion pump assembly 100. For example, voltage dividers 1812, 1814 may be configured to monitor the voltages (V1 & V2 respectively) sensed at distal ends of e.g., shape memory actuator 112. The value of voltages V1 & V2 in combination with the knowledge of the signals applied to relay/switch assemblies 1804, 1810 may allow for diagnostics to be performed on various

components of the circuit shown in FIG. 116 (in a manner similar to that shown in illustrative diagnostic table 1816).

As discussed above and as illustrated in FIGS. 115-116, to enhance the safety of infusion pump assembly 100, electrical control assembly 110 may include a plurality of microprocessors (e.g., supervisor processor 1800 and command processor 1802), each of which may be required to interact and concur in order to effectuate the delivery of a dose of the infusible fluid. In the event that the microprocessors fail to interact/concur, the delivery of the dose of infusible fluid may fail and one or more alarms may be triggered, thus enhancing the safety and reliability of infusion pump assembly 100.

A master alarm may be utilized that tracks the volume error over time. Accordingly, if the sum of the errors becomes too large, the master alarm may be initiated, indicating that something may be wrong with the system. Accordingly, the master alarm may be indicative of a total volume comparison being performed and a discrepancy being noticed. A typical value of the discrepancy required to initiate the master alarm may be 1.00 milliliters. The master alarm may monitor the sum in a leaky fashion (i.e., Inaccuracies have a time horizon).

Referring also to FIGS. 117A-117B, there is shown one such illustrative example of such interaction amongst multiple microprocessors during the delivery of a dose of the infusible fluid. Specifically, command processor 1802 may first determine 1900 the initial volume of infusible fluid within volume sensor chamber 620. Command processor 1802 may then provide 1902 a "pump power request" message to supervisor processor 1800. Upon receiving 1904 the "pump power request" message, supervisor processor 1800 may e.g., energize 1906 relay/switch 1810 (thus energizing shape memory actuator 112) and may send 1908 a "pump power on" message to command processor 1802. Upon receiving 1910 the "pump power on" message, command processor 1802 may actuate 1912 e.g., pump assembly 106 (by energizing relay/switch 1804), during which time supervisor processor 1800 may monitor 1914 the actuation of e.g., pump assembly 106.

Once actuation of pump assembly 106 is complete, command processor 1802 may provide 1914 a "pump power off" message to supervisor processor 1800. Upon receiving 1916 the "pump power off" message, supervisor processor 1800 may deenergize 1918 relay/switch 1810 and provide 1920 a "pump power off" message to command processor 1802. Upon receiving 1922 the "pump power off" message, command processor 1802 may measure 1924 the quantity of infusible fluid pumped by pump assembly 106. This may be accomplished by measuring the current quantity of fluid within volume sensor chamber 620 and comparing it with the quantity determined above (in step 1900). Once determined 1924, command processor 1802 may provide 1926 a "valve open power request" message to supervisor processor 1800. Upon receiving 1928 the "valve open power request" message, supervisor processor 1800 may energize 1930 relay/switch 1810 (thus energizing shape memory actuator 632) and may send 1932 a "valve open power on" message to command processor 1802. Upon receiving 1934 the "valve open power on" message, command processor 1802 may actuate 1936 e.g., measurement valve assembly 610 (by energizing relay/switch 1806), during which time supervisor processor 1800 may monitor 1938 the actuation of e.g., measurement valve assembly 610.

Once actuation of measurement valve assembly 610 is complete, command processor 1802 may provide 1940 a "valve power off" message to supervisor processor 1800.

Upon receiving 1942 the “valve power off” message, supervisor processor 1800 may deenergize 1944 relay/switch 1810 and provide 1946 a “valve power off” message to command processor 1802.

Upon receiving 1948 the “valve power off” message, command processor 1802 may provide 1950 a “valve close power request” message to supervisor processor 1800. Upon receiving 1952 the “valve close power request” message, supervisor processor 1800 may energize 1954 relay/switch 1810 (thus energizing shape memory actuator 652) and may send 1956 a “power on” message to command processor 1802. Upon receiving 1958 the “power on” message, command processor 1802 may actuate 1960 an energizing relay/switch (not shown) that is configured to energize shape memory actuator 652, during which time supervisor processor 1800 may monitor 1962 the actuation of e.g., shape memory actuator 652.

As discussed above (and referring temporarily to FIGS. 26A, 26B, 27A, 27B & 28), shape memory actuator 652 may be anchored on a first end using electrical contact 654. The other end of shape memory actuator 652 may be connected to bracket assembly 656. When shape memory actuator 652 is activated, shape memory actuator 652 may pull bracket assembly 656 forward and release valve assembly 634. As such, measurement valve assembly 610 may be activated via shape memory actuator 632. Once measurement valve assembly 610 has been activated, bracket assembly 656 may automatically latch valve assembly 610 in the activated position. Actuating shape memory actuator 652 may pull bracket assembly 656 forward and release valve assembly 634. Assuming shape memory actuator 632 is no longer activated, measurement valve assembly 610 may move to a de-activated state once bracket assembly 656 has released valve assembly 634. Accordingly, by actuating shape memory actuator 652, measurement valve assembly 610 may be deactivated.

Once actuation of shape memory actuator 652 is complete, command processor 1802 may provide 1964 a “power off” message to supervisor processor 1800. Upon receiving 1966 the “power off” message, supervisor processor 1800 may deenergize 1968 relay/switch 1810 and may provide 1970 a “power off” message to command processor 1802. Upon receiving 1972 the “power off” message, command processor 1802 may determine the quantity of infusible fluid within volume sensor chamber 620, thus allowing command processor 1802 to compare this measured quantity to the quantity determined above (in step 1924) to determine 1974 the quantity of infusible fluid delivered to the user.

In the event that the quantity of infusible fluid delivered 1974 to the user is less than the quantity of infusible fluid specified for the basal/bolus infusion event, the above-described procedure may be repeated (via loop 1976).

Referring also to FIG. 118, there is shown another illustrative example of the interaction amongst processors 1800, 1802, this time during the scheduling of a dose of infusible fluid. Command processor 1802 may monitor 2000, 2002 for the receipt of a basal scheduling message or a bolus request message (respectively). Upon receipt 2000, 2002 of either of these messages, command processor 1802 may set 2004 the desired delivery volume and may provide 2006 a “delivery request” message to supervisor processor 1800. Upon receiving 2008 the “delivery request” message, supervisor processor 1800 may verify 2010 the volume defined 2004 by command processor 1802. Once verified 2010, supervisor processor 1800 may provide 2012 a “delivery accepted” message to command processor 1802. Upon receipt 2014 of the “delivery accepted” message, command processor 1802

may update 2016 the controller (e.g., the controller discussed above and illustrated in FIG. 110) and execute 2018 delivery of the basal/bolus dose of infusible fluid. Command processor 1808 may monitor and update 2022 the total quantity of infusible fluid delivered to the user (as discussed above and illustrated in FIGS. 117A-117B). Once the appropriate quantity of infusible fluid is delivered to the user, command processor 1802 may provide 2024 a “delivery done” message to supervisor processor 1800. Upon receipt 2026 of the “delivery done” message, supervisor processor 1800 may update 2028 the total quantity of infusible fluid delivered to the user. In the event that the total quantity of infusible fluid delivered 2018 to the user is less than the quantity defined above (in step 2004), the infusion process discussed above may be repeated (via loop 2030).

Referring also to FIG. 119, there is shown an example of the manner in which supervisor processor 1800 and command processor 1802 may interact while effectuating a volume measurements via volume sensor assembly 148 (as described above).

Specifically, command processor 1802 may initialize 2050 volume sensor assembly 148 and begin collecting 2052 data from volume sensor assembly 148, the process of which may be repeated for each frequency utilized in the above-described sine sweep. Each time that data is collected for a particular sweep frequency, a data point message may be provided 2054 from command processor 1802, which may be received 2056 by supervisor processor 1800.

Once data collection 2052 is completed for the entire sine sweep, command processor 1802 may estimate 2058 the volume of infusible fluid delivered by infusion pump assembly 100. Command processor 1802 may provide 2060 a volume estimate message to supervisor processor 1800. Upon receiving 2062 this volume estimate message, supervisor processor 1800 may check (i.e., confirm) 2064 the volume estimate message. Once checked (i.e., confirmed), supervisor processor 1800 may provide 2066 a verification message to command processor 1802. Once received 2068 from supervisor processor 1800, command processor 1802 may set the measurement status for the dose of infusible fluid delivered by volume sensor assembly 148.

As discussed above and referring temporarily to FIG. 11), the various embodiments of the infusion pump assembly (e.g., infusion pump assembly 100, 100', 400, 500) discussed above may be configured via a remote control assembly 300. When configurable via remote control assembly 300, the infusion pump assembly may include telemetry circuitry (not shown) that allows for communication (e.g., wired or wireless) between the infusion pump assembly and e.g., remote control assembly 300, thus allowing remote control assembly 300 to remotely control the infusion pump assembly. Remote control assembly 300 (which may also include telemetry circuitry (not shown) and may be capable of communicating with the infusion pump assembly) may include display assembly 302 and input assembly 304. Input assembly 304 may include slider assembly 306 and switch assemblies 308, 310. In other embodiments, the input assembly may include a jog wheel, a plurality of switch assemblies, or the like. Remote control assembly 300 may allow the user to program basal and bolus delivery events.

Remote control assembly 300 may include two processors, one processor (e.g., which may include, but is not limited to a CC2510 microcontroller/RF transceiver, available from Chipcon AS, of Oslo, Norway) may be dedicated to radio communication, e.g., for communicating with infusion pump assembly 100, 100', 400, 500. The second processor included within remote control assembly (which may

include but are not limited to an ARM920T and an ARM922T manufactured by ARM Holdings PLC of the United Kingdom) may be a command processor and may perform data processing tasks associated with e.g., configuring infusion pump assembly **100**, **100'**, **400**, **500**.

Further and as discussed above, one embodiment of electrical control assembly **816** may include three micro-processors. One processor (e.g., which may include, but is not limited to a CC2510 microcontroller/RF transceiver, available from Chipcon AS, of Oslo, Norway) may be dedicated to radio communication, e.g., for communicating with a remote control assembly **300**. Two additional micro-processors (e.g., supervisor processor **1800** and command processor **1802**) may effectuate the delivery of the infusible fluid (as discussed above). Examples of supervisor processor **1800** and command processor **1802** may include, but is not limited to an MSP430 microcontroller, available from Texas Instruments Inc. of Dallas, Texas.

The OS may be a non-preemptive scheduling system, in that all tasks may run to completion before the next task is allowed to run regardless of priority. Additionally, context switches may not be performed. When a task completes executing, the highest priority task that is currently scheduled to run may then be executed. If no tasks are scheduled to execute, the OS may place the processor (e.g., supervisor processor **1800** and/or command processor **1802**) into a low power sleep mode and may wake when the next task is scheduled. The OS may only be used to manage main loop code and may leave interrupt-based functionality unaffected.

The OS may be written to take advantage of the C++ language. Inheritance as well as virtual functions may be key elements of the design, allowing for easy creation, scheduling and managing of tasks.

At the base of the OS infrastructure may be the ability to keep track of system time and controlling the ability to place the processor in Low Power Mode (LPM; also known as sleep mode). This functionality along with the control and configuration of all system clocks may be encapsulated by the SysClocks class.

The SysClocks class may contain the functionality to place the processor (e.g., supervisor processor **1800** and/or command processor **1802**) into LPM to reduce energy consumption. While in LPM, the slow real time clock may continue to run while the fast system clock that runs the CPU core and most peripherals may be disabled.

Placing the processor into LPM may always be done by the provided SysClocks function. This function may contain all required power down and power up sequences resulting in consistency whenever entering or exiting LPM. Waking from LPM may be initiated by any interrupts based on the slow clock.

The OS may keep track of three aspects of time: seconds, milliseconds and the time of day. Concerning seconds, SysClocks may count seconds starting when the processor comes out of reset. The second counter may be based on the slow system clocks and, therefore, may increment regardless of whether the processor is in LPM or at full power. As a result, it is the boundary at which the processor may wake from sleep to execute previously scheduled tasks. If a task is scheduled to run immediately from an interrupt service routine (ISR), the ISR may wake the processor from LPM on exit and the task may be executed immediately. Concerning milliseconds, in addition to counting the seconds since power on, SysClocks may also count milliseconds while the processor is in full power mode. Since the fast clock is stopped during LPM, the millisecond counter may not increment. Accordingly, whenever a task is scheduled to

execute based on milliseconds, the processor may not enter LPM. Concerning time of day, the time of day may be represented within SysClocks as seconds since a particular point time (e.g., seconds since 1 Jan. 2004).

The SysClocks class may provide useful functionality to be used throughout the Command and Supervisor project code base. The code delays may be necessary to allow hardware to settle or actions to be completed. SysClocks may provide two forms of delays, a delay based on seconds or a delay based on milliseconds. When a delay is used, the processor may simply wait until the desired time has passed before continue with its current code path. Only ISRs may be executed during this time. SysClocks may provide all of the required functionality to set or retrieve the current time of day.

The word "task" may be associated with more complex scheduling systems; therefore within the OS, task may be represented by and referred to as Managed Functions. The ManagedFunc class may be an abstract base class that provides all the necessary control members and functionality to manage and schedule the desired functionality.

The ManagedFunc base class may have five control members, two scheduling manipulation member functions, and one pure virtual execute function that may contain the managed functionality. All of the ManagedFunc control members may be hidden from the derived class and may only be directly set by the derived class during creation, thus simplifying the use and enhancing the safety of infusion pump assembly **100**, **100'**, **400**, **500**.

The Function ID may be set at the time of creation and may never be changed. All Function IDs may be defined within a single .h file, and the base ManagedFunc constructor may strongly enforce that the same ID may not be used for more than one managed function. The ID may also define the priority of a function (with respect to other functions) based upon the function ID assigned, wherein higher priority functions are assigned lower function IDs. The highest priority task that is currently scheduled to execute may execute before lower priority tasks.

All other control members may be used to represent the function's current scheduled state, when it should be executed, and if (upon execution) the function should be rescheduled to execute in a previously set amount of time. Manipulation of these controls and states may be allowed but only through the public member functions (thus enforcing safety controls on all settings).

To control the scheduling of a managed function, the set start and set repeat functions may be used. Each of these member functions may be a simple interface allowing the ability to configure or disable repeat settings as well as control whether a managed function is inactive, scheduled by seconds, milliseconds, or time of day.

Through inheritance, creating a Managed Function may be done by creating a derived class and defining the pure virtual 'execute' function containing the code that needs to be under scheduling control. The ManagedFunc base class constructor may be based upon the unique ID of a function, but may also be used to set default control values to be used at start up.

For example to create a function that runs thirty seconds after start up and every 15 seconds thereafter, the desired code is placed into the virtual execute function and the function ID, scheduled by second state, thirty second start time, and repeat setting of fifteen seconds is provided to the constructor.

The following is an illustrative code example concerning the creation of a managed function. In this particular

example, a “heartbeat” function is created that is scheduled to execute for the first time one second after startup of infusion pump assembly **100, 100', 400, 500** and execute every ten seconds thereafter:

```
#include "ManagedFunc.h"
// The SendGoodFunc is a "heartbeat" status message
class SendGoodFunc : public ManagedFunc
{
public:
    // Initialize the managed func to run 2 seconds after start up
    // and repeat every second.
    SendGoodFunc( ) :
        ManagedFunc(IPC_SEND_GOOD, SCHEDULED_SEC, 1, true,
            10) { };
    ~SendGoodFunc( ) { };
protected:
    void execute(void);
};
void SendGoodFunc::execute(void)
{
    // << code to send the heartbeat >>
}
SendGoodFunc g_sendGoodFunc;
// to manipulate the heartbeat timing simply call:
// g_sendGoodFunc.setFuncStart(...) or
// g_sendGoodFunc.setRepeat( ... )
```

The actual execution of the Managed Functions may be controlled and performed by the SleepManager class. The SleepManager may contain the actual prioritized list of managed functions. This prioritized list of functions may automatically be populated by the managed function creation process and may ensure that each function is created properly and has a unique ID.

The main role of the SleepManager class may be to have its ‘manage’ function called repeatedly from the processors main loop and/or from a endless while loop. Upon each call of manage, the SleepManager may execute all functions that are scheduled to run until the SleepManager has exhausted all scheduled functions; at which time the SleepManager may place the processor in LPM. Once the processor wakes from LPM, the manage function may be reentered until the processor is again ready to enter LPM (this process may be repeated until stopped, e.g., by a user or by the system).

If the processor has to be kept in full power mode for an extended period of time (e.g., while an analog-to-digital conversion is being sampled), the SleepManager may provide functionality to disable entering LPM. While LPM is disabled, the manage function may continuously search for a scheduled task.

The SleepManager may also provide an interface to manipulate the scheduling and repeat settings of any managed function through the use of the unique ID of the function, which may allow any section of code to perform any required scheduling without having direct access to or unnecessary knowledge of the desired ManagedFunc object.

Radio circuitry included within each of infusion pump assembly **100, 100', 400, 500** and remote control assembly **300** may effectuate wireless communication between remote control assembly **300** and infusion pump assembly **100, 100', 400, 500**. A 2.4 GHz radio communications chip (e.g., a Texas Instruments CC2510 radio transceiver) with an internal 8051 microcontroller may be used for radio communications.

The radio link may balance the following three objectives: link availability; latency; and energy.

Concerning link availability, remote control assembly **300** may provide the primary means for controlling the infusion pump assembly **100, 100', 400, 500** and may provide

detailed feedback to the user via the graphical user interface (GUI) of remote control assembly **300**. Concerning latency, the communications system may be designed to provide for low latency to deliver data from remote control assembly **300** to the infusion pump assembly **100, 100', 400, 500** (and vice versa). Concerning energy, both remote control assembly **300** and infusion pump assembly **100, 100', 400, 500** may have a maximum energy expenditure for radio communications.

The radio link may support half-duplex communications. Remote control assembly **300** may be the master of the radio link, initiating all communications. Infusion pump assembly **100, 100', 400, 500** may only respond to communications and may never initiate communications. The use of such a radio communication system may provide various benefits, such as: increased security; a simplified design (e.g., for airplane use); and coordinated control of the radio link.

Referring also to FIG. **120A**, there is shown one illustrative example of the various software layers of the radio communication system discussed above.

The radio processors included within remote control assembly **300** and infusion pump assembly **100, 100', 400, 500** may transfer messaging packets between an SPI port and a 2.4 GHz radio link (and vice versa). The radio may always be the SPI slave. On infusion pump assembly **100, 100', 400, 500**, radio processor (PRP) **1818** (See FIGS. **115-116**) may service two additional nodes over the SPI port that are upstream (namely command processor **1800** and supervisor processor **1802**. In some embodiments, on remote control assembly **300**, the radio processor (CRP) may service at least one additional node over the SPI port that may be either upstream or down stream, for example, in some embodiments, the above-described remote control processor (UI) and the Continuous Glucose Engine (CGE).

A messaging system may allow for communication of messages between various nodes in the network. The UI processor of remote control assembly **300** and e.g., supervisor processor **1800** may use the messaging system to configure and initiate some of the mode switching on the two system radios. It may be also used by the radios to convey radio and link status information to other nodes in the network.

When the radio of remote control assembly **300** wishes to gather channel statistics from the infusion pump assembly **100, 100', 400, 500** or update the master channel list of the radio of infusion pump assembly **100, 100', 400, 500**, the radio of remote control assembly **300** may use system messages. Synchronization for putting the new updated list into effect may use flags in the heartbeat messages to remove timing uncertainty.

The radio communication system may be written in C++ to be compatible with the messaging software. A four byte radio serial number may be used to address each radio node. A hash table may be used to provide a one-to-one translation between the device “readable” serial number string and the radio serial number. The hash table may provide a more randomized 8-bit logical address so that pumps (e.g., infusion pump assembly **100, 100', 400, 500**) or controllers with similar readable serial numbers are more likely to have unique logical addresses. Radio serial numbers may not have to be unique between pumps (e.g., infusion pump assembly **100, 100', 400, 500**) and controllers due to the unique roles each has in the radio protocol.

The radio serial number of remote control assembly **300** and the radio serial number of infusion pump assembly **100, 100', 400, 500** may be included in all radio packets except for the RF Pairing Request message that may only include

the radio serial number of remote control assembly **300**, thus ensuring that only occur with the remote control assembly/infusion pump assembly to which it is paired. The CC2510 may support a one byte logical node address and it may be advantageous to use one byte of the radio serial number as the logical node address to provide a level of filtering for incoming packets.

The Quiet_Radio signal may be used by the UI processor of remote control assembly **300** to prevent noise interference on the board of remote control assembly **300** by other systems on the board. When Quiet_Radio is asserted, the radio application of remote control assembly **300** may send a message to the radio of infusion pump assembly **100, 100', 400, 500** asserting Radio Quiet Mode for a pre-determined period of time. The Quiet_Radio feature may not be required based on noise interference levels measured on the PC board of remote control assembly **300**. During this period of time, the radio of remote control assembly **300** may stay in Sleep Mode 2 for up to a maximum of 100 ms. The radio of remote control assembly **300** may come out of Sleep Mode 2 when the Quiet_Radio signal is de-asserted or the maximum time period has expired. The UI processor of remote control assembly **300** may assert Quiet_Radio at least one radio communication's interval before the event needs to be asserted. The radio of remote control assembly **300** may inform the radio of infusion pump assembly **100, 100', 400, 500** that communications will be shutdown during this quiet period. The periodic radio link protocol may have status bits/bytes that accommodate the Quiet_Radio feature unless Quiet_Radio is not required.

The radio software may integrate with the messaging system and radio bootloader on the same processor, and may be verified using a throughput test. The radio software may integrate with the messaging system, SPI Driver using DMA, and radio bootloader, all on the same processor (e.g., the TI CC2510).

The radio of remote control assembly **300** may be configured to consume no more than 32 mAh in three days (assuming one hundred minutes of fast heartbeat mode communications per day). The radio of infusion pump assembly **100, 100', 400, 500** may be configured to consume no more than 25 mAh in three days (assuming one hundred minutes of fast heartbeat mode communications per day).

The maximum time to reacquire communications may be <6.1 seconds including connection request mode and acquisition mode. The radio of remote control assembly **300** may use the fast heartbeat mode or slow heartbeat mode setting to its advantage in order to conserve power and minimize latency to the user. The difference between the infusion pump assembly **100, 100', 400, 500** and remote control assembly **300** entering acquisition mode may be that the infusion pump assembly **100, 100', 400, 500** needs to enter acquisition mode often enough to ensure communications may be restored within the maximum latency period. However, the remote control assembly **300** may change how often to enter acquisition mode with the infusion pump assembly **100, 100', 400, 500** when in slow heartbeat mode and heartbeats are lost. The radio of remote control assembly **300** may have knowledge of the user GUI interaction, but the infusion pump assembly **100, 100', 400, 500** may not.

The radio of remote control assembly **300** may set the heartbeat period for both radios. The period may be selectable in order to optimize power and link latency depending on activity. The desired heartbeat period may be communicated in each heartbeat from the radio of remote control assembly **300** to the radio of infusion pump assembly **100, 100', 400, 500**. This may not exclusively establish the

heartbeat rate of infusion pump assembly **100, 100', 400, 500** due to other conditions that determine what mode to be in. When in fast heartbeat mode, the radio of remote control assembly **300** may set the heartbeat period to 20 ms if data packets are available to send or receive, thus providing low link latency communications when data is actively being exchanged.

When in fast heartbeat mode, the radio of remote control assembly **300** may set the heartbeat period to 60 ms four heartbeats after a data packet was last exchanged in either direction on the radio. Keeping the radio heartbeat period short after a data packet has been sent or received may assure that any data response packet may be also serviced using a low link latency. When in slow heartbeat mode, the heartbeat rate may be 2.00 seconds or 6.00 second, depending upon online or offline status respectively.

The infusion pump assembly **100, 100', 400, 500** may use the heartbeat rate set by the radio of remote control assembly **300**. The radio of remote control assembly **300** may support the following mode requests via the messaging system:

- Pairing Mode

- Connection Mode

- Acquisition Mode (includes the desired paired infusion pump assembly **100, 100', 400, 500** radio serial number)

- Sync Mode—Fast Heartbeat

- Sync Mode—Slow Heartbeat

- RF Off Mode

The radio of infusion pump assembly **100, 100', 400, 500** may support the following mode requests via the messaging system:

- Pairing Mode

- Acquisition Mode

- RF Off Mode

The radio may use a system message to obtain the local radio serial number. On remote control assembly **300**, the radio may get the serial number from the UI processor of remote control assembly **300**. The radio may use a system message to store the paired radio serial number.

Remote control assembly **300** and the radio of infusion pump assembly **100, 100', 400, 500** may issue a status message using the messaging system to the UI processor of remote control assembly **300** and command processor **1802** whenever the following status changes:

- Online Fast: Successful connection

- Online Fast: Change from Acquisition Mode to Fast Heartbeat Mode

- Online Slow: Successful request change from Fast Heartbeat to Slow Heartbeat

- Offline: Automatic change to Search Sync mode due to lack of heartbeat exchanges.

- Online Fast: Successful request change from Slow Heartbeat to Fast Heartbeat

- Offline: Bandwidth falls below 10% in Sync Mode

- Online: Bandwidth rises above 10% in Search Sync mode

- Offline: Successful request change to RF Off Mode

The radio configuration message may be used to configure the number of radio retries. This message may be sent over the messaging system. The UI processor of remote control assembly **300** will send this command to both the radio of remote control assembly **300** and the radio of infusion pump assembly **100, 100', 400, 500** to configure these radio settings.

There may be two parameters in the radio configuration message: namely the number of RF retries (e.g., the value may be from 0 to 10); and the radio offline parameters (e.g., the value may be from 1 to 100 in percent of bandwidth).

The radio application on both the remote control assembly **300** and infusion pump assembly **100, 100', 400, 500** may have an API that allows the messaging system to configure the number of RF retries and radio offline parameters.

The following parameters may be recommended for the radio hardware configuration:

- Base Radio Specifications
- MSK
- 250 kbps over air baud rate
- Up to 84 channels
- Channel spacing 1000 kHz
- Filter bandwidth 812 kHz
- No Manchester encoding
- Data whitening
- 4 byte preamble
- 4 byte sync (word)
- CRC appended to packet
- LQI (Link Quality Indicator) appended to packet
- Automatic CRC filtering enabled

Forward Error Correction (FEC) may or may not be utilized. Although Forward Error Correction (FEC) may be used to increase the effective signal dynamic range by approximately 3 dB, FEC requires fixed packet sizes and doubles the number of over the air bits for the same fixed size message.

The radio may function within 1.83 meters distance under nominal operating conditions (except in pairing mode). It may be a goal that the radio function within 7.32 meters distance under nominal operating conditions. The transmit power level may be 0 dBm (except in pairing mode) and the transmit power level in pairing mode may be -22 dBm. Since the desired radio node address of infusion pump assembly **100, 100', 400, 500** may be not known by the remote control assembly **300** in pairing mode, both infusion pump assembly **100, 100', 400, 500** and remote control assembly **300** may use a lower transmit power to reduce the likelihood of inadvertently pairing with another infusion pump assembly.

AES Encryption may be used for all packets but may not be required, as the Texas Instruments CC2510 radio transceiver includes this functionality. If AES encryption is used, fixed keys may be utilized, as fixed keys provide a quick way to enable encryption without passing keys. However, key exchange may be provided for in future versions of infusion pump assembly **100, 100', 400, 500**. The fixed keys may be contained in one separate header source file with no other variables but the fixed keys data, thus allowing for easier management of read access of the file.

The radio software may support the following eight modes:

- Pairing Mode
- RF Off Mode
- Connection Mode
- Acquisition Mode
- Fast Heartbeat Mode
- Slow Heartbeat Mode
- Search Sync Mode
- Sync'ed Acquisition Mode

which are graphically depicted in FIGS. 120B-120C.

Pairing may be the process of exchanging radio serial numbers between remote control assembly **300** and infusion pump assembly **100, 100', 400, 500**. Remote control assembly **300** may be "paired" with infusion pump assembly **100, 100', 400, 500** when infusion pump assembly **100, 100', 400, 500** knows its serial number. Infusion pump assembly **100,**

100', 400, 500 may be "paired" with remote control assembly **300** when remote control assembly **300** knows its serial number.

Pairing mode (which is graphically depicted in FIG. 120D) may require that four messages to be exchanged over the RF link:

RF Pairing Request (broadcast from Remote control assembly **300** to any Infusion pump assembly **100, 100', 400, 500**)

RF Pairing Acknowledge (from Infusion pump assembly **100, 100', 400, 500** to Remote control assembly **300**)

RF Pairing Confirm Request (from Remote control assembly **300** to Infusion pump assembly **100, 100', 400, 500**)

RF Pairing Confirm Acknowledge (from Infusion pump assembly **100, 100', 400, 500** to Remote control assembly **300**)

Additionally, remote control assembly **300** may cancel the pairing process at any time via the RF pairing abort message (from remote control assembly **300** to infusion pump assembly **100, 100', 400, 500**). Pairing mode may not support messaging system data transfers.

The radio of infusion pump assembly **100, 100', 400, 500** may enter pairing mode upon receiving a pairing mode request message. It may be the responsibility of supervisor processor **1800** on infusion pump assembly **100, 100', 400, 500** to request the radio to enter pairing mode if there is no disposable attached to infusion pump assembly **100, 100', 400, 500** and the user has pressed the button of infusion pump assembly **100, 100', 400, 500** for six seconds. The radio of infusion pump assembly **100, 100', 400, 500** may set the appropriate transmit power level for pairing mode. Infusion pump assembly **100, 100', 400, 500** may only be paired with one remote control assembly **300** at a time.

Upon receiving the first valid RF pairing request message while in pairing mode, the radio of infusion pump assembly **100, 100', 400, 500** may use the serial number of remote control assembly **300** for the duration of pairing mode and respond with an RF pairing acknowledge message containing the radio serial number infusion pump assembly **100, 100', 400, 500**.

The radio of infusion pump assembly **100, 100', 400, 500** may timeout of pairing mode automatically after 2.0 ± 0.2 seconds if no RF pairing request is received. The radio of infusion pump assembly **100, 100', 400, 500** may issue a pairing request received message after transmitting the RF pairing acknowledge. This message to supervisor processors will allow feedback to the user during the pairing confirm process. The radio of infusion pump assembly **100, 100', 400, 500** may automatically timeout of pairing mode in 1.0 ± 0.1 minutes after sending an RF pairing acknowledge unless an RF pairing confirm request is received. The radio of infusion pump assembly **100, 100', 400, 500** may issue a store paired radio serial number message if an RF pairing confirm request message is received after receiving a RF pairing request message. This action may store the radio serial number of remote control assembly **300** in the non-volatile memory of infusion pump assembly **100, 100', 400, 500** and may overwrite the existing pairing data for the infusion pump assembly **100, 100', 400, 500**.

The radio of infusion pump assembly **100, 100', 400, 500** may transmit an RF pairing confirm acknowledge and exit pairing mode after the acknowledgment from the store paired radio serial number message is received. This may be the normal exit of pairing mode on infusion pump assembly **100, 100', 400, 500** and may result in infusion pump

assembly 100, 100', 400, 500 powering down until connection mode or pairing mode entered by the user.

If the radio of infusion pump assembly 100, 100', 400, 500 exits pairing mode upon successfully receiving a pairing confirm request message, then the radio of infusion pump assembly 100, 100', 400, 500 may revert to the newly paired remote control assembly 300 and may send a pairing completion success message to command processor 1802. The radio of infusion pump assembly 100, 100', 400, 500 may exit pairing mode upon receiving an RF pairing abort message. The radio of infusion pump assembly 100, 100', 400, 500 may exit pairing mode upon receiving a pairing abort request message addressed to it. This may allow command processor 1802 or supervisor processor 1800 to abort the pairing process locally on the infusion pump assembly 100, 100', 400, 500.

The radio of remote control assembly 300 may enter pairing mode upon receiving a pairing mode request message. It may be the responsibility of the UI processor of remote control assembly 300 to request that the radio enter pairing mode under the appropriate conditions. The radio of remote control assembly 300 may set the appropriate transmit power level for pairing mode. The radio of remote control assembly 300 may transmit RF pairing requests until an RF pairing acknowledge is received or pairing is aborted.

The radio of remote control assembly 300 may automatically abort pairing mode if the RF pairing acknowledge message is not received within 30.0 ± 1.0 seconds after entering pairing mode. Upon receiving the first valid RF pairing acknowledge message while in pairing mode, the radio of remote control assembly 300 may send a pairing success message to the UI processor of remote control assembly 300 that includes the serial number of infusion pump assembly 100, 100', 400, 500 and may use that serial number for the duration of pairing mode. This message may provide a means for the UI processor of remote control assembly 300 to have the user confirm the serial number of the desired infusion pump assembly 100, 100', 400, 500. If the radio of remote control assembly 300 receives multiple responses (concerning a single pairing request) from infusion pump assembly 100, 100', 400, 500, the first valid one may be used.

The Radio of remote control assembly 300 may only accept an RF pairing confirm acknowledge messages after an RF pairing acknowledge is received while in pairing mode. The radio of remote control assembly 300 may transmit the RF pairing confirm message upon receiving a pair confirm request message from the UI processor of remote control assembly 300.

The radio of remote control assembly 300 may check that infusion pump assembly 100, 100', 400, 500 confirms the pairing before adding infusion pump assembly 100, 100', 400, 500 to the pairing list. The radio of remote control assembly 300 may issue a store paired radio serial number message if an RF pairing complete message is received. This action may allow the UI processor of remote control assembly 300 to store the new serial number of infusion pump assembly 100, 100', 400, 500 and provide user feedback of a successful pairing. It may be the responsibility of the UI processor of remote control assembly 300 to manage the list of paired infusion pump assemblies.

The radio of remote control assembly 300 may send an RF pairing abort message and exit pairing mode upon receiving a pairing abort request message. This may allow the UI processor of the remote control assembly 300 to abort the pairing process on both the remote control assembly 300 and acknowledged infusion pump assembly 100, 100', 400, 500.

In connection request mode, the radio of remote control assembly 300 may attempt to acquire each infusion pump assembly 100, 100', 400, 500 in its paired infusion pump assembly list and retrieve its "connection ready" status. The "connection" process (which is graphically depicted in FIG. 120E) may allow remote control assembly 300 to quickly identify one of its paired infusion pump assemblies that may be ready to be used. The radio of remote control assembly 300 may be capable of performing the connection request mode with up to six paired infusion pump assemblies. Connection request mode may be only supported on remote control assembly 300 and may be a special form of acquisition mode. In connection request mode, remote control assembly 300 may connect with the first infusion pump assembly to respond. However, each message may be directed to a specific infusion pump assembly serial number.

The radio of remote control assembly 300 may obtain the latest paired infusion pump assembly serial number list upon entering connection mode. The radio of remote control assembly 300 may enter connection mode upon receiving a connection mode request message. It may be the responsibility of the UI processor of remote control assembly 300 to request that the radio enter connection mode when it desires communications with a paired infusion pump assembly. The radio of remote control assembly 300 may issue a connection assessment message to the UI processor of remote control assembly 300 containing the radio serial number of the first infusion pump assembly, if any, that is "connection ready". The radio of remote control assembly 300 may generate the connection assessment message within thirty seconds of entering connection request mode. The radio of remote control assembly 300 may exit connection request mode upon receipt of the connection assessment acknowledgement and transition to fast heartbeat mode. The radio of remote control assembly 300 may exit connection request mode upon receipt of a connection request abort message from the UI processor of remote control assembly 300.

On remote control assembly 300, acquisition mode may be used to find a particular paired infusion pump assembly. The radio of remote control assembly 300 may send RF RUT (aRe yoU There) packets to the desired paired infusion pump assembly. If the infusion pump assembly receives the RF RUT message, it may respond to the radio of remote control assembly 300. Multiple channels may be used in the acquisition mode algorithm to improve the opportunity for the radio of remote control assembly 300 to find the paired infusion pump assembly.

The radio of remote control assembly 300 may enter acquisition mode upon receiving an acquisition mode request or fast heartbeat mode request message while in RF Off Mode. The radio of remote control assembly 300 may enter sync'd acquisition mode upon receiving an acquisition mode request or fast heartbeat mode request message while in search sync mode. It may be the responsibility of the UI processor of remote control assembly 300 to request that the radio enter acquisition mode when the RF link is off-line and remote control assembly 300 desires communications with infusion pump assembly 100, 100', 400, 500.

The radio of remote control assembly 300 may only communicate with one paired infusion pump assembly 100, 100', 400, 500 (except in pairing and connection modes). When communications are lost, the UI processor of remote control assembly 300 may use acquisition mode (at some periodic rate limited by the power budget) to attempt to restore communications.

Infusion pump assembly 100, 100', 400, 500 may enter acquisition mode under the following conditions:

When in Radio Off Mode and Acquisition Mode may be requested

When Search Sync Mode times out due to lack of heartbeats

Upon entering acquisition mode, the radio of infusion pump assembly **100, 100', 400, 500** may obtain the serial number of the last stored paired remote control assembly **300**. The radio of infusion pump assembly **100, 100', 400, 500** may only communicate with the remote control assembly to which it has been "paired" (except while in the "pairing request" mode). The radio of infusion pump assembly **100, 100', 400, 500** may transition from acquisition mode to fast heartbeat mode upon successfully acquiring synchronization with the remote control assembly **300**. The acquisition mode of infusion pump assembly **100, 100', 400, 500** may be capable of acquiring synchronization within 6.1 seconds, which may implies that the infusion pump assembly **100, 100', 400, 500** may always be listening at least every ~6 seconds when in acquisition mode.

Data packets may be sent between two paired devices when the two devices are in sync mode and online. The two devices may sync via a heartbeat packet before data packets are exchanged. Each radio may send data packets at known time intervals after the heartbeat exchange. The infusion pump assembly **100, 100', 400, 500** may adjust its timing to anticipate reception of a packet. The radio may support one data packet in each direction on each heartbeat. The radio may provide a negative response to a fast heartbeat mode request if the radio is offline. The radio of remote control assembly **300** may change to fast heartbeat mode if a system request for fast heartbeat mode is received while in slow heartbeat mode and the radio is online.

Upon transitioning to fast heartbeat mode from acquisition mode, the radio of remote control assembly **300** may send the master channel list message. The master channel list may be built by the radio of remote control assembly **300** and sent to the radio of infusion pump assembly **100, 100', 400, 500** to allow a selection of frequency hopping channels based on historical performance. When in fast heartbeat mode or slow heartbeat mode, periodic heartbeat messages may be exchanged between the radio of remote control assembly **300** and the radio of infusion pump assembly **100, 100', 400, 500**. The periodicity of these messages may be at the heartbeat rate. The heartbeat messages may allow data packet transfers to take place and may also exchange status information. The two radios may exchange the following status information: Quiet Mode, data availability, buffer availability, heartbeat rate, and prior channel performance. It may be a goal to keep the packet size of the heartbeat messages small in order to conserve power. The radio may provide for a maximum data packet size of eighty-two bytes when in Sync Mode. The messaging system may be designed to support packet payload sizes up to sixty-four bytes. This maximum size was selected as an optimal trade-off between minimum messages types and non-fragmented messages. The eighty-two bytes may be the maximum packet size of the messaging system including packet overhead.

The messaging system has an API that may allow the radio protocol to send an incoming radio packet to it. The messaging system may also have an API that allows the radio protocol to get a packet for transmission over the radio network. The messaging system may be responsible for packet routing between the radio protocol and the SPI port. Data packets may be given to the messaging system for processing. The messaging system may have an API that allows the radio protocol to obtain a count of the number of

data packets waiting to be sent over the radio network. The radio protocol may query the messaging system on each heartbeat to determine if data packets are available to send over the radio network. It may be desirable for the software to check the availability of a message just before the heartbeat is sent to minimize round trip message latency.

The radio protocol may be capable of buffering one incoming radio data packet and passing the packet to the messaging system. The radio protocol may send the data packet to the messaging system upon receipt of the data packet. The message system may be responsible for routing radio data packets to the proper destination node. The radio protocol may be capable of buffering one packet from the messaging system.

The radio protocol may be responsible for acknowledging receipt of valid data packets over the RF link via an RF ACK reply packet to the sending radio. The RF ACK packet may contain the source and destination radio serial numbers, RF ACK command identification, and sequence number of the data packet being acknowledged.

The radio transmitting a radio data packet may retransmit that radio data packet on the next heartbeat with the same sequence number if an RF ACK is not received and the retry count is within the maximum RF retries allowed. It may be expected that, from time to time, interference will corrupt a transmission on a particular frequency. An RF retry allows the same packet to be retransmitted at the next opportunity at a different frequency. The sequence number provides a means of uniquely identifying the packet over a short time window. The number of radio packet retries may be configurable using the radio configuration command. Allowing more retries may increase the probability of a packet being exchanged but introduces more latency for a round trip messages. The default number of radio retries at power up may be ten (i.e., the maximum transmission attempts before dropping the message).

A one byte (modulo 256) radio sequence number may be included in all radio data packets over the RF link. Since the radio may be responsible for retrying data packet transmission if not acknowledged, the sequence number may provide a way for the two radios to know if a data packet is a duplicate. The transmitted sequence number may be incremented for each new radio data packet and may be allowed to rollover. When a data packet is successfully received with the same sequence number as the previous successfully received data packet (and in the same direction), the data packet may be ACK'd and the received data packet discarded. This may remove duplicate packets generated by the RF protocol before they are introduced into the network. Note that it may be possible that multiple data packets in a row may need to be dropped with the same sequence number under extreme situations.

If a heartbeat is missed, the radio of remote control assembly **300** and the radio of infusion pump assembly **100, 100', 400, 500** may attempt to send and listen respectively for subsequent heartbeats. The radio of remote control assembly **300** and the radio of infusion pump assembly **100, 100', 400, 500** may automatically change from fast heartbeat mode or slow heartbeat mode to search sync mode if heartbeats are missed for two seconds. This may minimize power consumption when the link is lost by allowing the radios to continue to use their synchronization information, as two seconds allows sufficient time to hop through all channels.

The radio may be considered online while in the following modes:

Fast Heartbeat mode
Slow Heartbeat mode

as these are the only conditions where messaging system traffic may be exchanged. All other conditions may be considered offline.

The radio may initialize to radio off mode at the start of code execution from reset. When code first executes on the radio processor, the initial state may be the radio off mode to allow other processors to perform self-tests before requesting the radio to be active. This requirement does not intend to define the mode when waking from sleep mode. The radio may cease RF communications when set to radio off mode. On remote control assembly **300**, this mode may be intended for use on an airplane to suppress RF emissions. Since infusion pump assembly **100**, **100'**, **400**, **500** only responds to transmissions from remote control assembly **300** (which will have ceased transmitting in airplane mode), radio off mode may only be used on infusion pump assembly **100**, **100'**, **400**, **500** when charging.

Command processor **1802** may be informed of airplane mode and that, therefore, the RF was intentionally turned off on remote control assembly **300** so that it does not generate walk-away alerts. However, this may be completely hidden from the radio of infusion pump assembly **100**, **100'**, **400**, **500**.

The radio of remote control assembly **300** and the radio of infusion pump assembly **100**, **100'**, **400**, **500** may periodically attempt to exchange heartbeats in order to reestablish data bandwidth while in search sync mode. The radio of remote control assembly **300** may transition to radio off mode after twenty minutes of search sync mode with no heartbeats successfully exchanged.

The radio of infusion pump assembly **100**, **100'**, **400**, **500** may transition to acquisition mode after twenty minutes of search sync mode with no heartbeats successfully exchanged. Listening during pre-agreed time slots may be the most efficient use of power for infusion pump assembly **100**, **100'**, **400**, **500** to re-establish the RF link. After a loss of communications, the crystal tolerance and temperature drift may make it necessary to expand the receive window of infusion pump assembly **100**, **100'**, **400**, **500** over time. Staying in search sync mode for extended periods (e.g., 5-20 minutes) after communications loss may cause the instantaneous power consumed to exceed the average power budgeted for the radio of infusion pump assembly **100**, **100'**, **400**, **500**. The radio of remote control assembly **300** may not be forced to expand its window, so staying in search sync mode may be very power efficient. Acquisition mode may consume more power for remote control assembly **300**. Twenty minutes may be used as a compromise to balance power consumption on both the radio of remote control assembly **300** and the radio of infusion pump assembly **100**, **100'**, **400**, **500**.

The radio of remote control assembly **300** and the radio of infusion pump assembly **100**, **100'**, **400**, **500** may transition to slow heartbeat mode if they successfully exchange three of the last five heartbeats. Approximately every six seconds, a burst of five heartbeats may be attempted. If three of these are successful, the bandwidth may be assumed to be sufficient to transition to slow heartbeat mode. The radio of infusion pump assembly **100**, **100'**, **400**, **500** may be acquirable while in search sync mode with a latency of 6.1 seconds. This may imply that the infusion pump assembly **100**, **100'**, **400**, **500** may always be listening at least every ~6 seconds when in search sync mode.

Radio protocol performance statistics may be necessary to promote troubleshooting of the radio and to assess radio performance. The following radio performance statistics may be maintained by the radio protocol in a data structure:

NAME	SIZE	DESCRIPTION
TX Heartbeat Count	32 Bits	Total transmitted heartbeats
RX Heartbeat Count	32 bits	Total valid received heartbeats
CRC Errors	16 bits	Total packets received over the RF link which were dropped due to bad CRC. This may be a subset of RX Packets Nacked.
First Retry Count	32 bits	Total number of packets which were successfully acknowledged after 1 retry
Second Retry Count	32 bits	Total number of packets which were successfully acknowledged after 2 retries
Third Retry Count	32 bits	Total number of packets which were successfully acknowledged after 3 retries
Fourth Retry Count	32 bits	Total number of packets which were successfully acknowledged after 4 retries
Fifth Retry Count	16 bits	Total number of packets which were successfully acknowledged after 5 retries
Sixth Retry Count	16 bits	Total number of packets which were successfully acknowledged after 6 retries
Seventh Retry Count	16 bits	Total number of packets which were successfully acknowledged after 7 retries
Eighth Retry Count	16 bits	Total number of packets which were successfully acknowledged after 8 retries
Ninth Retry Count	16 bits	Total number of packets which were successfully acknowledged after 9 retries
Tenth Retry Count	16 bits	Total number of packets which were successfully acknowledged after 10 retries
Dropped Retry Count	16 bits	Total number of packets which were dropped after maximum retries attempts
Duplicate Packet Count	16 bits	Total number of received packets dropped due to duplicate packet
1 to 5 Missed Fast Mode Hops	16 bits	Count of 1 to 5 consecutive missed hops in Fast mode (i.e. not received)
6 to 16 Missed Fast Mode Hops	16 bits	Count of 6 to 16 consecutive missed hops in Fast mode.
17 to 33 Missed Fast Mode Hops	16 bits	Count of 17 to 33 consecutive missed hops in Fast mode
34+ Missed Fast Mode Hops	16 bits	Count of 34 or more consecutive missed hops in Fast mode
1 to 2 Missed Slow Mode Hops	16 bits	Count of 1 to 2 consecutive missed hops in Slow mode (i.e. not received)
3 to 5 Missed Slow Mode Hops	16 bits	Count of 3 to 5 consecutive missed hops in Slow mode
5 to 7 Missed Slow Mode Hops	16 bits	Count of 5 to 7 consecutive missed hops in Slow mode
8+ Missed Slow Mode Hops	16 bits	Count of 8 or more consecutive missed hops in Slow mode

-continued

NAME	SIZE	DESCRIPTION
Destination Radio Serial Number Mismatch	16 bits	Count of received packets in which the destination made it past the hardware filtering but does not match this radio's serial number. This may be not an error but indicates that the radio may be waking up and receiving (but not processing) packets intended for other radios
Total Walkaway Time (minutes)	16 bits	
Total Walkaway Events	16 bits	Together with total walkaway time provides an average walkaway time
Number of Pairing Attempts	16 bits	
Total Time in Acquisition Mode (Infusion pump assembly 100, 100', 400, 500 Only)	16 bits	
Total Acquisition Mode Attempts (Remote control assembly 300 Only)	16 bits	Successful Acquisition Count 16 bits Count of transitions from Connect or Acquisition Mode to Fast Heartbeat Mode
Requested Slow Heartbeat Mode Transitions	16 bits	
Automatic Slow Heartbeat Mode Transitions	16 bits	
Radio offline messages sent	16 bits	
Radio online messages sent	16 bits	

A #define DEBUG option (compiler option may be use to gather the following additional radio performance statistics per each channel (16 bit numbers):

- Number of missed hops
- CCA good count
- CCA bad count
- Average RSSI (accumulated for good RX packets only)
- Dropped from Frequency Hop List count
- Acquisition Mode count (found pair on this channel)

The debug option may be used to gather engineering only statistics. If processor performance, power, and memory allow, it may be desirable to keep this information at runtime. The radio statistics may be made available to the messaging system.

Link quality may be intended to be used on remote control assembly 300 to provide a bar indicator, similar to a cell phone, of the radio link quality. Link quality may be made available to both remote control assembly 300 and infusion pump assembly 100, 100', 400, 500. It may be anticipated that the link quality status will consist of a one byte indicator of the quality of the radio link.

The radio may change frequency for each heartbeat. An adaptive pseudo random frequency hopping algorithm may be used for sync mode and heartbeat attempts in search sync mode. It may be a goal to use sixty-four channels for frequency hopping. An algorithm may be developed to adaptively generate a channel list on remote control assembly 300 for frequency hopping. The radio of remote control assembly 300 may build, maintain, and distribute the master channel list. Prior channel statistics and historical performance information may be obtained from the radio of infusion pump assembly 100, 100', 400, 500 by the radio of remote control assembly 300 using the messaging system as needed to meet performance requirements. By building the channel list from the perspective of both units, the radio interference environment of both units may be considered. The radios may adaptively select hopping channels to meet the round trip message latency, while operating in a desirable RF environment.

Occlusions and/or leaks may occur anywhere along the fluid delivery path of infusion pump assembly 100. For example and referring to FIG. 121, occlusions/leaks may occur: in the fluid path between reservoir 118 and reservoir valve assembly 614; in the fluid path between reservoir valve assembly 614 and pump assembly 106; in the fluid path between pump assembly 106 and volume sensor valve assembly 612; in the fluid path between volume sensor valve assembly 612 and volume sensor chamber 620; in the fluid path between volume sensor chamber 620 and measurement valve assembly 610; and in the fluid path between measurement valve assembly 610 and the tip of disposable cannula 138. Infusion pump assembly 100 may be configured to execute one or more occlusion/leak detection algorithms that detect and locate such occlusions/leaks and enhance the safety/reliability of infusion pump assembly 100.

As discussed above, when administering the infusible fluid, infusion pump assembly 100 may first determine the volume of infusible fluid within volume sensor chamber 620 prior to the administration of the dose of infusible fluid and may subsequently determine the volume of infusible fluid within volume sensor chamber 620 after the administration of the dose of infusible fluid. By monitoring these values, the occurrence of occlusions/leaks may be detected.

Occlusion Type—Total: When a total occlusion is occurring, the difference between the initial measurement prior to the administration of the dose of infusible fluid and the final measurement after the administration of the dose of infusible fluid will be zero (or essentially zero), indicating a large residual quantity of infusible fluid within volume sensor chamber 620. Accordingly, no fluid may be leaving volume sensor chamber 620.

Specifically, if the tip of disposable cannula is occluded, the fluid path down stream of volume sensor chamber 620 will fill with fluid and eventually become pressurized to a level equivalent to the mechanical pressure exerted by spring diaphragm 628. Accordingly, upon measurement valve assembly 610 opening, zero (or essentially zero) fluid will be dispensed and, therefore, the value of the initial and final measurements (as made by volume sensor assembly 148) will essentially be equal.

Upon detecting the occurrence of such a condition, a total occlusion flag may be set and infusion pump assembly 100 may e.g., trigger an alarm, thus indicating that the user needs to seek alternative means for receiving their therapy.

Occlusion Type—Partial: When a partial occlusion is occurring, the difference between the initial measurement prior to the administration of the dose of infusible fluid and the final measurement after the administration of the dose of infusible fluid will indicate that less than a complete dose of infusible fluid was delivered. For example, assume that at the end of a particular pumping cycle, volume sensor assembly 148 indicated that 0.10 microliters of infusible fluid were present in volume sensor chamber 620. Further, assume that measurement valve assembly 610 is subsequently closed and pump assembly 106 is subsequently actuated, resulting in volume sensor chamber 620 being filled with the infusible fluid. Further assume that volume sensor assembly 148 determines that volume sensor chamber 620 is now filled with 1.00 microliters of infusible fluid (indicating a pumped volume of 0.90 microliters).

Accordingly, upon the opening of measurement valve assembly 610, the quantity of infusible fluid included within volume sensor chamber would be expected to drop to 0.10 microliters (or reasonably close thereto). However, in the event of a partial occlusion, due to a slower-than-normal flow rate from volume sensor chamber 620, the quantity of

infusible fluid within volume sensor chamber 620 may only be reduced to 0.40 microliters (indicating a delivered volume of 0.60 microliters). Accordingly, by monitoring the difference between the pumped volume (0.90 microliters) and the delivered volume (0.60 microliters), the residual volume may be defined and the occurrence of a partial occlusion may be detected.

Upon detecting the occurrence of such a condition, a partial occlusion flag may be set and infusion pump assembly 100 may e.g., trigger an alarm, thus indicating that the user needs to seek alternative means for receiving their therapy. However, as this is indicative of a partial occlusion (as opposed to a complete occlusion), the issuance of an alarm may be delayed, as the partial occlusion may clear itself.

Alternatively, infusion pump assembly 100 may: calculate a pump ontime to volume delivered ratio; track it through time; and track by using a fast moving and a slow moving exponential average of the pump ontime. The exponential average may be tracked, in a fashion similar to the leaky sum integrator. The infusion pump assembly 100 may filter signal and look for a fast change. The rate of fluid outflow and/or residual volume may be monitored. If the residual volume does not change, then there may be a total occlusion. If the residual volume changed, they may be a partial occlusion. Alternatively still, the residual values may be summed. If the number of valve actuations or the latch time is being varied, the fluid flow rate may be examined, even if you build up pressure in volume sensor assembly 148.

Total/Partial Empty Reservoir: When reservoir 118 is becoming empty, it will become more difficult to fill volume sensor chamber 620 to the desired level. Typically, pump assembly 106 is capable of pumping 1.0 microliters per millisecond. For example, assume that an “empty” condition for volume sensor chamber 620 is 0.10 microliters and a “full” condition for volume sensor chamber 620 is 1.00 microliters. However, as reservoir 118 begins to empty, it may become harder for pump assembly 106 to fill volume sensor chamber 620 to the “full” condition and may consistently miss the goal. Accordingly, during normal operations, it may take one second for pump assembly 106 to fill volume sensor chamber 620 to the “full” condition and, as reservoir 118 empties, it may take three seconds to fill volume sensor chamber 620 to the “full” condition. Eventually, if reservoir 118 completely empties, volume sensor chamber 620 may never be able to achieve a “full condition”. Accordingly, the inability of pump assembly 106 to fill volume sensor chamber 620 to a “full” condition may be indicative of reservoir 118 being empty. Alternatively, the occurrence of such a condition may be indicative of other situations (e.g., the failure of pump assembly 106 or an occlusion in the fluid path prior to volume sensor chamber 620). Infusion pump assembly 100 may determine the difference between the “full” condition and the amount actually pumped. These differences may be summed and the made up for once the reservoir condition is addressed.

Upon detecting the occurrence of such a condition, an empty flag may be set and infusion pump assembly 100 may e.g., trigger an alarm, thus indicating that the user needs to e.g., replace disposable housing assembly 114.

Additionally, as reservoir 118 empties, reservoir 118 will eventually result in a “vacuum” condition and the ability of pump assembly 106 to deliver fluid to volume sensor chamber 620 may be compromised. As discussed above, volume controller 1602 may include feed forward controller 1652 for setting an initial “guess” concerning “on-time” signal 1606, wherein this initial guess is based upon a pump

calibration curve. For example, in order for pump assembly 106 to deliver 0.010 units of infusible fluid, feed forward controller 1652 may define an initial “on-time” of e.g., one millisecond. However, as reservoir 118 begins to empty, due to compromised pumping conditions, it may take two milliseconds to deliver 0.010 units of infusible fluid. Further, as reservoir 118 approaches a fully empty condition, it may take ten milliseconds to deliver 0.010 units of infusible fluid. Accordingly, the occurrence of reservoir 118 approaching an empty condition may be detected by monitoring the level at which the actual operation of pump assembly 106 (e.g., two milliseconds to deliver 0.010 units of infusible fluid) differs from the anticipated operation of pump assembly 106 (e.g., one millisecond to deliver 0.010 units of infusible fluid).

Upon detecting the occurrence of such a condition, a reserve flag may be set and infusion pump assembly 100 may e.g., trigger an alarm, thus indicating that the user will need to e.g., replace disposable housing assembly 114 shortly.

Leak Detection: In the event of a leak (e.g., a leaky valve or a rupture/perforation) within the fluid path, the ability of the fluid path to retain fluid pressure may be compromised. Accordingly, in order to check for leaks within the fluid path, a bleed down test may be performed in which pump assembly 106 is used to pressurize volume sensor chamber 620. Volume sensor assembly 148 may then perform a first volume measurement (as described above) to determine the volume of infusible fluid within volume sensor chamber 620. Infusion pump assembly 100 may then wait a defined period of time to allow for bleed down in the event of a leak. For example, after a sixty second bleed down period, volume sensor assembly 148 may perform a second volume measurement (as described above) to determine the volume of infusible fluid within volume sensor chamber 620. If there are no leaks, the two volume measurements should be essentially the same. However, in the event of a leak, the second measurement may be less than the first measurement. Additionally, depending on the severity of the leak, pump assembly 106 may be incapable of filling volume sensor chamber 620. Typically, a leak check may be performed as part of a delivery of infusible fluid.

In the event that the difference between the first volume measurement and the second volume measurement exceeds an acceptable threshold, a leak flag may be set and infusion pump assembly 100 may e.g., trigger an alarm, thus indicating that the user needs to seek alternative means for receiving their therapy.

As discussed above, infusion pump assembly 100 may include supervisor processor 1800, command processor 1802, and radio processor 1818. Unfortunately, once assembled, access to electrical control assembly 110 within infusion pump assembly 100 very limited. Accordingly, the only means to access electrical control assembly 110 (e.g., for upgrading flash memories) may be through the communication channel established between infusion pump assembly 100, 100', 400, 500 and remote control assembly 300, or via electrical contacts 834 used by battery charger 1200.

Electrical contacts 834 may be directly coupled to radio processor 1818 and may be configured to provide I2C communication capability for erasing/programming any flash memory (not shown) included within radio processor 1818. The process of loading a program into radio processor 1818 may provide a means for erasing/programming of the flash memories in both the supervisor processor 1800 and command processor 1802.

When programming supervisor processor 1800 or command processor 1802, the program (i.e., data) to be loaded

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into flash memory accessible by supervisor processor **1800** or command processor **1802** may be provided in a plurality of data blocks. This is because the radio processor **1818** may not have enough memory to hold the entire flash image of the software as one block.

Referring also to FIG. **122**, there is shown one illustrative example of the manner in which the various systems within infusion pump assembly **100, 100', 400, 500** may be interconnected. For example, battery charger **1200** may be coupled to computing device **2100** (e.g., a personal computer) via bus translator **2102**, which converts e.g., RS232 formatted data to e.g., I2C formatted data. Bus translator **2102** may execute a pass-through program that effectuates the above-described translation. Battery charger **1200** may be coupled to radio processor **181** via electrical contacts **834** (described above). Radio processor **1818** may then be coupled to supervisor processor **1800** and command processor **1802** via e.g., an RS232 bus. Radio processor **1818** may execute an update program that allows radio processor **1818** to control/orchestrate the updating of the flash memories accessible by supervisor processor **1800** and command processor **1802**. Accordingly, through the use of the above-described coupling, software updates obtained by computing device **2100** may be uploaded to flash memory (not shown) accessible by supervisor processor **1800** and command processor **1802**. The above-described software updates may be command line program that may be automatically invoked by a script process.

As discussed above, infusion pump assembly **100, 100' 400, 500** may be configured to deliver an infusible fluid to a user. Further and as discussed above, infusion pump assembly **100, 100' 400, 500** may deliver the infusible fluid via sequential, multi-part, infusion events (that may include a plurality of discrete infusion events) and/or one-time infusion events. However, in some embodiments, infusion pump assembly **100, 100' 400, 500** may deliver stacking bolus infusion events. For example, a user may request the delivery of a bolus, e.g., 6 units. While the 6 units are in the process of being delivered to the user, the user may request a second bolus, e.g., 3 units. In some embodiments of infusion pump assembly **100, 100' 400, 500** may deliver the second bolus at the completion of the first bolus.

Examples of other such sequential, multi-part, infusion events may include but are not limited to a basal infusion event and an extended-bolus infusion event. As is known in the art, a basal infusion event refers to the repeated injection of small (e.g. 0.05 unit) quantities of infusible fluid at a predefined interval (e.g. every three minutes) that may be repeated until stopped, e.g., by a user or by the system. Further, the basal infusion rates may be pre-programmed and may include specified rates for pre-programmed time-frames, e.g., a rate of 0.50 units per hour from 6:00 am-3:00 pm; a rate of 0.40 units per hour from 3:00 pm-10:00 pm; and a rate of 0.35 units per hour from 10:00 pm-6:00 am. However, the basal rate may be 0.025 units per hour, and may not change according to pre-programmed time-frames. The basal rates may be repeated regularly/daily until otherwise changed.

Further and as is known in the art, an extended-bolus infusion event may refer to the repeated injection of small (e.g. 0.05 unit) quantities of infusible fluid at a predefined interval (e.g. every three minutes) that is repeated for a defined number of intervals (e.g., three intervals) or for a defined period of time (e.g., nine minutes). An extended-bolus infusion event may occur simultaneously with a basal infusion event.

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If multiple infusion events conflict with each other, infusion pump assembly **100, 100' 400, 500** may prioritize the infusion event in the follow manner.

Referring also to FIG. **123**, assume for illustrative purposes only that the user configures infusion pump assembly **100, 100' 400, 500** to administer a basal dose (e.g. 0.05 units) of infusible fluid every three minutes. The user may utilize remote control assembly **300** to define a basal infusion event for the infusible fluid (e.g., 1.00 units per hour). Infusion pump assembly **100, 100' 400, 500** may then determine an infusion schedule based upon the basal infusion event defined. Once determined, infusion pump assembly **100, 100' 400, 500** may administer the sequential, multi-part, infusion event (e.g., 0.05 units of infusible fluid every three minutes). Accordingly, while administering the sequential, multi-part, infusion event, infusion pump assembly **100, 100' 400, 500** may infuse a first 0.05 unit dose **2200** of the infusible fluid at t=0:00 (i.e., a first discrete infusion event), may infuse a second 0.05 unit dose **2202** of the infusible fluid at t=3:00 (i.e., a second discrete infusion event); may infuse a third 0.05 unit dose **2204** of the infusible fluid at t=6:00 (i.e., a third discrete infusion event); may infuse a fourth 0.05 unit dose **2206** of the infusible fluid at t=9:00 (i.e., a fourth discrete infusion event); and may infuse a fifth 0.05 unit dose **2208** of the infusible fluid at t=12:00 (i.e., a fifth discrete infusion event). As discussed above, this pattern of infusing 0.05 unit doses of the infusible fluid every three minutes may be repeated until stopped, e.g., by a user or by the system, in this example, as this is an illustrative example of a basal infusion event.

Further, assume for illustrative purposes that the infusible fluid is insulin and sometime after the first 0.05 unit dose **2200** of infusible fluid is administered (but before the second 0.05 unit dose **2202** of infusible fluid is administered), the user checks their blood glucose level and realizes that their blood glucose level is running a little higher than normal. Accordingly, the user may define an extended bolus infusion event via remote control assembly **300**. An extended bolus infusion event may refer to the continuous infusion of a defined quantity of infusible fluid over a finite period of time. However, as such an infusion methodology is impractical/undesirable for an infusion pump assembly, when administered by such an infusion pump assembly, an extended bolus infusion event may refer to the infusion of additional small doses of infusible fluid over a finite period of time.

Accordingly, the user may utilize remote control assembly **300** to define an extended bolus infusion event for the infusible fluid (e.g., 0.20 units over the next six minutes), which may be confirmed in a manner discussed above. While, in this example, the extended bolus infusion event is described as 0.20 units over the next six minutes, this is for illustrative purposes only and is not intended to be a limitation of this disclosure, as either or both of the unit quantity and total time interval may be adjusted upward or downward. Once defined and/or confirmed, infusion pump assembly **100, 100' 400, 500** may determine an infusion schedule based upon the extended bolus infusion event defined; and may administer the infusible fluid. For example, infusion pump assembly **100, 100' 400, 500** may deliver 0.10 units of infusible fluid every three minutes for the next two interval cycles (or six minutes), resulting in the delivery of the extended bolus dose of infusible fluid defined by the user (i.e., 0.20 units over the next six minutes).

Accordingly, while administering the second, sequential, multi-part, infusion event, infusion pump assembly **100, 100' 400, 500** may infuse a first 0.10 unit dose **2210** of the

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infusible fluid at $t=3:00$ (e.g., after administering the second 0.05 unit dose **2202** of infusible fluid). Infusion pump assembly **100, 100' 400, 500** may also infuse a second 0.10 unit dose **2212** of the infusible fluid at $t=6:00$ (e.g., after administering the third 0.05 unit dose **2204** of infusible fluid).

Assume for illustrative purposes only that after the user programs infusion pump assembly **100, 100' 400, 500** via remote control assembly **300** to administer the first sequential, multi-part, infusion event (i.e., 0.05 units infused every three minute interval repeated continuously) and administer the second sequential, multi-part, infusion event (i.e., 0.10 units infused every three minute interval for two intervals), the user decides to eat a very large meal. Predicting that their blood glucose level might increase considerably, the user may program infusion pump assembly **100, 100' 400, 500** (via remote control assembly **300**) to administer a one-time infusion event. An example of such a one-time infusion event may include but is not limited to a normal bolus infusion event. As is known in the art, a normal bolus infusion event refers to a one-time infusion of the infusible fluid.

For illustrative purposes only, assume that the user wishes to have infusion pump assembly **100, 100' 400, 500** administer a bolus dose of thirty-six units of the infusible fluid. Infusion pump assembly **100, 100' 400, 500** may monitor the various infusion events being administered to determine whether a one-time infusion event is available to be administered. If a one-time infusion event is available for administration, infusion pump assembly **100, 100' 400, 500** may delay the administration of at least a portion of the sequential, multi-part, infusion event.

Continuing with the above-stated example, once the user completes the programming of infusion pump assembly **100, 100' 400, 500** to deliver one-time infusion event **2214** (i.e., the thirty-six unit bolus dose of the infusible fluid), upon infusion pump assembly **100, 100' 400, 500** determining that the one-time infusion event is available for administration, infusion pump assembly **100, 100' 400, 500** may delay the administration of each sequential, multi-part infusion event and administer the available one-time infusion event.

Specifically and as discussed above, prior to the user programming infusion pump assembly **100, 100' 400, 500** to deliver one-time infusion event **2214**, infusion pump assembly **100, 100' 400, 500** was administering a first sequential, multi-part, infusion event (i.e., 0.05 units infused every three minute interval repeated continuously) and administering a second sequential, multi-part, infusion event (i.e., 0.10 units infused every three minute interval for two intervals).

For illustrative purposes only, the first sequential, multi-part, infusion event may be represented within FIG. 123 as 0.05 unit dose **2200** @ $t=0:00$, 0.05 unit dose **2202** @ $t=3:00$, 0.05 unit dose **2204** @ $t=6:00$, 0.05 unit dose **2206** @ $t=9:00$, and 0.05 unit dose **2208** @ $t=12:00$. As the first sequential, multi-part, infusion event as described above is a basal infusion event, infusion pump assembly **100, 100' 400, 500** may continue to infuse 0.05 unit doses of the infusible fluid at three minute intervals indefinitely (i.e., until the procedure is cancelled by the user).

Further and for illustrative purposes only, the second sequential, multi-part, infusion event may be represented within FIG. 123 as 0.10 unit dose **2210** @ $t=3:00$ and 0.10 unit dose **2212** @ $t=6:00$. As the second sequential, multi-part, infusion event is described above as an extended bolus infusion event, infusion pump assembly **100, 100' 400, 500** may continue to infuse 0.10 unit doses of the infusible fluid

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at three minute intervals for exactly two intervals (i.e., the number of intervals defined by the user).

Continuing with the above-stated example, upon infusion pump assembly **100, 100' 400, 500** determining that the thirty-six unit normal bolus dose of the infusible fluid (i.e., one-time infusion event **2214**) is available for administration, infusion pump assembly **100, 100' 400, 500** may delay the administration of each sequential, multi-part infusion event and may start administering one-time infusion event **2214** that is available for administration.

Accordingly and for illustrative purposes only, assume that upon completion of the programming of infusion pump assembly **100, 100' 400, 500** to deliver the thirty-six unit normal bolus dose of the infusible fluid (i.e., the one-time infusion event), infusion pump assembly **100, 100' 400, 500** begins administering one-time infusion event **2214**. Being that one-time infusion event **2214** is comparatively large, it may take longer than three minutes (i.e., the time interval between individual infused doses of the sequential, multi-part, infusion events) and one or more of the individual infused doses of the sequential, multi-part, infusion events may need to be delayed.

Specifically, assume that it will take infusion pump assembly **100, 100' 400, 500** greater than six minutes to infuse thirty-six units of the infusible fluid. Accordingly, infusion pump assembly **100, 100' 400, 500** may delay 0.05 unit dose **2202** (i.e., scheduled to be infused @ $t=3:00$), 0.05 unit dose **2204** (i.e., scheduled to be infused @ $t=6:00$), and 0.05 unit dose **2206** (i.e., scheduled to be infused @ $t=9:00$) until after one-time infusion event **2214** (i.e., the thirty-six unit normal bolus dose of the infusible fluid) is completely administered. Further, infusion pump assembly **100, 100' 400, 500** may delay 0.10 unit dose **2210** (i.e., scheduled to be infused @ $t=3:00$ and 0.10 unit dose **2212** (i.e., scheduled to be infused @ $t=6:00$) until after one-time infusion event **2214**.

Once administration of one-time infusion event **2214** is completed by infusion pump assembly **100, 100' 400, 500**, any discrete infusion events included within the sequential, multi-part, infusion event that were delayed may be administered by infusion pump assembly **100, 100' 400, 500**. Accordingly, once one-time infusion event **2214** (i.e., the thirty-six unit normal bolus dose of the infusible fluid) is completely administered, infusion pump assembly **100, 100' 400, 500** may administer 0.05 unit dose **2202**, 0.05 unit dose **2204**, 0.05 unit dose **2206**, 0.10 unit dose **2210**, and 0.10 unit dose **2212**.

While infusion pump assembly **100, 100' 400, 500** is shown to administer 0.05 unit dose **2202**, then 0.10 unit dose **2210**, then 0.05 unit dose **2204**, then 0.10 unit dose **2212**, and then 0.05 unit dose **2206**, this is for illustrative purposes only and is not intended to be a limitation of this disclosure, as other configurations are possible and are considered to be within the scope of this disclosure. For example, upon infusion pump assembly **100, 100' 400, 500** completing the administration of one-time infusion event **2214** (i.e., the thirty-six unit normal bolus dose of the infusible fluid), infusion pump assembly **100, 100' 400, 500** may administer all of the delayed discrete infusion events associated with the first sequential, multi-part infusion event (i.e., namely 0.05 unit dose **2202**, 0.05 unit dose **2204**, and 0.05 unit dose **2206**). Infusion pump assembly **100, 100' 400, 500** may then administer all of the delayed discrete infusion events associated with the second sequential, multi-part infusion event (i.e., 0.10 unit dose **2210**, and 0.10 unit dose **2212**).

While one-time infusion event **2214** (i.e., the thirty-six unit normal bolus dose of the infusible fluid) is shown as

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being infused beginning at t=3:00, this is for illustrative purposes only and is not intended to be a limitation of this disclosure. Specifically, infusion pump assembly **100**, **100'**, **400**, **500** may not need to begin infusing one-time infusion event **2214** at one of the three-minute intervals (e.g., t=0:00, t=3:00, t=6:00, t=9:00, or t=12:00) and may begin administering one-time infusion event **2214** at any time.

While each discrete infusion event (e.g., 0.05 unit dose **2202**, 0.05 unit dose **2204**, 0.05 unit dose **2206**, 0.10 unit dose **2210**, and 0.10 unit dose **2212**) and one-time infusion event **2214** are shown as being a single event, this is for illustrative purposes only and is not intended to be a limitation of this disclosure. Specifically, at least one of the plurality of discrete infusion events e.g., 0.05 unit dose **2202**, 0.05 unit dose **2204**, 0.05 unit dose **2206**, 0.10 unit dose **2210**, and 0.10 unit dose **2212**) may include a plurality of discrete infusion sub-events. Further, one-time infusion event **2214** may include a plurality of one-time infusion sub-events.

Referring also to FIG. **124** and for illustrative purposes, 0.05 unit dose **2202** is shown to include ten discrete infusion sub-events (e.g., infusion sub-events **2216**₁₋₁₀), wherein a 0.005 unit dose of the infusible fluid is infused during each of the ten discrete infusion sub-events. Additionally, 0.10 unit dose **2210** is shown to include ten discrete infusion sub-events (e.g., infusion sub-events **2218**₁₋₁₀), wherein a 0.01 unit dose of the infusible fluid is delivered during each of the ten discrete infusion sub-events. Further, one-time infusion event **2214** may include e.g., three-hundred-sixty one-time infusion sub-events (not shown), wherein a 0.1 unit dose of the infusible fluid is delivered during each of the three-hundred-sixty one-time infusion sub-events. The number of sub-events defined above and the quantity of the infusible fluid delivered during each sub-event is solely for illustrative purposes only and is not intended to be a limitation of this disclosure, as the number of sub-events and/or the quantity of the infusible fluid delivered during each sub-event may be increased or decreased depending upon e.g., the design criteria of infusion pump assembly **100**, **100'**, **400**, **500**.

Before, after, or in between the above-described infusion sub-events, infusion pump assembly **100**, **100'**, **400**, **500** may confirm the proper operation of infusion pump assembly **100**, **100'**, **400**, **500** through the use of any of the above-described safety features (e.g., occlusion detection methodologies and/or failure detection methodologies).

In the exemplary embodiments, the infusion pump assembly may be wirelessly controlled by a remote control device. In the exemplary embodiments, a split ring resonator antenna may be used for wireless communication between the infusion pump assembly and the remote control device (or other remote device). The term “wirelessly controlled” refers to any device that may receive input, instructions, data, or other, wirelessly. Further, a wirelessly controlled insulin pump refers to any insulin pump that may wirelessly transmit and/or receive data from another device. Thus, for example, an insulin pump may both receive instructions via direct input by a user and may receive instructions wirelessly from a remote controller.

Referring to FIG. **127** and FIG. **131**, an exemplary embodiment of a split ring resonator antenna adapted for use in a wirelessly controlled medical device, and is used in the exemplary embodiment of the infusion pump assembly, includes at least one split ring resonator antenna (hereinafter “SRR antenna”) **2508**, a wearable electric circuit, such as a wirelessly controlled medical infusion apparatus (hereinaf-

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ter “infusion apparatus”) **2514**, capable of powering the antenna, and a control unit **2522**.

In various embodiments, a SRR antenna **2508** may reside on the surface of a non-conducting substrate base **2500**, allowing a metallic layer (or layers) to resonate at a predetermined frequency. The substrate base **2500** may be composed of standard printed circuit board material such as Flame Retardant 2 (FR-2), FR-3, FR-4, FR-5, FR-6, G-10, CEM-1, CEM-2, CEM-3, CEM-4, CEM-5, Polyimide, Teflon, ceramics, or flexible Mylar. The metallic resonating bodies comprising a SRR antenna **2508** may be made of two rectangular metallic layers **2502**, **2504**, made of, for example, platinum, iridium, copper, nickel, stainless steel, silver or other conducting materials. In other various embodiments, a SRR antenna **2508** may contain only one metallic resonating body.

In the exemplary embodiment, a gold-plated copper outer layer **2502**, surrounds, without physically contacting, a gold-plated copper inner ring **2504**. That is, the inner ring **2504** resides in the cavity **2510** (or aperture) formed by the outer layer **2502**. The inner ring **2504** may contain a gap, or split **2506**, along its surface completely severing the material to form an incomplete ring shape. Both metallic resonating bodies **2502**, **2504** may reside on the same planar surface of the substrate base **2500**. In such a configuration, the outer layer **2502** may be driven via a transmission line **2512** coupled to the outer layer **2502**, for example. Additionally, in various other embodiments, a transmission line **2512** may be coupled to the inner ring **2504**.

Antenna design software, such as AWR Microwave Office, capable of simulating electromagnetic geometries, such as, antenna performance, may significantly decrease the time required to produce satisfactory dimensions compared to physically fabricating and testing antennas. Accordingly, with aid of such software, the SRR antenna **2508** may be designed such that the geometric dimensions of the resonant bodies **2502**, **2504** facilitate an operational frequency of 2.4 GHz. FIG. **132** depicts the exemplary dimensions of the inner ring **2504** and outer layer **2502**, and the positioning of the cavity **2510** in which the inner ring **2504** resides. The distance in between the outer layer **2502** and the inner ring **2504** is a constant 0.005 inches along the perimeter of the cavity **2510**. However, in other embodiments, the distance between the outer layer and the inner ring may vary and in some embodiments, the operational frequency may vary.

In various embodiments, a SRR antenna **2508** may have dimensions such that it could be categorized as electrically small, that is, the greatest dimension of the antenna being far less than one wavelength at operational frequency.

In various other embodiments, a SRR antenna **2508** may be composed of one or more alternatively-shaped metallic outer layers, such as circular, pentagonal, octagonal, or hexagonal, surrounding one or more metallic inner layers of similar shape. Further, in various other embodiments, one or more metallic layers of a SRR antenna **2508** may contain gaps in the material, forming incomplete shapes.

Referring to FIG. **130**, a SRR antenna **2508** having the exemplary geometry exhibits acceptable return loss and frequency values when placed in contact with human skin. As shown in FIG. **130**, focusing on the band of interest denoted by markers **1** and **2** on the graph, return loss prior to contact with human skin is near -15 dB while monitoring a frequency band centered around 2.44 GHz. Return loss during contact with human skin, as shown in FIG. **130A**, remains a suitable value near -25 dB at the same frequency, yielding approximately 97% transmission power.

These results are favorable especially as compared with a non-split ring resonator antenna type, such as the Inverted-F. Return loss of an Inverted-F antenna may exhibit a difference when the antenna contacts human skin, resulting in a low percentage of power transmitted outward from the antenna. By way of example, as shown in FIG. 133, and again focusing on the band of interest denoted by markers 1 and 2 on the graph, return loss of an Inverted-F antenna prior to contact with human skin is near -25 dB at a frequency centered around 2.44 GHz. Return loss during contact with human skin is nearly -2 dB at the same frequency, yielding approximately 37% power transmission.

Integration with a Wireless Medical Device

In the exemplary embodiment, referring to FIG. 132 and FIG. 128, one application of a SRR antenna 2508 may be integration into a wearable infusion apparatus 2514 capable of delivering fluid medication to a user/patient 2524. In such an application, the safety of the user/patient is dependent on fluid operation between these electrical components, thus reliable wireless transmission to and from a control unit 2522 is of great importance.

An infusion apparatus 2514 may be worn directly on the human body. By way of example, such a device may be attached on or above the hip joint in direct contact with human skin, placing the SRR antenna 2508 at risk of unintended dielectric loading causing a frequency shift in electrical operation. However, in such an application, electrical characteristics of the SRR antenna 2508 which allow it to be less sensitive to nearby parasitic objects are beneficial in reducing or eliminating degradation to the performance. A controlling component, such as a control unit 2522 (generally shown in FIG. 131), may be paired with an infusion apparatus 2514, and may be designed to transmit and receive wireless signals to and from the infusion apparatus 2514 at a predetermined frequency, such as 2.4 GHz. In the exemplary embodiment, the control unit 2522 serves as the main user interface through which a patient or third party may manage insulin delivery. In other embodiments, infusion apparatus 2514 may utilize a SRR antenna 2508 to communicate with one or more control units 2522.

In various embodiments, a number of different wireless communication protocols may be used in conjunction with the SRR antenna 2508, as the protocol and data types to be transferred are independent of the electrical characteristics of the antenna. However, in the exemplary embodiment, a bi-directional master/slave means of communication organizes the data transfer through the SRR antenna 2508. The control unit 2522 may act as the master by periodically polling the infusion apparatus 2514, or slave, for information. In the exemplary embodiment, only when the slave is polled, the slave may send signals to the control unit 2522 only when the slave is polled. However, in other embodiments, the slave may send signals before being polled. Signals sent by way of this system may include, but are not limited to, control, alarm, status, patient treatment profile, treatment logs, channel selection and negotiation, handshaking, encryption, and check-sum. In some embodiments, transmission through the SRR antenna 2508 may also be halted during certain infusion operations as an added precaution against electrical disruption of administration of insulin to the patient.

In the exemplary embodiment, the SRR antenna 2508 may be coupled to electrical source circuitry via one or more pins 2516 on a transmission line 2512. In various other embodiments a transmission line may comprise a wire, pairs of wire, or other controlled impedance methods providing a channel by which the SRR antenna 2508 is able to resonate

at a certain frequency. The transmission line 2512 may reside on the surface of the substrate base 2500 and may be composed of the same material as the SRR antenna 2508, such as gold-plated copper. Additionally, a ground plane may be attached to the surface of the substrate base opposite the transmission line 2512.

The electrical circuitry coupled to the SRR antenna 2508 may apply an RF signal to the end of the transmission line 2512 nearest the circuitry, creating an electromagnetic field throughout, and propagating from, the SRR antenna 2508. The electrical circuitry coupled to the SRR antenna 2508 facilitates resonance at a predetermined frequency, such as 2.4 GHz. Preferably, transmission line 2512 and SRR antenna 2508 both have impedances of 50 Ohms to simplify circuit simulation and characterization. However, in other various embodiments, the transmission line and split ring resonator antenna may have other impedance values, or a different resonating frequency.

Referring to FIG. 129, a signal processing component(s) 2518, such as, a filter, amplifier, or switch, may be integrated into the transmission line 2512, or at some point between the signal source connection pins 2516 and the SRR antenna 2508. In the exemplary embodiment, the signal processing component 2518 is a band-pass filter to facilitate desired signal processing, such as, allowing only the exemplary frequency to be transmitted to the antenna, and rejecting frequencies outside that range. In the exemplary embodiment, a Combline band-pass filter 2518 may be included in the transmission line 2512 between the antenna and the signal source. However in other embodiments, any other signal processing device, for example, but not limited to, filters, amplifiers, or any other signal processing devices known in the art.

In various embodiments, a SRR antenna 2508 may be composed of metallic bodies capable of resonating on a flexible or rigid substrate. As shown in FIG. 128 and FIG. 3, the exemplary embodiment incorporates a curved SRR antenna on a flexible Polyimide substrate 2520. Polyimide may be the exemplary material because it tends to be more flexible than alternative substrates. This configuration may allow for simplified integration into circular-shaped devices (such as a wirelessly controlled medical infusion apparatus 2514), devices with irregular-shaped external housing, or devices in which saving space is paramount.

In various embodiments, both control unit 2522 and base unit 2514 may incorporate a split SRR antenna 2508. This configuration may prove beneficial where the control unit is meant to be handheld, in close proximity to human skin, or is likely to be in close proximity to a varying number of materials with varying dielectric constants.

In various other embodiments, a SRR antenna 2508 may be integrated into a human or animal limb replacement. As prosthetic limbs are becoming more sophisticated the electrical systems developed to control and simulate muscle movements require much more wiring and data transfer among subsystems. Wireless data transfer within a prosthetic limb may reduce weight through reduced physical wiring, conserve space, and allow greater freedom of movement. However, common antennas in such a system may be susceptible to dielectric loading. Similar to the previously mentioned benefits of integrating a SRR antenna 2508 into a wirelessly controlled medical infusion apparatus, a prosthetic limb, such as a robotic arm, may also come into contact with human skin or other dielectric materials and benefit from the reduction of electrical disturbances associated with such an antenna. In other various embodiments, the SRR antenna 2508 may be integrated into any device

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comprised of the electrical components capable of powering and transmitting/receiving data to an antenna and susceptible to electrical disturbances associated with proximity to dielectric materials.

In various embodiments, a SRR antenna **2508** may be integrated into a configuration of medical components in which one or more implantable medical devices, operating within the human body, communicate wirelessly to a handheld, body-mounted, or remote control unit. In certain embodiments, both body-mounted and in-body wireless devices may utilize a SRR antenna **2508** for wireless communication. Additionally, one or more of the components utilizing a SRR antenna **2508** may be completely surrounded by human skin, tissue or other dielectric material. By way of example, such a configuration may be used in conjunction with a heart monitoring/control system where stability and consistency of wireless data transmission are of fundamental concern.

In various other embodiments, a SRR antenna **2508** may be integrated into the embodiments of the infusion pump assembly. configuration of medical components in which one or more electrical sensors positioned on, or attached to, the human body wirelessly communicate to a remote transceiving unit. By way of example, a plurality of electrodes positioned on the body may be coupled to a wireless unit employing a SRR antenna **2508** for wireless transmission to a remotely located electrocardiogram machine. By way of further example, a wireless temperature sensor in contact with human skin may employ SRR antenna **2508** for wireless communication to a controller unit for temperature regulation of the room in which the sensor resides.

A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made. Accordingly, other embodiments are within the scope of the following claims.

What is claimed is:

1. A wearable infusion pump assembly comprising:

a reusable housing assembly;

a disposable housing assembly including a reservoir for receiving an infusible fluid;

a releasable engagement assembly configured to allow the reusable housing assembly to releasably engage the disposable housing assembly;

a bolus switch assembly configured to effectuate a bolus infusion event;

at least one processor; and

a computer readable medium coupled to the at least one processor, the computer readable medium including a plurality of instructions stored thereon which, when executed by the at least one processor, cause the at least one processor to perform operations comprising:

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receiving an initiation signal from the bolus switch assembly indicative of a bolus infusion event;

receiving a dose signal from the bolus switch assembly indicative of at least a portion of a bolus quantity of the infusible fluid;

rendering an audible quantity signal in response to the dose signal; and

receiving an approval signal from the bolus switch assembly indicative of a concurrence with the audible quantity signal.

2. The wearable infusion pump assembly of claim 1 wherein the computer readable medium further includes instructions for:

administering the bolus quantity of the infusible fluid.

3. The wearable infusion pump assembly of claim 1 wherein the bolus switch assembly is configured to check a battery life of a battery of the reusable housing assembly.

4. The wearable infusion pump assembly of claim 3 wherein the bolus switch assembly is configured to check the battery life of the battery of the reusable housing assembly when the reusable housing assembly is disconnected from the disposable housing assembly and the bolus switch assembly is depressed for a defined period of time.

5. The wearable infusion pump assembly of claim 4 wherein the wearable infusion pump assembly renders a battery pass tone when the check of the battery life of the battery is determined to have a charge above a desired threshold.

6. The wearable infusion pump assembly of claim 4 wherein the wearable infusion pump wearable pump assembly renders a battery fail tone when the check of the battery life of the battery is determined to have a charge below a desired threshold.

7. The wearable infusion pump assembly of claim 1 wherein the bolus switch assembly is configured to cause pairing of the reusable housing assembly with a remote control assembly.

8. The wearable infusion pump assembly of claim 7 wherein the wearable infusion pump assembly is configured such that the reusable housing assembly is required to be uncoupled from the disposable housing assembly in order for the bolus switch assembly to cause the pairing of the reusable housing assembly with the remote control assembly.

9. The wearable infusion pump assembly of claim 1 wherein the bolus switch assembly is configured to cause aborting of administration of a bolus dose of the infusible fluid.

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